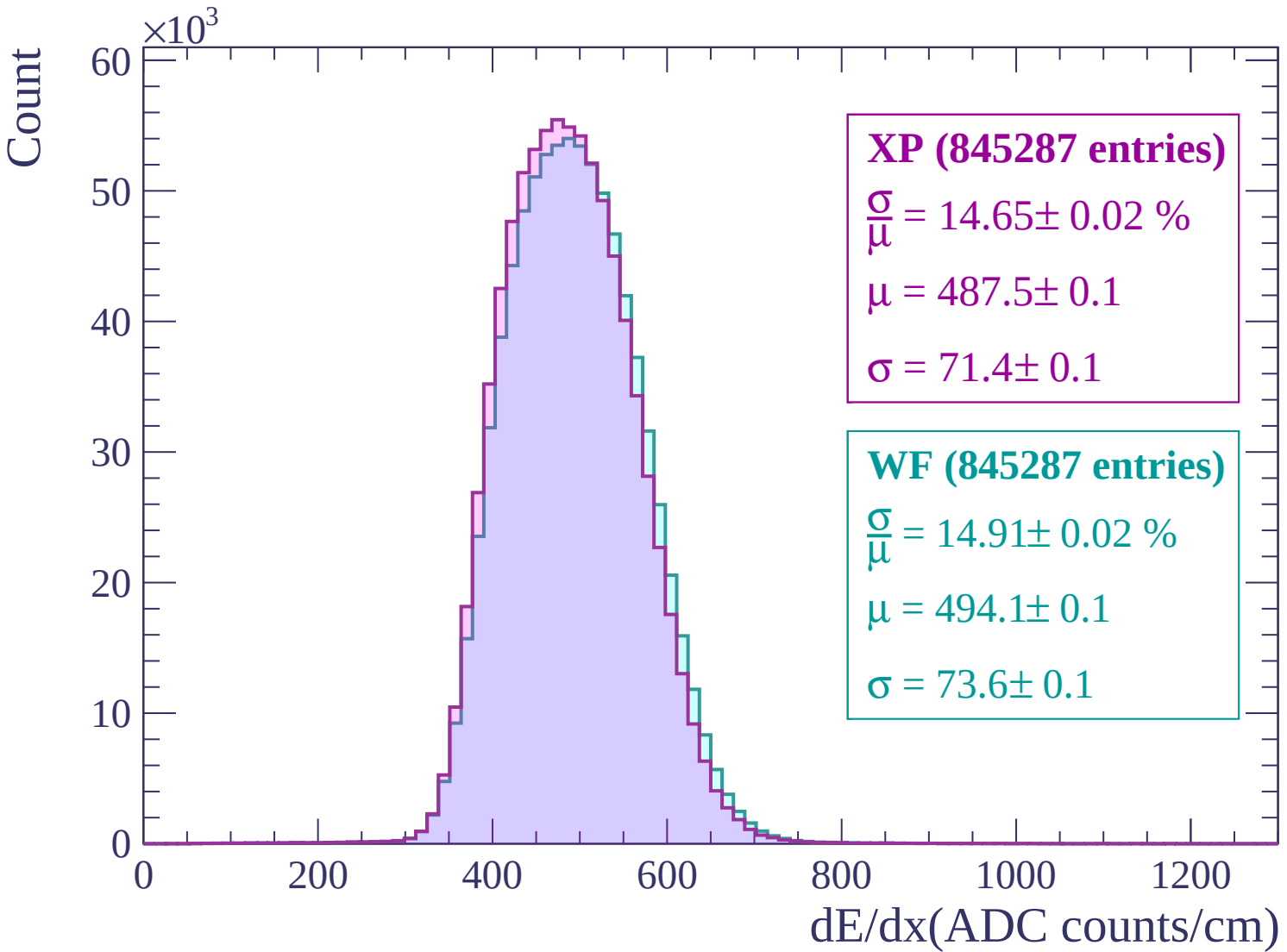
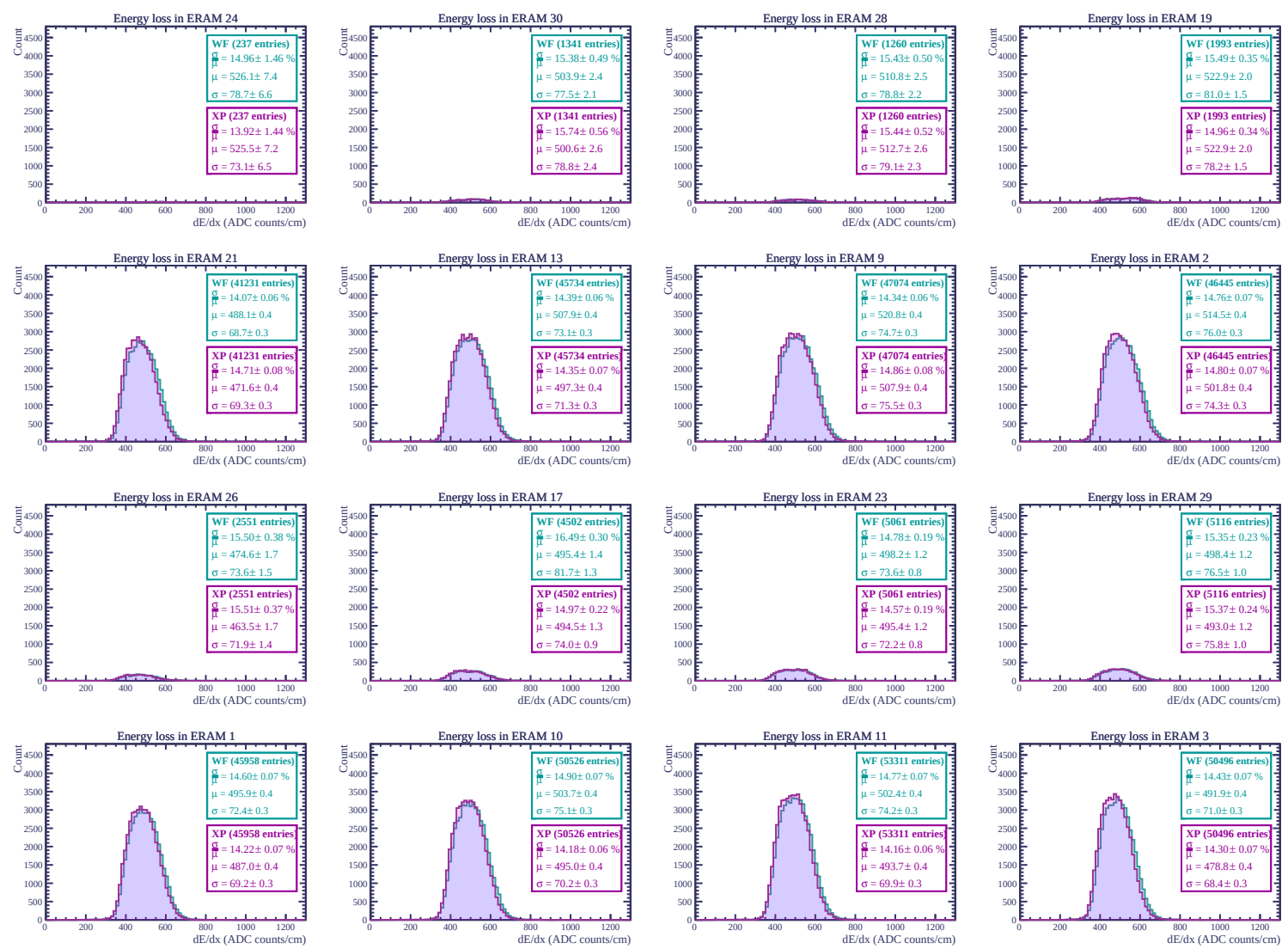
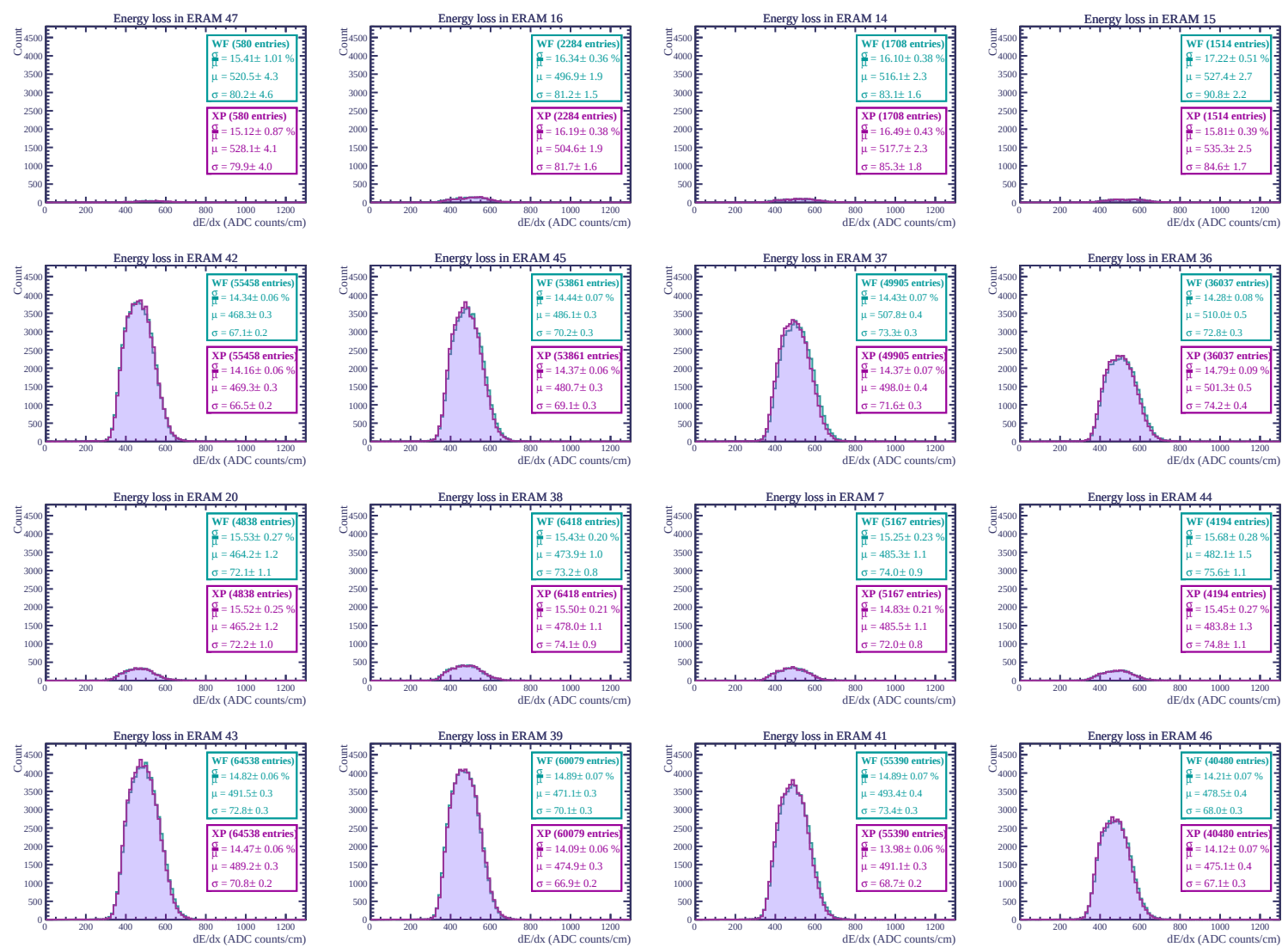
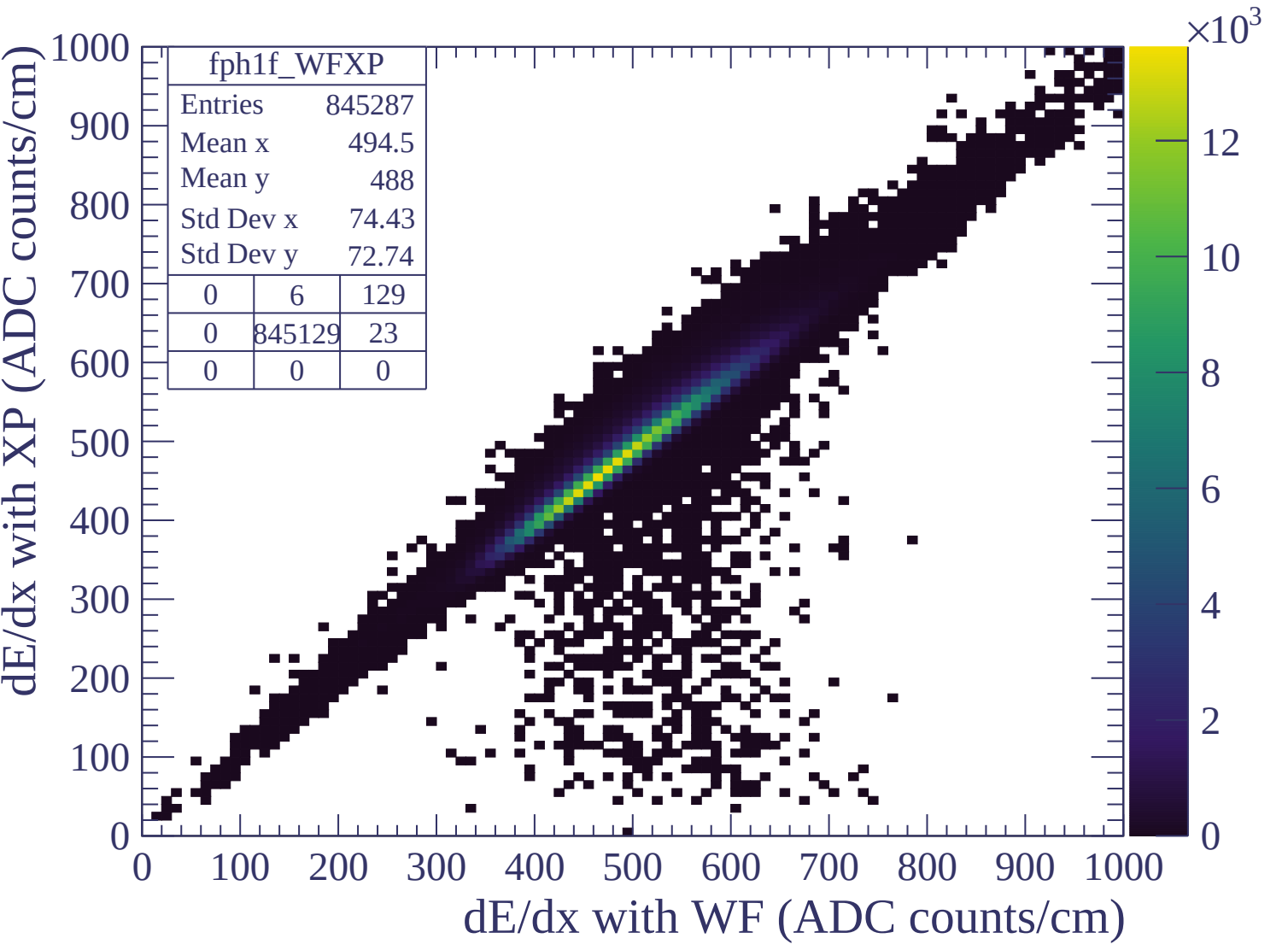
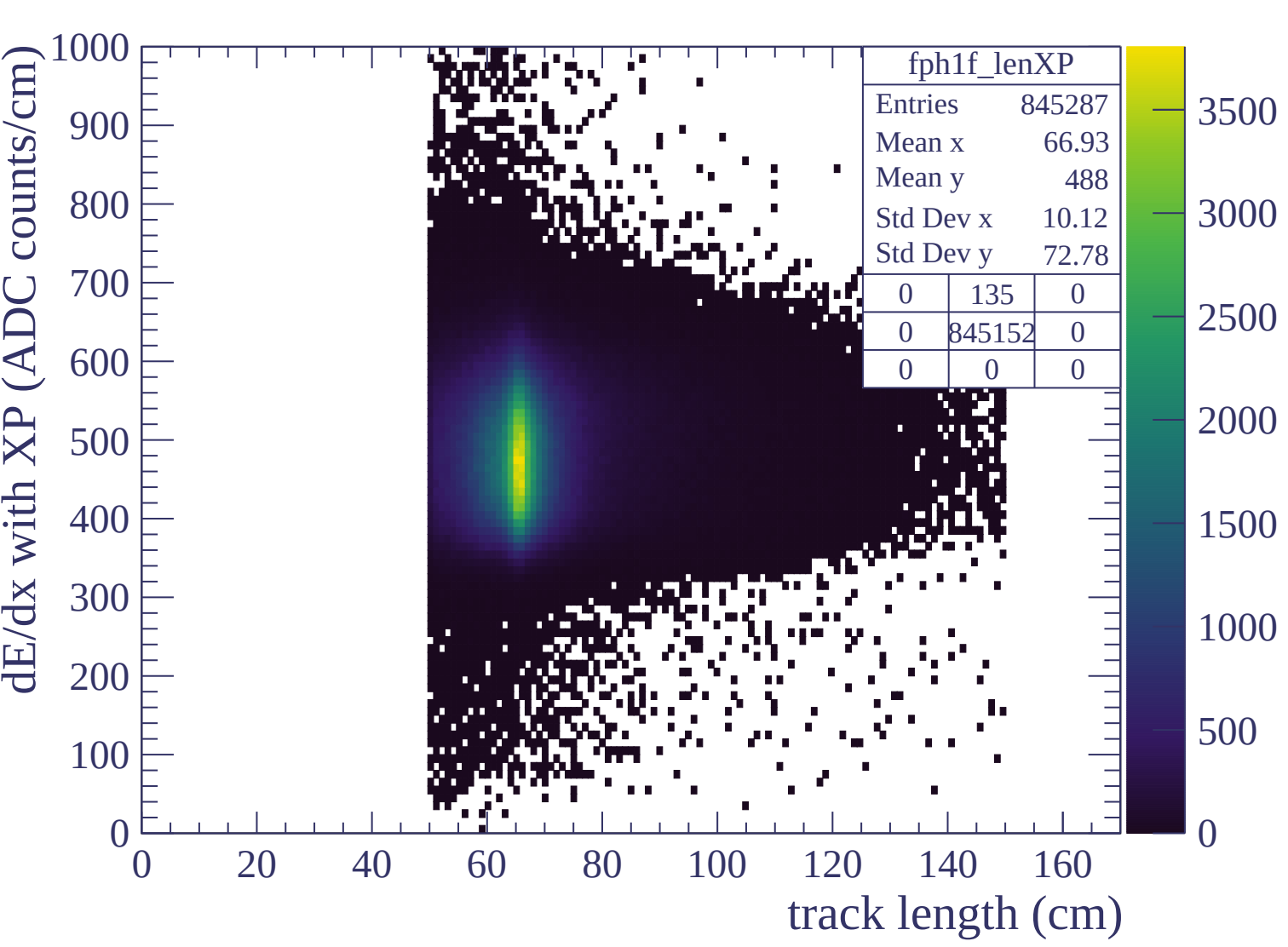


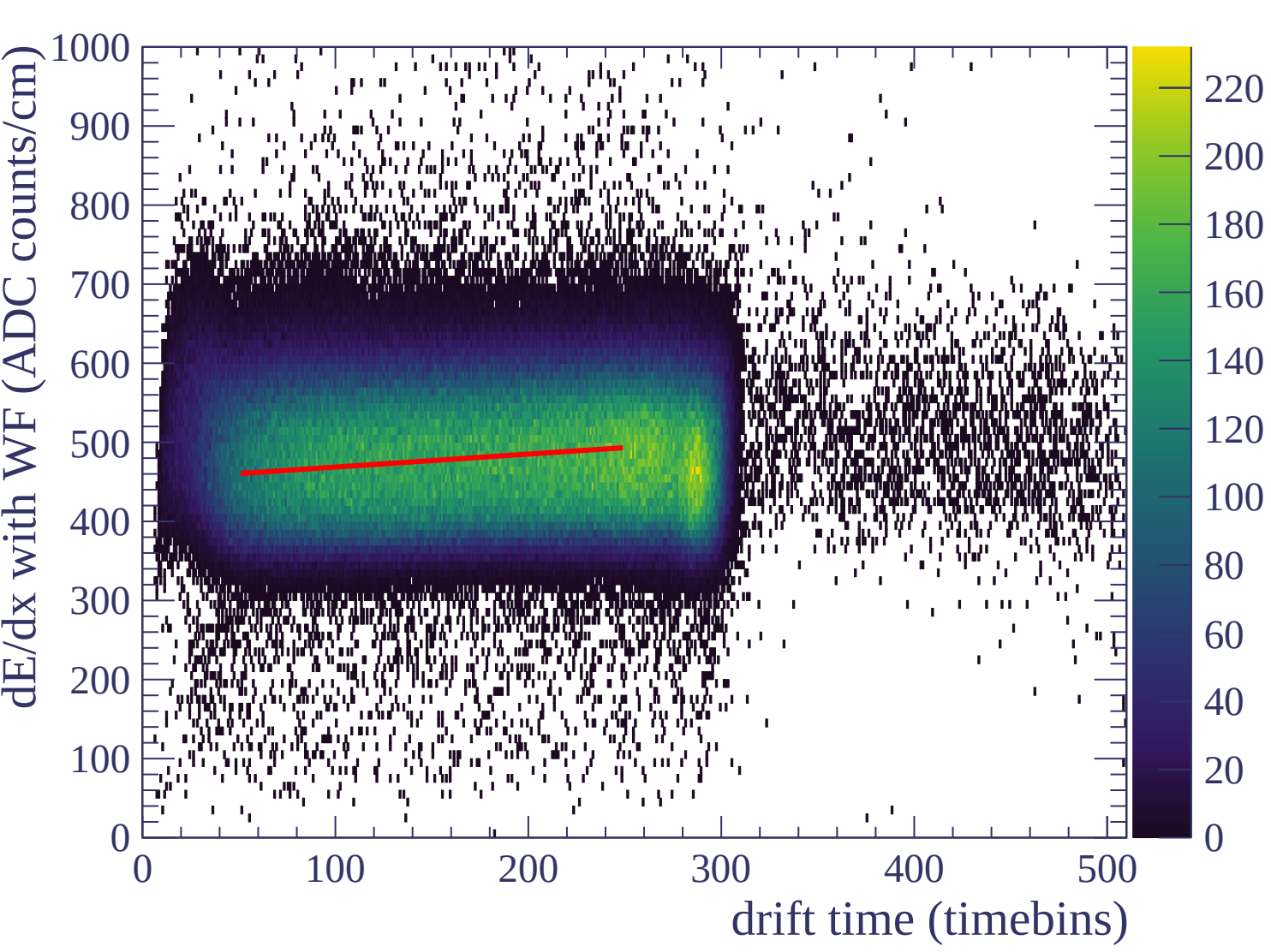


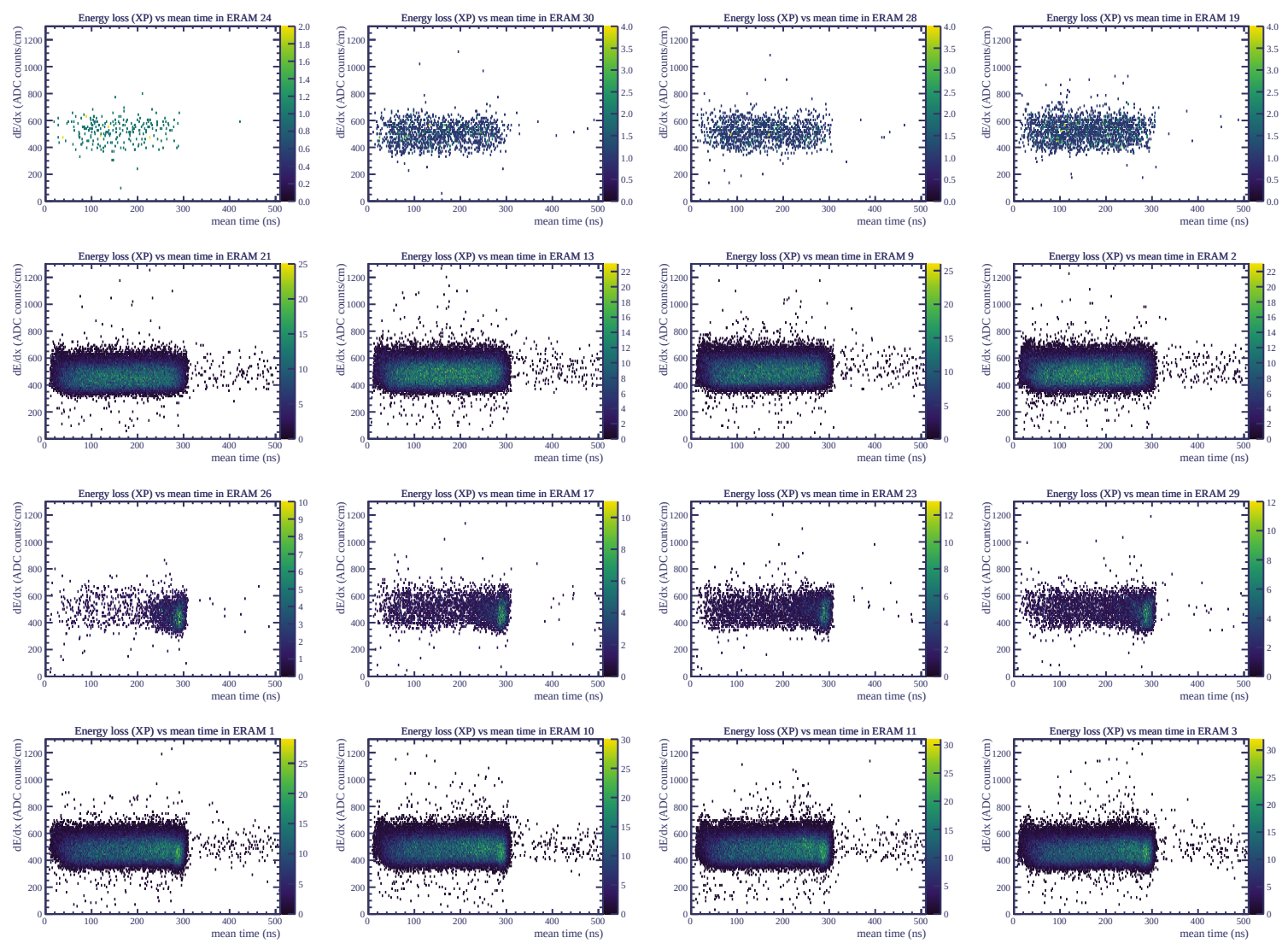


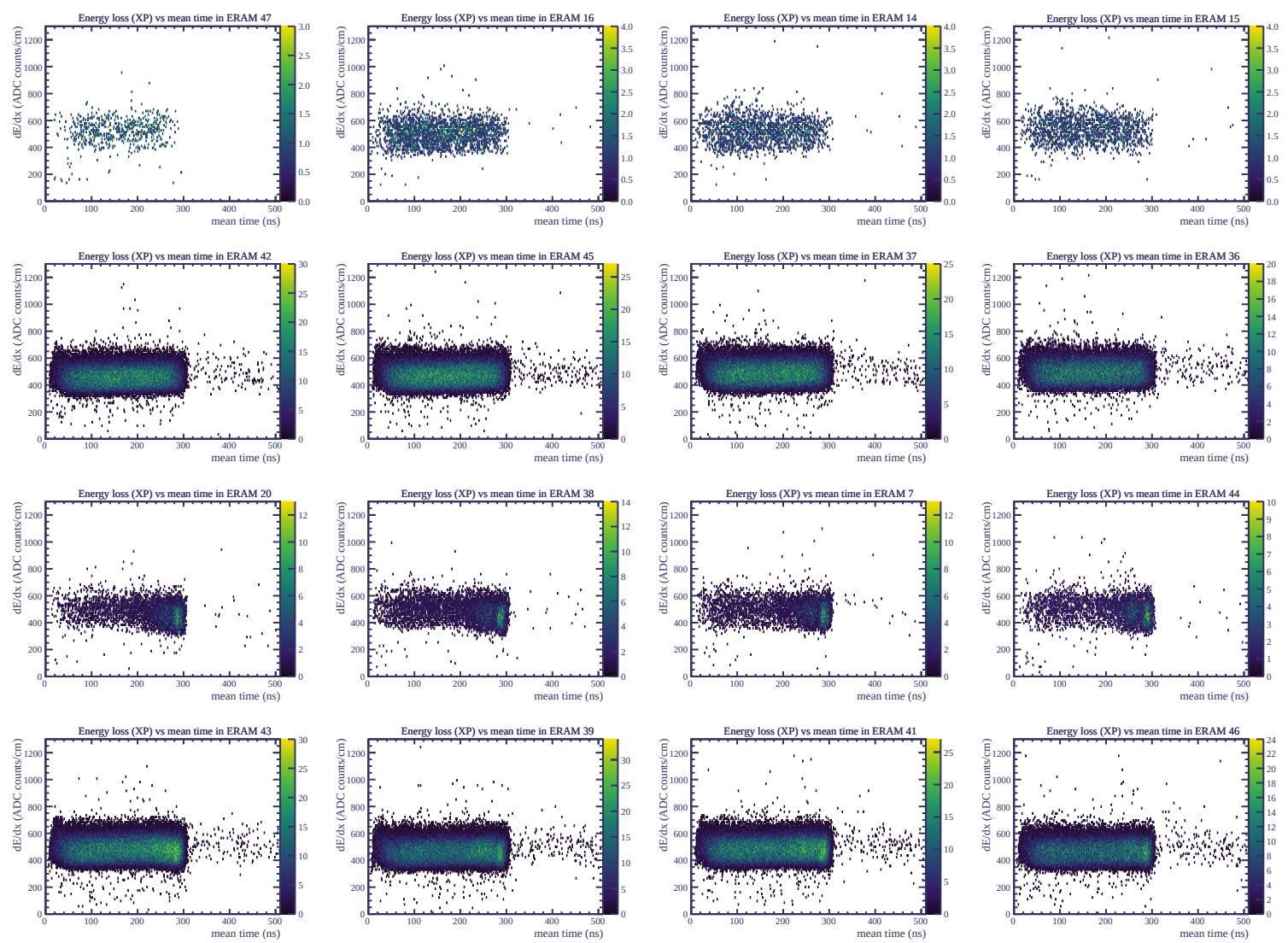






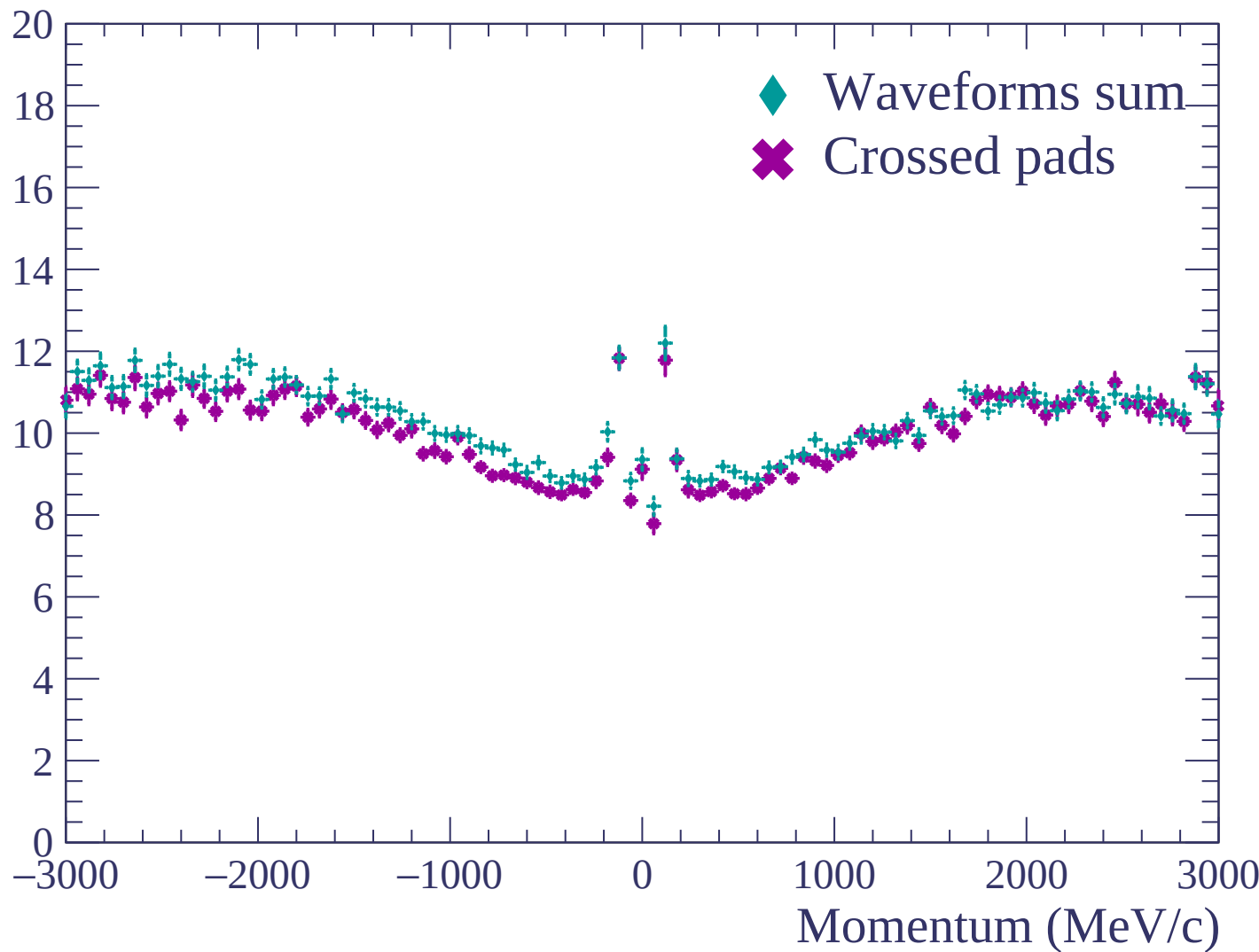


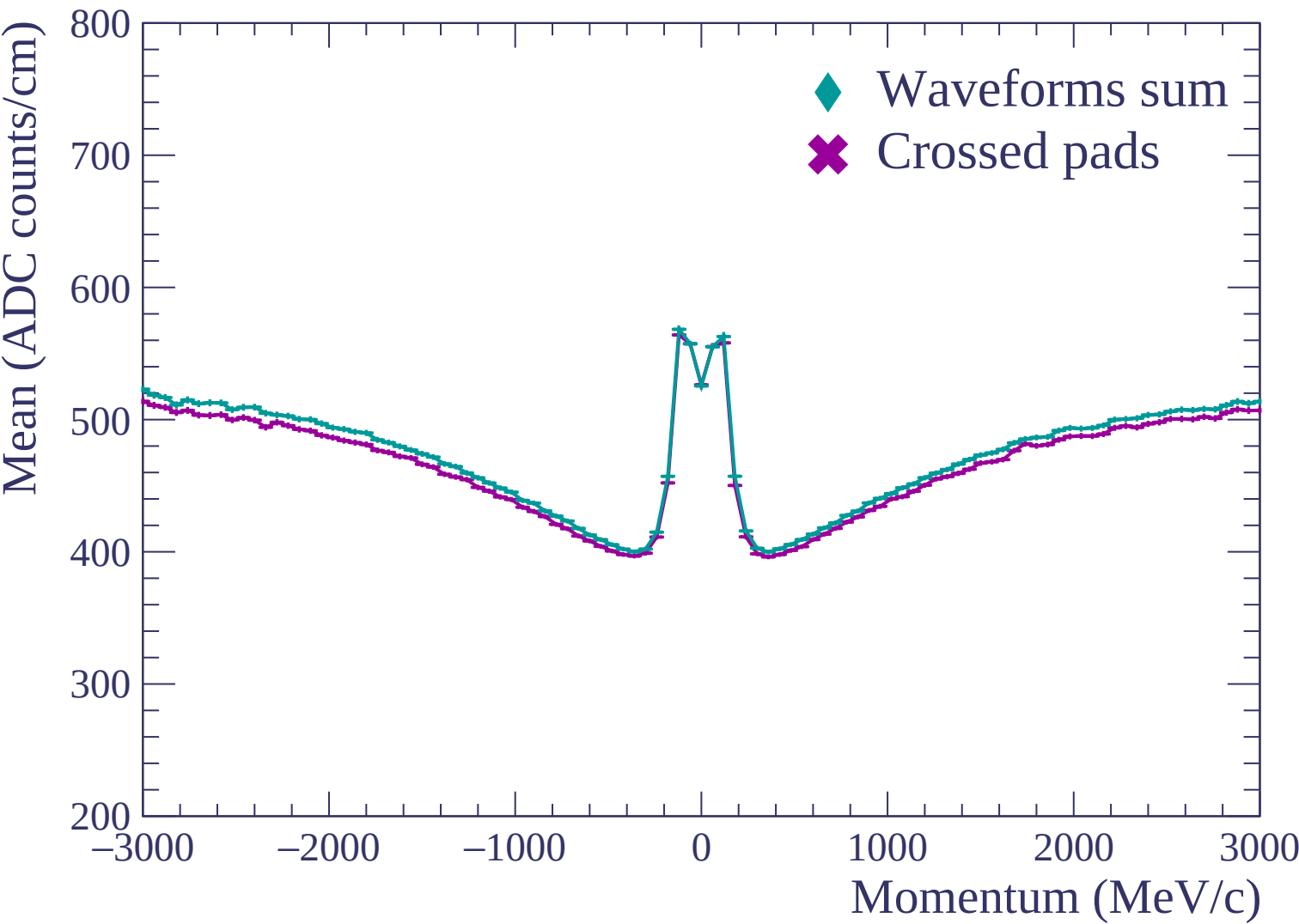


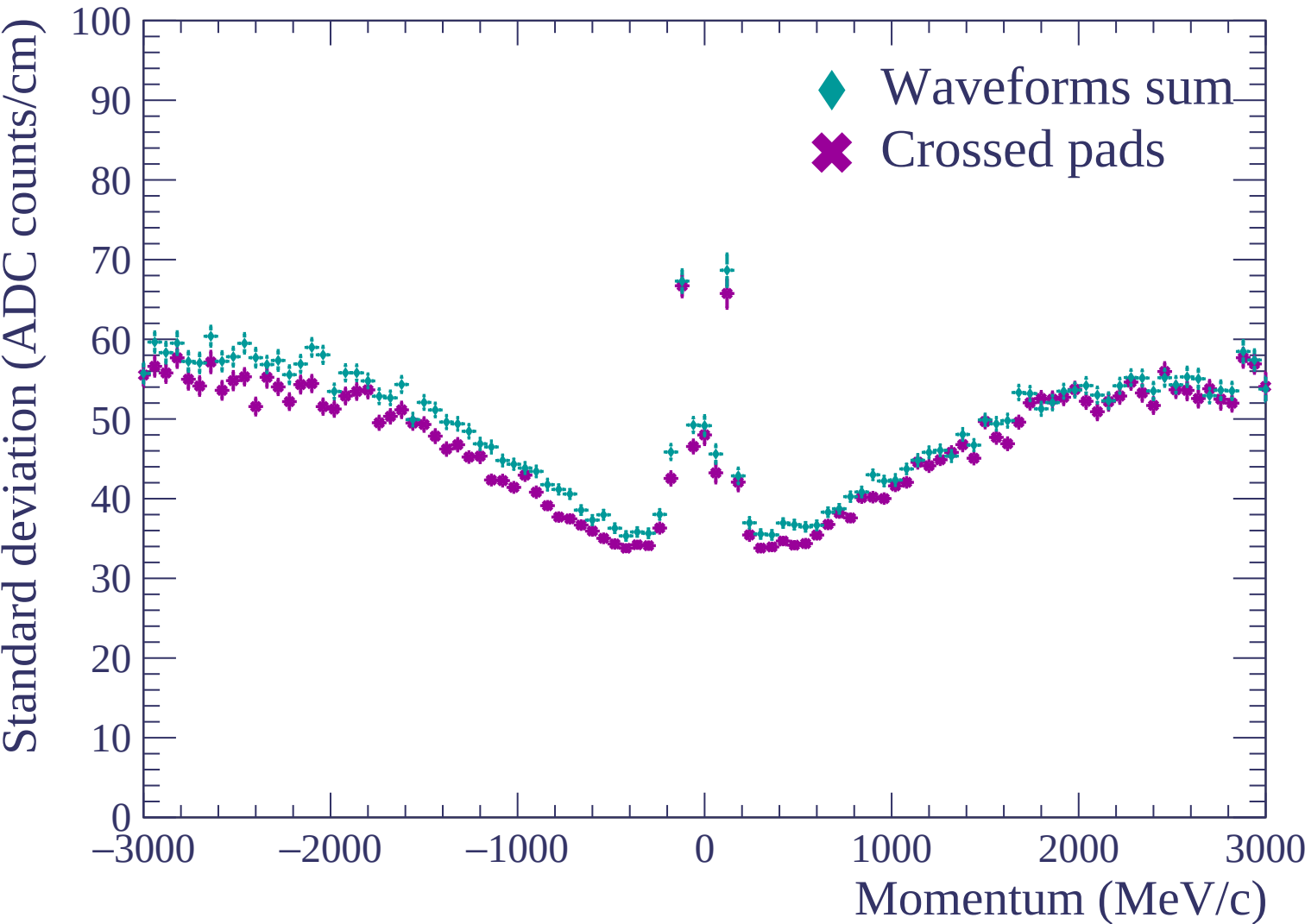


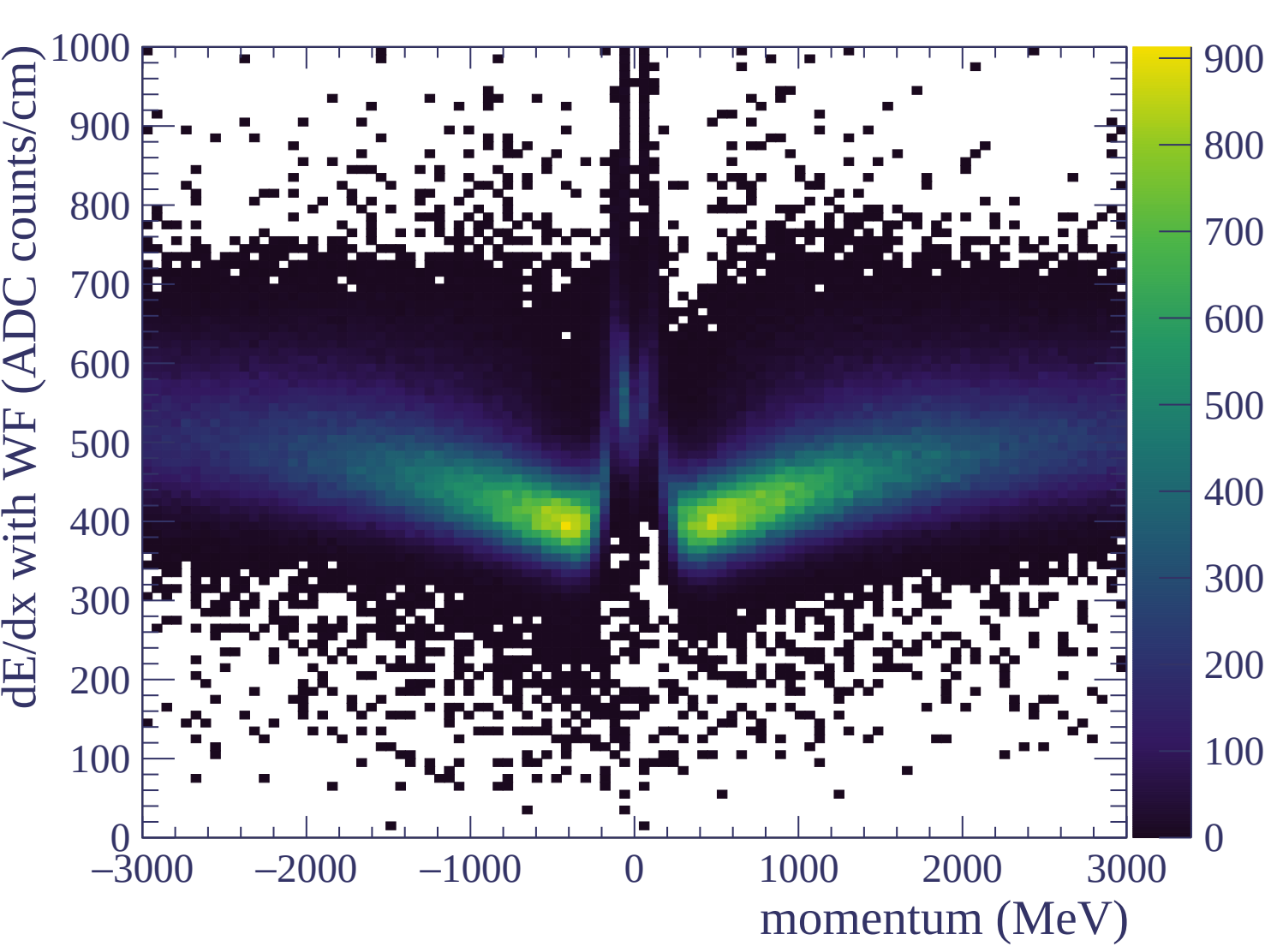


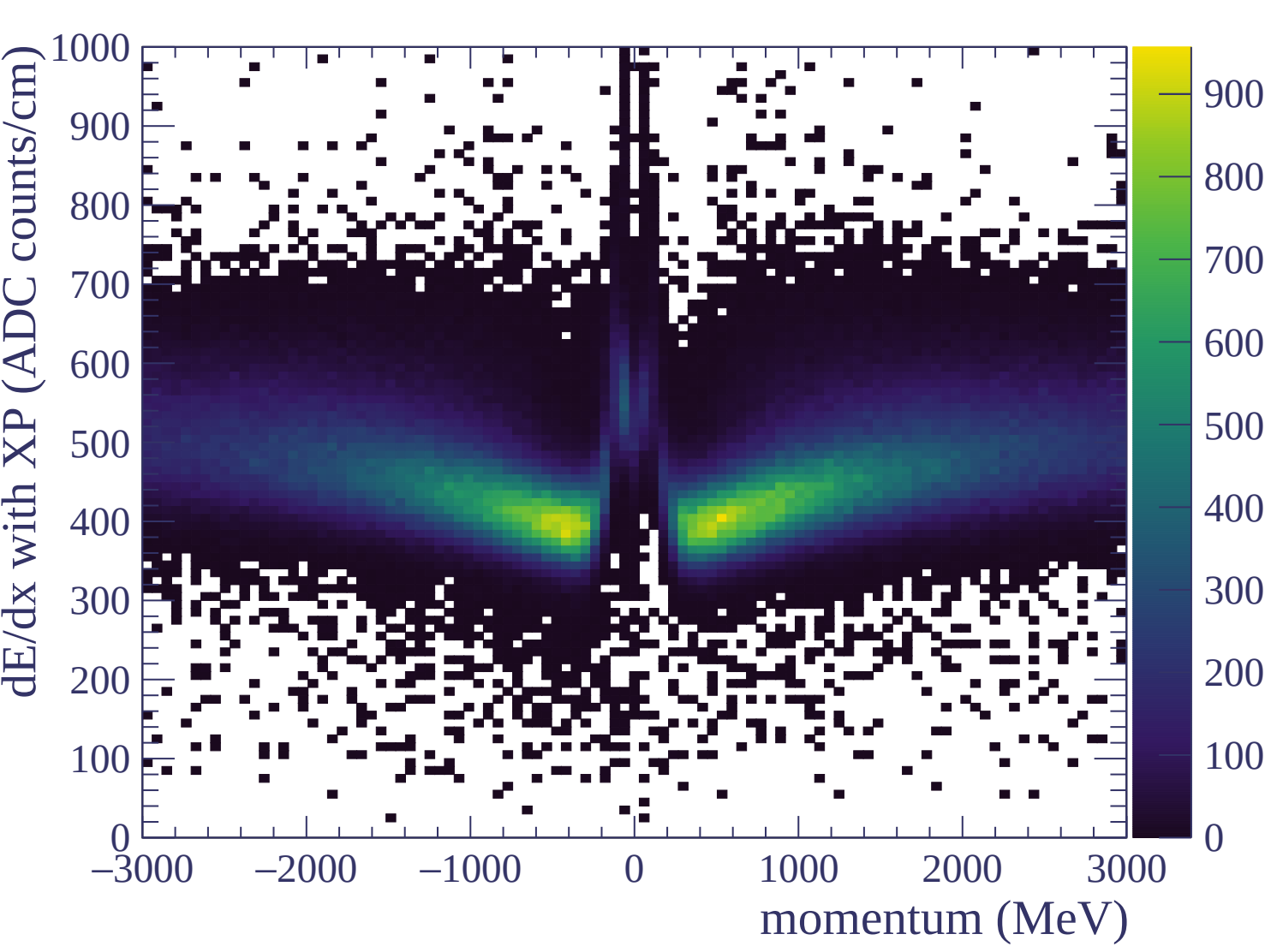
Resolution (%)

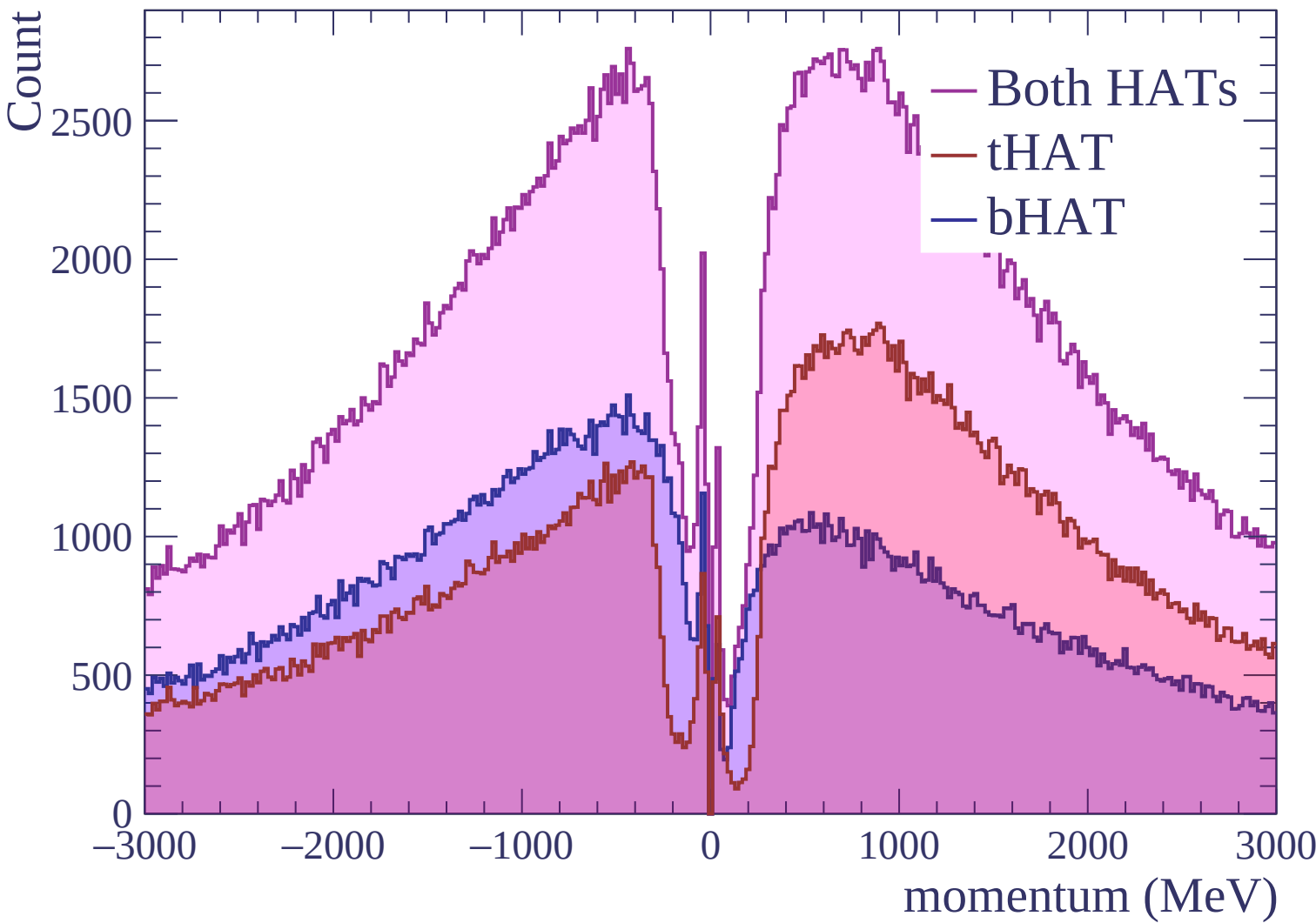






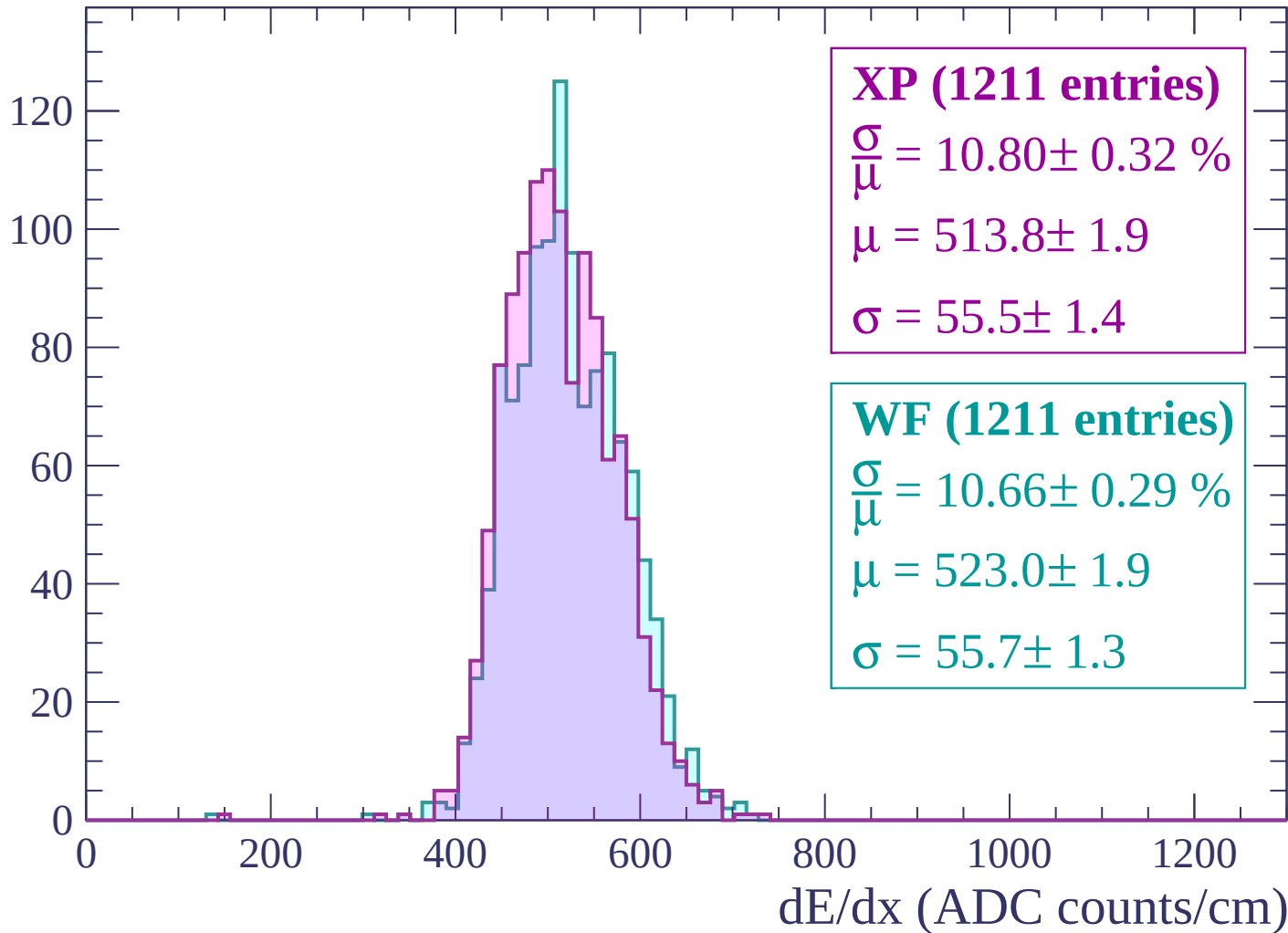






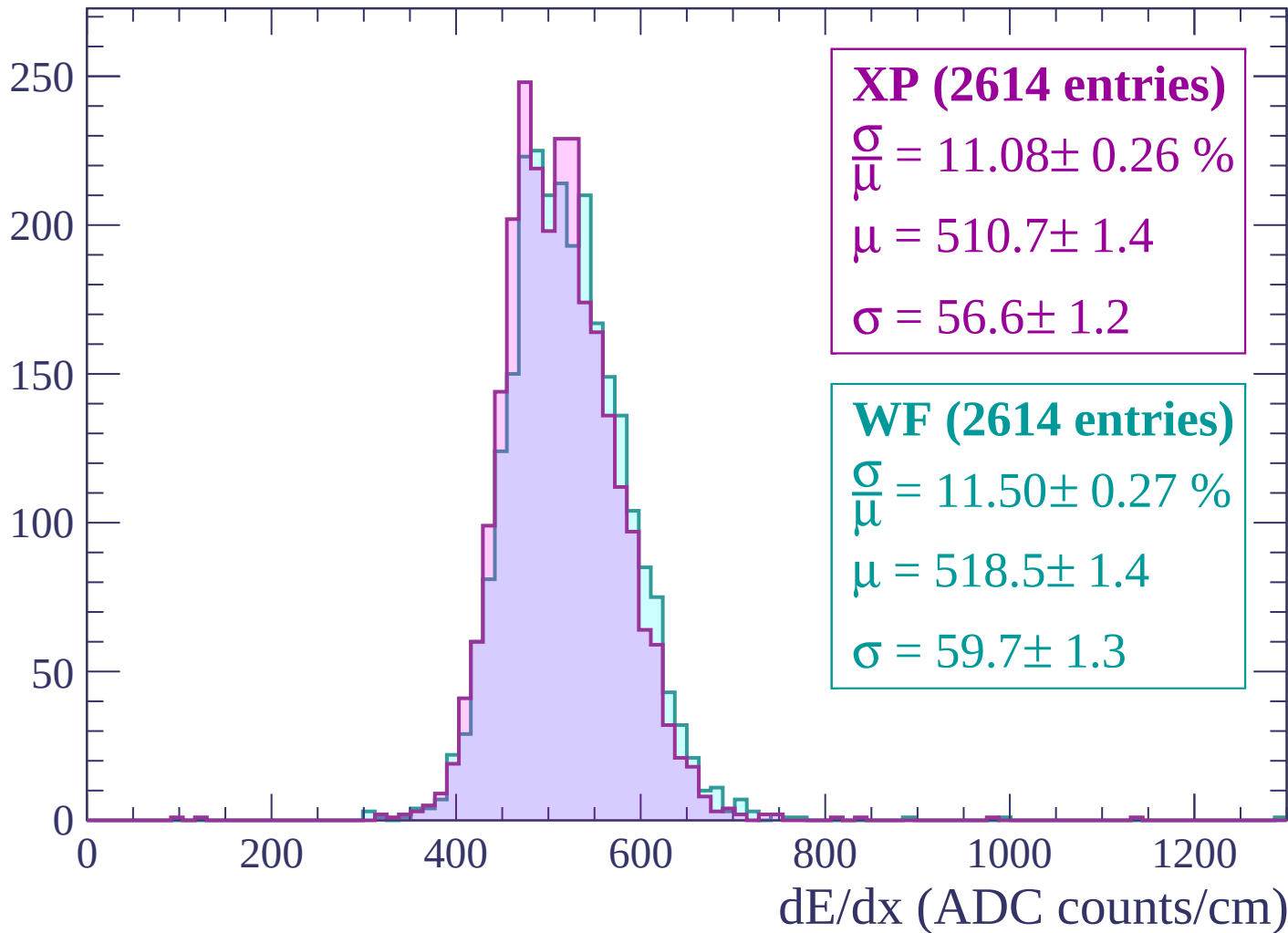
# Energy loss | $-3000 < p < -2940$

Count



Energy loss |  $-2940 < p < -2880$

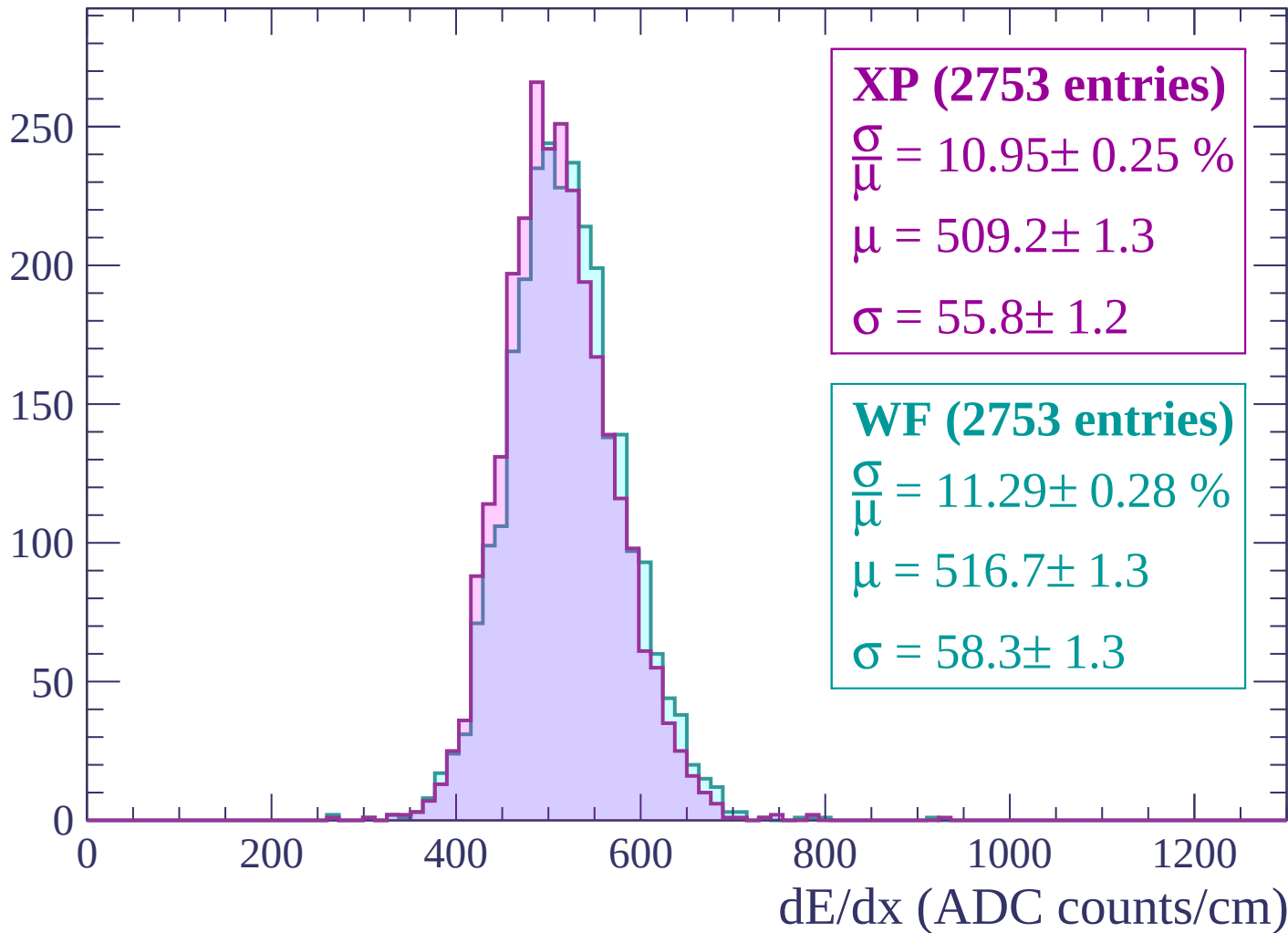
Count





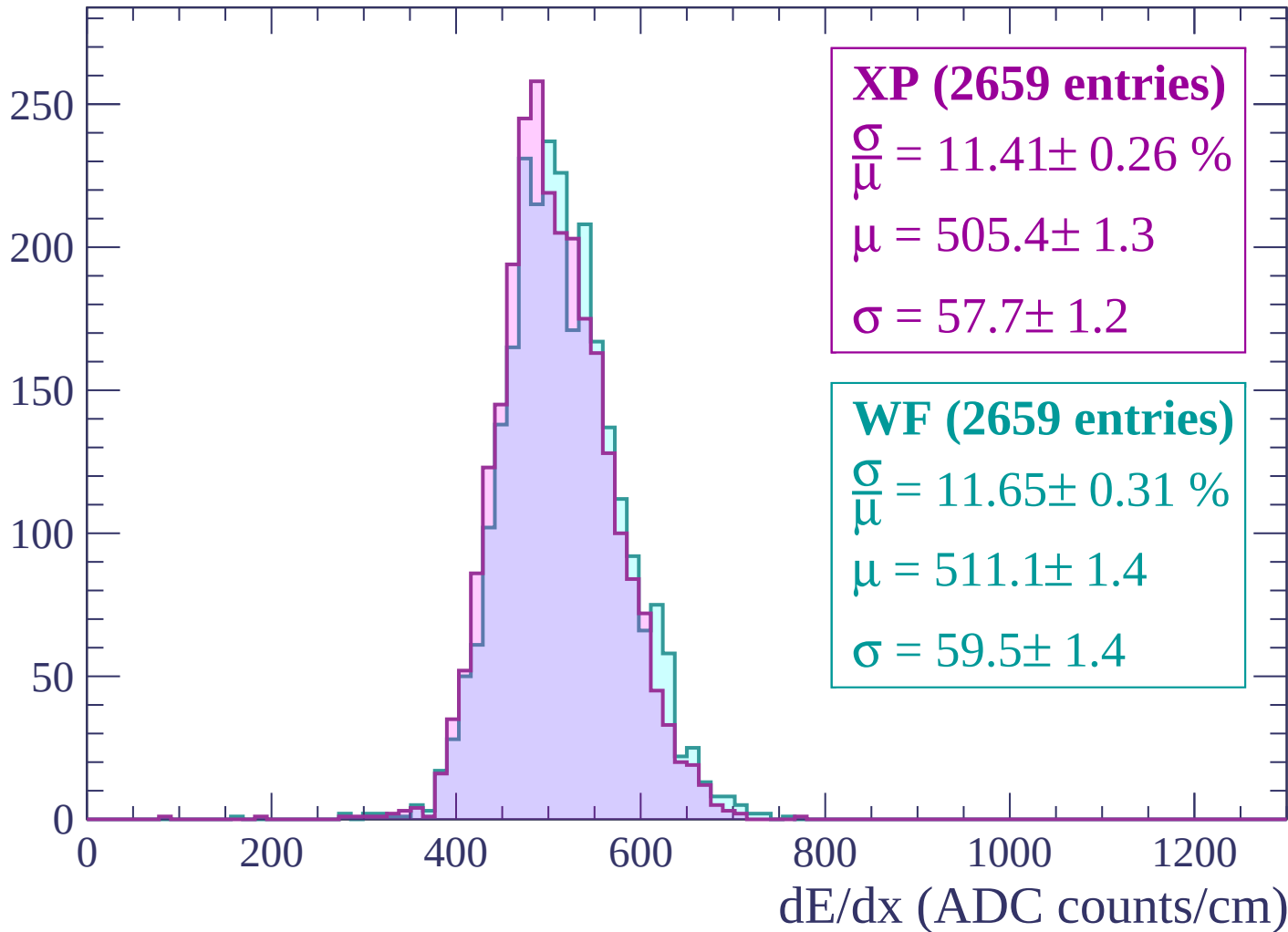
Energy loss |  $-2880 < p < -2820$

Count



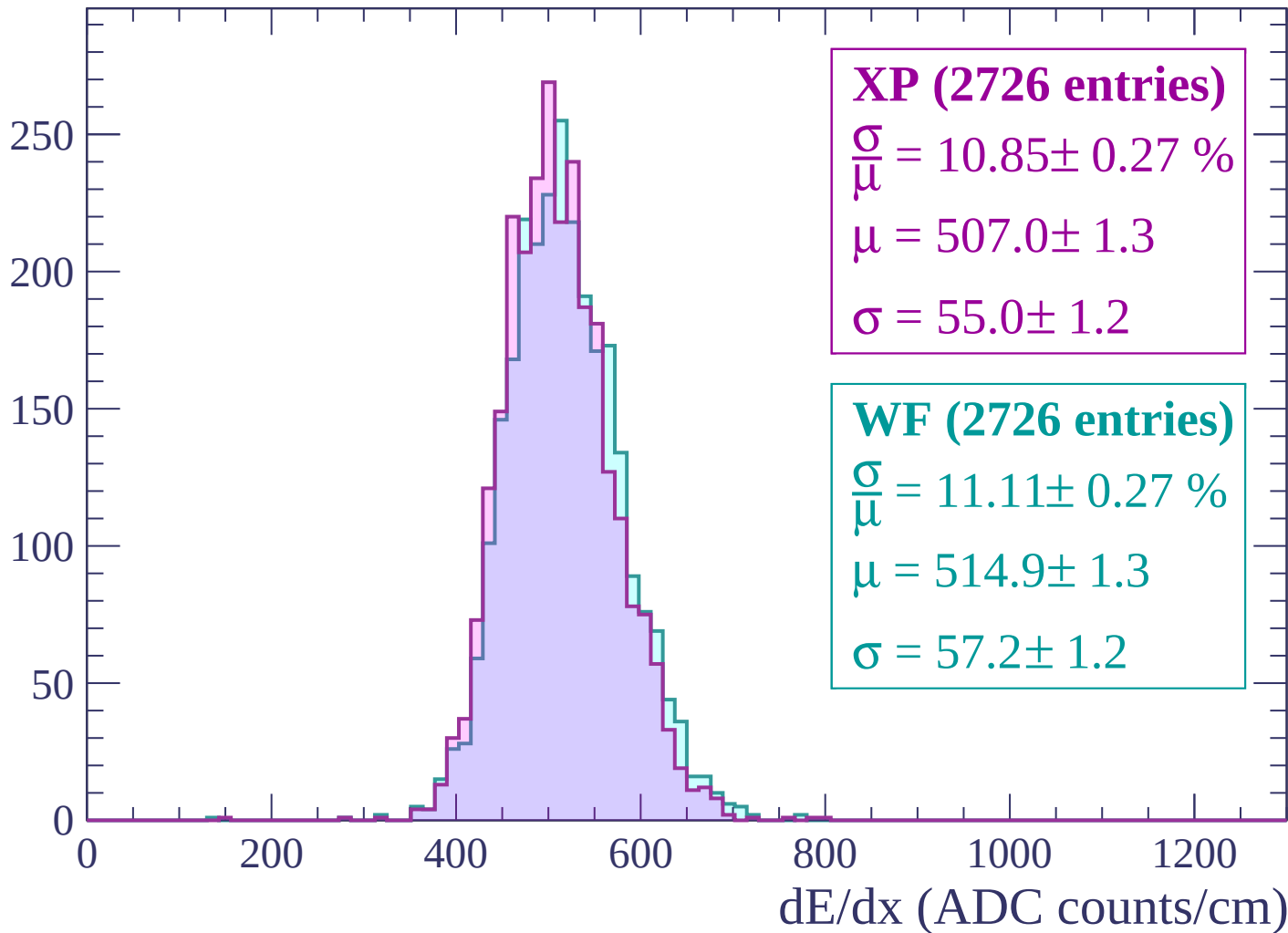
# Energy loss | -2820 < p < -2760

Count



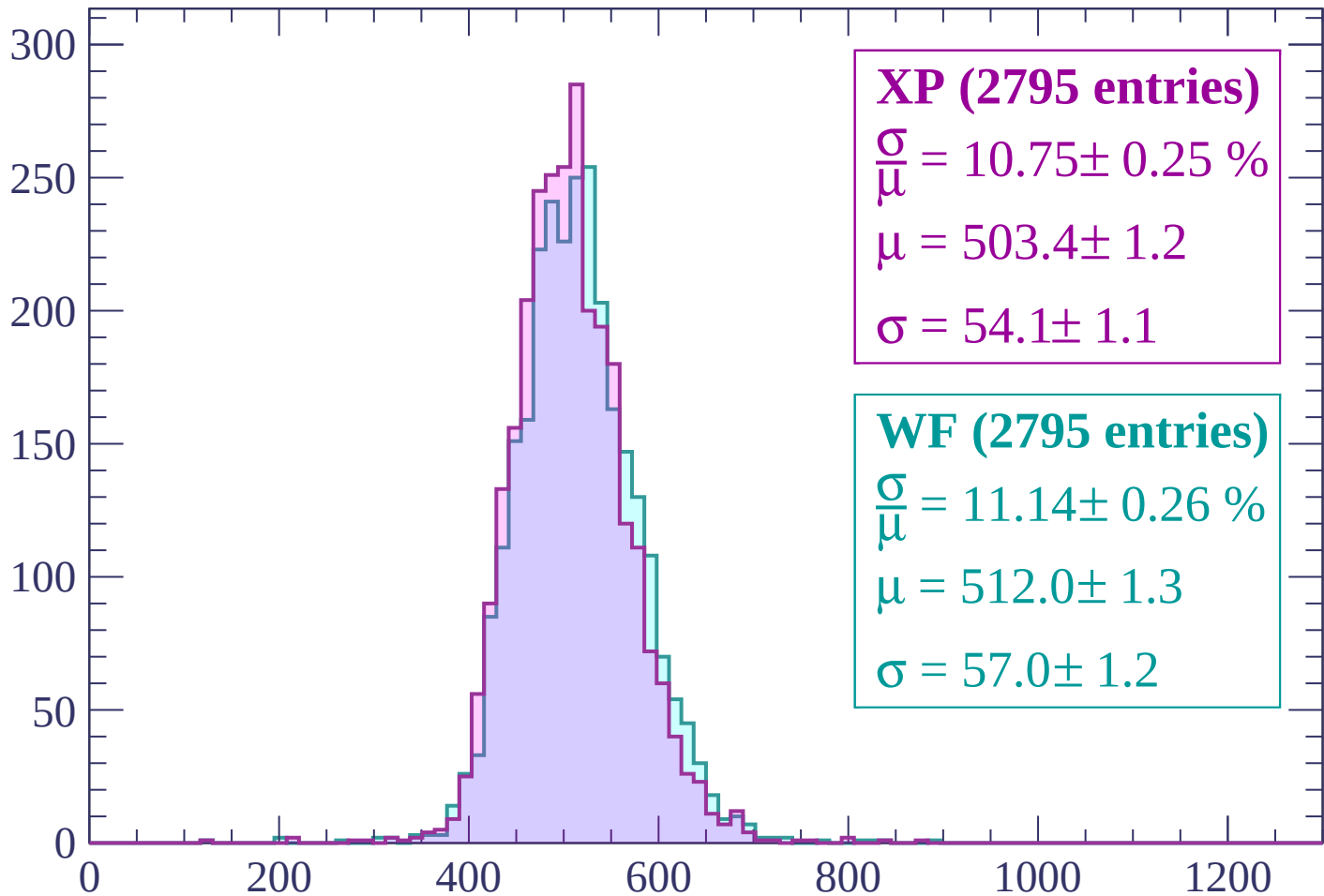
# Energy loss | -2760 < p < -2700

Count



# Energy loss | -2700 < p < -2640

Count



**XP (2795 entries)**

$\frac{\sigma}{\mu} = 10.75 \pm 0.25 \%$

$\mu = 503.4 \pm 1.2$

$\sigma = 54.1 \pm 1.1$

**WF (2795 entries)**

$\frac{\sigma}{\mu} = 11.14 \pm 0.26 \%$

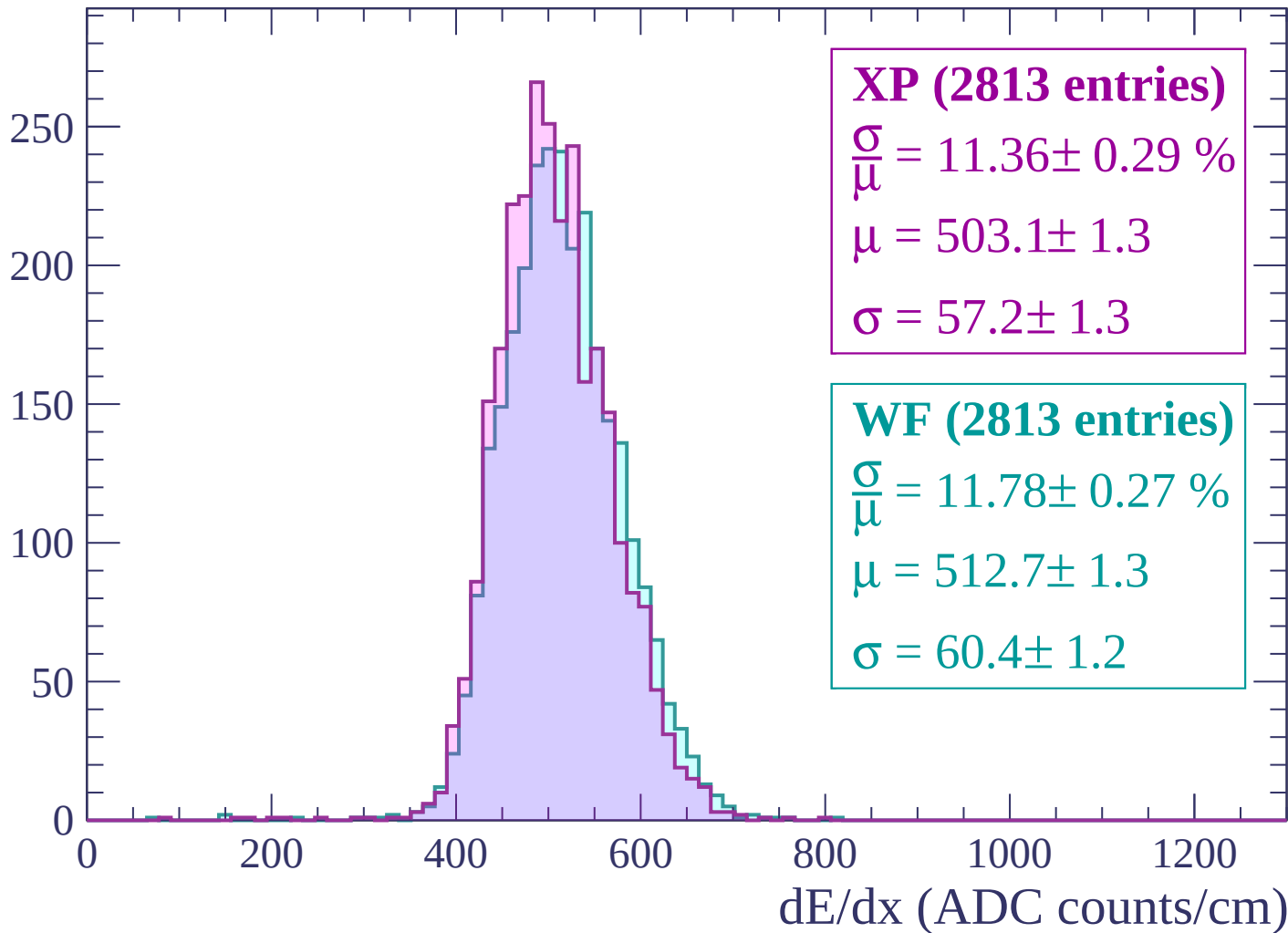
$\mu = 512.0 \pm 1.3$

$\sigma = 57.0 \pm 1.2$

dE/dx (ADC counts/cm)

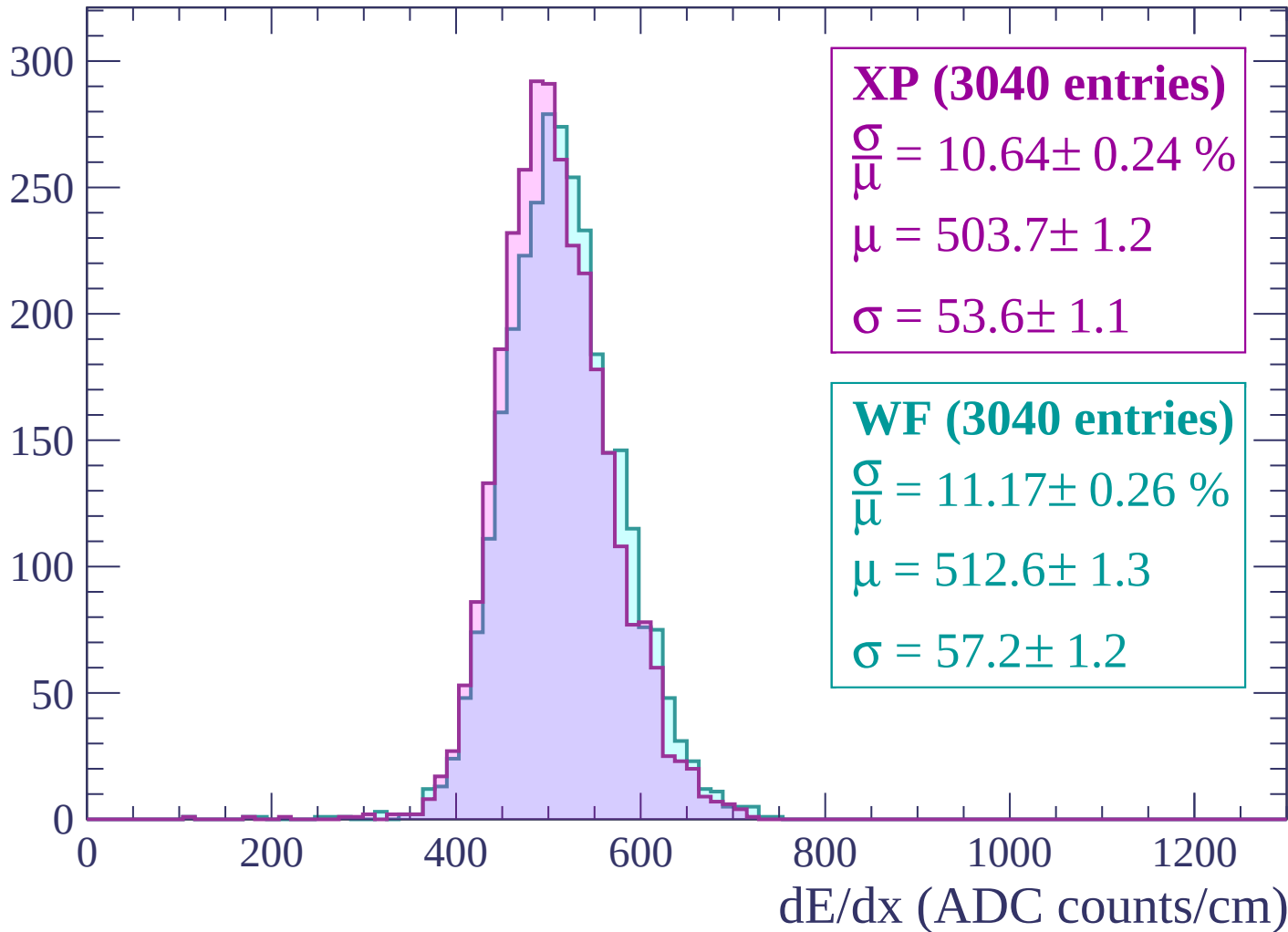
Energy loss |  $-2640 < p < -2580$

Count



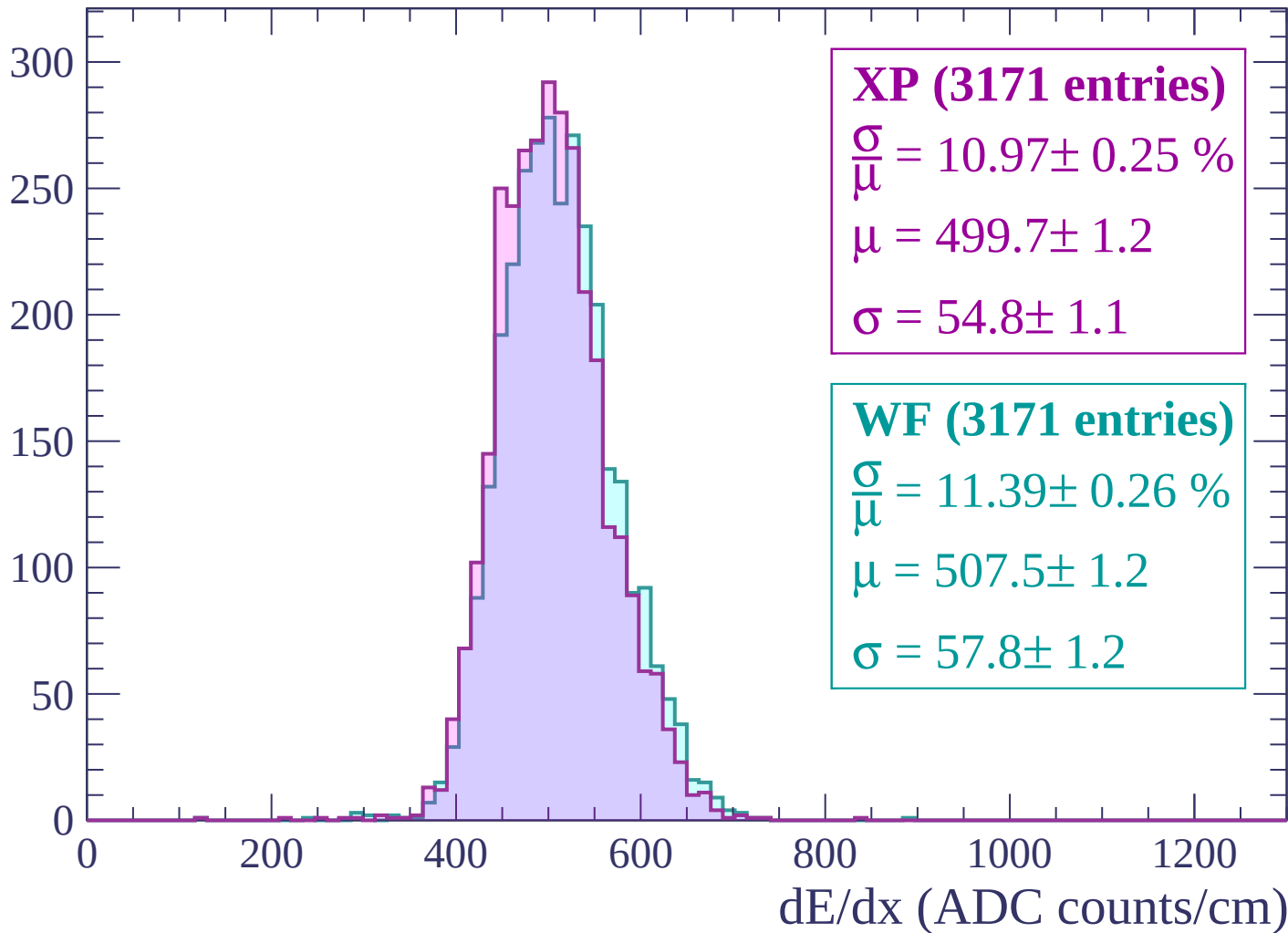
# Energy loss | -2580 < p < -2520

Count



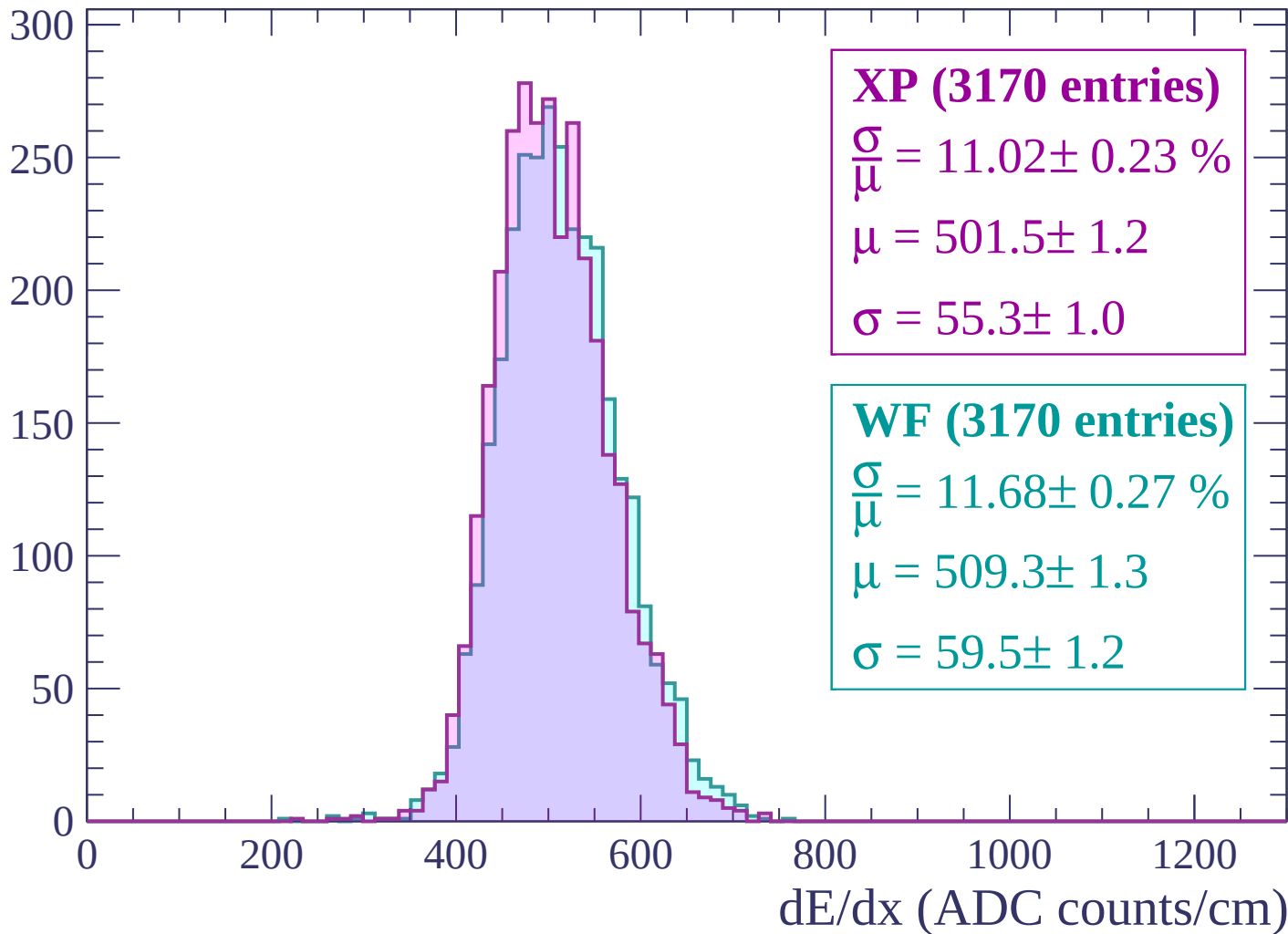
# Energy loss | -2520 < p < -2460

Count



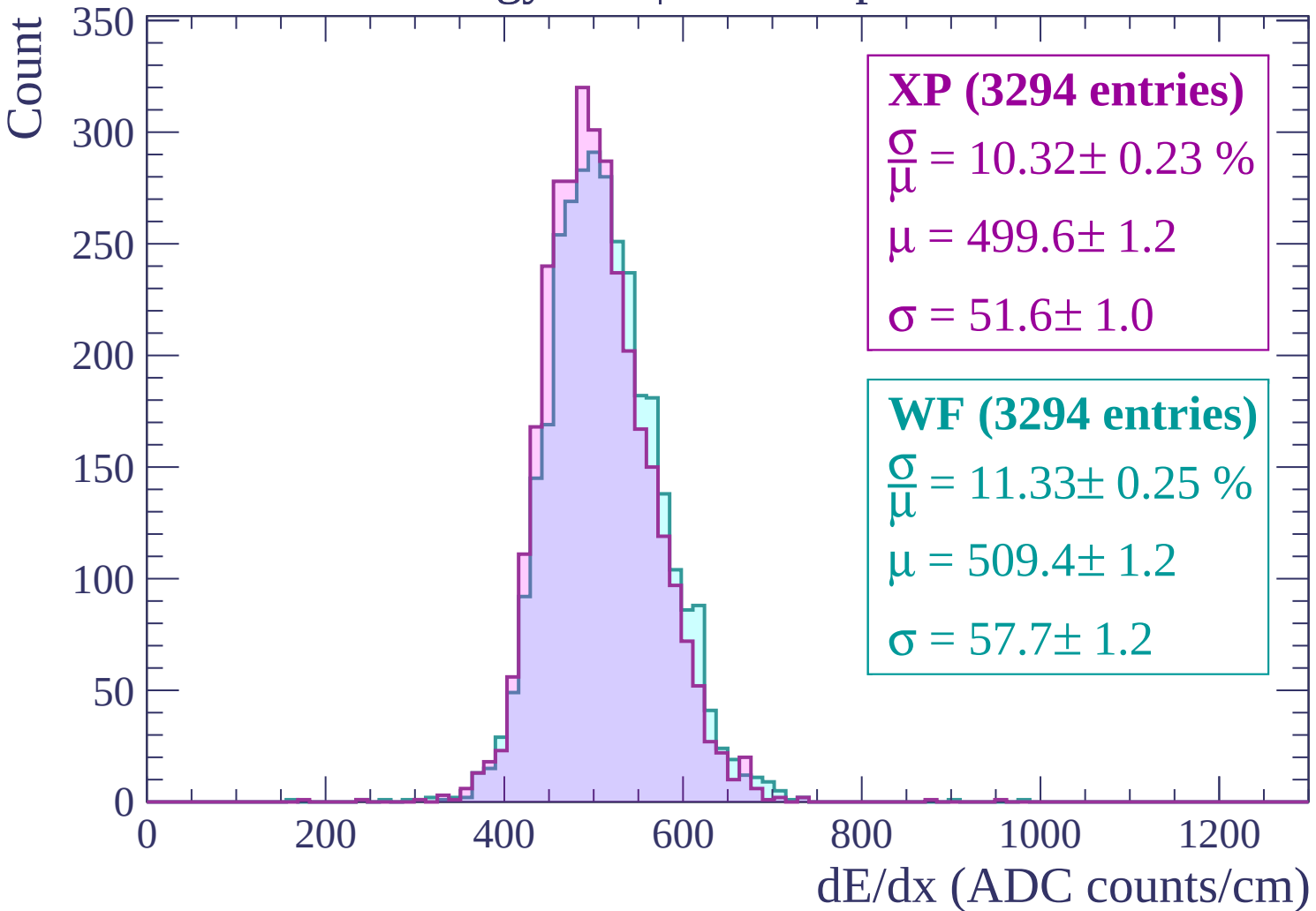
Energy loss |  $-2460 < p < -2400$

Count



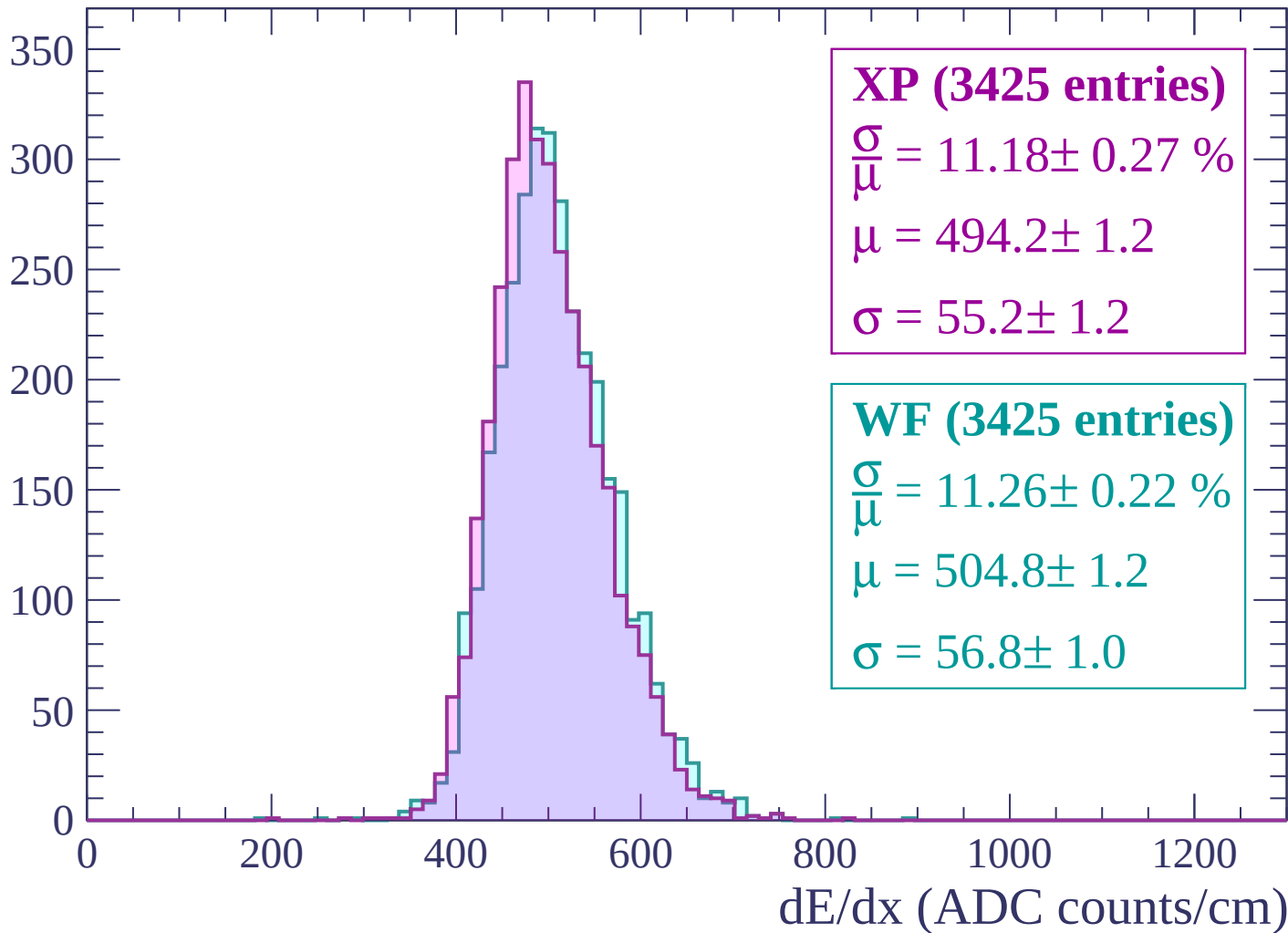


# Energy loss | -2400 < p < -2340



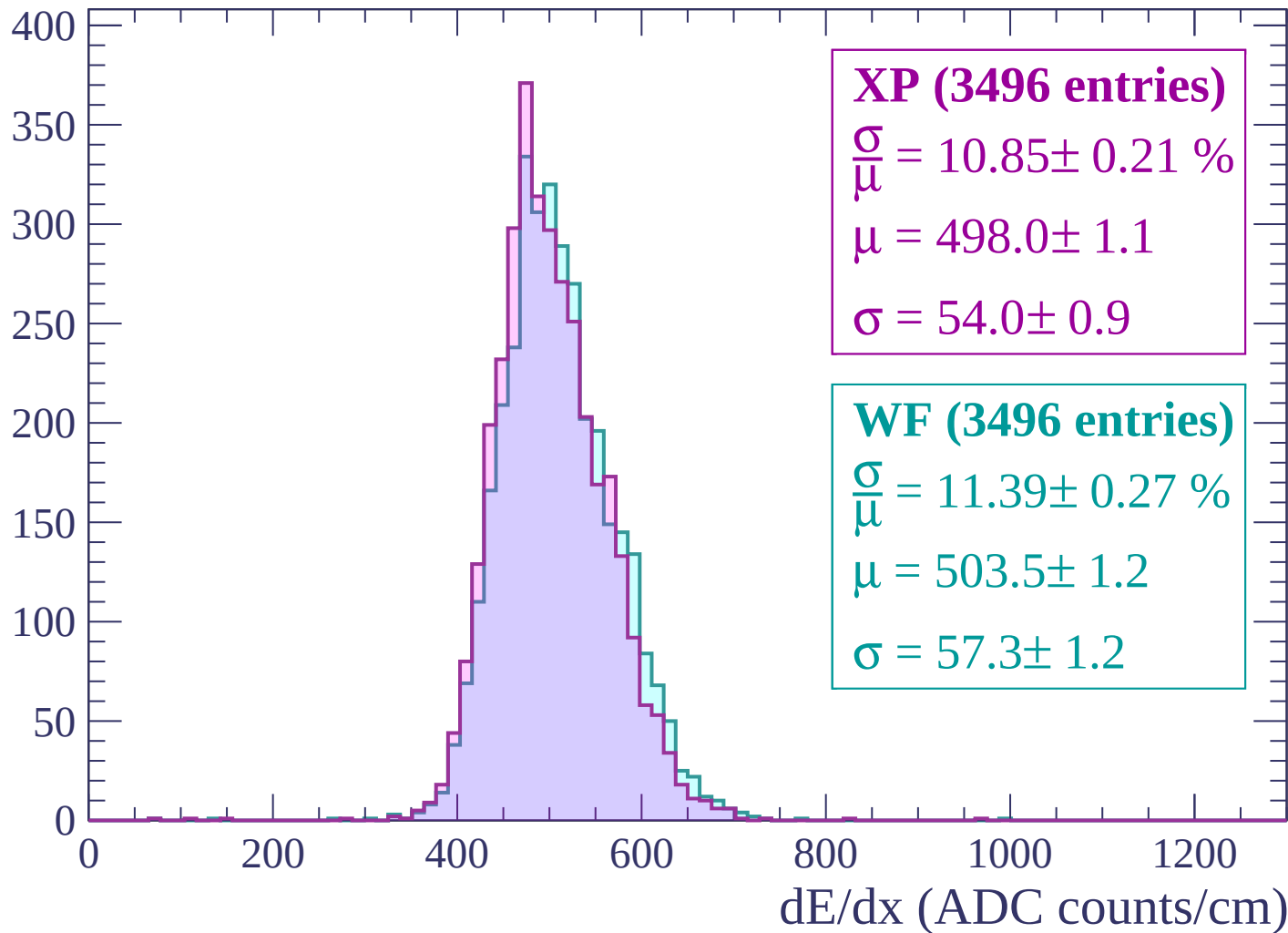
Energy loss |  $-2340 < p < -2280$

Count



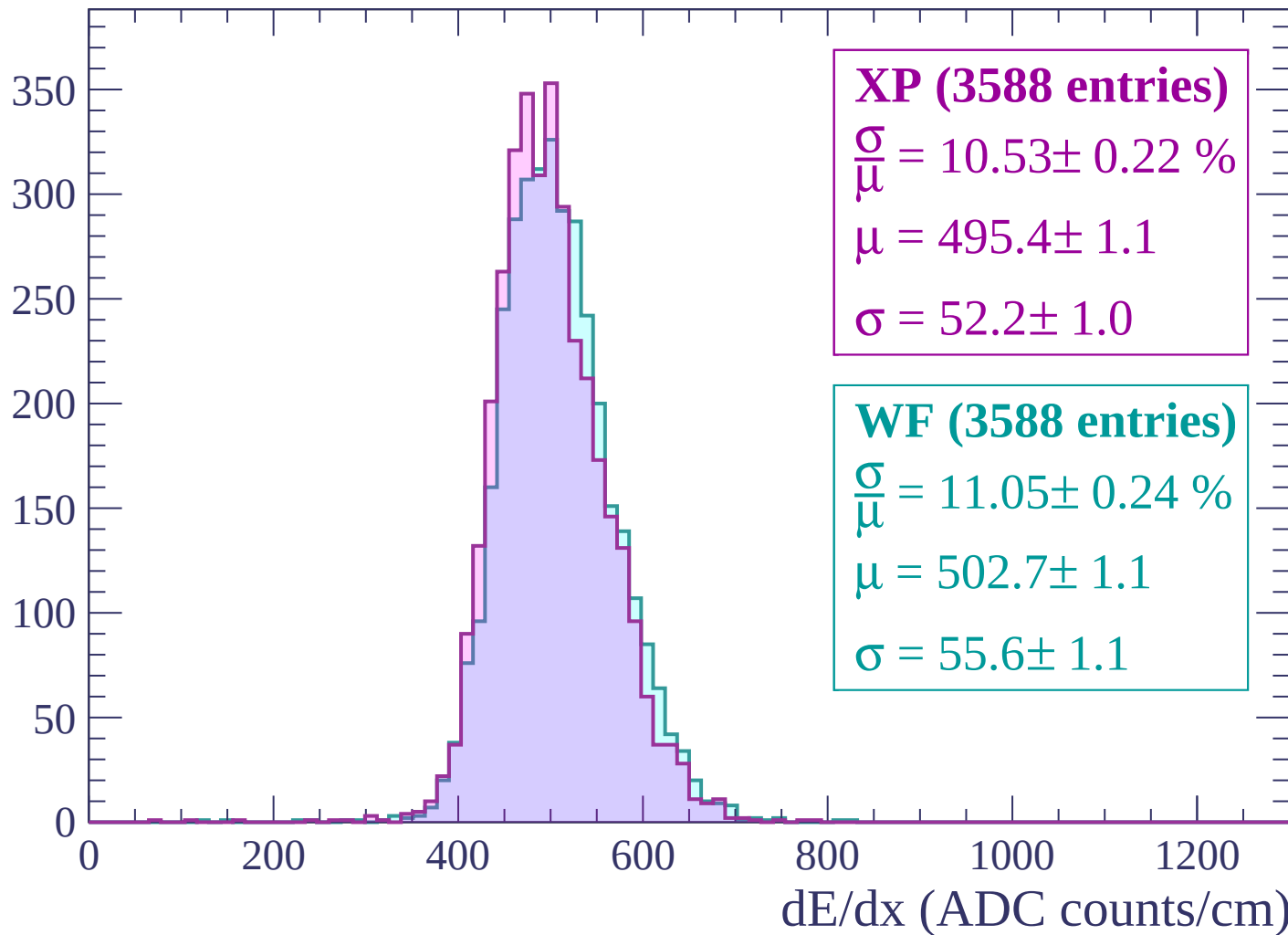
# Energy loss | -2280 < p < -2220

Count



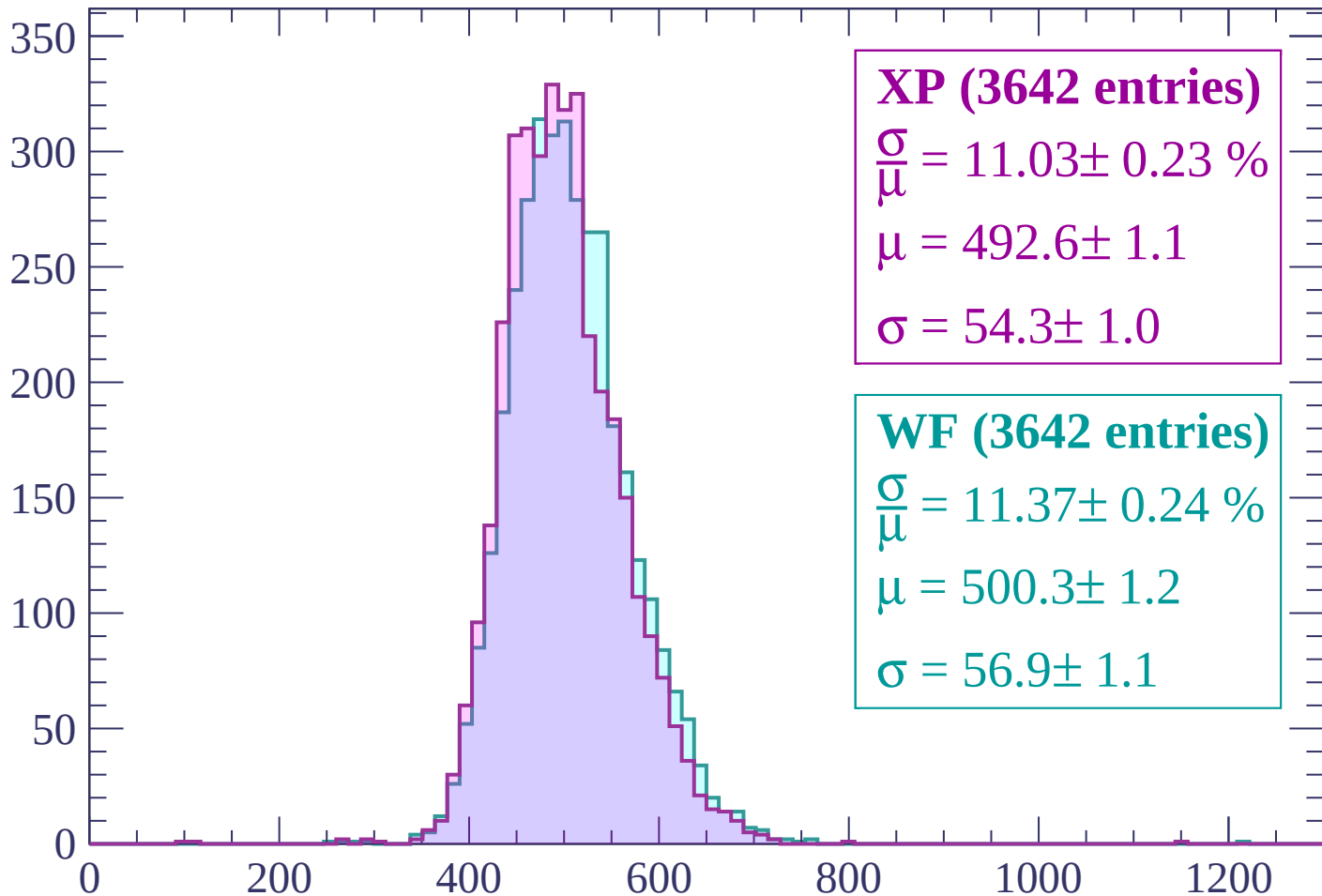
Energy loss |  $-2220 < p < -2160$

Count



# Energy loss | $-2160 < p < -2100$

Count



**XP (3642 entries)**

$\frac{\sigma}{\mu} = 11.03 \pm 0.23 \%$

$\mu = 492.6 \pm 1.1$

$\sigma = 54.3 \pm 1.0$

**WF (3642 entries)**

$\frac{\sigma}{\mu} = 11.37 \pm 0.24 \%$

$\mu = 500.3 \pm 1.2$

$\sigma = 56.9 \pm 1.1$

dE/dx (ADC counts/cm)

# Energy loss | -2100 < p < -2040

Count

400

350

300

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (3959 entries)**

$\frac{\sigma}{\mu} = 11.08 \pm 0.23 \%$

$\mu = 491.6 \pm 1.1$

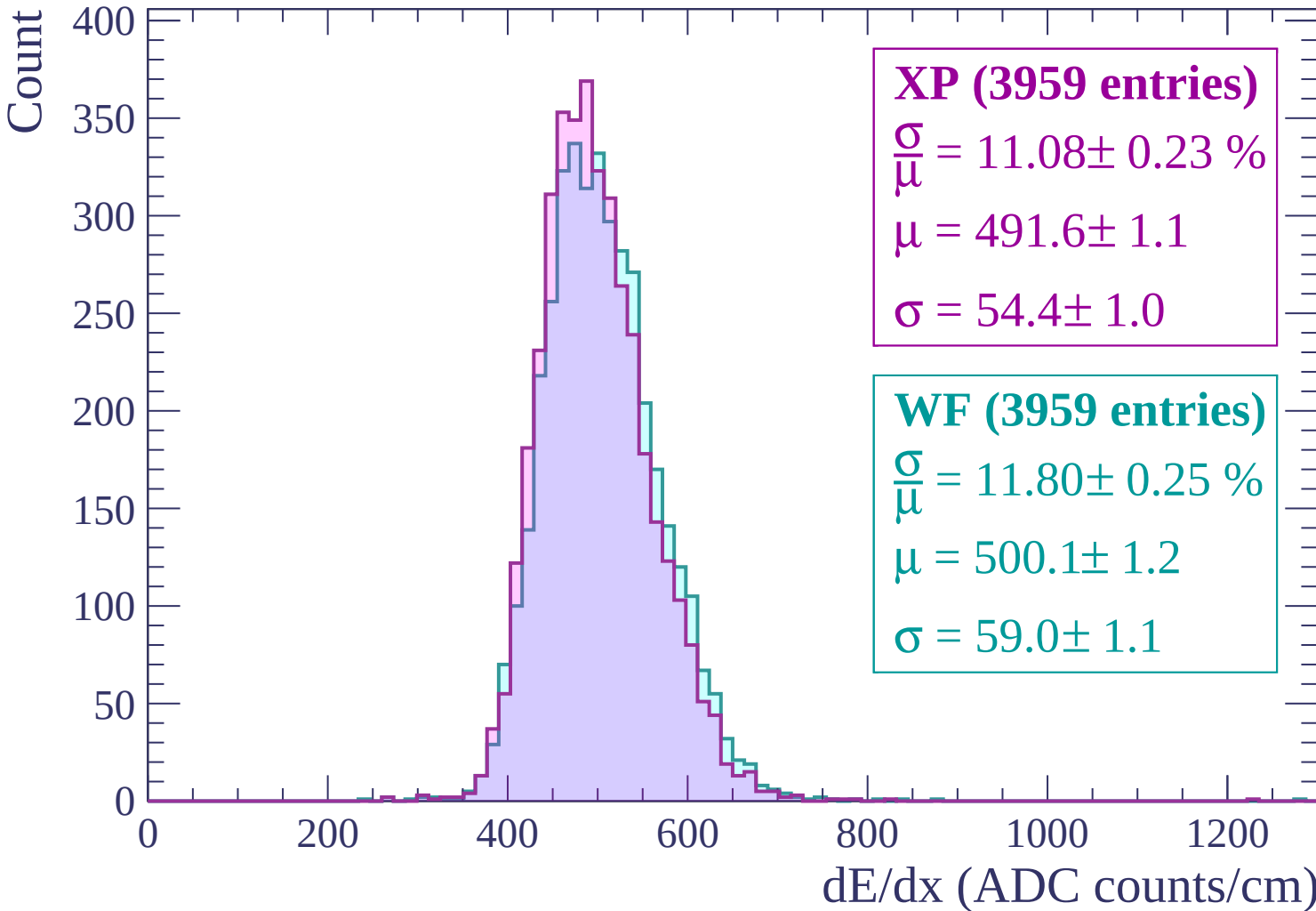
$\sigma = 54.4 \pm 1.0$

**WF (3959 entries)**

$\frac{\sigma}{\mu} = 11.80 \pm 0.25 \%$

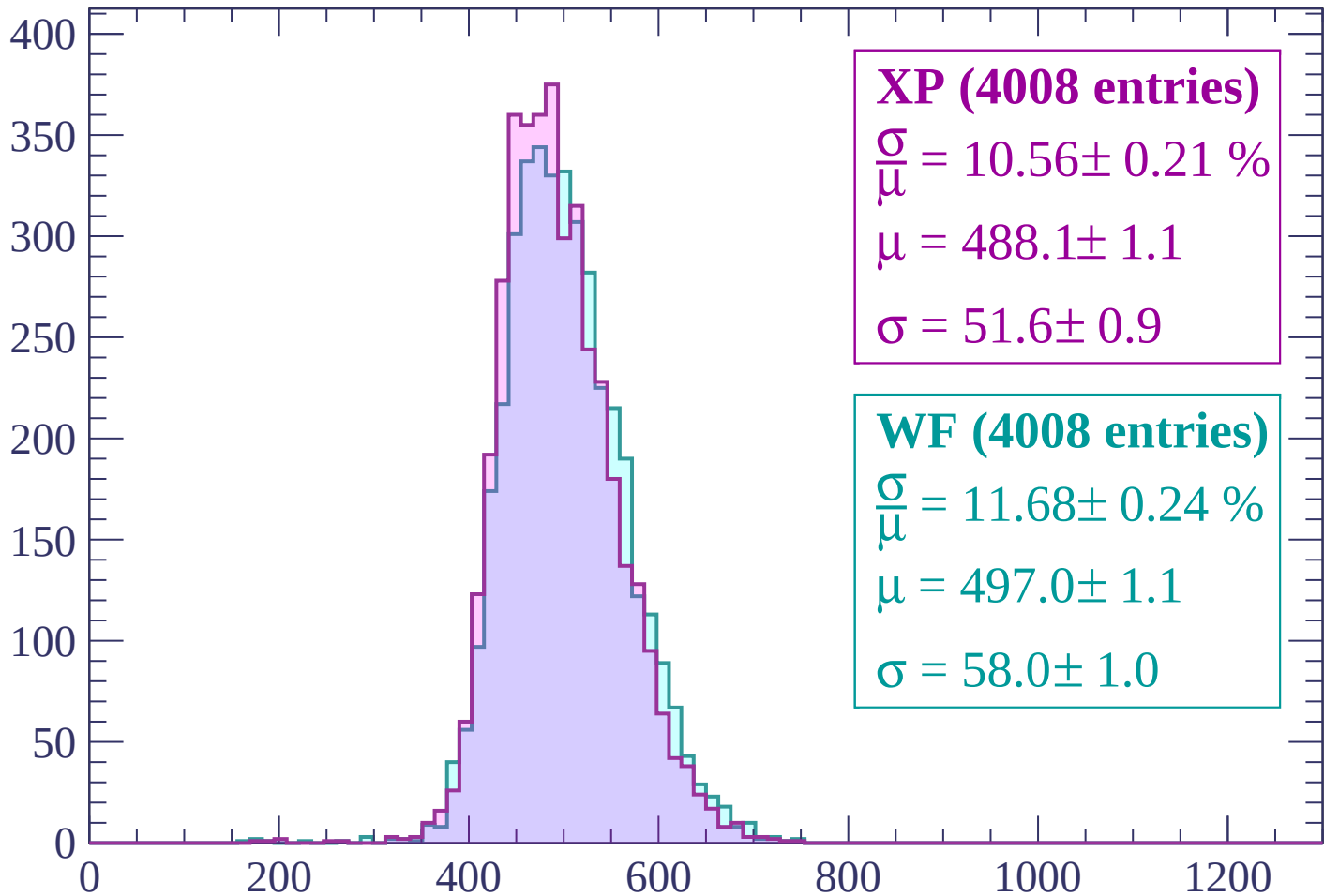
$\mu = 500.1 \pm 1.2$

$\sigma = 59.0 \pm 1.1$



# Energy loss | -2040 < p < -1980

Count



**XP (4008 entries)**

$$\frac{\sigma}{\mu} = 10.56 \pm 0.21 \%$$

$$\mu = 488.1 \pm 1.1$$

$$\sigma = 51.6 \pm 0.9$$

**WF (4008 entries)**

$$\frac{\sigma}{\mu} = 11.68 \pm 0.24 \%$$

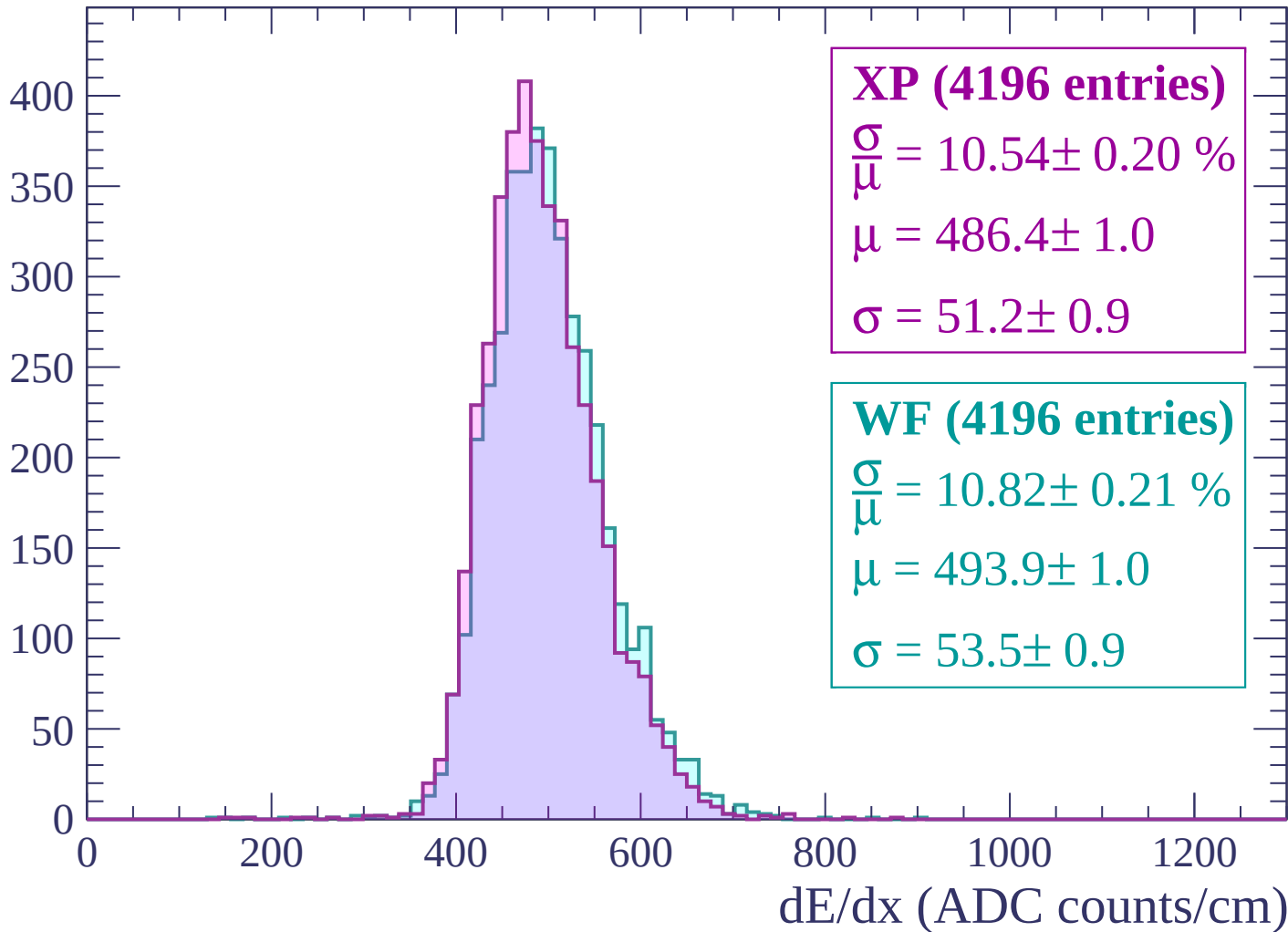
$$\mu = 497.0 \pm 1.1$$

$$\sigma = 58.0 \pm 1.0$$

dE/dx (ADC counts/cm)

# Energy loss | -1980 < p < -1920

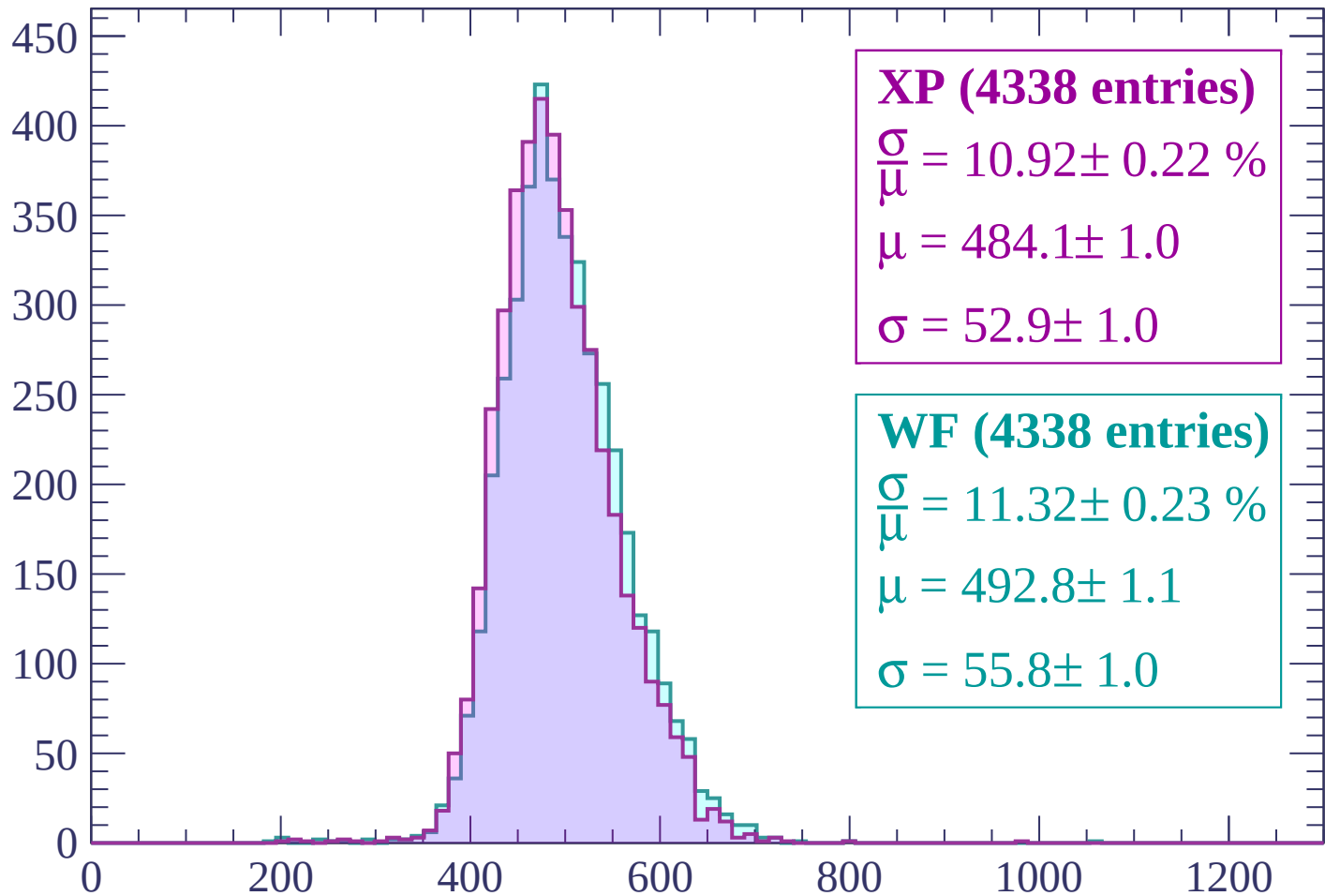
Count





# Energy loss | -1920 < p < -1860

Count



**XP (4338 entries)**

$$\frac{\sigma}{\mu} = 10.92 \pm 0.22 \%$$

$$\mu = 484.1 \pm 1.0$$

$$\sigma = 52.9 \pm 1.0$$

**WF (4338 entries)**

$$\frac{\sigma}{\mu} = 11.32 \pm 0.23 \%$$

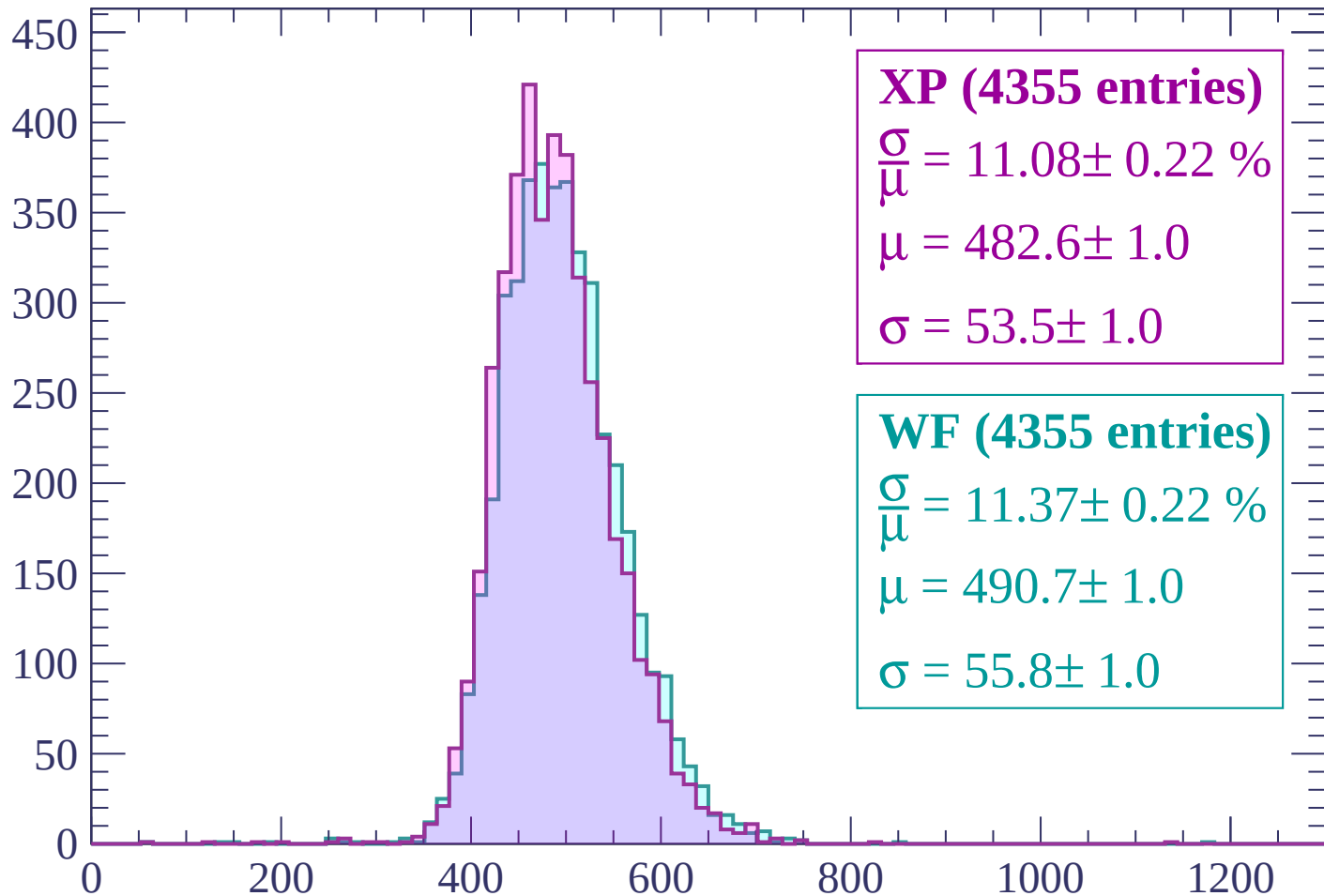
$$\mu = 492.8 \pm 1.1$$

$$\sigma = 55.8 \pm 1.0$$

dE/dx (ADC counts/cm)

# Energy loss | -1860 < p < -1800

Count



**XP (4355 entries)**

$$\frac{\sigma}{\mu} = 11.08 \pm 0.22 \%$$

$$\mu = 482.6 \pm 1.0$$

$$\sigma = 53.5 \pm 1.0$$

**WF (4355 entries)**

$$\frac{\sigma}{\mu} = 11.37 \pm 0.22 \%$$

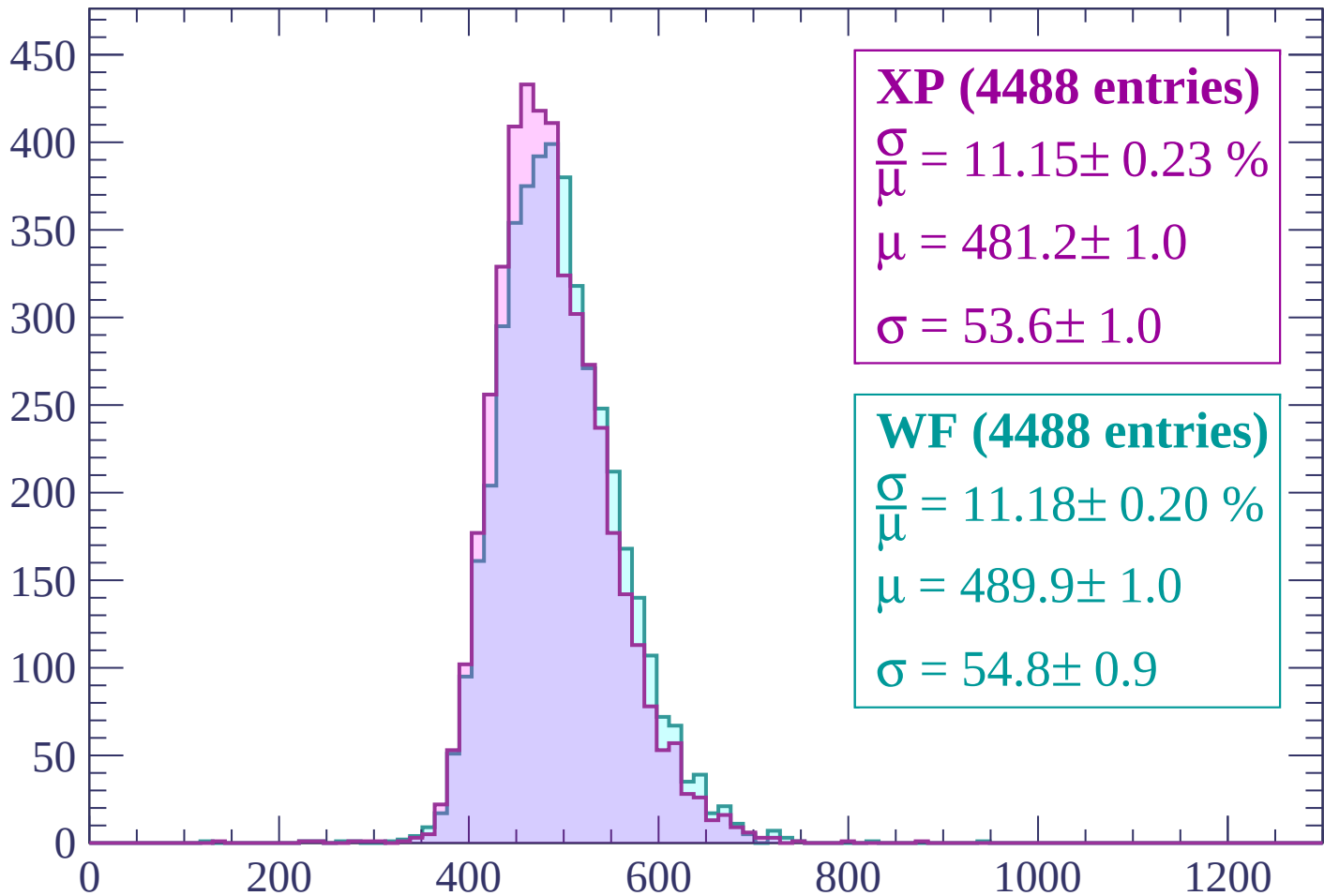
$$\mu = 490.7 \pm 1.0$$

$$\sigma = 55.8 \pm 1.0$$

dE/dx (ADC counts/cm)

# Energy loss | -1800 < p < -1740

Count



**XP (4488 entries)**

$$\frac{\sigma}{\mu} = 11.15 \pm 0.23 \%$$

$$\mu = 481.2 \pm 1.0$$

$$\sigma = 53.6 \pm 1.0$$

**WF (4488 entries)**

$$\frac{\sigma}{\mu} = 11.18 \pm 0.20 \%$$

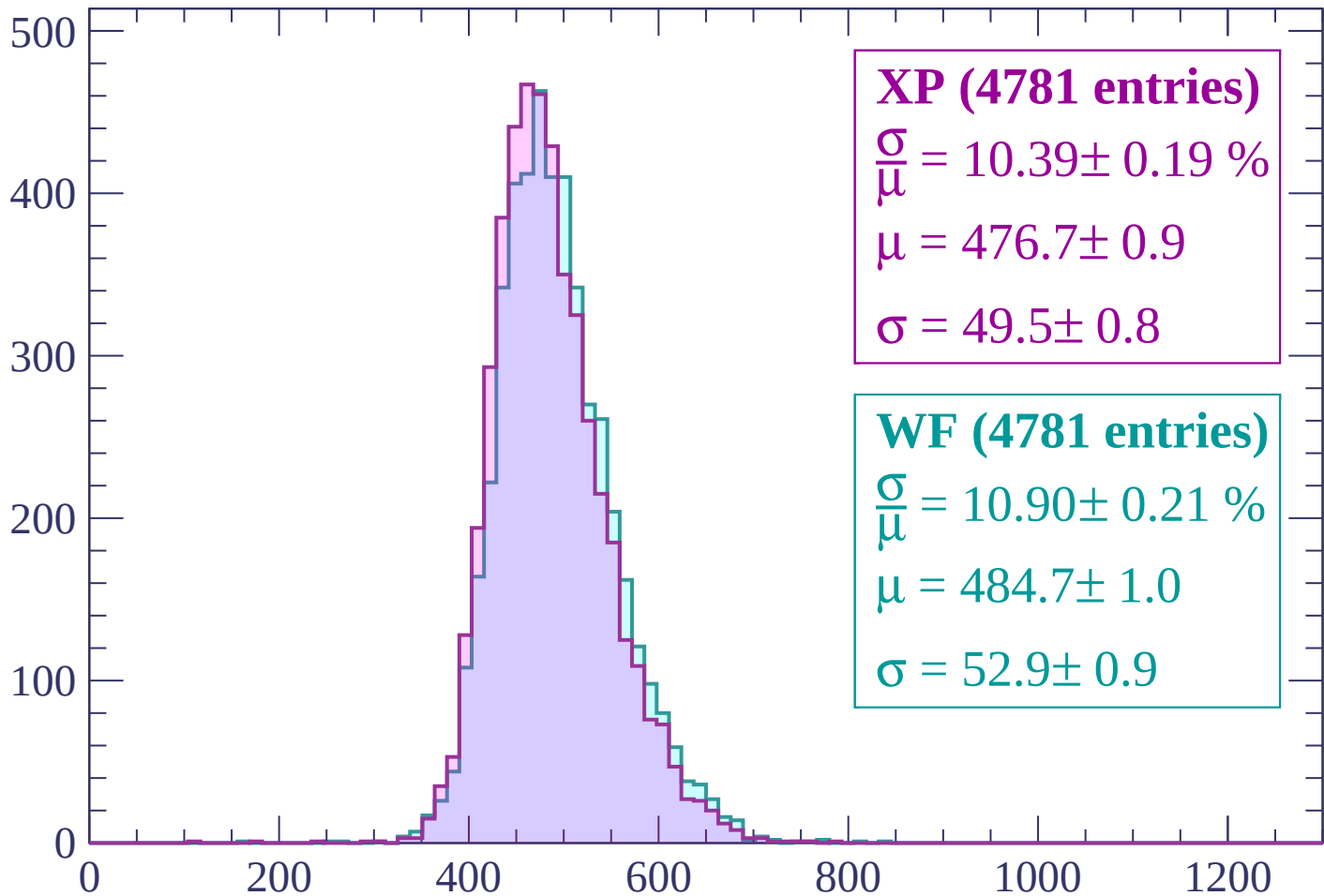
$$\mu = 489.9 \pm 1.0$$

$$\sigma = 54.8 \pm 0.9$$

dE/dx (ADC counts/cm)

# Energy loss | -1740 < p < -1680

Count



**XP (4781 entries)**

$$\frac{\sigma}{\mu} = 10.39 \pm 0.19 \%$$

$$\mu = 476.7 \pm 0.9$$

$$\sigma = 49.5 \pm 0.8$$

**WF (4781 entries)**

$$\frac{\sigma}{\mu} = 10.90 \pm 0.21 \%$$

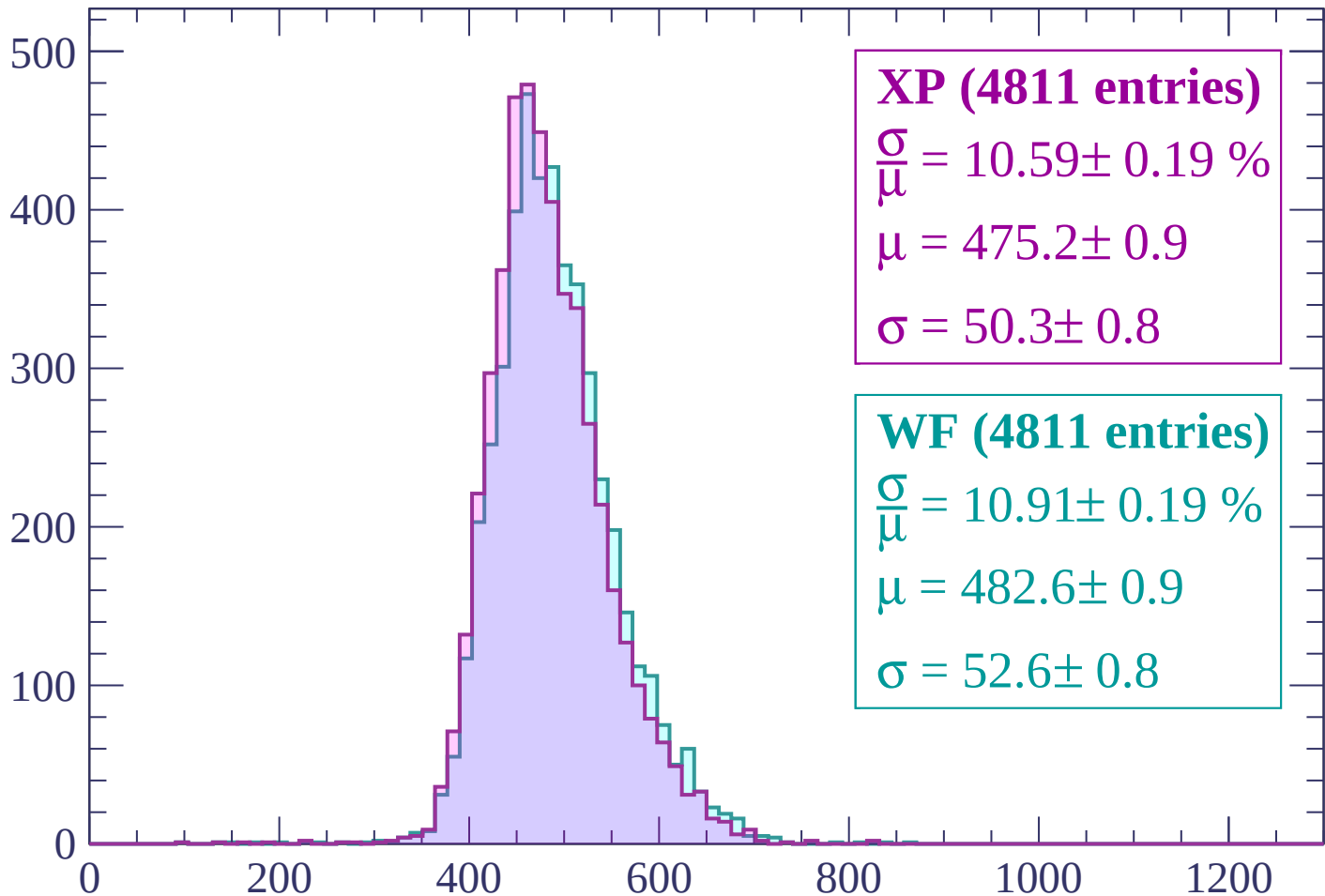
$$\mu = 484.7 \pm 1.0$$

$$\sigma = 52.9 \pm 0.9$$

dE/dx (ADC counts/cm)

# Energy loss | -1680 < p < -1620

Count



**XP (4811 entries)**

$$\frac{\sigma}{\mu} = 10.59 \pm 0.19 \%$$

$$\mu = 475.2 \pm 0.9$$

$$\sigma = 50.3 \pm 0.8$$

**WF (4811 entries)**

$$\frac{\sigma}{\mu} = 10.91 \pm 0.19 \%$$

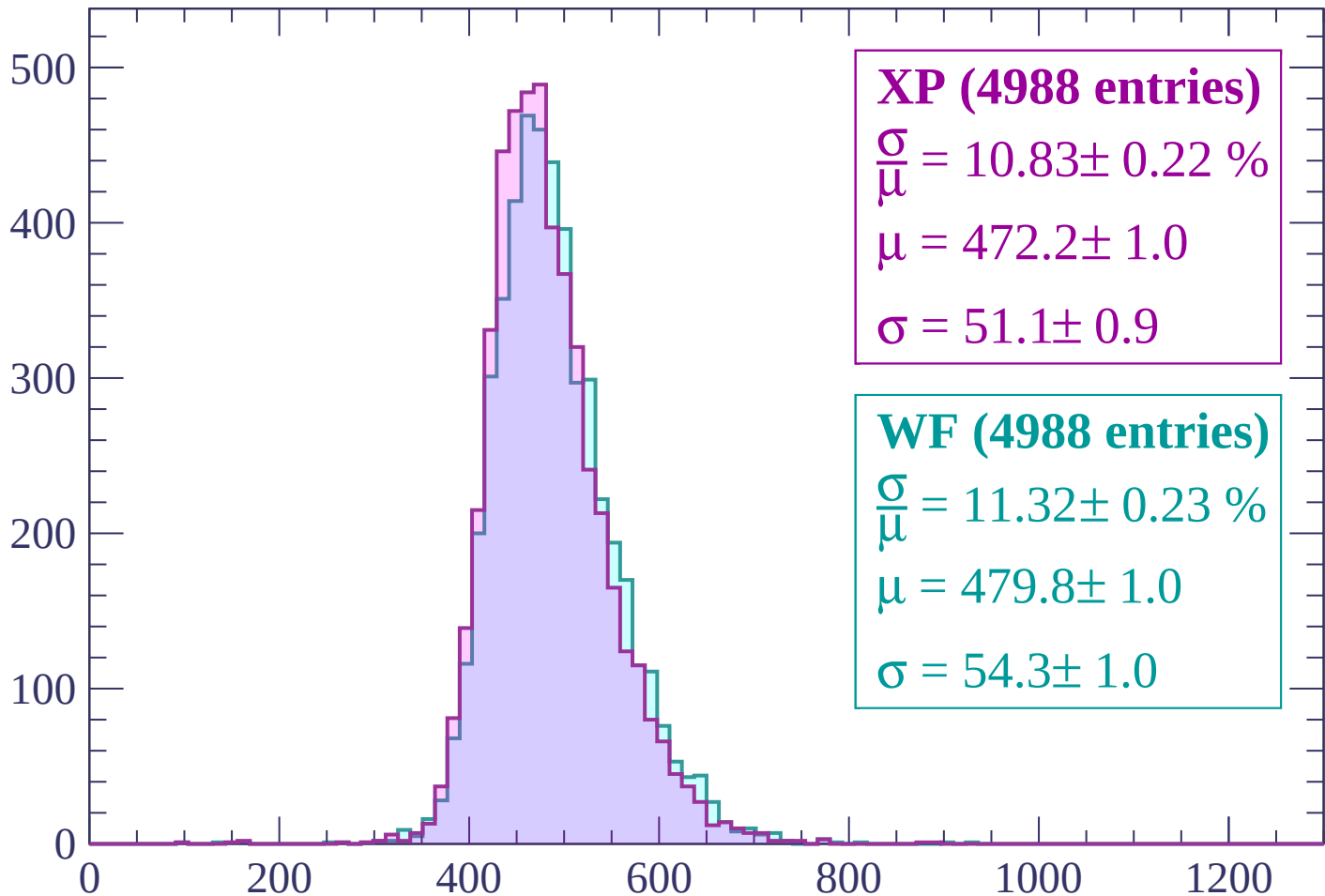
$$\mu = 482.6 \pm 0.9$$

$$\sigma = 52.6 \pm 0.8$$

dE/dx (ADC counts/cm)

# Energy loss | -1620 < p < -1560

Count



**XP (4988 entries)**

$$\frac{\sigma}{\mu} = 10.83 \pm 0.22 \%$$

$$\mu = 472.2 \pm 1.0$$

$$\sigma = 51.1 \pm 0.9$$

**WF (4988 entries)**

$$\frac{\sigma}{\mu} = 11.32 \pm 0.23 \%$$

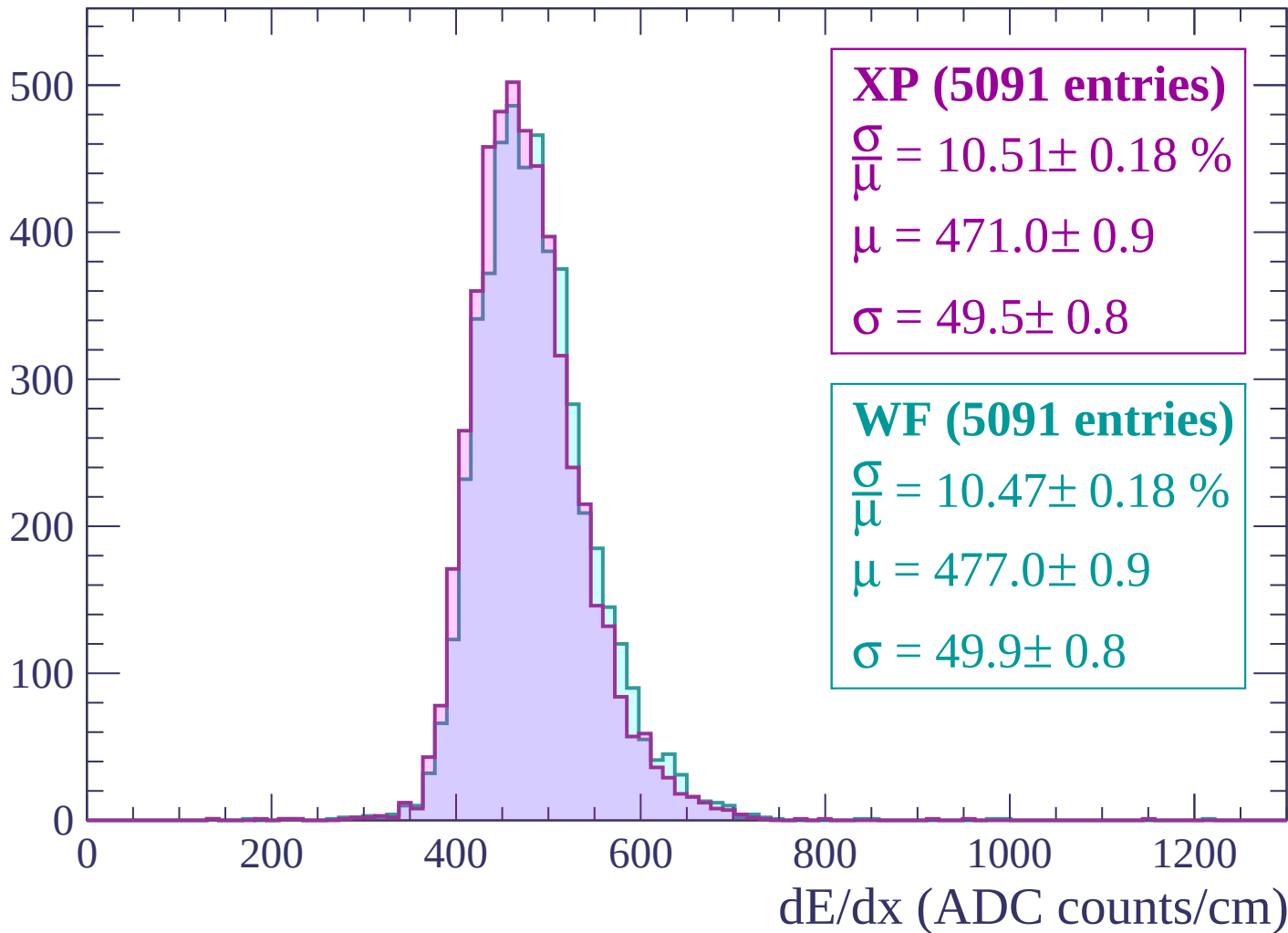
$$\mu = 479.8 \pm 1.0$$

$$\sigma = 54.3 \pm 1.0$$

dE/dx (ADC counts/cm)

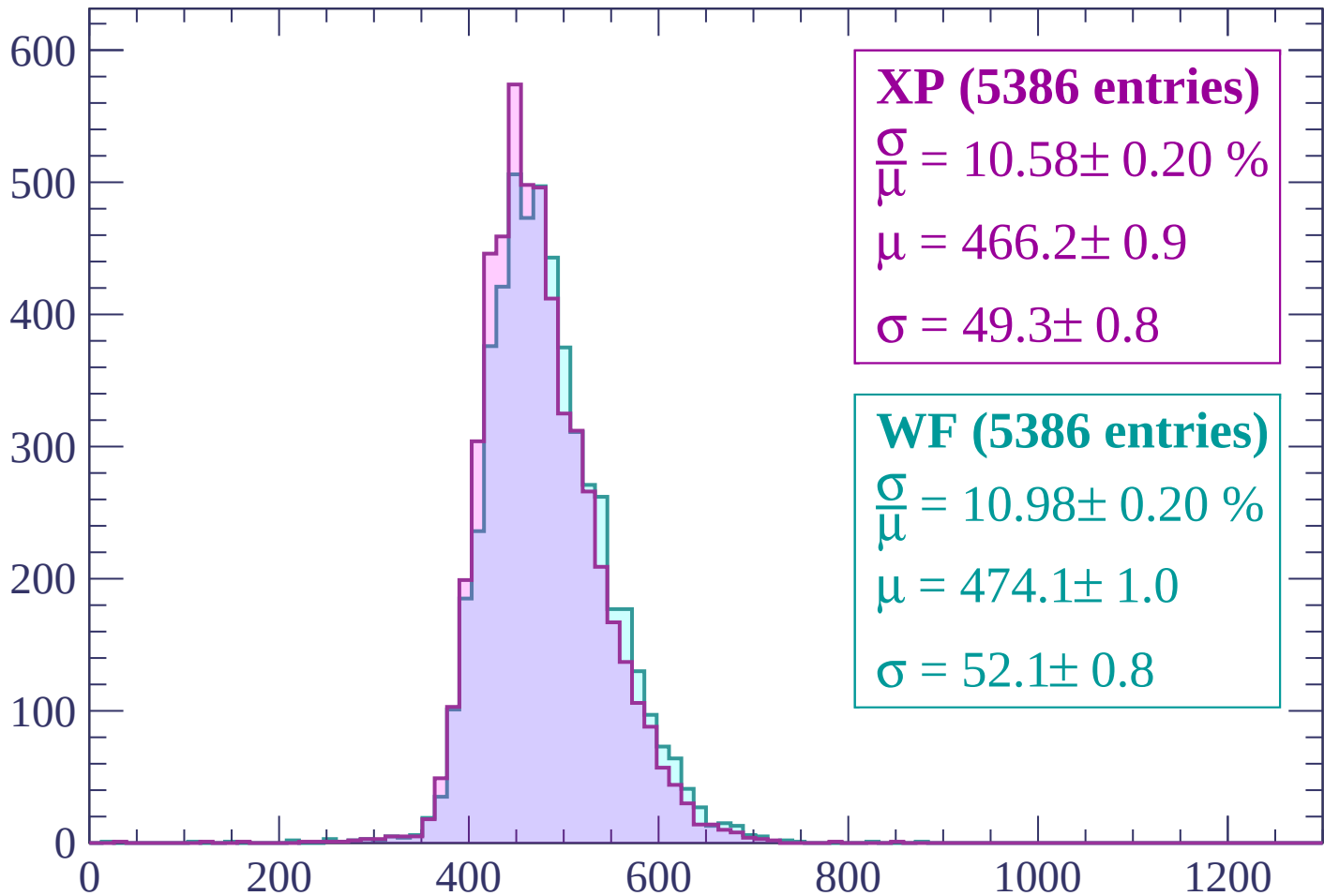
# Energy loss | -1560 < p < -1500

Count



# Energy loss | $-1500 < p < -1440$

Count



**XP (5386 entries)**

$\frac{\sigma}{\mu} = 10.58 \pm 0.20 \%$

$\mu = 466.2 \pm 0.9$

$\sigma = 49.3 \pm 0.8$

**WF (5386 entries)**

$\frac{\sigma}{\mu} = 10.98 \pm 0.20 \%$

$\mu = 474.1 \pm 1.0$

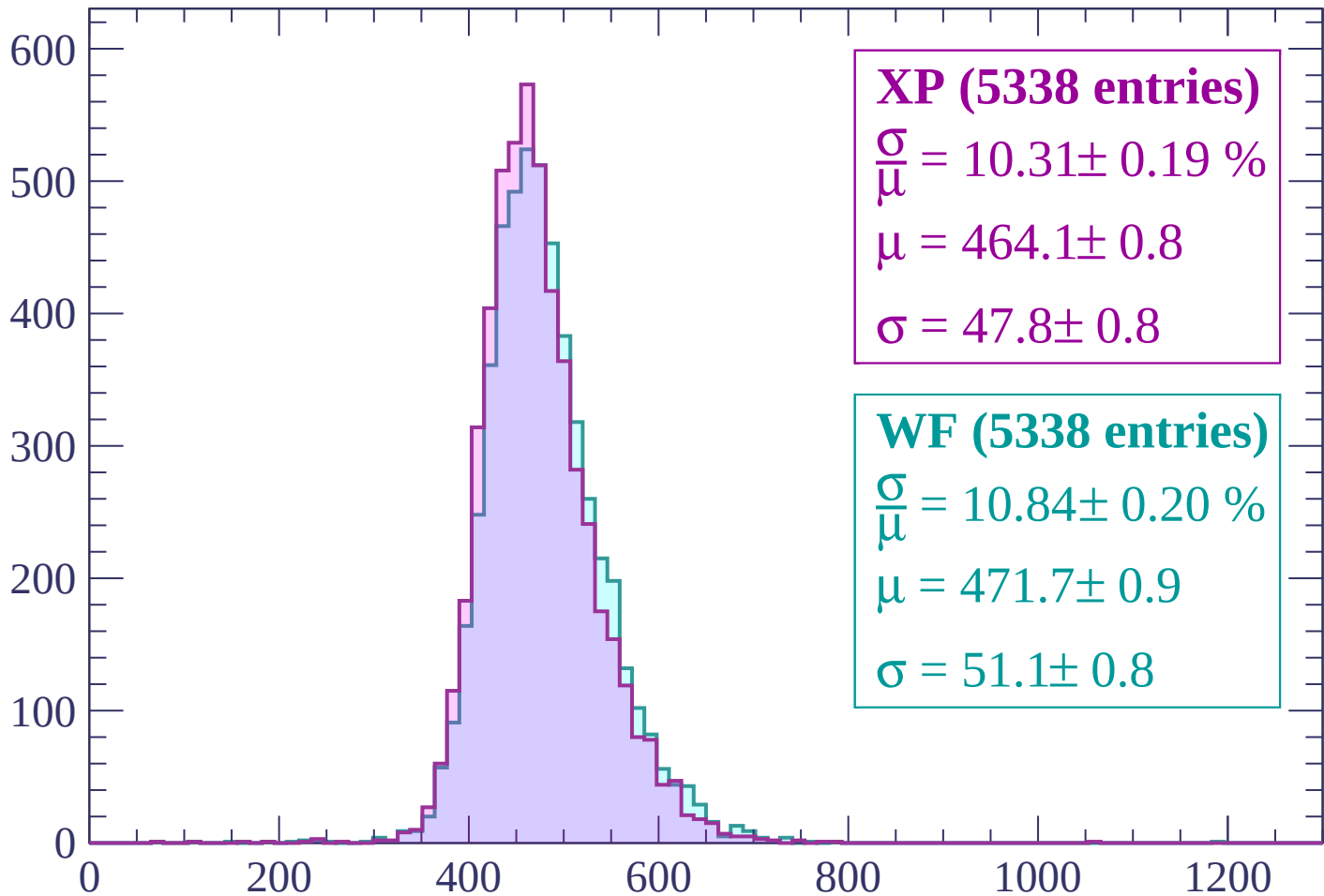
$\sigma = 52.1 \pm 0.8$

dE/dx (ADC counts/cm)



Energy loss |  $-1440 < p < -1380$

Count



**XP (5338 entries)**

$\frac{\sigma}{\mu} = 10.31 \pm 0.19 \%$

$\mu = 464.1 \pm 0.8$

$\sigma = 47.8 \pm 0.8$

**WF (5338 entries)**

$\frac{\sigma}{\mu} = 10.84 \pm 0.20 \%$

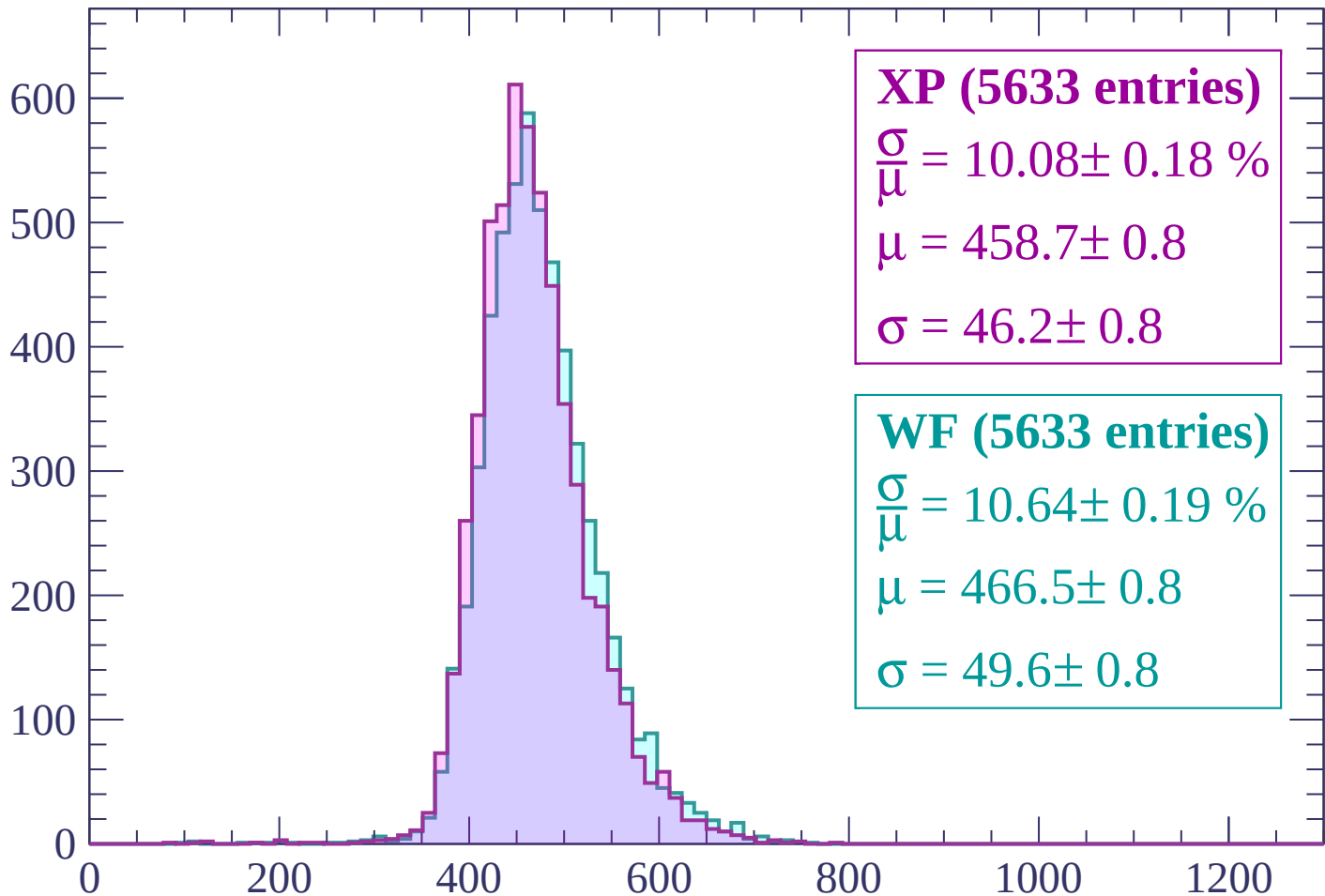
$\mu = 471.7 \pm 0.9$

$\sigma = 51.1 \pm 0.8$

dE/dx (ADC counts/cm)

# Energy loss | -1380 < p < -1320

Count



**XP (5633 entries)**

$$\frac{\sigma}{\mu} = 10.08 \pm 0.18 \%$$

$$\mu = 458.7 \pm 0.8$$

$$\sigma = 46.2 \pm 0.8$$

**WF (5633 entries)**

$$\frac{\sigma}{\mu} = 10.64 \pm 0.19 \%$$

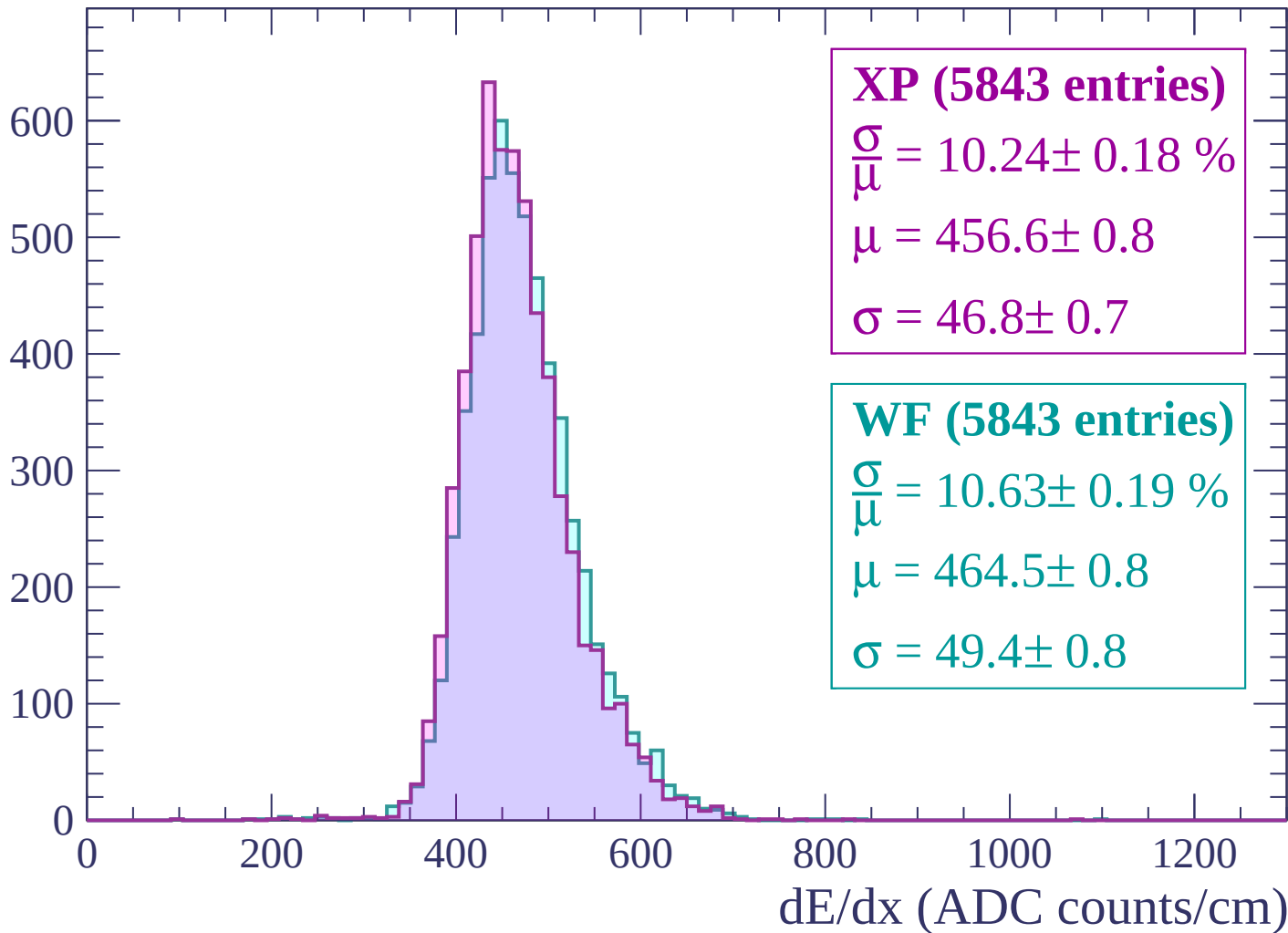
$$\mu = 466.5 \pm 0.8$$

$$\sigma = 49.6 \pm 0.8$$

dE/dx (ADC counts/cm)

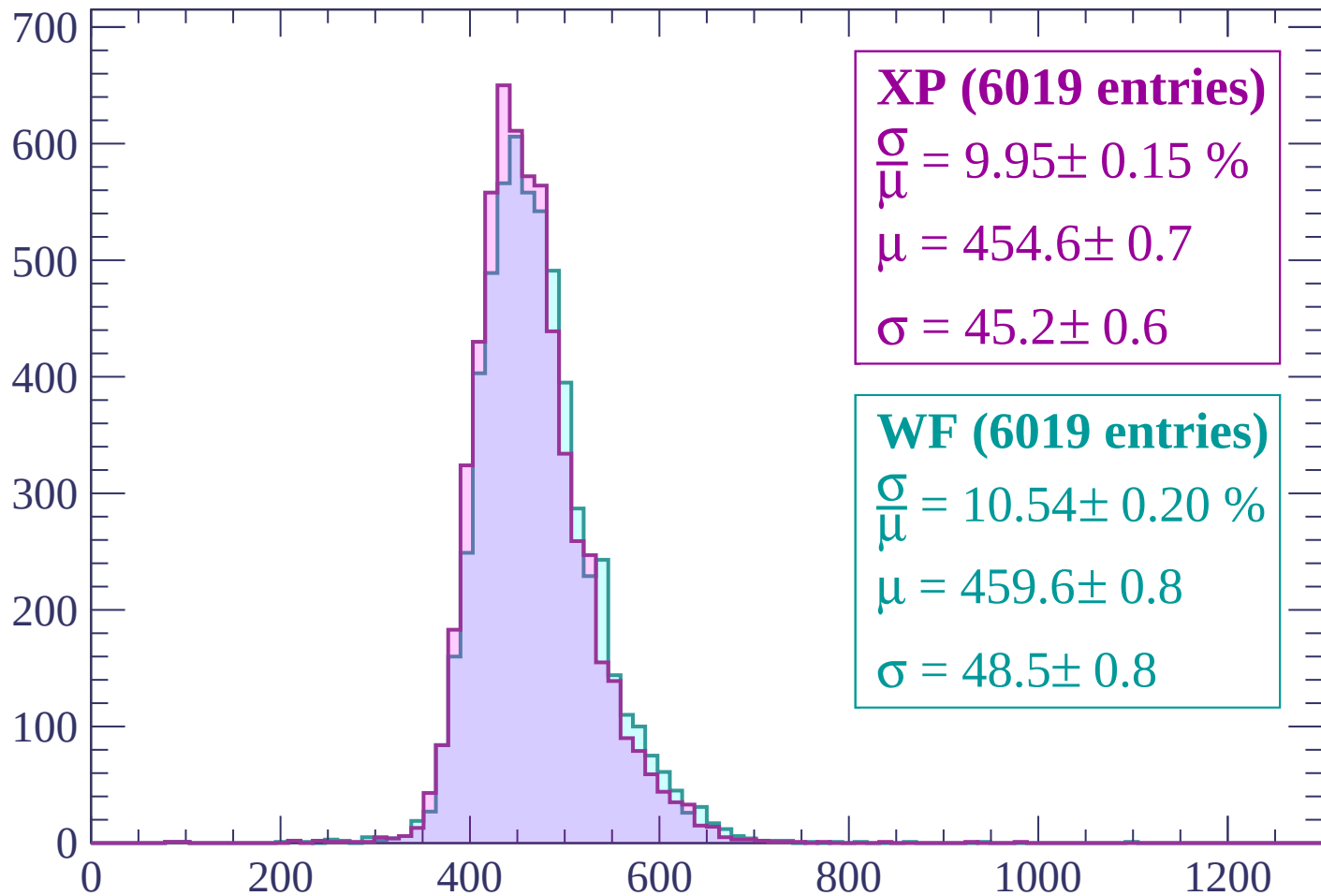
# Energy loss | -1320 < p < -1260

Count



Energy loss |  $-1260 < p < -1200$

Count



**XP (6019 entries)**

$$\frac{\sigma}{\mu} = 9.95 \pm 0.15 \%$$

$$\mu = 454.6 \pm 0.7$$

$$\sigma = 45.2 \pm 0.6$$

**WF (6019 entries)**

$$\frac{\sigma}{\mu} = 10.54 \pm 0.20 \%$$

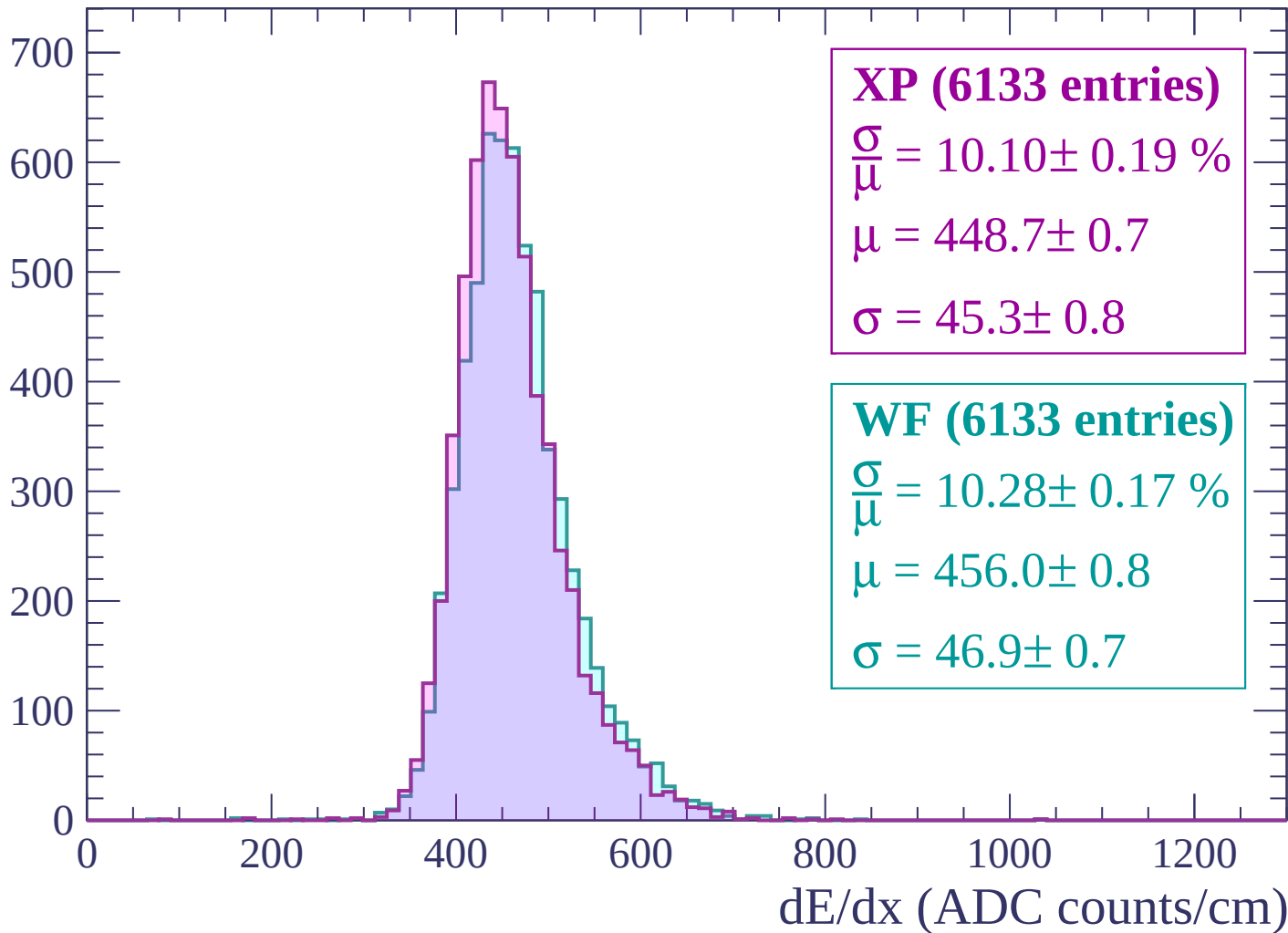
$$\mu = 459.6 \pm 0.8$$

$$\sigma = 48.5 \pm 0.8$$

dE/dx (ADC counts/cm)

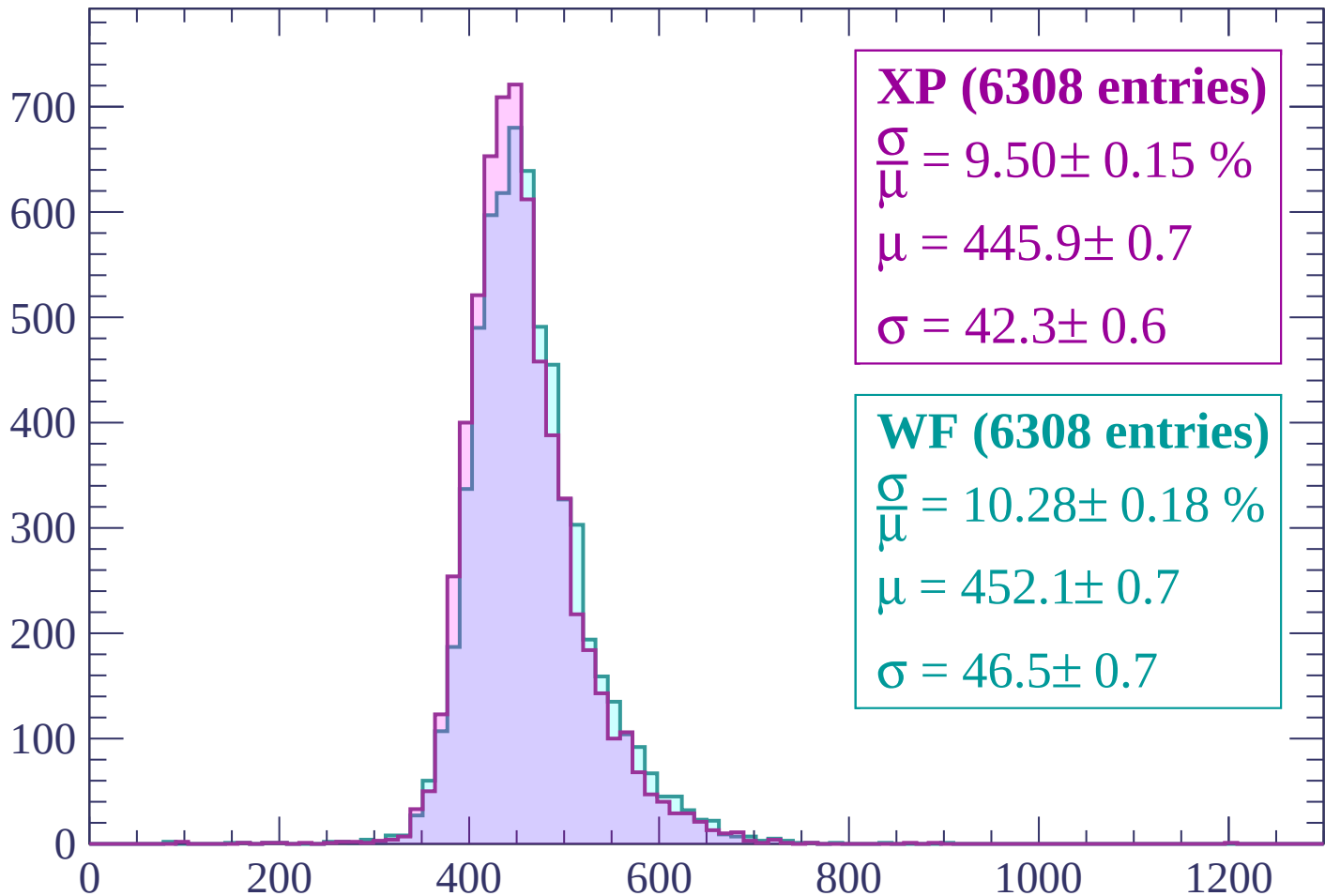
# Energy loss | $-1200 < p < -1140$

Count



Energy loss |  $-1140 < p < -1080$

Count



**XP (6308 entries)**

$$\frac{\sigma}{\mu} = 9.50 \pm 0.15 \%$$

$$\mu = 445.9 \pm 0.7$$

$$\sigma = 42.3 \pm 0.6$$

**WF (6308 entries)**

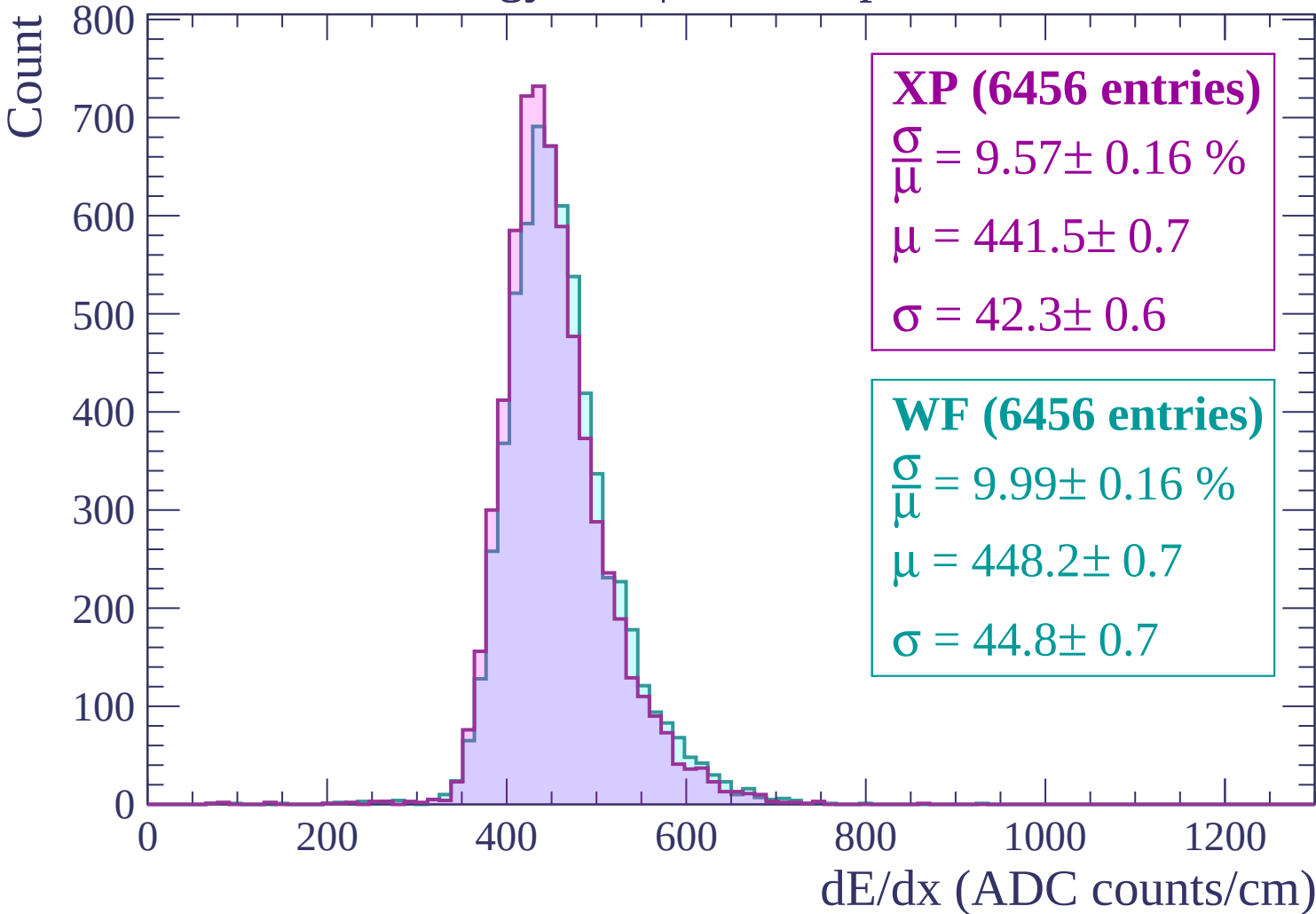
$$\frac{\sigma}{\mu} = 10.28 \pm 0.18 \%$$

$$\mu = 452.1 \pm 0.7$$

$$\sigma = 46.5 \pm 0.7$$

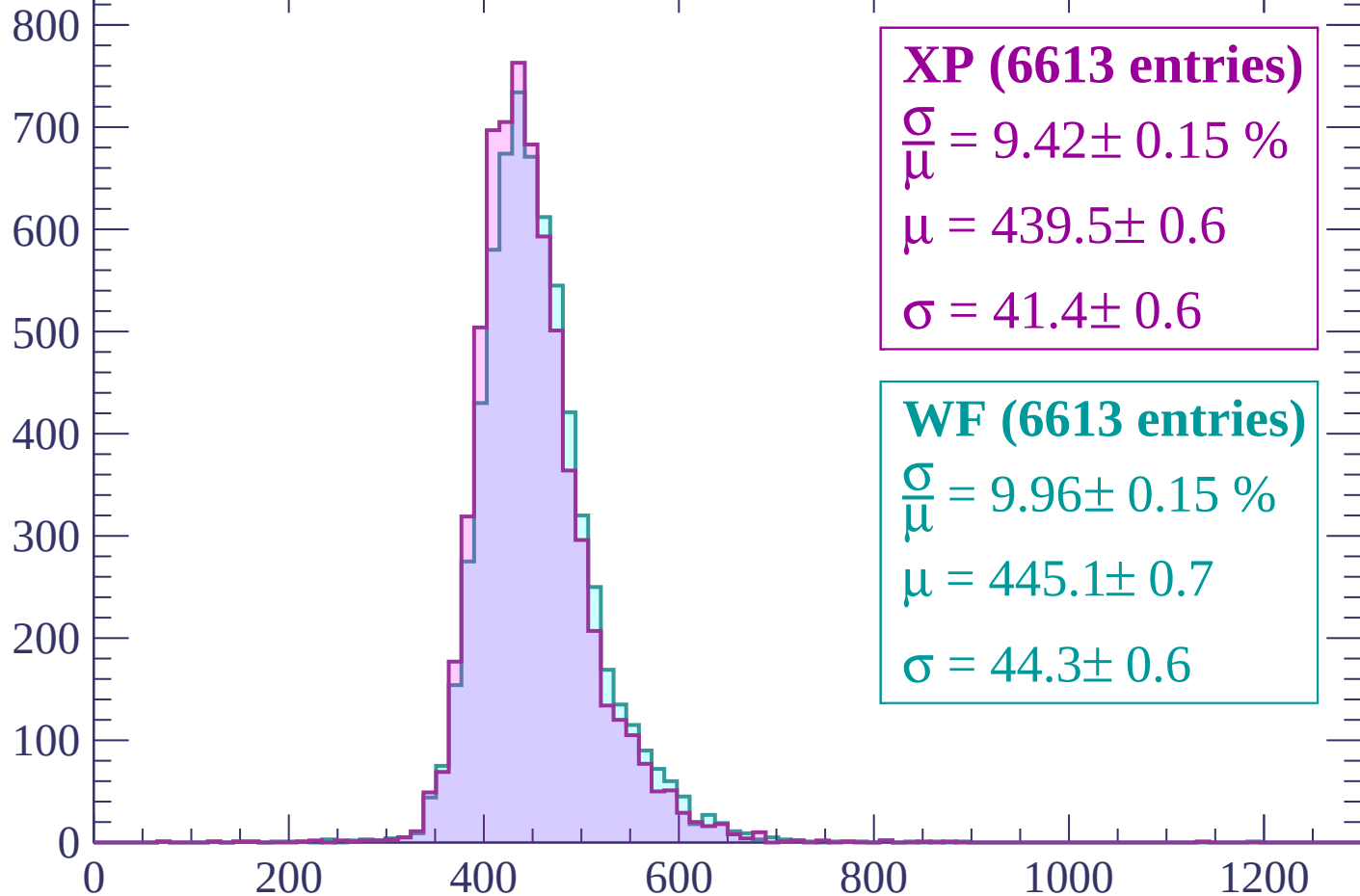
dE/dx (ADC counts/cm)

# Energy loss | $-1080 < p < -1020$



Energy loss |  $-1020 < p < -960$

Count



**XP (6613 entries)**

$$\frac{\sigma}{\mu} = 9.42 \pm 0.15 \%$$

$$\mu = 439.5 \pm 0.6$$

$$\sigma = 41.4 \pm 0.6$$

**WF (6613 entries)**

$$\frac{\sigma}{\mu} = 9.96 \pm 0.15 \%$$

$$\mu = 445.1 \pm 0.7$$

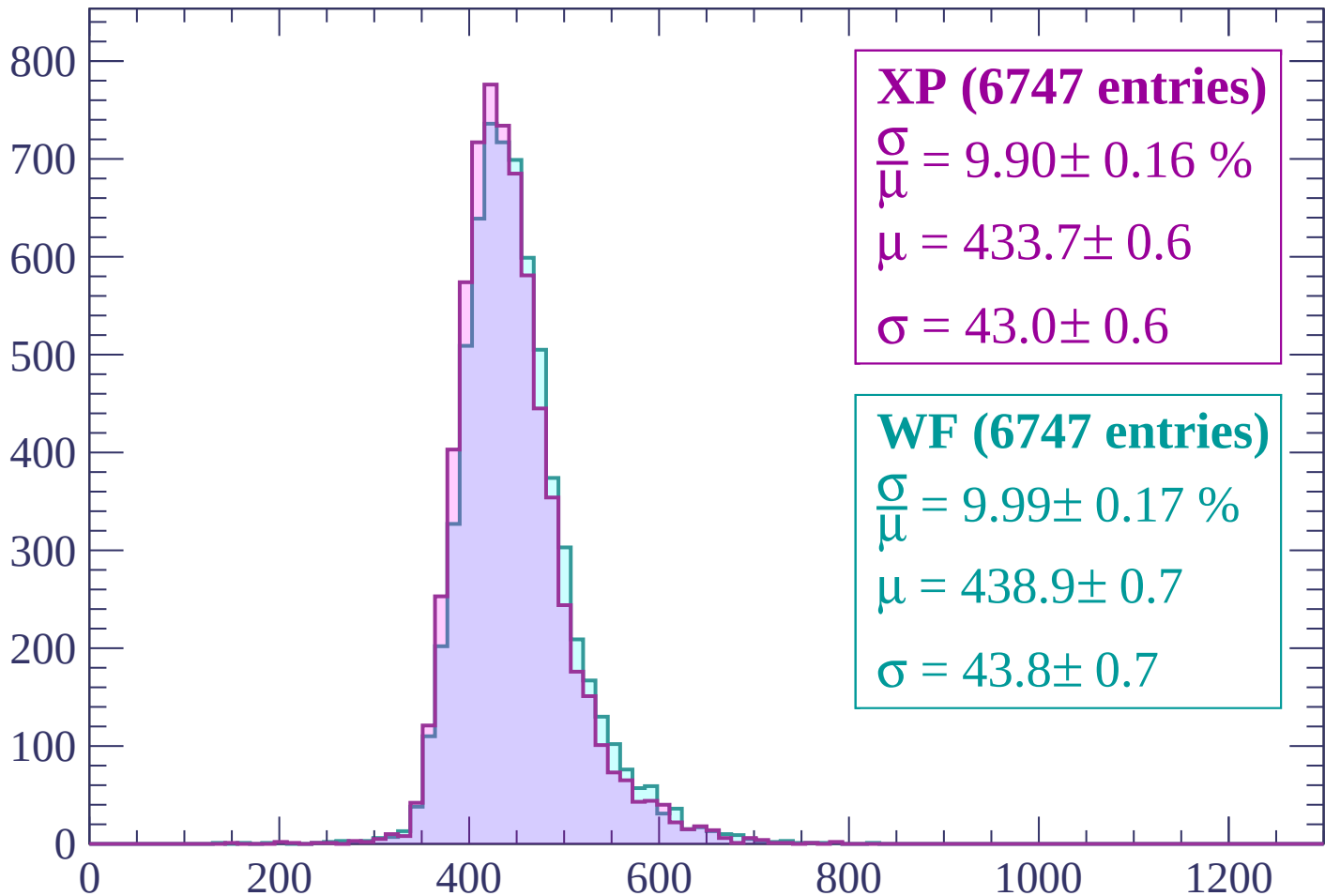
$$\sigma = 44.3 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss |  $-960 < p < -900$

Count



**XP (6747 entries)**

$$\frac{\sigma}{\mu} = 9.90 \pm 0.16 \%$$

$$\mu = 433.7 \pm 0.6$$

$$\sigma = 43.0 \pm 0.6$$

**WF (6747 entries)**

$$\frac{\sigma}{\mu} = 9.99 \pm 0.17 \%$$

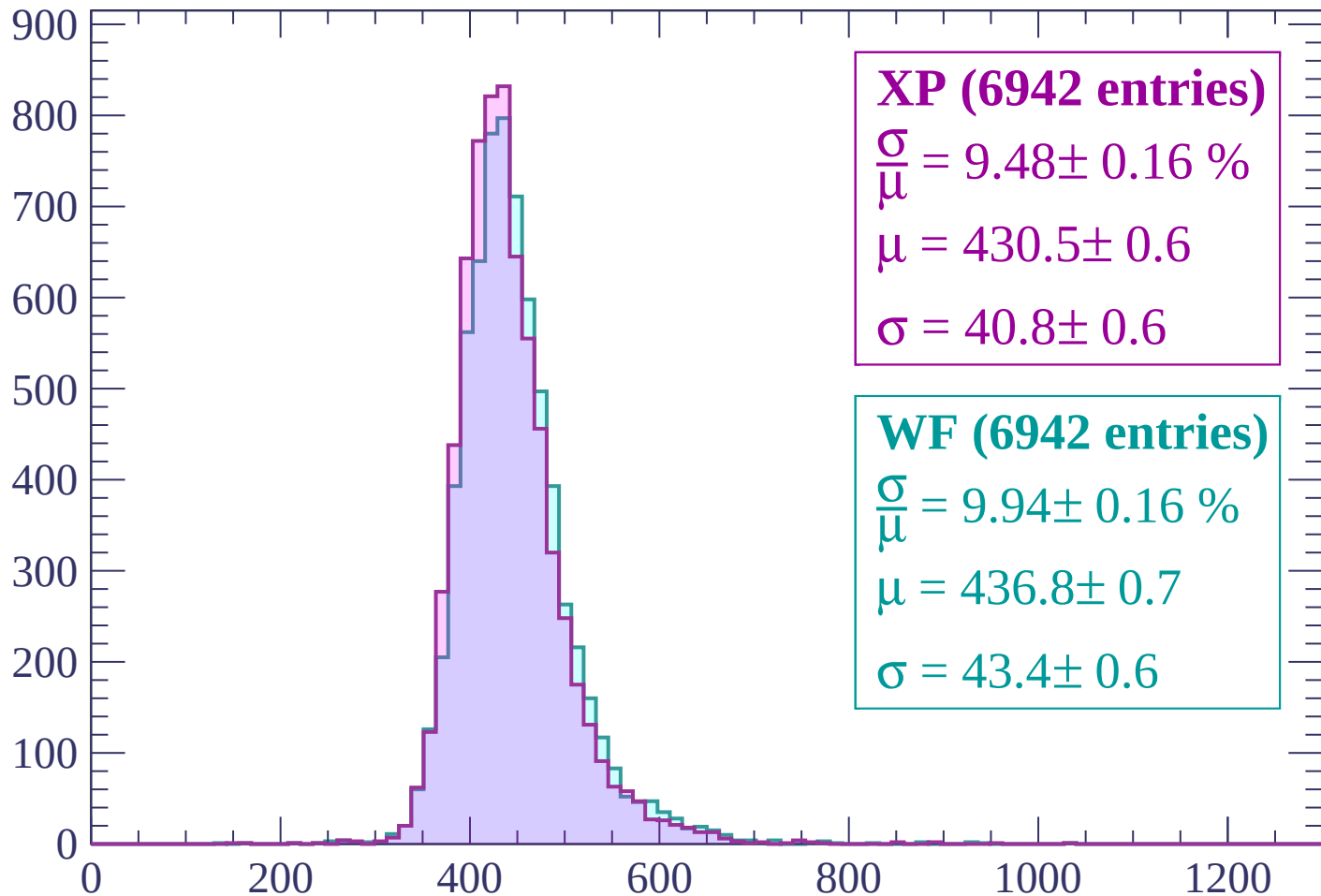
$$\mu = 438.9 \pm 0.7$$

$$\sigma = 43.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-900 < p < -840$

Count



**XP (6942 entries)**

$$\frac{\sigma}{\mu} = 9.48 \pm 0.16 \%$$

$$\mu = 430.5 \pm 0.6$$

$$\sigma = 40.8 \pm 0.6$$

**WF (6942 entries)**

$$\frac{\sigma}{\mu} = 9.94 \pm 0.16 \%$$

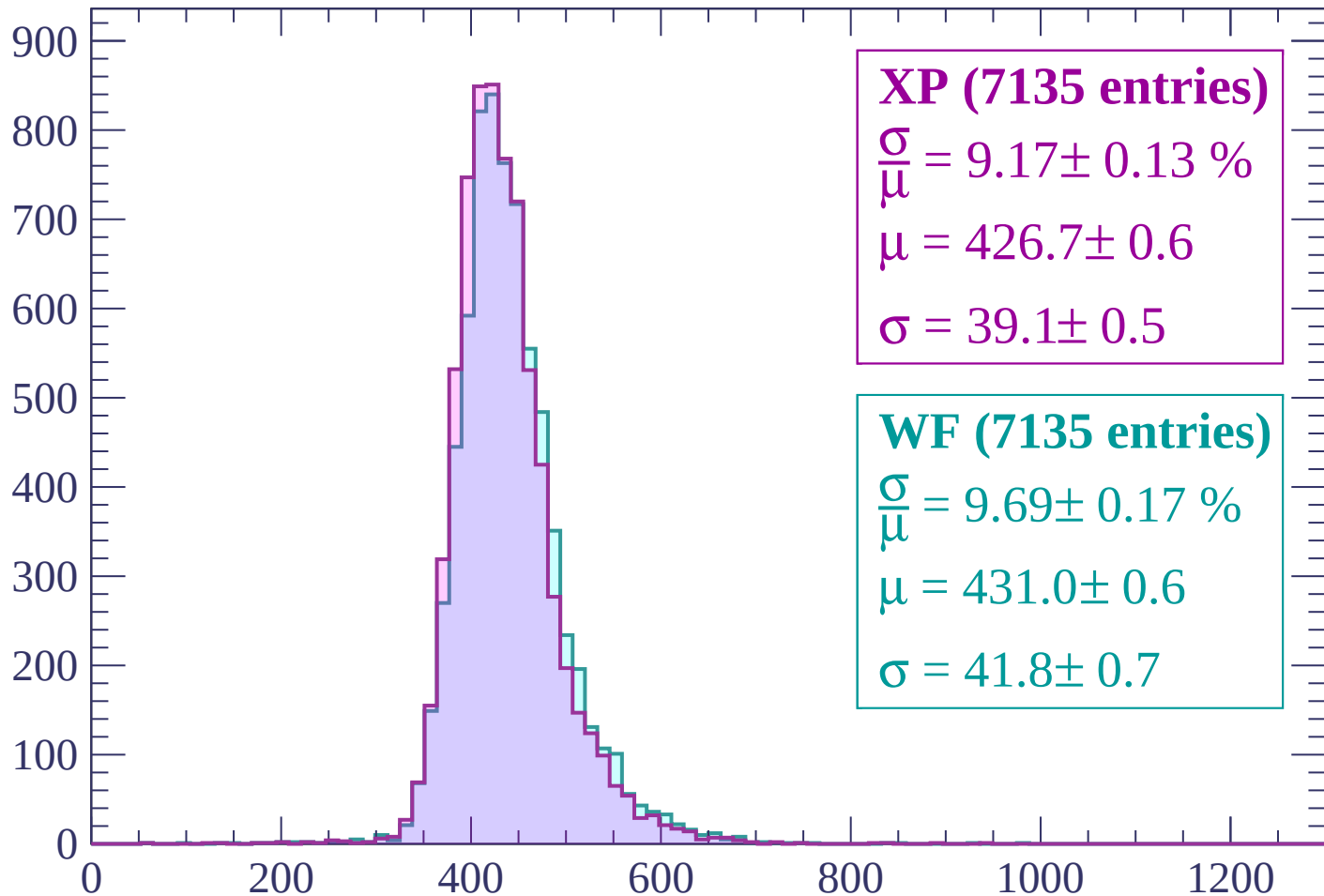
$$\mu = 436.8 \pm 0.7$$

$$\sigma = 43.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $-840 < p < -780$

Count



**XP (7135 entries)**

$\frac{\sigma}{\mu} = 9.17 \pm 0.13 \%$

$\mu = 426.7 \pm 0.6$

$\sigma = 39.1 \pm 0.5$

**WF (7135 entries)**

$\frac{\sigma}{\mu} = 9.69 \pm 0.17 \%$

$\mu = 431.0 \pm 0.6$

$\sigma = 41.8 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss |  $-780 < p < -720$

Count

1000

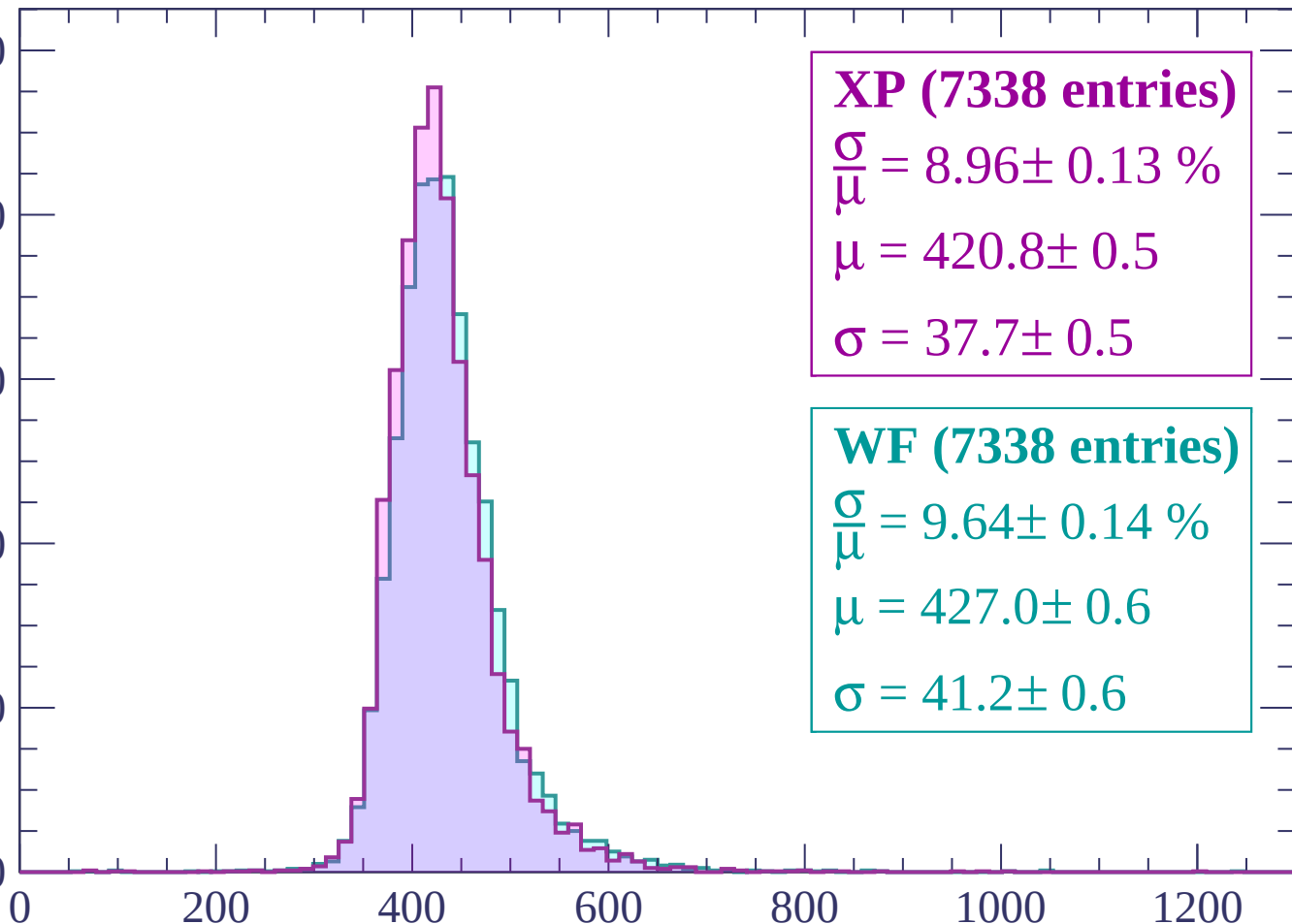
800

600

400

200

0



**XP (7338 entries)**

$\frac{\sigma}{\mu} = 8.96 \pm 0.13 \%$

$\mu = 420.8 \pm 0.5$

$\sigma = 37.7 \pm 0.5$

**WF (7338 entries)**

$\frac{\sigma}{\mu} = 9.64 \pm 0.14 \%$

$\mu = 427.0 \pm 0.6$

$\sigma = 41.2 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $-720 < p < -660$

Count

1000

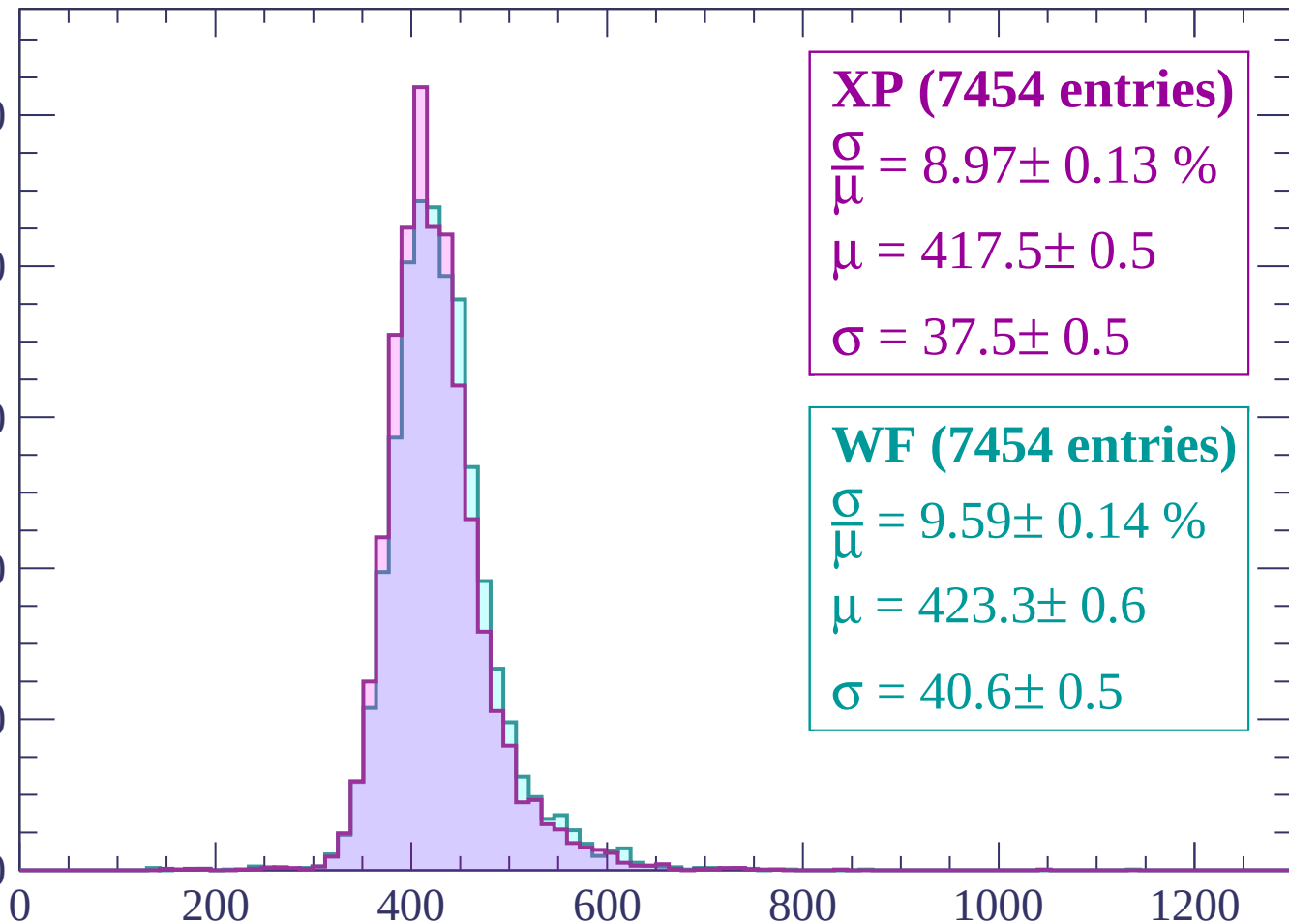
800

600

400

200

0



**XP (7454 entries)**

$\frac{\sigma}{\mu} = 8.97 \pm 0.13 \%$

$\mu = 417.5 \pm 0.5$

$\sigma = 37.5 \pm 0.5$

**WF (7454 entries)**

$\frac{\sigma}{\mu} = 9.59 \pm 0.14 \%$

$\mu = 423.3 \pm 0.6$

$\sigma = 40.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss |  $-660 < p < -600$

Count

1000

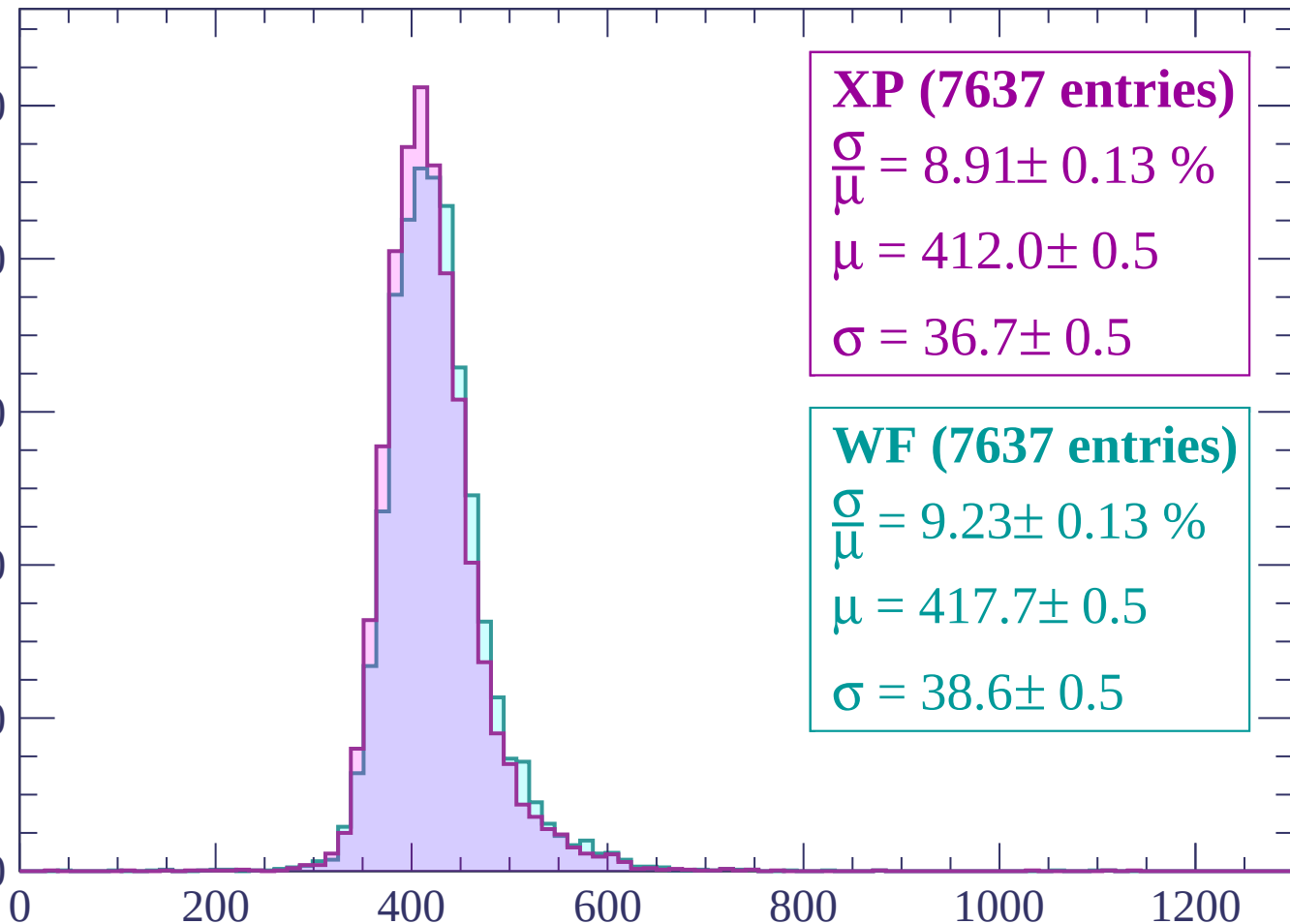
800

600

400

200

0



**XP (7637 entries)**

$\frac{\sigma}{\mu} = 8.91 \pm 0.13 \%$

$\mu = 412.0 \pm 0.5$

$\sigma = 36.7 \pm 0.5$

**WF (7637 entries)**

$\frac{\sigma}{\mu} = 9.23 \pm 0.13 \%$

$\mu = 417.7 \pm 0.5$

$\sigma = 38.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss |  $-600 < p < -540$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (7590 entries)**

$$\frac{\sigma}{\mu} = 8.80 \pm 0.13 \%$$

$$\mu = 408.2 \pm 0.5$$

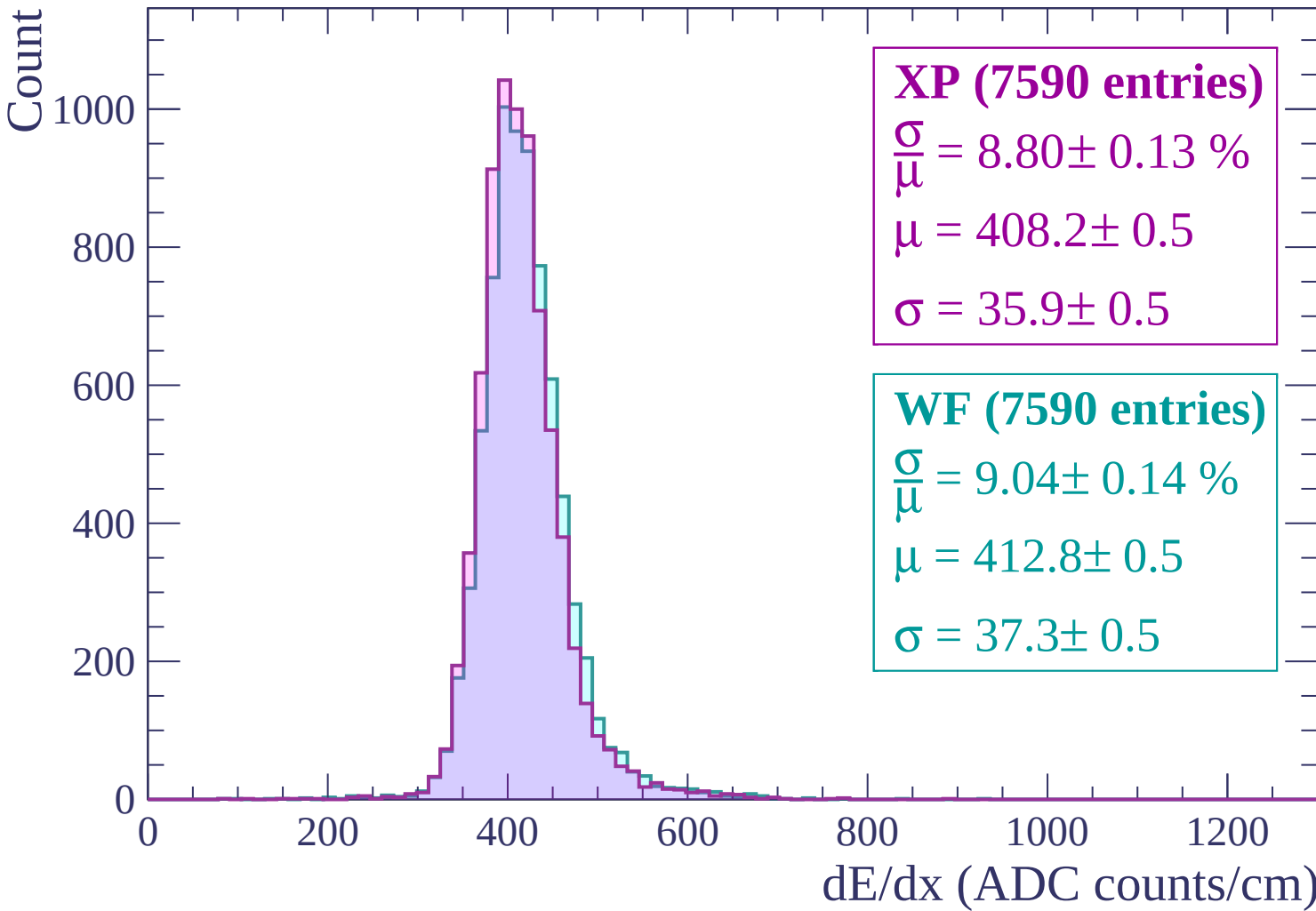
$$\sigma = 35.9 \pm 0.5$$

**WF (7590 entries)**

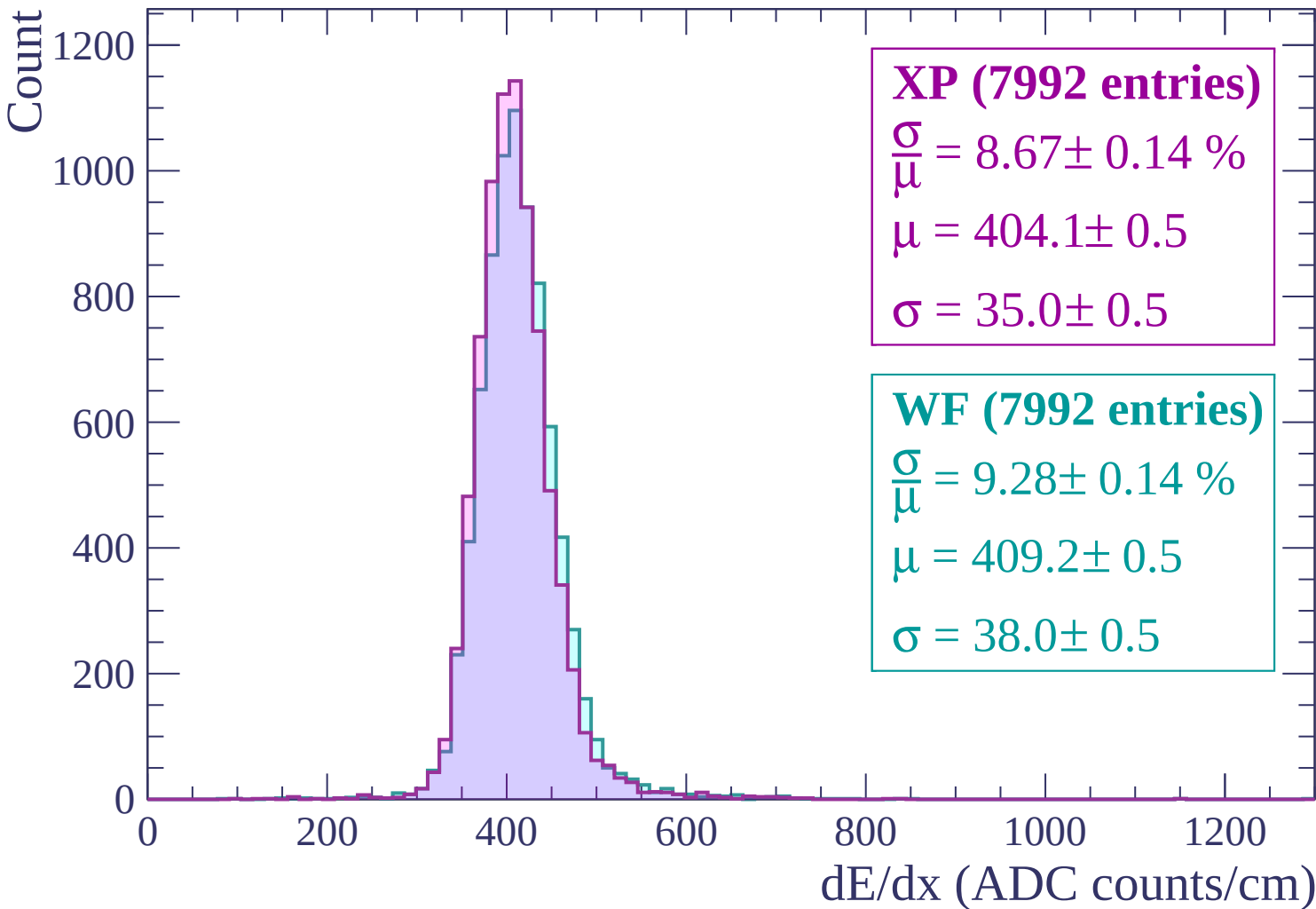
$$\frac{\sigma}{\mu} = 9.04 \pm 0.14 \%$$

$$\mu = 412.8 \pm 0.5$$

$$\sigma = 37.3 \pm 0.5$$



Energy loss |  $-540 < p < -480$





Energy loss |  $-480 < p < -420$

Count

1200

1000

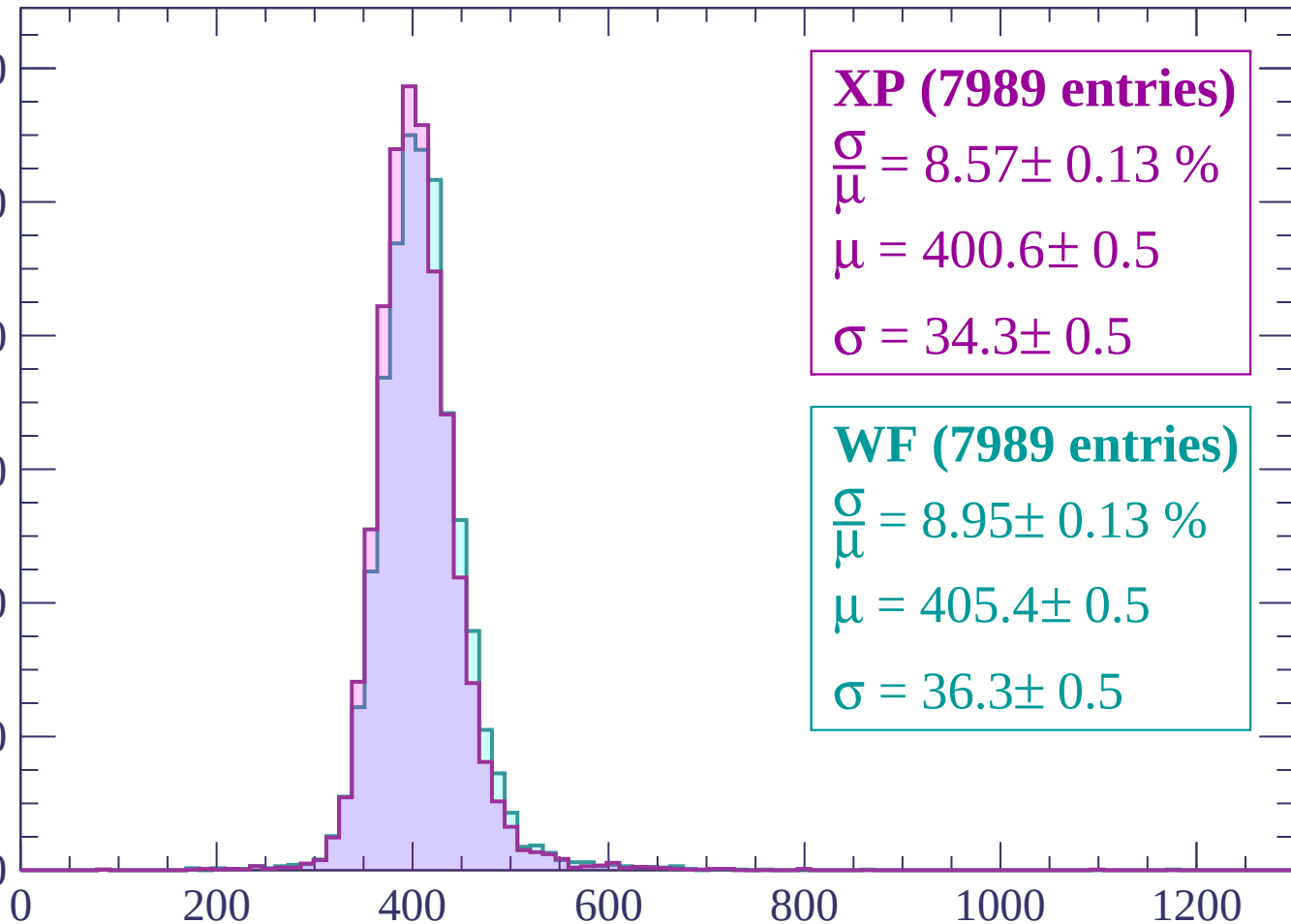
800

600

400

200

0



**XP (7989 entries)**

$\frac{\sigma}{\mu} = 8.57 \pm 0.13 \%$

$\mu = 400.6 \pm 0.5$

$\sigma = 34.3 \pm 0.5$

**WF (7989 entries)**

$\frac{\sigma}{\mu} = 8.95 \pm 0.13 \%$

$\mu = 405.4 \pm 0.5$

$\sigma = 36.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss |  $-420 < p < -360$

Count

1200

1000

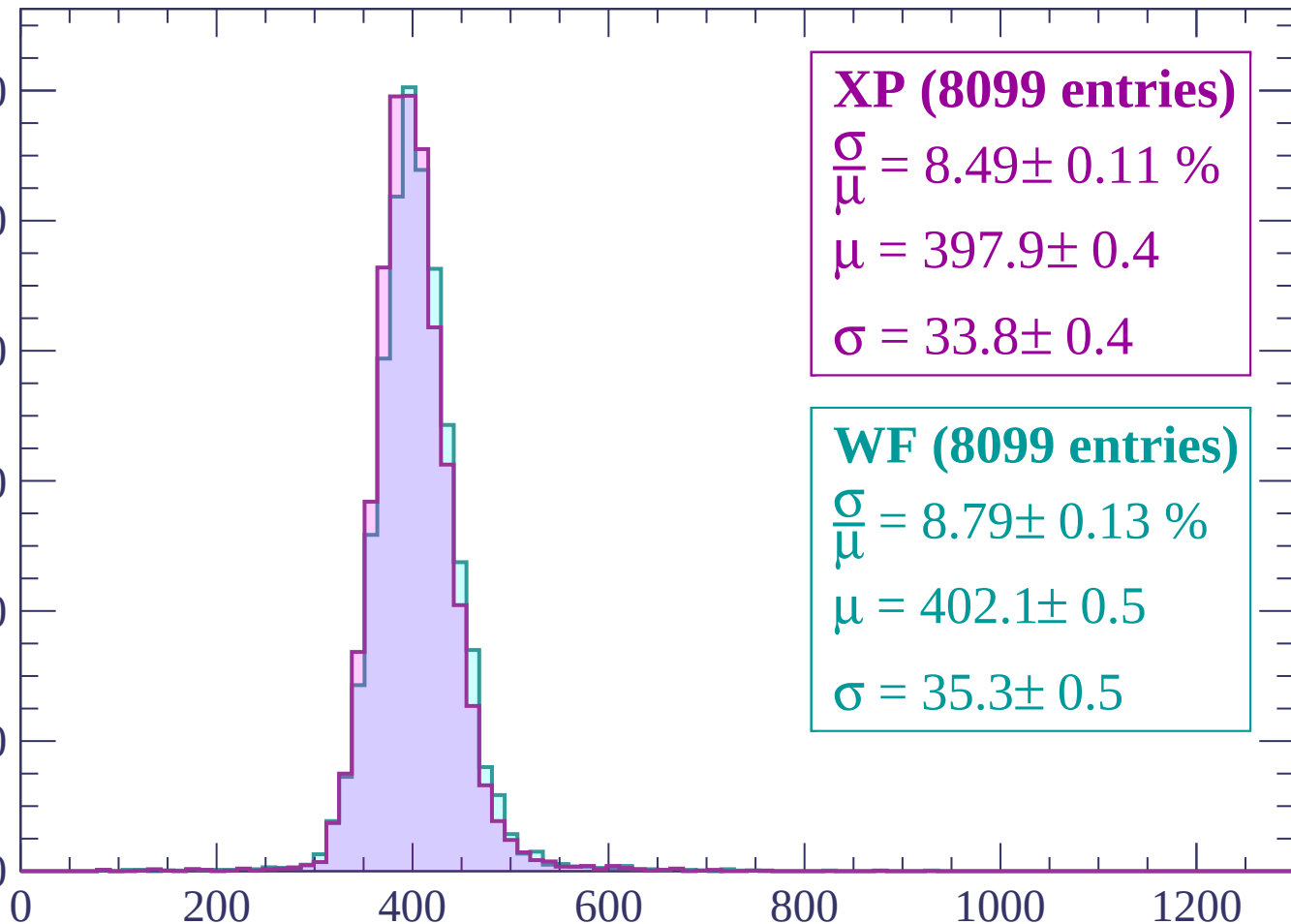
800

600

400

200

0



**XP (8099 entries)**

$\frac{\sigma}{\mu} = 8.49 \pm 0.11 \%$

$\mu = 397.9 \pm 0.4$

$\sigma = 33.8 \pm 0.4$

**WF (8099 entries)**

$\frac{\sigma}{\mu} = 8.79 \pm 0.13 \%$

$\mu = 402.1 \pm 0.5$

$\sigma = 35.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss |  $-360 < p < -300$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

**XP (7985 entries)**

$$\frac{\sigma}{\mu} = 8.62 \pm 0.11 \%$$

$$\mu = 396.8 \pm 0.4$$

$$\sigma = 34.2 \pm 0.4$$

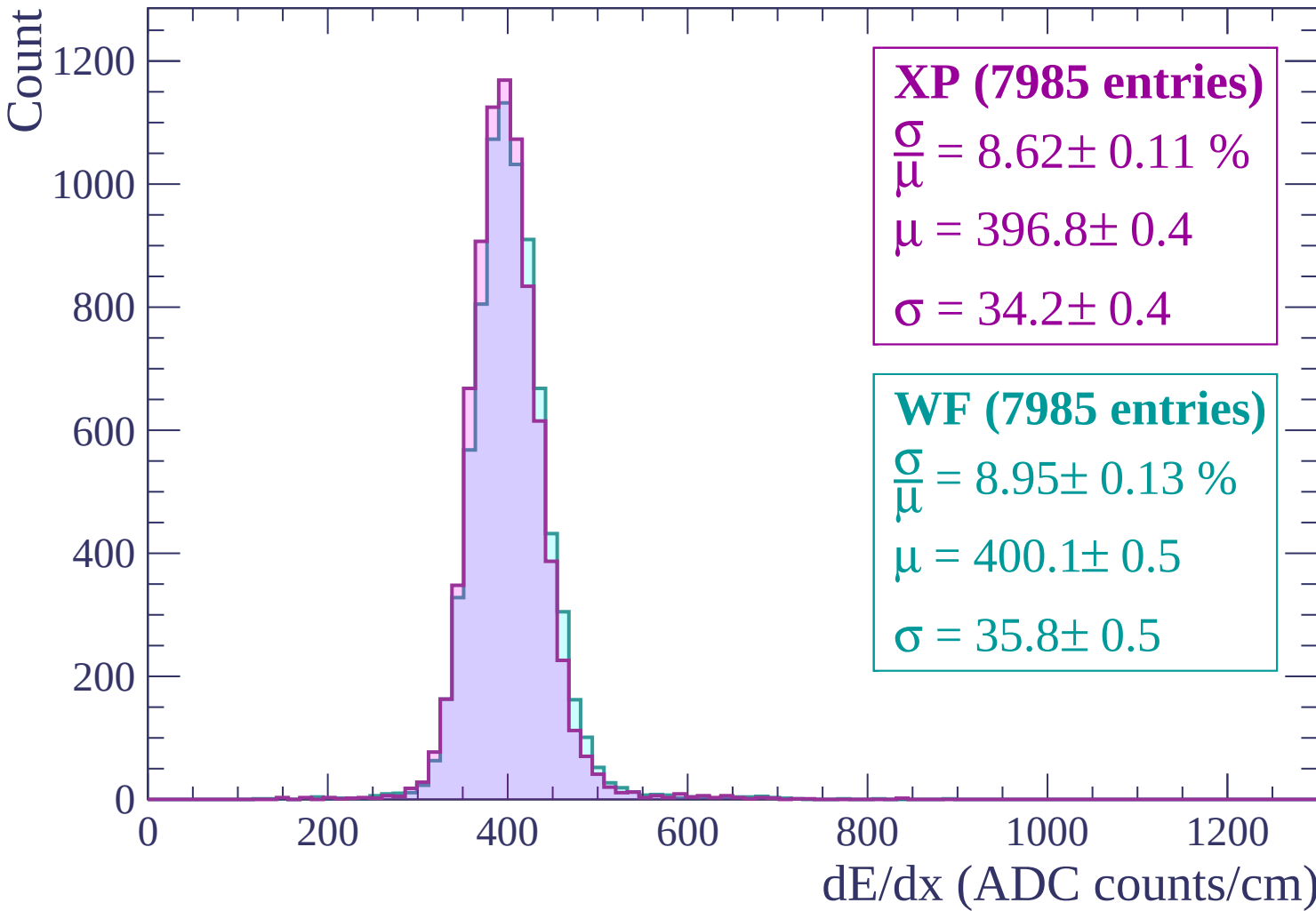
**WF (7985 entries)**

$$\frac{\sigma}{\mu} = 8.95 \pm 0.13 \%$$

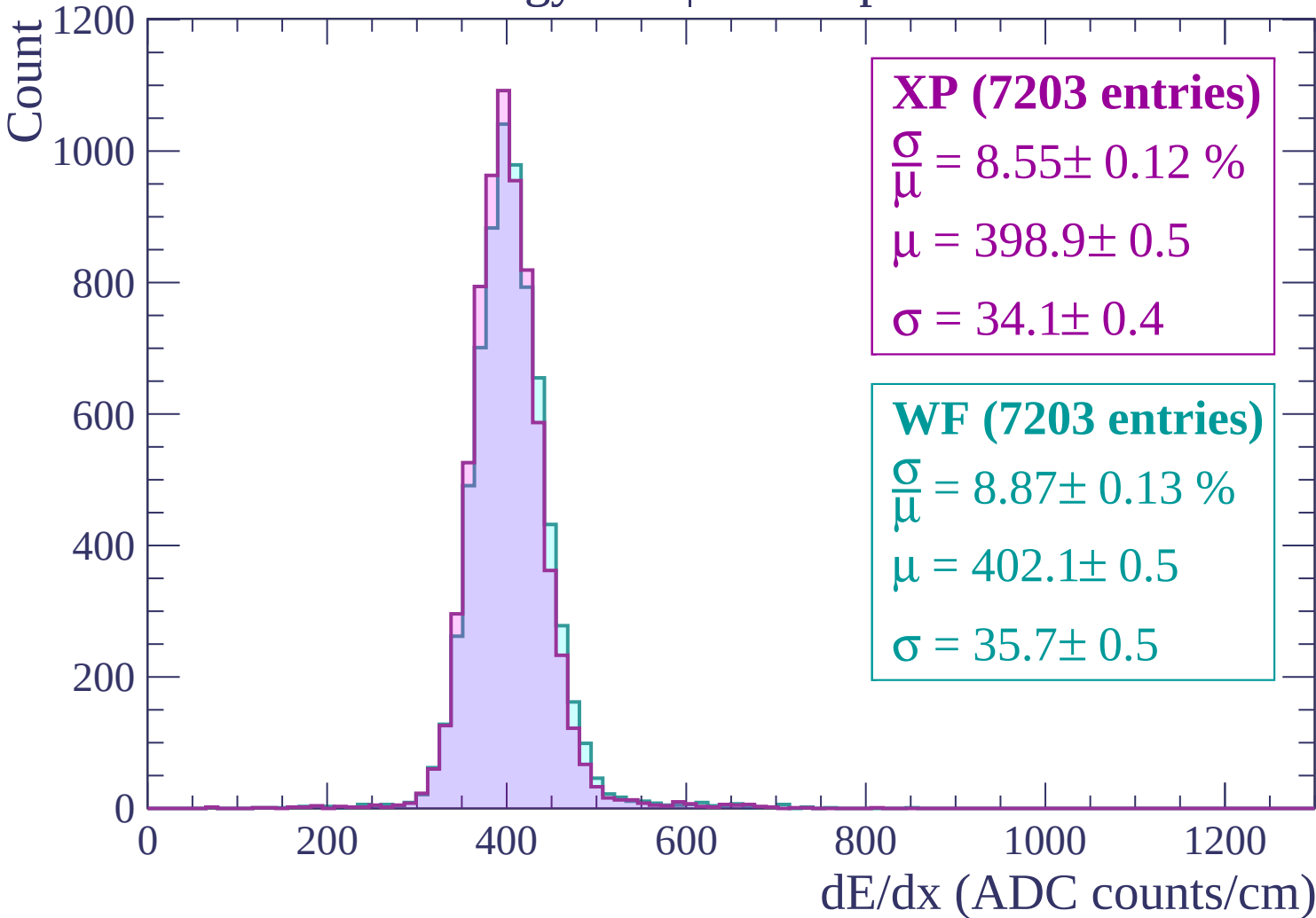
$$\mu = 400.1 \pm 0.5$$

$$\sigma = 35.8 \pm 0.5$$

dE/dx (ADC counts/cm)

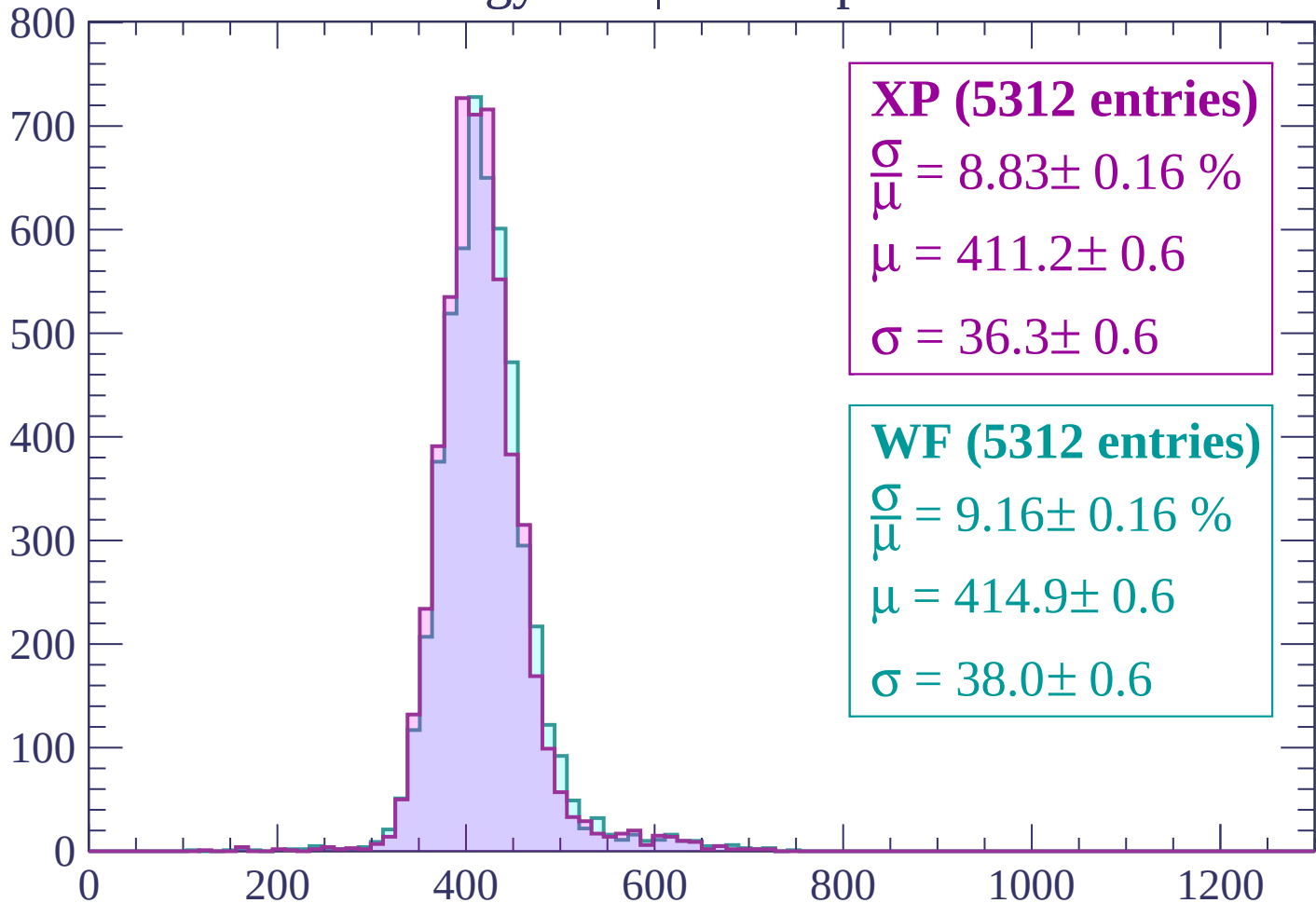


Energy loss |  $-300 < p < -240$



Energy loss |  $-240 < p < -180$

Count



**XP (5312 entries)**

$$\frac{\sigma}{\mu} = 8.83 \pm 0.16 \%$$

$$\mu = 411.2 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

**WF (5312 entries)**

$$\frac{\sigma}{\mu} = 9.16 \pm 0.16 \%$$

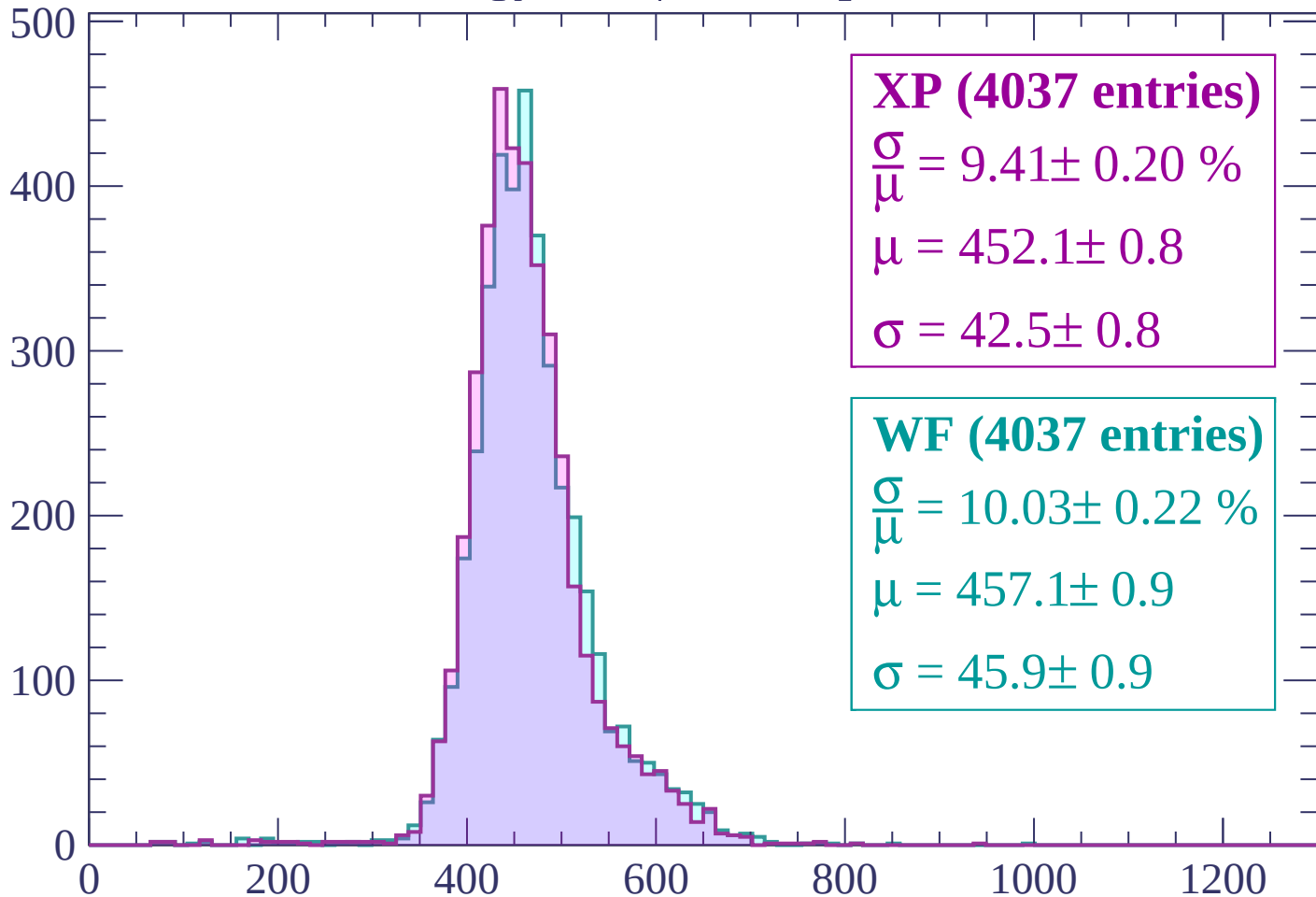
$$\mu = 414.9 \pm 0.6$$

$$\sigma = 38.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $-180 < p < -120$

Count



**XP (4037 entries)**

$$\frac{\sigma}{\mu} = 9.41 \pm 0.20 \%$$

$$\mu = 452.1 \pm 0.8$$

$$\sigma = 42.5 \pm 0.8$$

**WF (4037 entries)**

$$\frac{\sigma}{\mu} = 10.03 \pm 0.22 \%$$

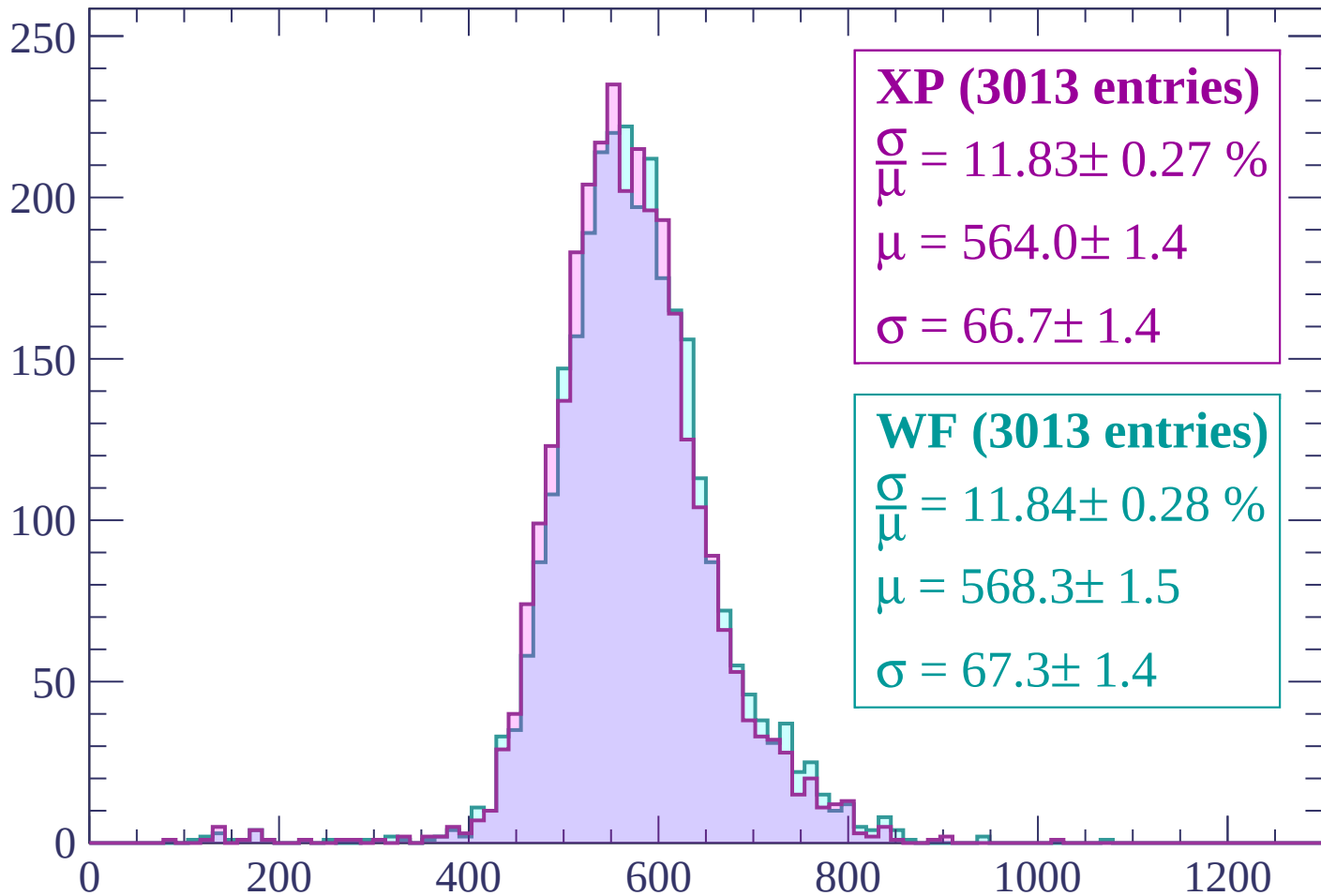
$$\mu = 457.1 \pm 0.9$$

$$\sigma = 45.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $-120 < p < -60$

Count



**XP (3013 entries)**

$$\frac{\sigma}{\mu} = 11.83 \pm 0.27 \%$$

$$\mu = 564.0 \pm 1.4$$

$$\sigma = 66.7 \pm 1.4$$

**WF (3013 entries)**

$$\frac{\sigma}{\mu} = 11.84 \pm 0.28 \%$$

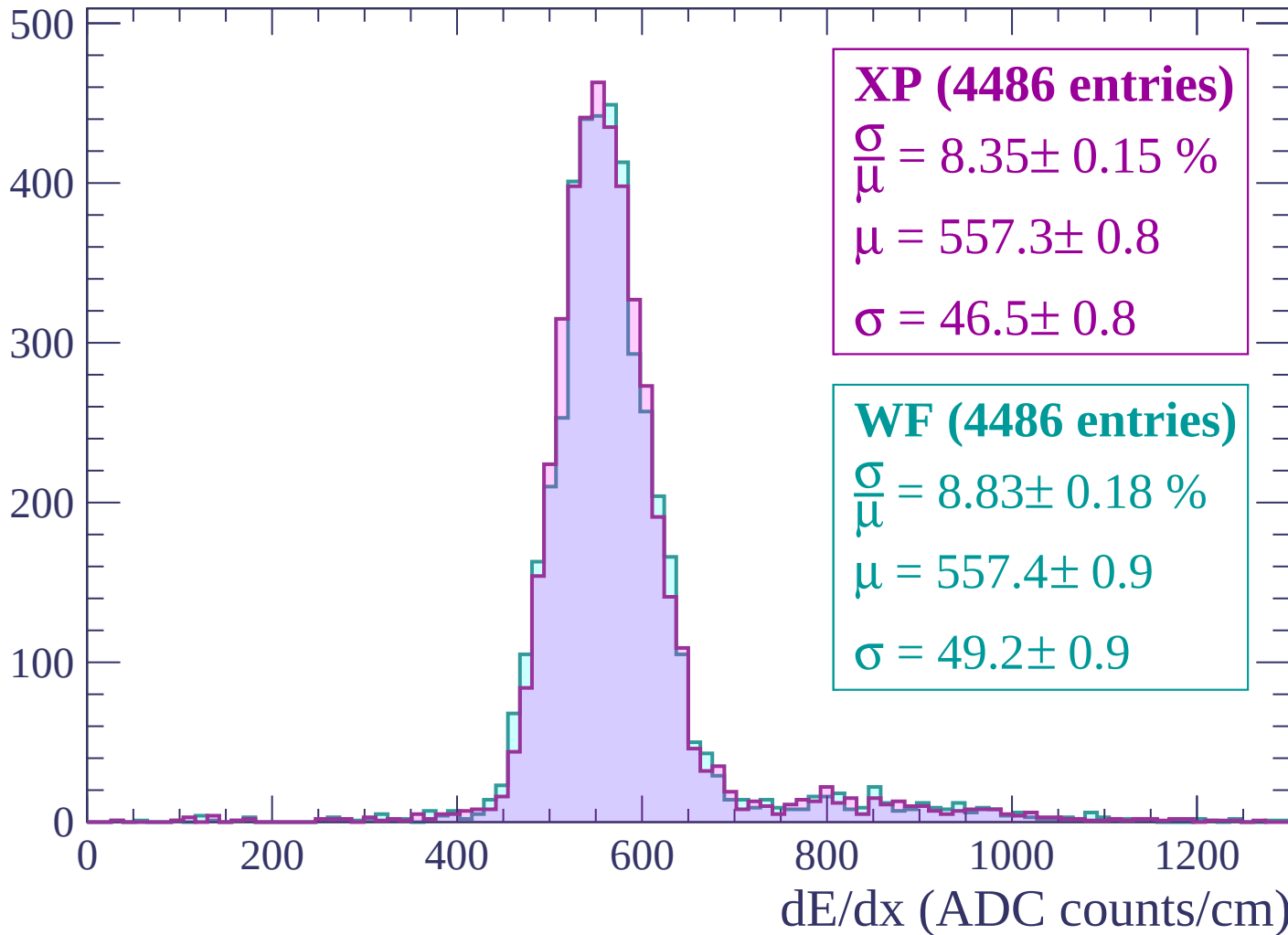
$$\mu = 568.3 \pm 1.5$$

$$\sigma = 67.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss |  $-60 < p < 0$

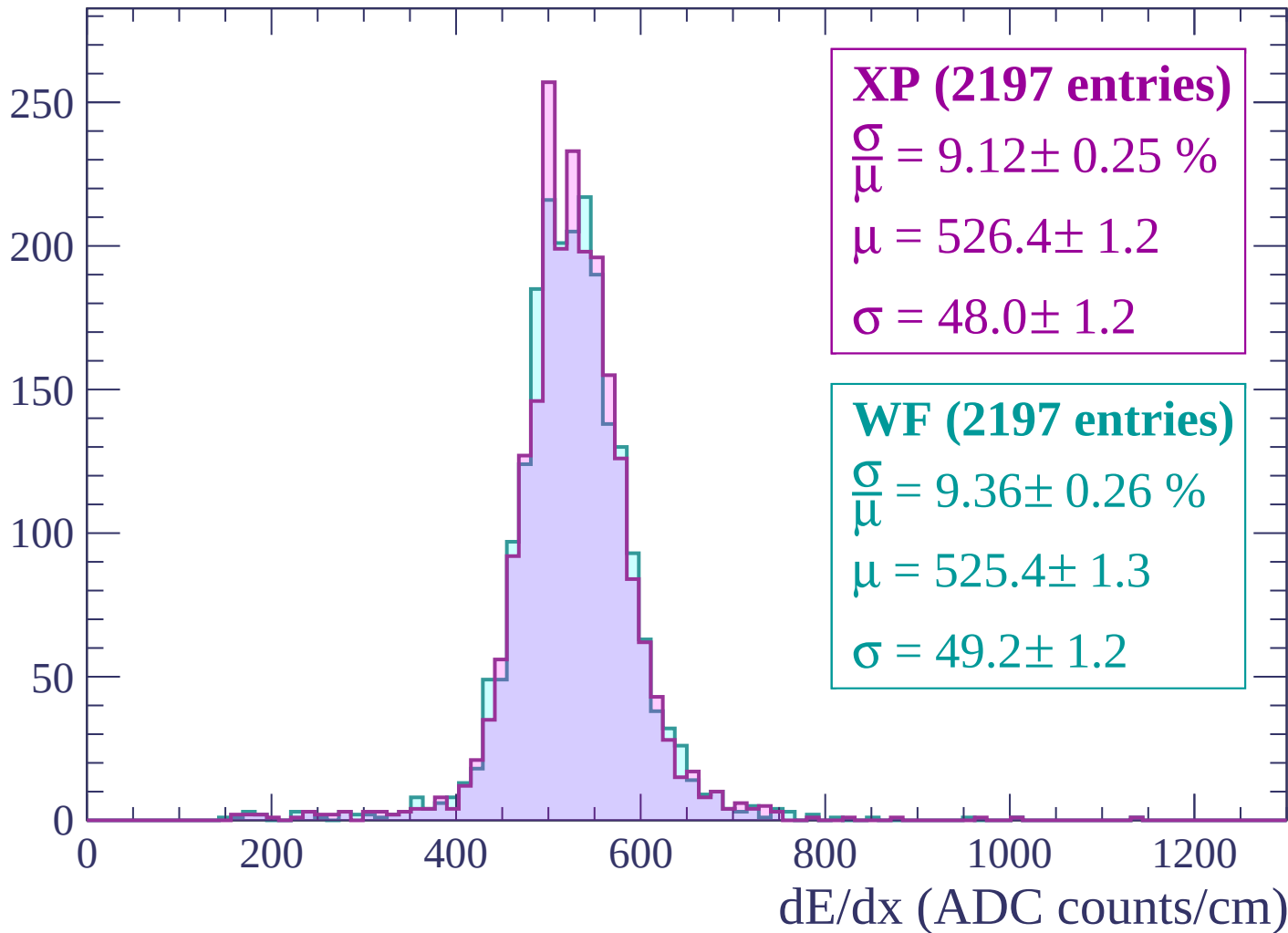
Count





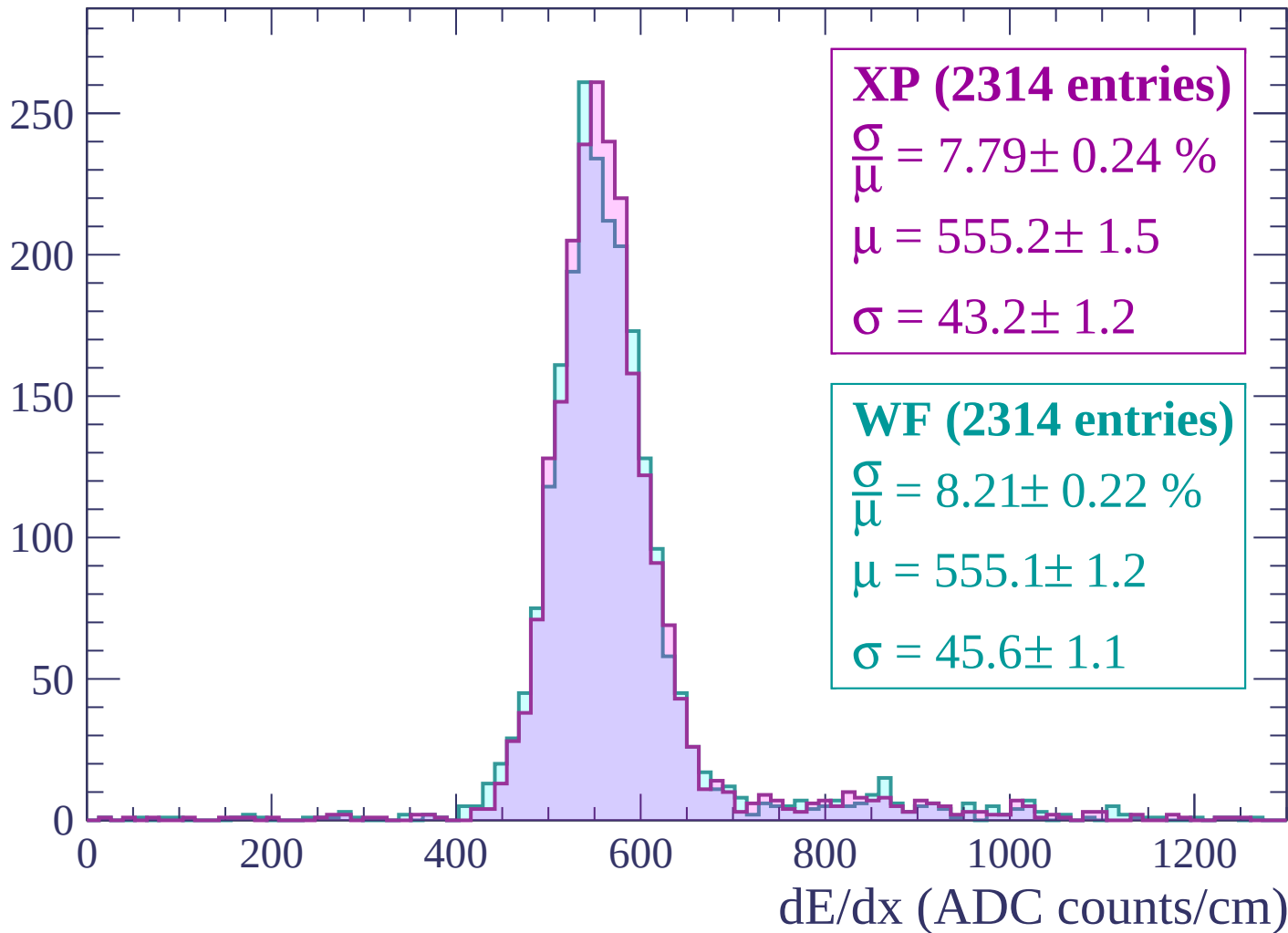
Energy loss |  $0 < p < 60$

Count



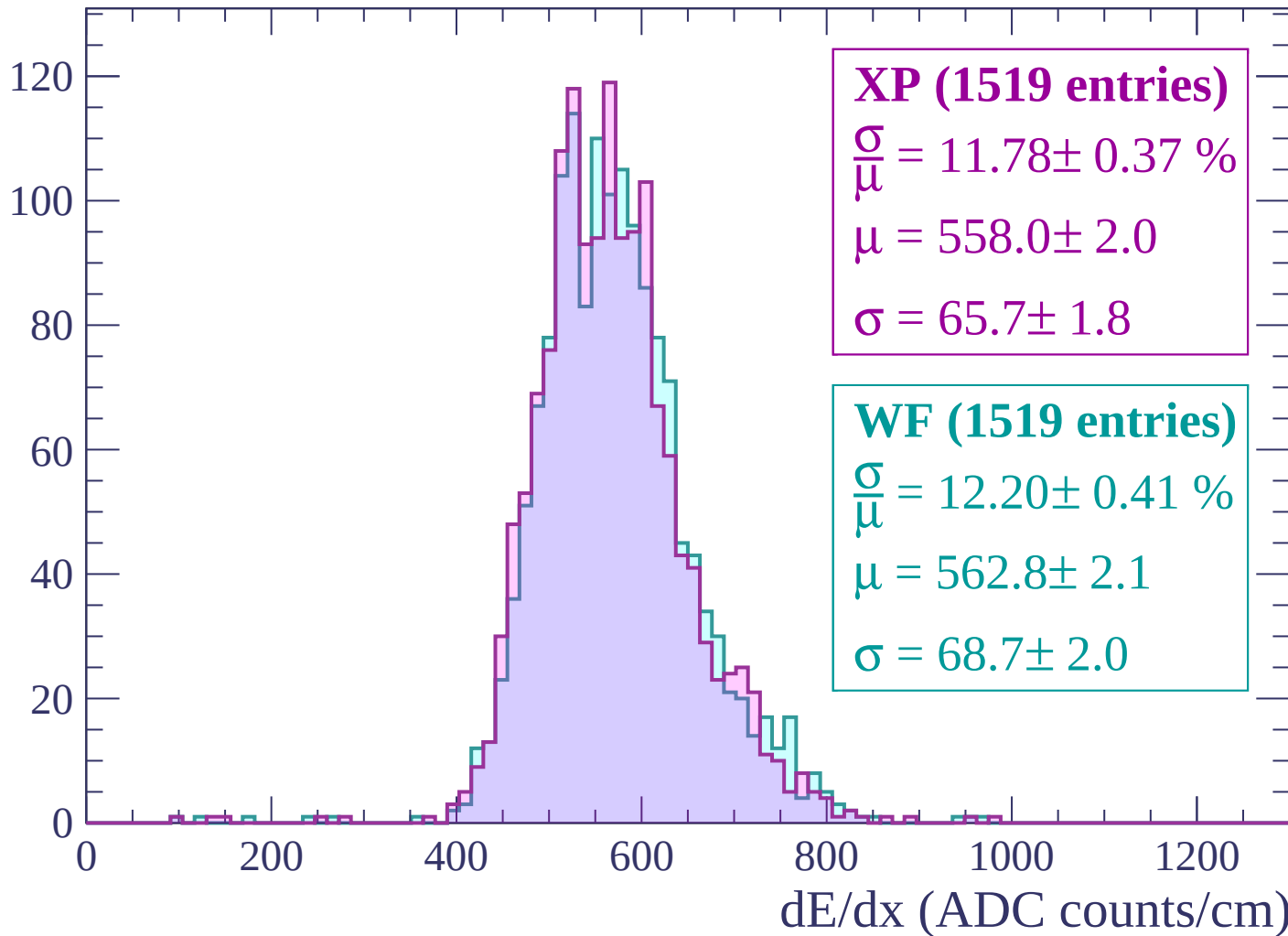
Energy loss |  $60 < p < 120$

Count



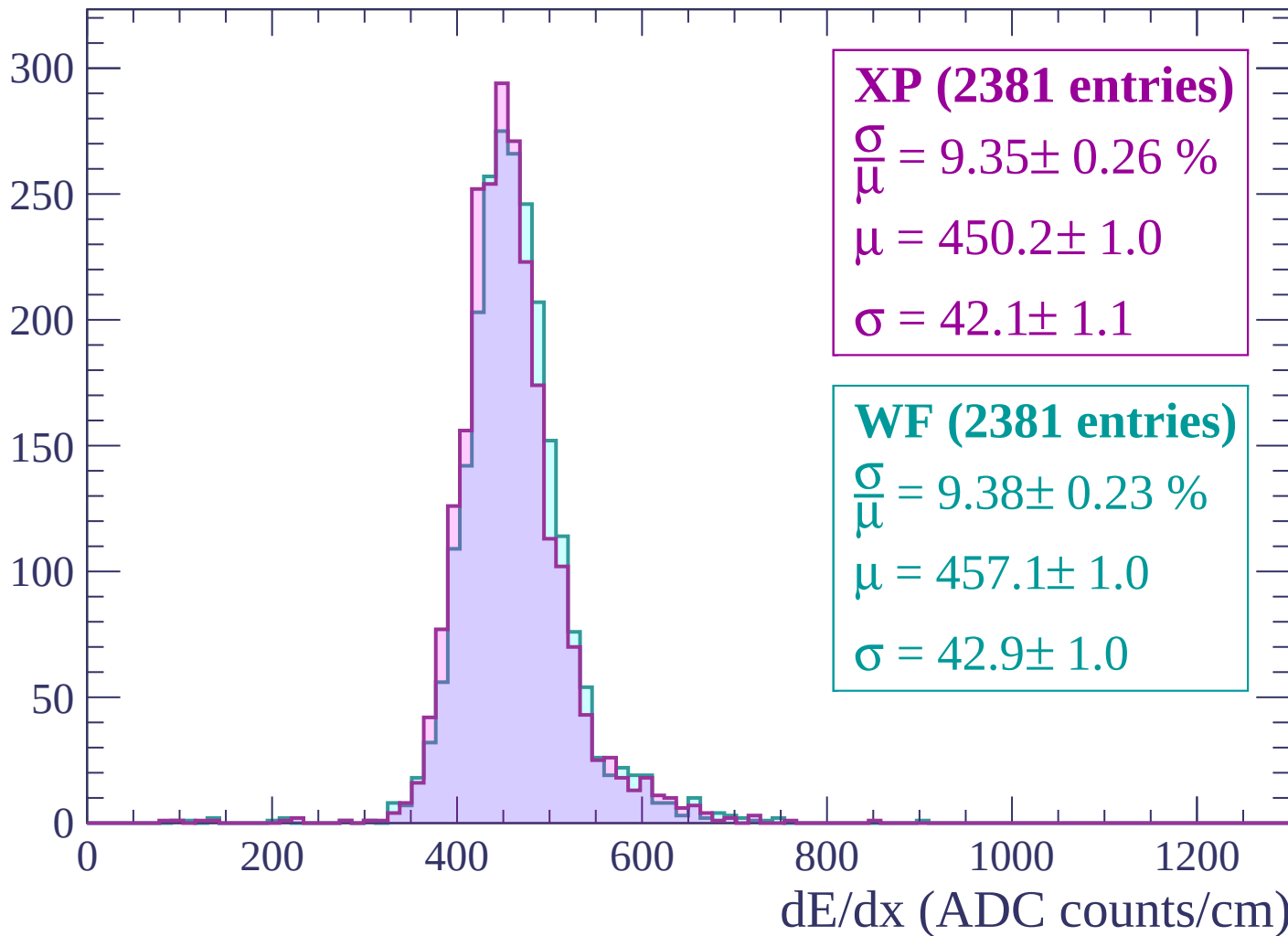
Energy loss |  $120 < p < 180$

Count



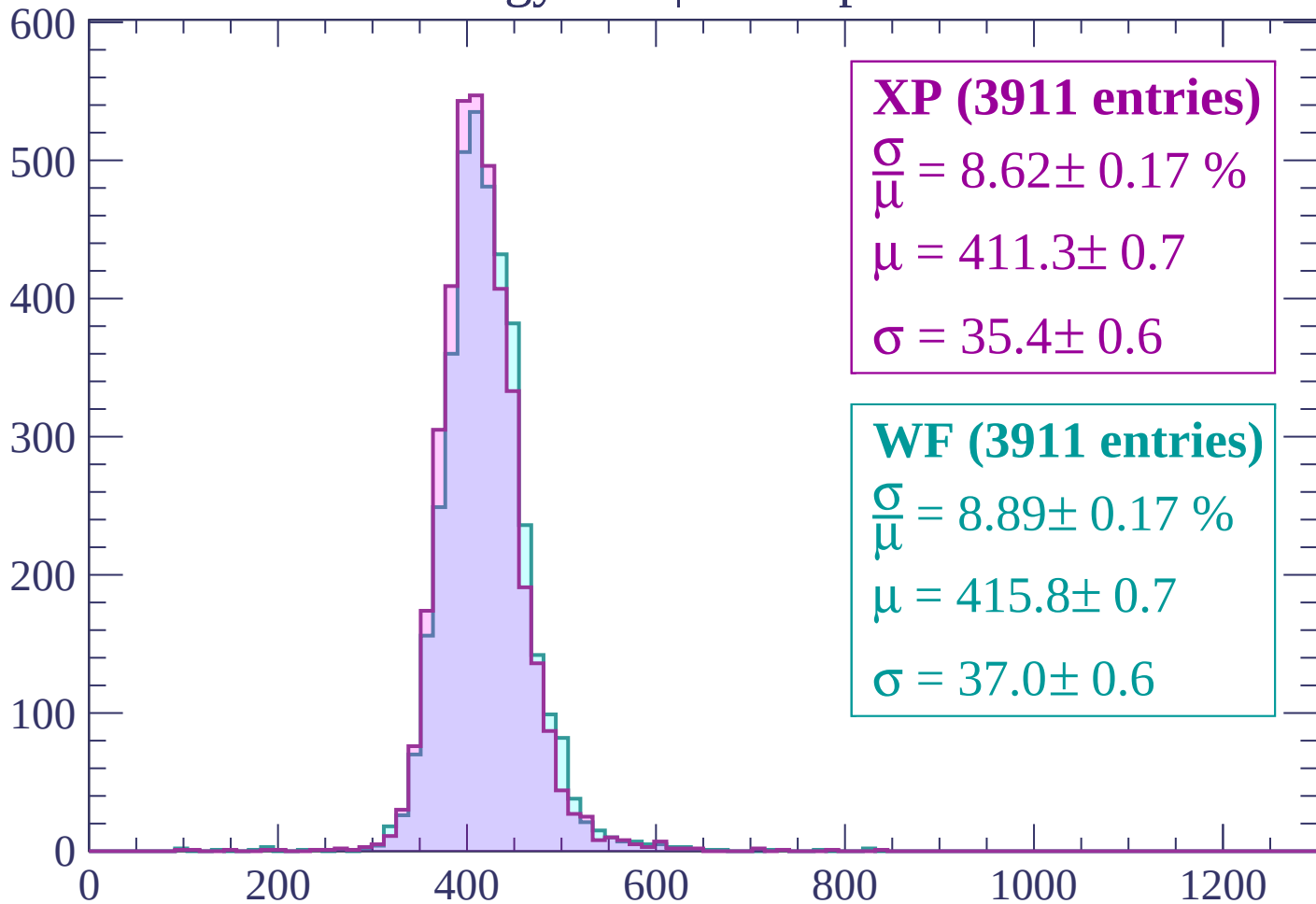
Energy loss |  $180 < p < 240$

Count



Energy loss |  $240 < p < 300$

Count



**XP (3911 entries)**

$$\frac{\sigma}{\mu} = 8.62 \pm 0.17 \%$$

$$\mu = 411.3 \pm 0.7$$

$$\sigma = 35.4 \pm 0.6$$

**WF (3911 entries)**

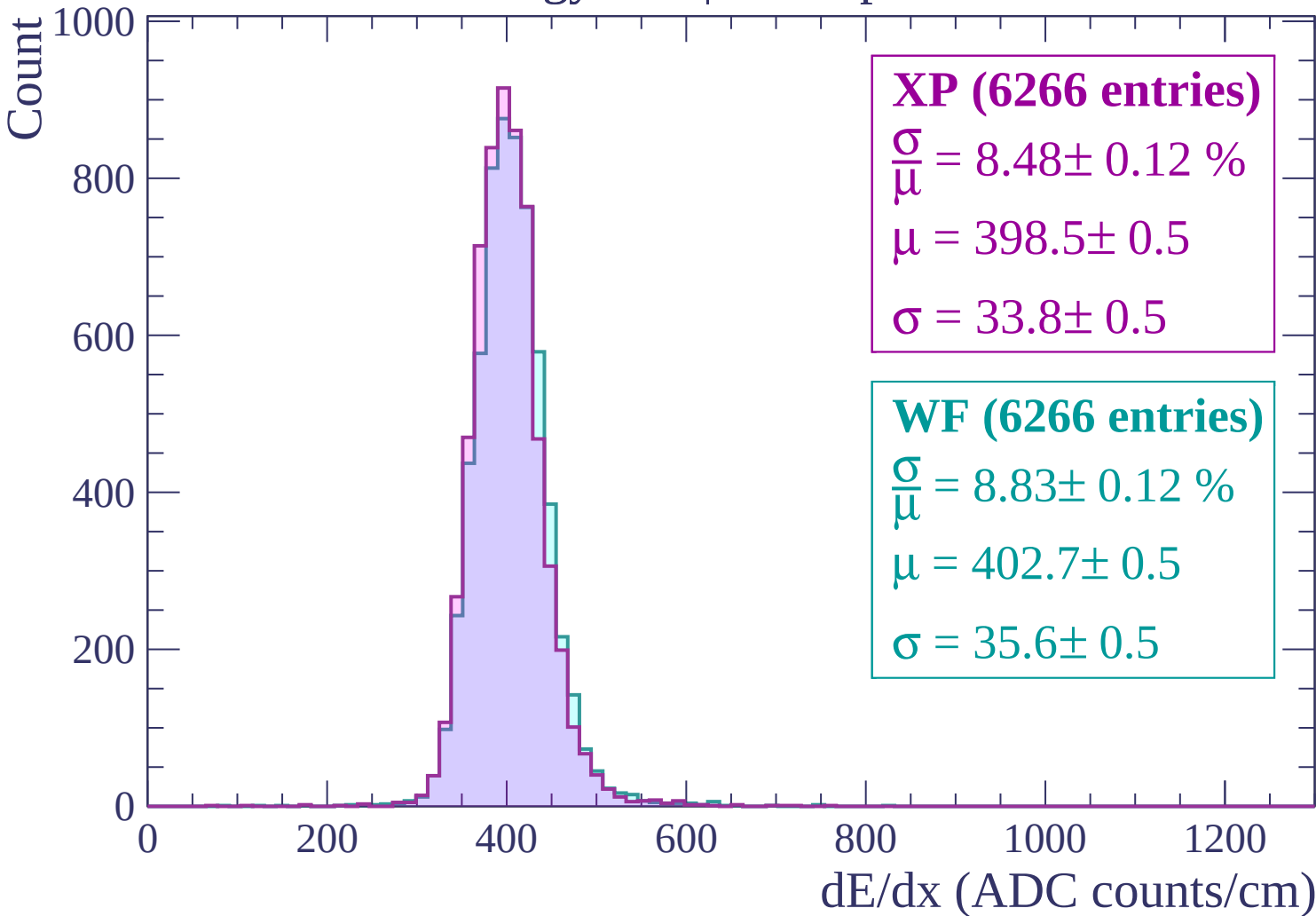
$$\frac{\sigma}{\mu} = 8.89 \pm 0.17 \%$$

$$\mu = 415.8 \pm 0.7$$

$$\sigma = 37.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $300 < p < 360$



Energy loss |  $360 < p < 420$

Count

1000

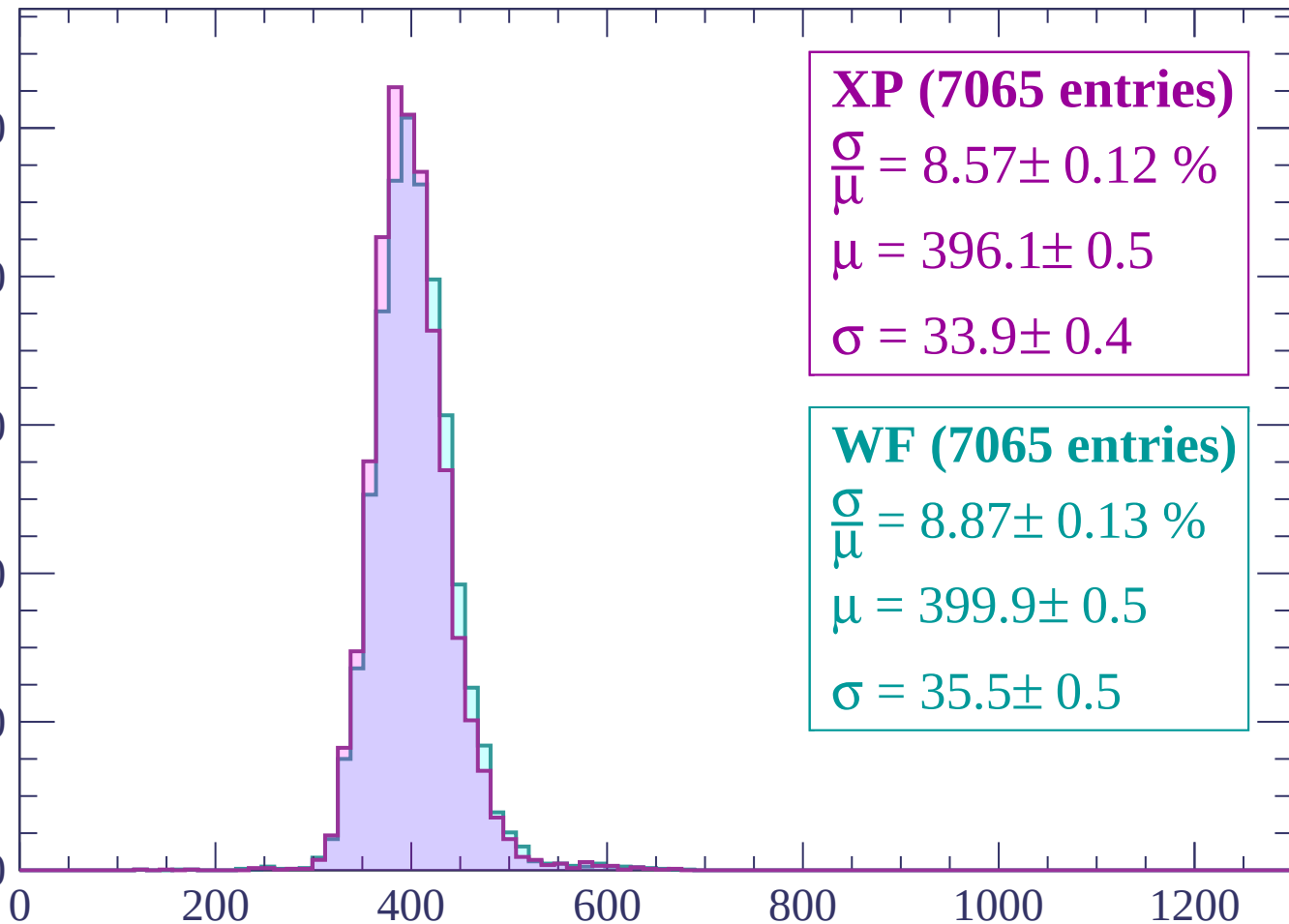
800

600

400

200

0



**XP (7065 entries)**

$\frac{\sigma}{\mu} = 8.57 \pm 0.12 \%$

$\mu = 396.1 \pm 0.5$

$\sigma = 33.9 \pm 0.4$

**WF (7065 entries)**

$\frac{\sigma}{\mu} = 8.87 \pm 0.13 \%$

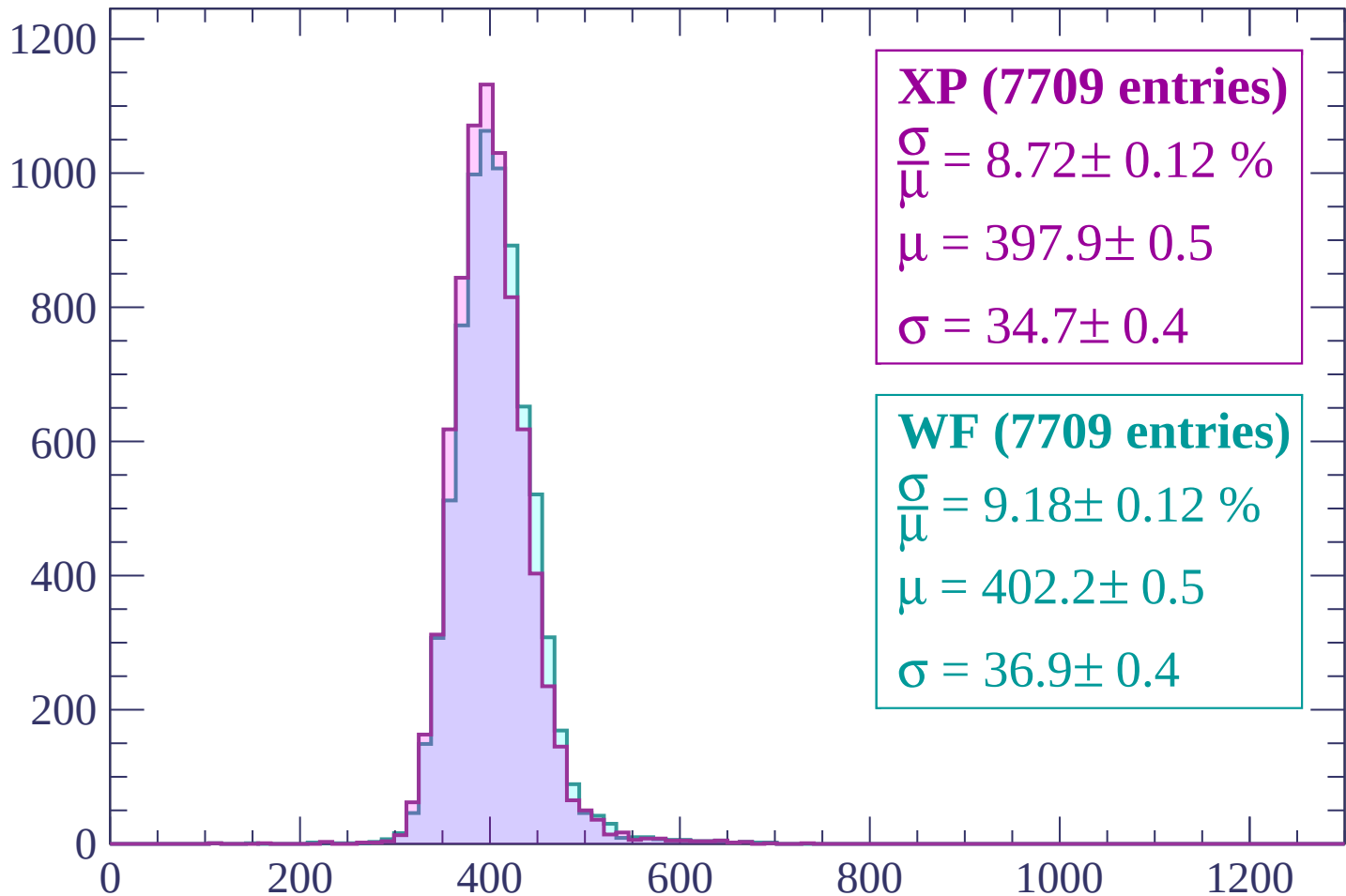
$\mu = 399.9 \pm 0.5$

$\sigma = 35.5 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss |  $420 < p < 480$

Count



**XP (7709 entries)**

$$\frac{\sigma}{\mu} = 8.72 \pm 0.12 \%$$

$$\mu = 397.9 \pm 0.5$$

$$\sigma = 34.7 \pm 0.4$$

**WF (7709 entries)**

$$\frac{\sigma}{\mu} = 9.18 \pm 0.12 \%$$

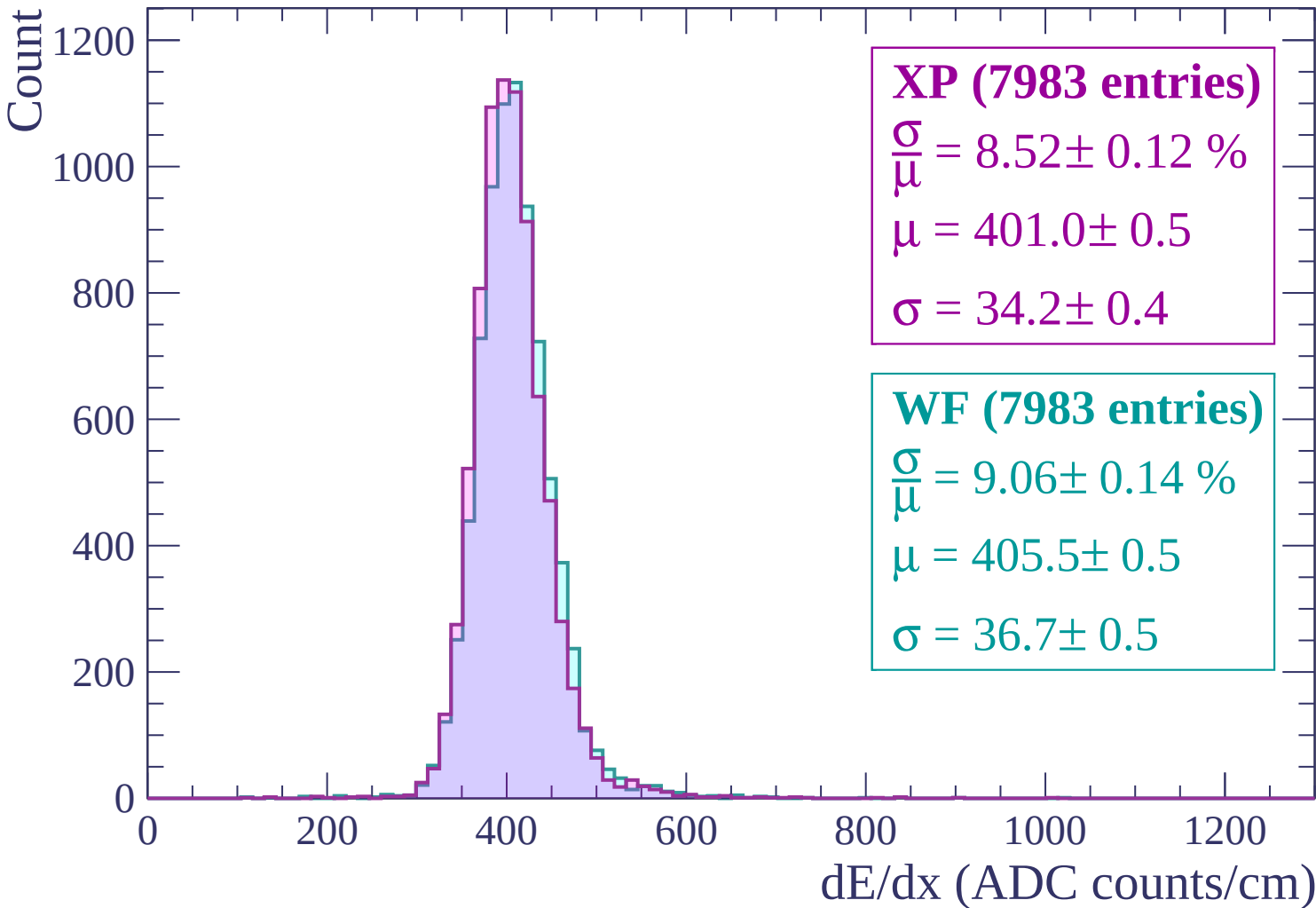
$$\mu = 402.2 \pm 0.5$$

$$\sigma = 36.9 \pm 0.4$$

dE/dx (ADC counts/cm)



Energy loss |  $480 < p < 540$



Energy loss |  $540 < p < 600$

Count

1200

1000

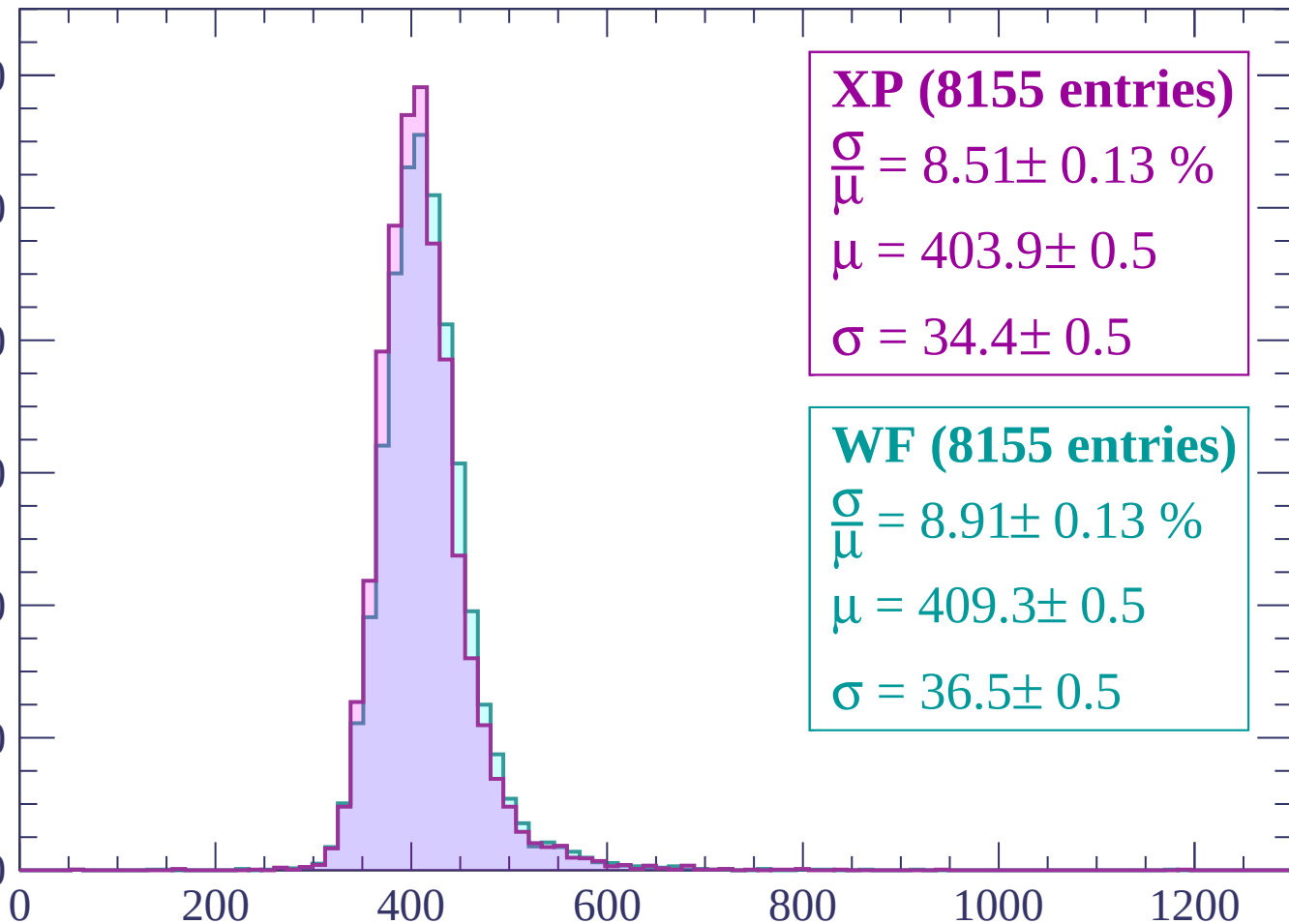
800

600

400

200

0



**XP (8155 entries)**

$\frac{\sigma}{\mu} = 8.51 \pm 0.13 \%$

$\mu = 403.9 \pm 0.5$

$\sigma = 34.4 \pm 0.5$

**WF (8155 entries)**

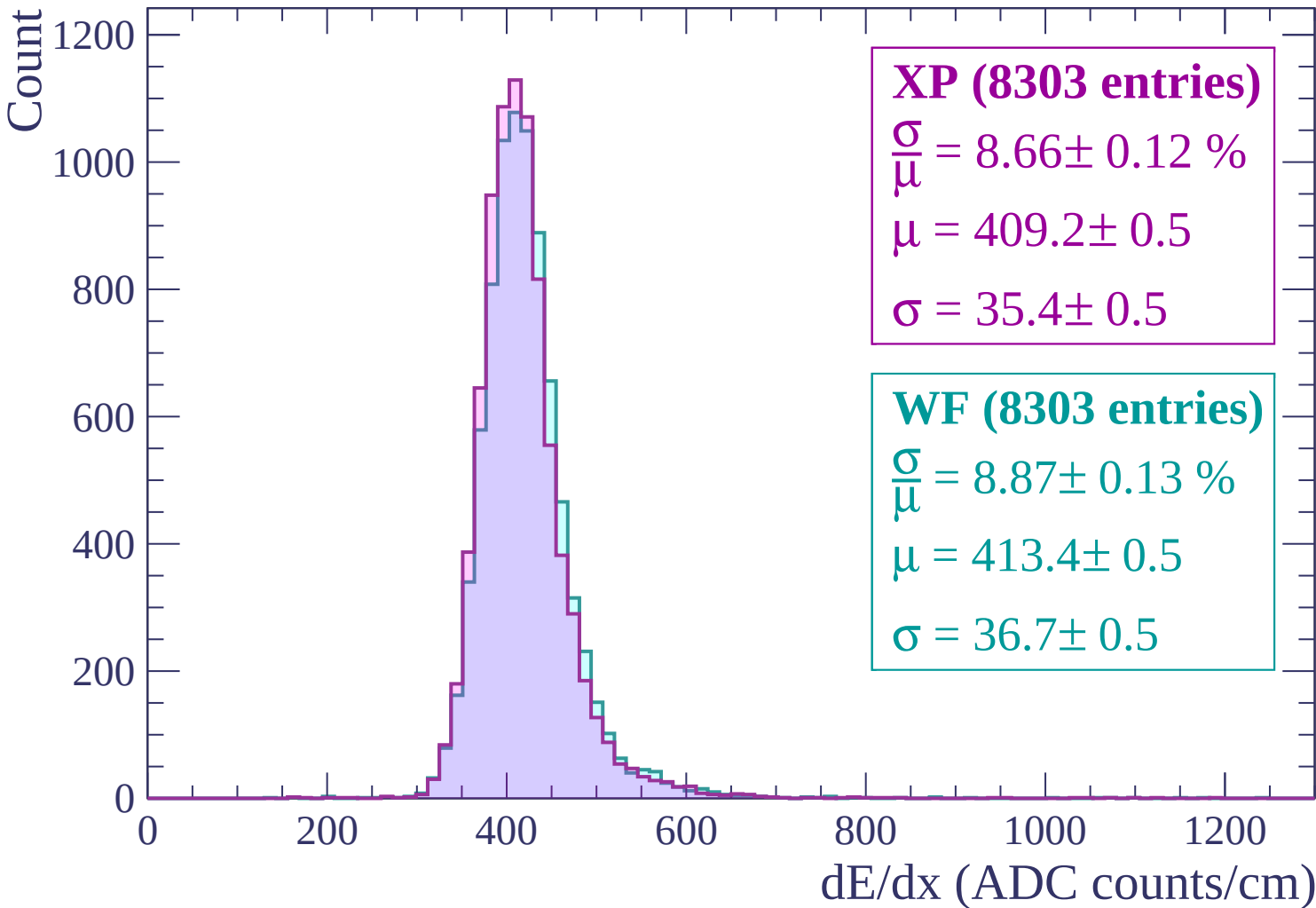
$\frac{\sigma}{\mu} = 8.91 \pm 0.13 \%$

$\mu = 409.3 \pm 0.5$

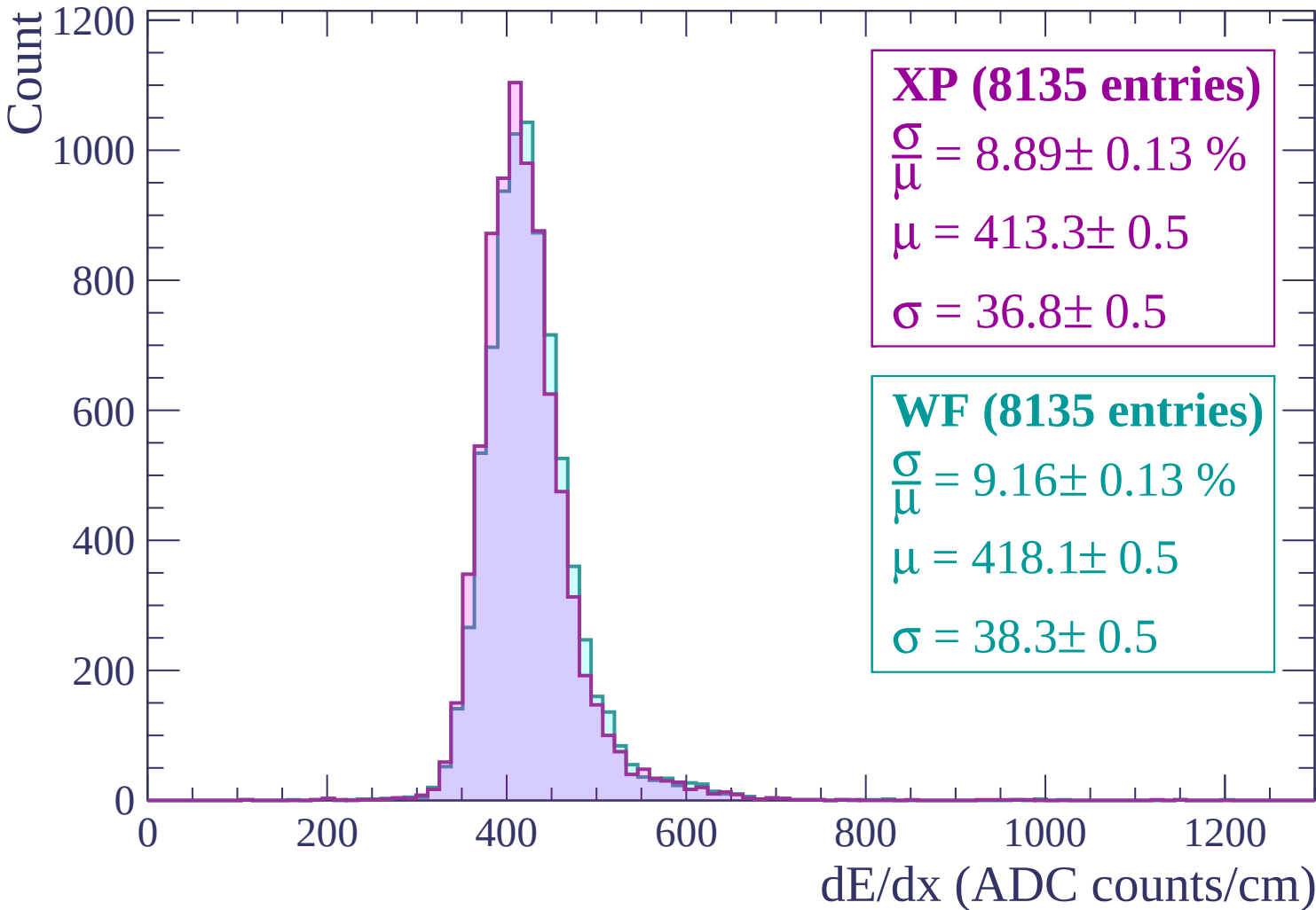
$\sigma = 36.5 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss |  $600 < p < 660$



Energy loss |  $660 < p < 720$



Energy loss |  $720 < p < 780$

Count

1000

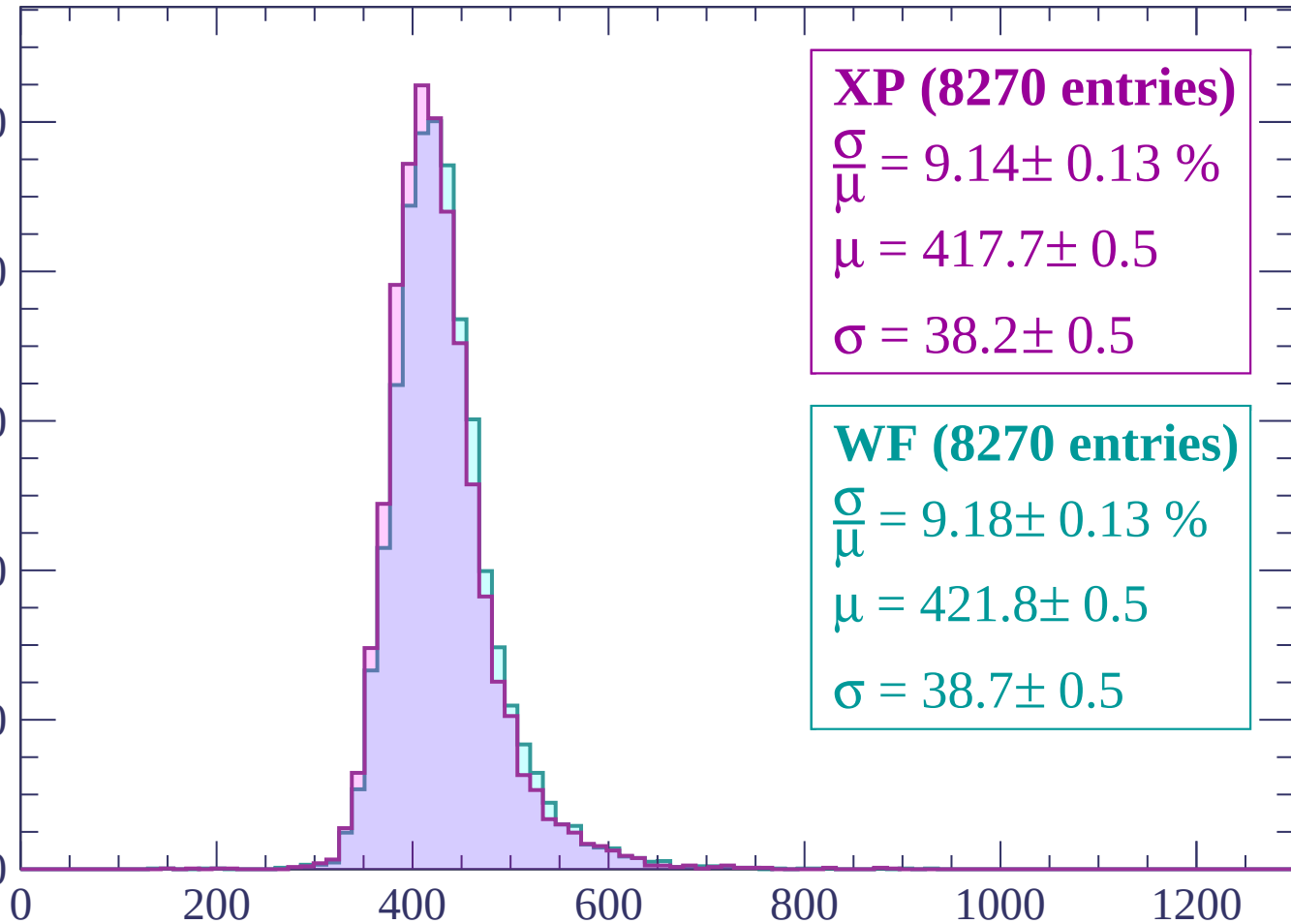
800

600

400

200

0



**XP (8270 entries)**

$\frac{\sigma}{\mu} = 9.14 \pm 0.13 \%$

$\mu = 417.7 \pm 0.5$

$\sigma = 38.2 \pm 0.5$

**WF (8270 entries)**

$\frac{\sigma}{\mu} = 9.18 \pm 0.13 \%$

$\mu = 421.8 \pm 0.5$

$\sigma = 38.7 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss |  $780 < p < 840$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (8044 entries)**

$$\frac{\sigma}{\mu} = 8.90 \pm 0.12 \%$$

$$\mu = 422.5 \pm 0.5$$

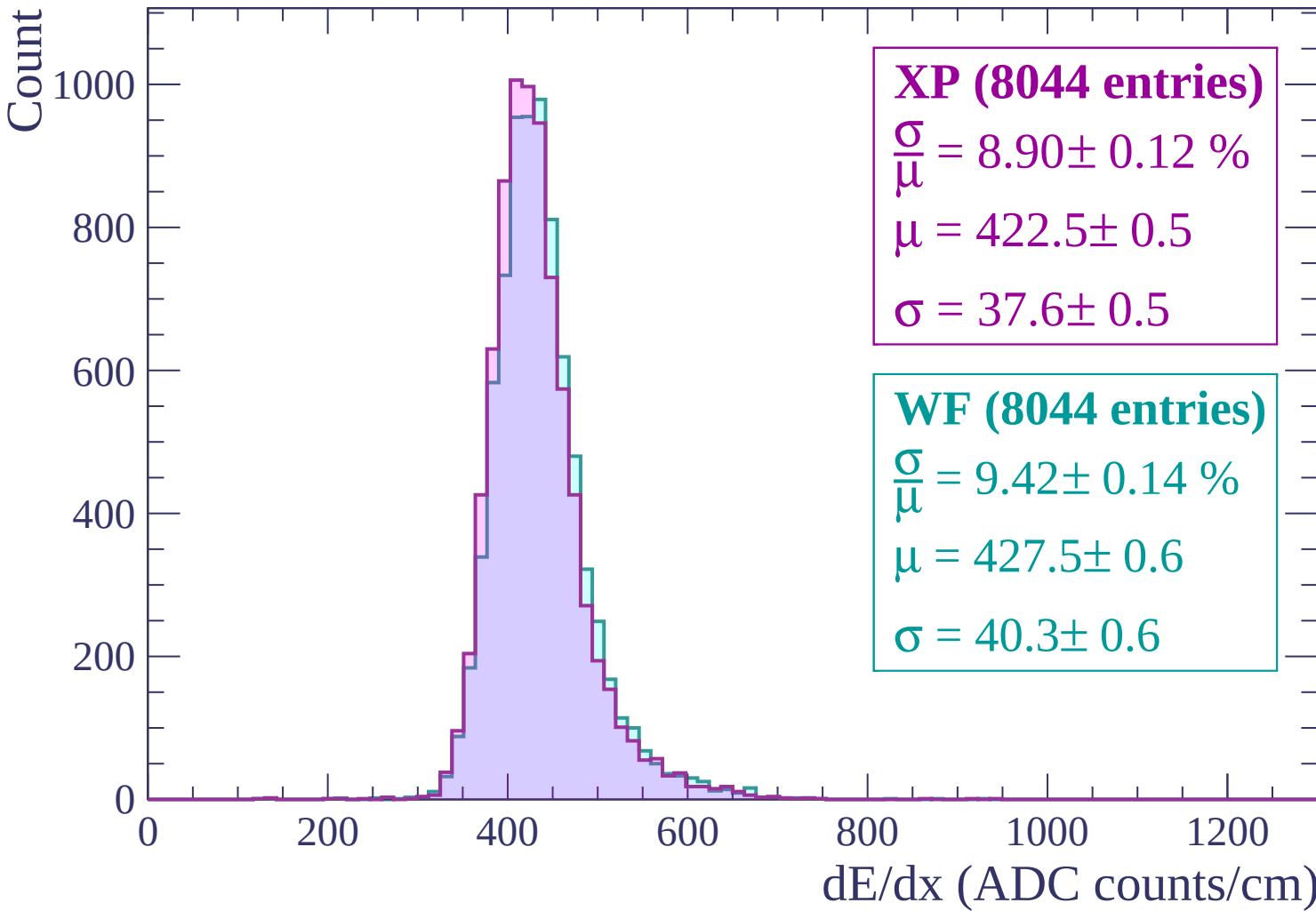
$$\sigma = 37.6 \pm 0.5$$

**WF (8044 entries)**

$$\frac{\sigma}{\mu} = 9.42 \pm 0.14 \%$$

$$\mu = 427.5 \pm 0.6$$

$$\sigma = 40.3 \pm 0.6$$



Energy loss |  $840 < p < 900$

Count

1000

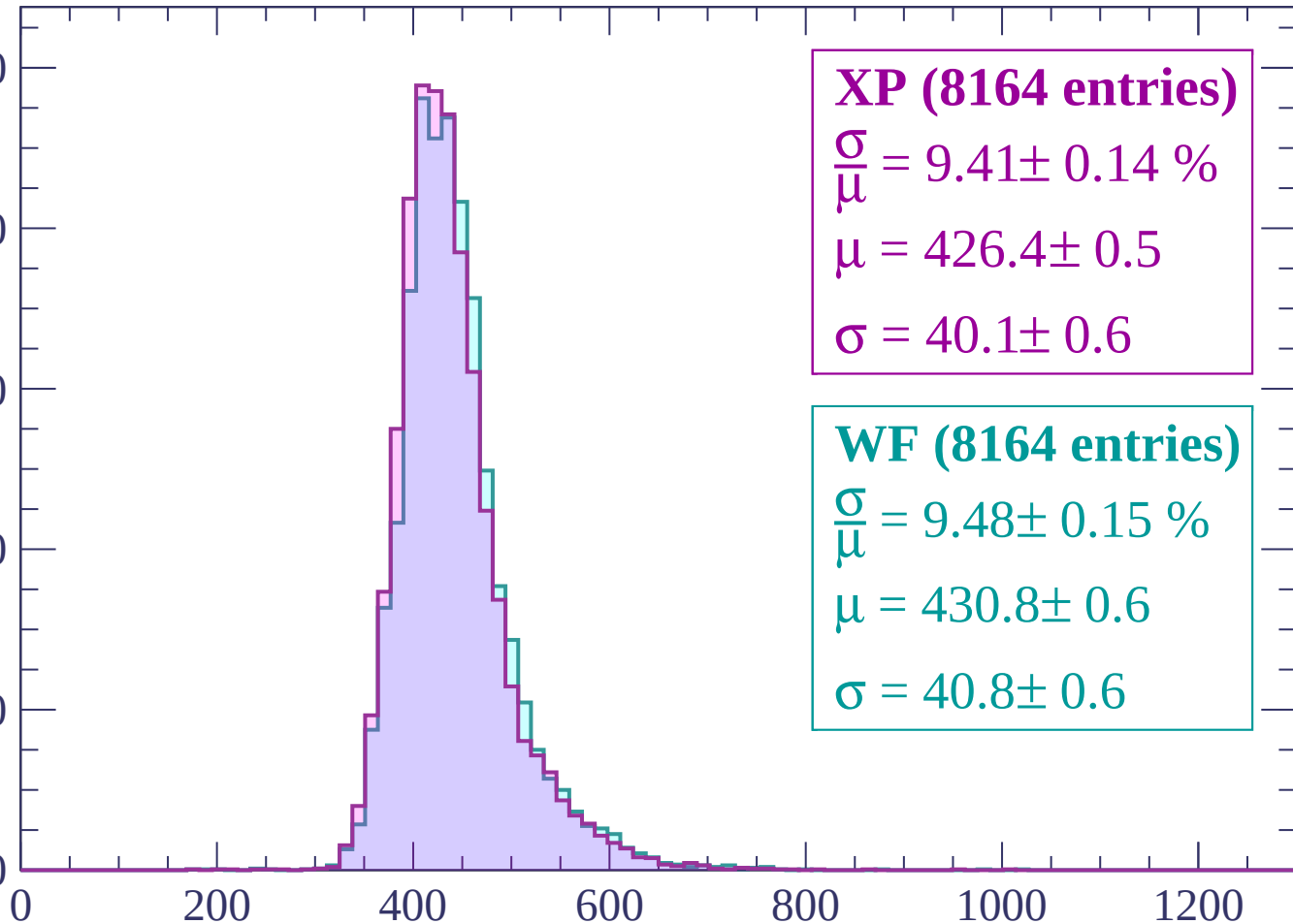
800

600

400

200

0



**XP (8164 entries)**

$\frac{\sigma}{\mu} = 9.41 \pm 0.14 \%$

$\mu = 426.4 \pm 0.5$

$\sigma = 40.1 \pm 0.6$

**WF (8164 entries)**

$\frac{\sigma}{\mu} = 9.48 \pm 0.15 \%$

$\mu = 430.8 \pm 0.6$

$\sigma = 40.8 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $900 < p < 960$

Count

1000

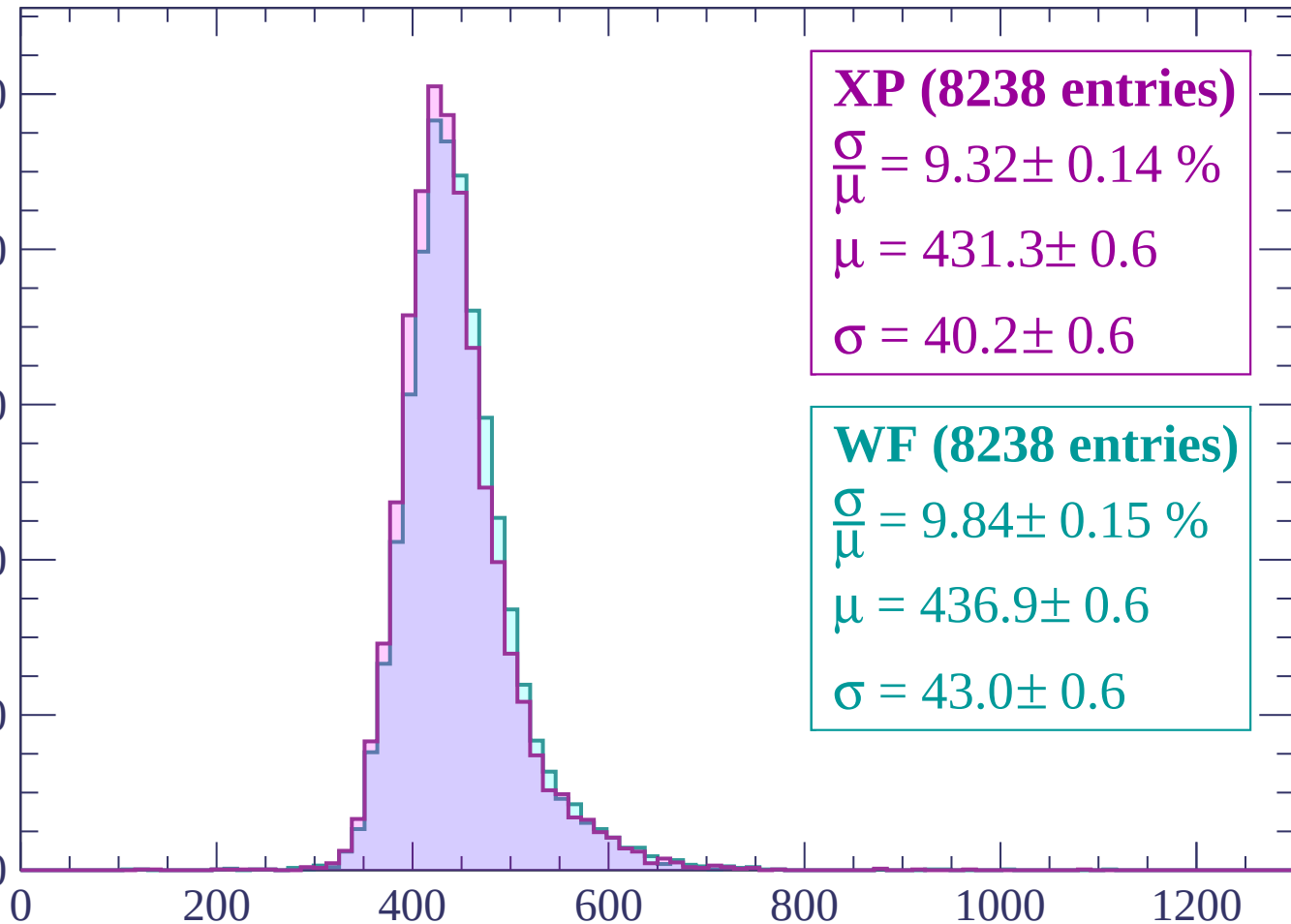
800

600

400

200

0



**XP (8238 entries)**

$\frac{\sigma}{\mu} = 9.32 \pm 0.14 \%$

$\mu = 431.3 \pm 0.6$

$\sigma = 40.2 \pm 0.6$

**WF (8238 entries)**

$\frac{\sigma}{\mu} = 9.84 \pm 0.15 \%$

$\mu = 436.9 \pm 0.6$

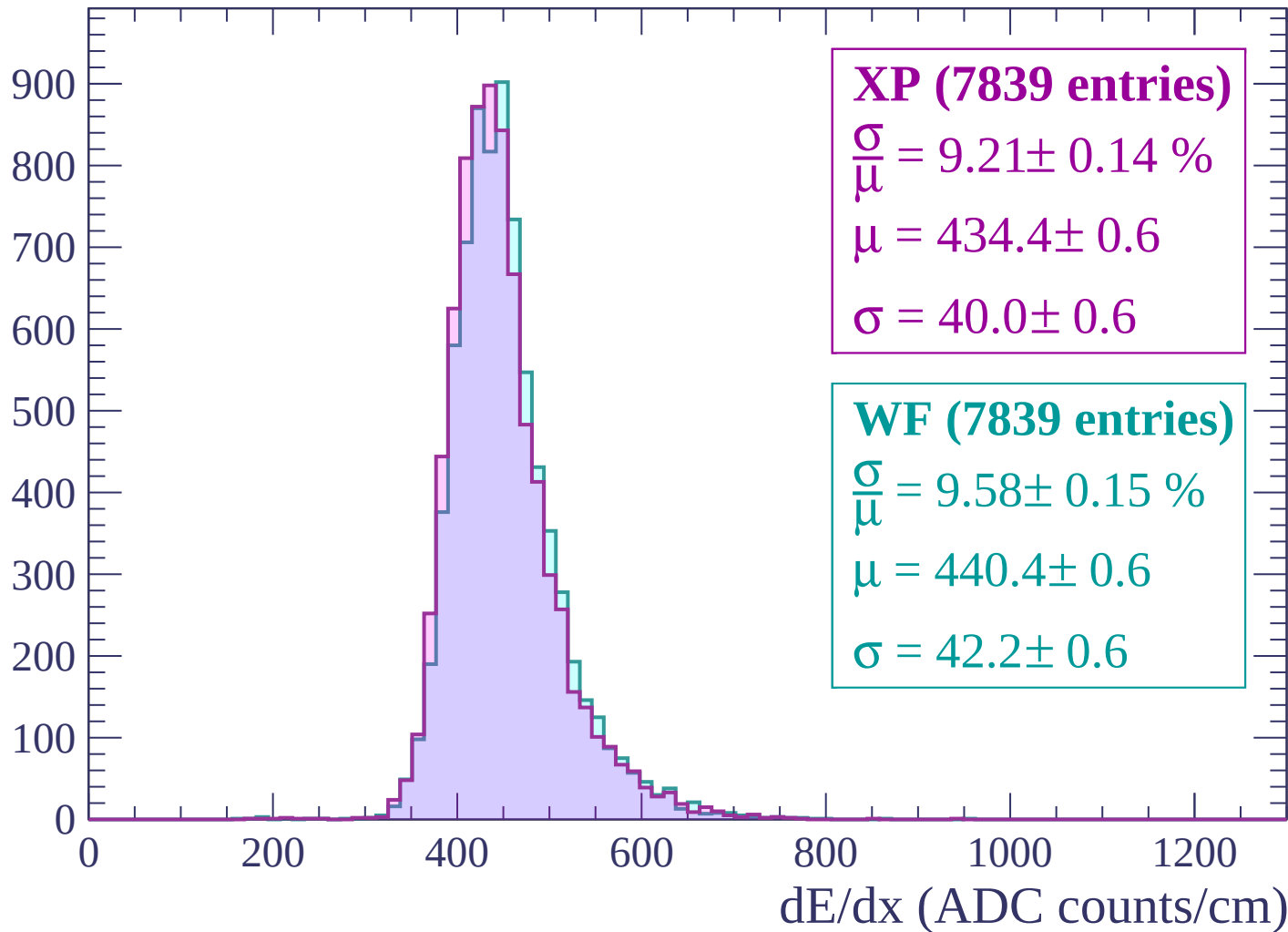
$\sigma = 43.0 \pm 0.6$

dE/dx (ADC counts/cm)



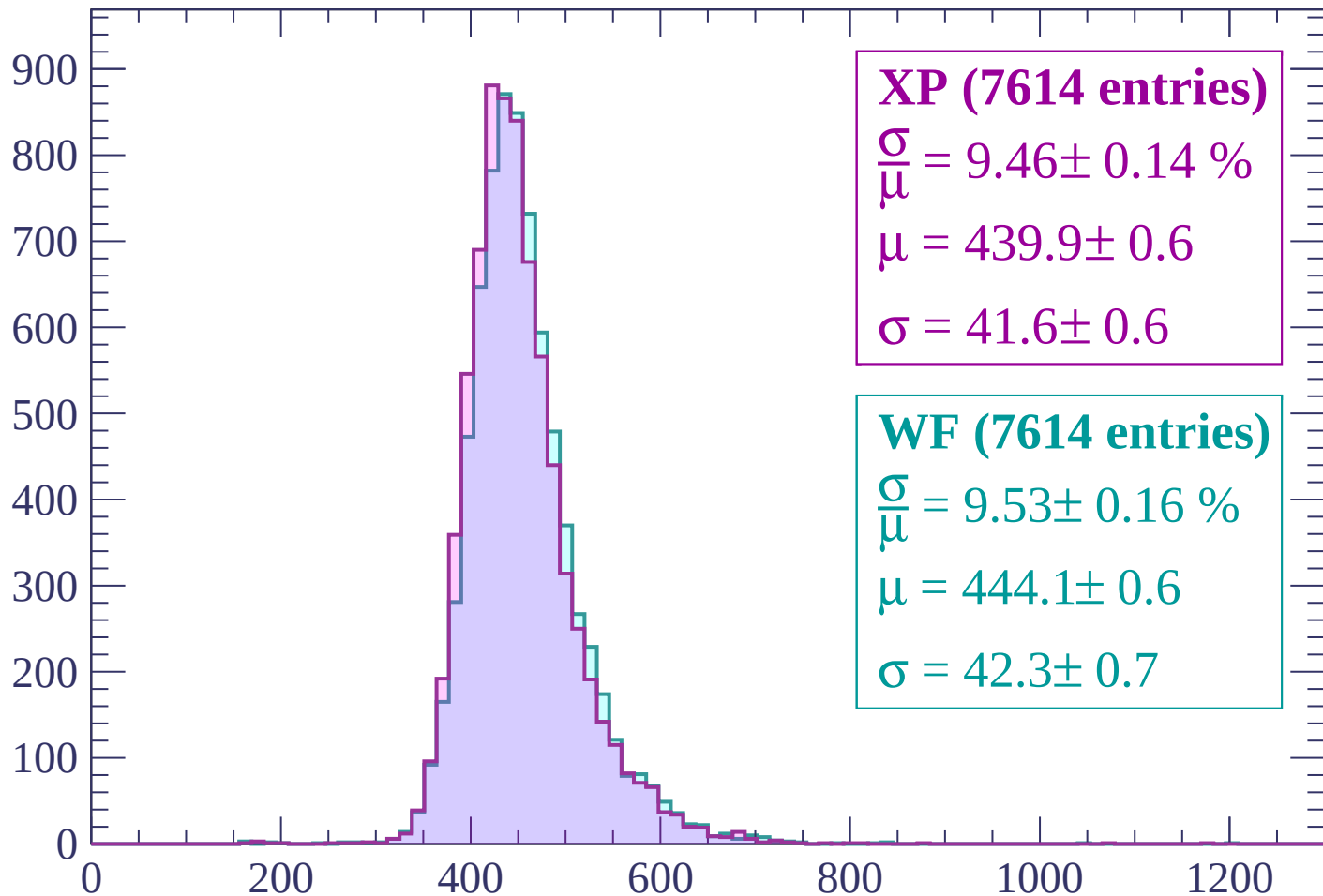
Energy loss |  $960 < p < 1020$

Count



Energy loss |  $1020 < p < 1080$

Count



**XP (7614 entries)**

$\frac{\sigma}{\mu} = 9.46 \pm 0.14 \%$

$\mu = 439.9 \pm 0.6$

$\sigma = 41.6 \pm 0.6$

**WF (7614 entries)**

$\frac{\sigma}{\mu} = 9.53 \pm 0.16 \%$

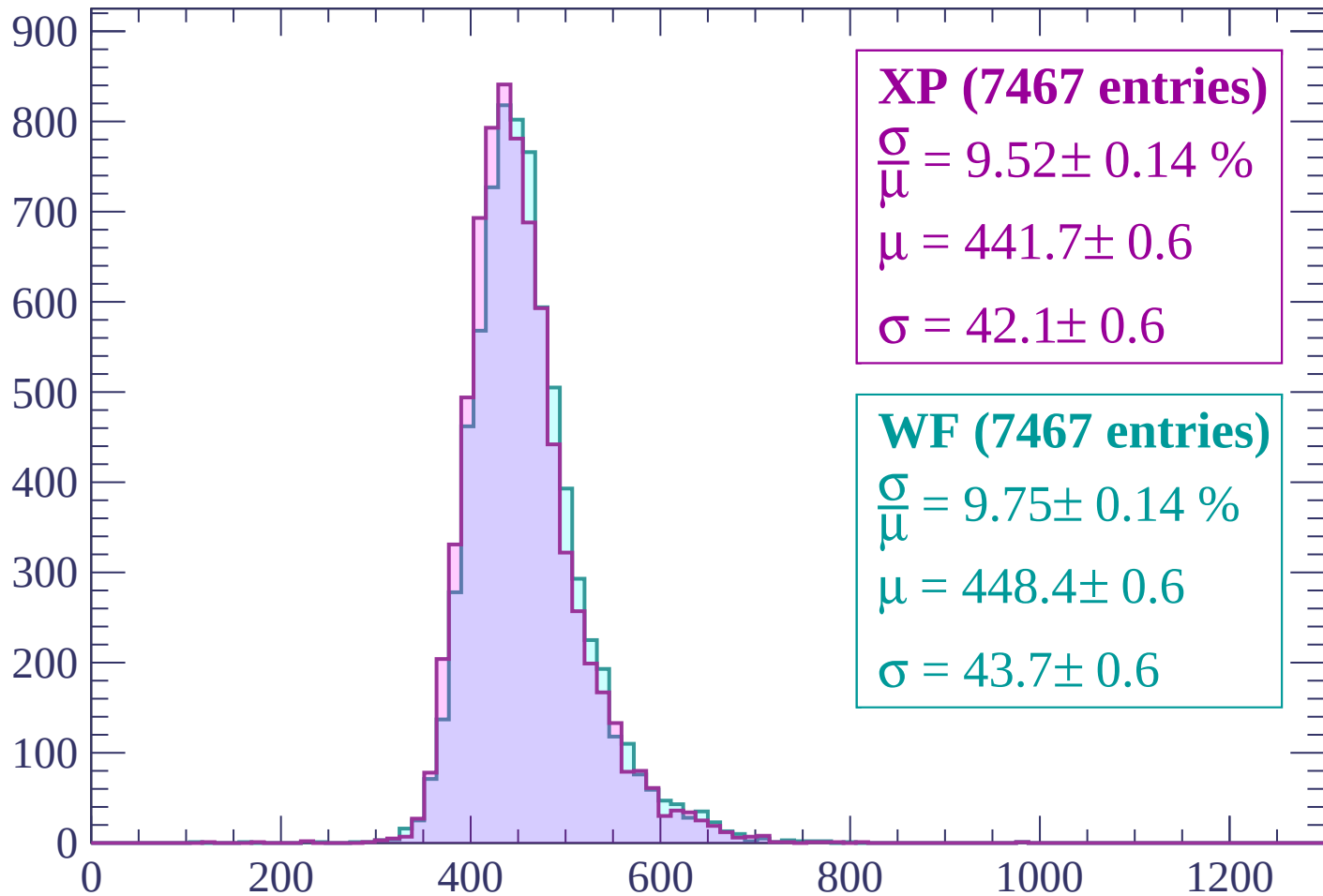
$\mu = 444.1 \pm 0.6$

$\sigma = 42.3 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss |  $1080 < p < 1140$

Count



**XP (7467 entries)**

$$\frac{\sigma}{\mu} = 9.52 \pm 0.14 \%$$

$$\mu = 441.7 \pm 0.6$$

$$\sigma = 42.1 \pm 0.6$$

**WF (7467 entries)**

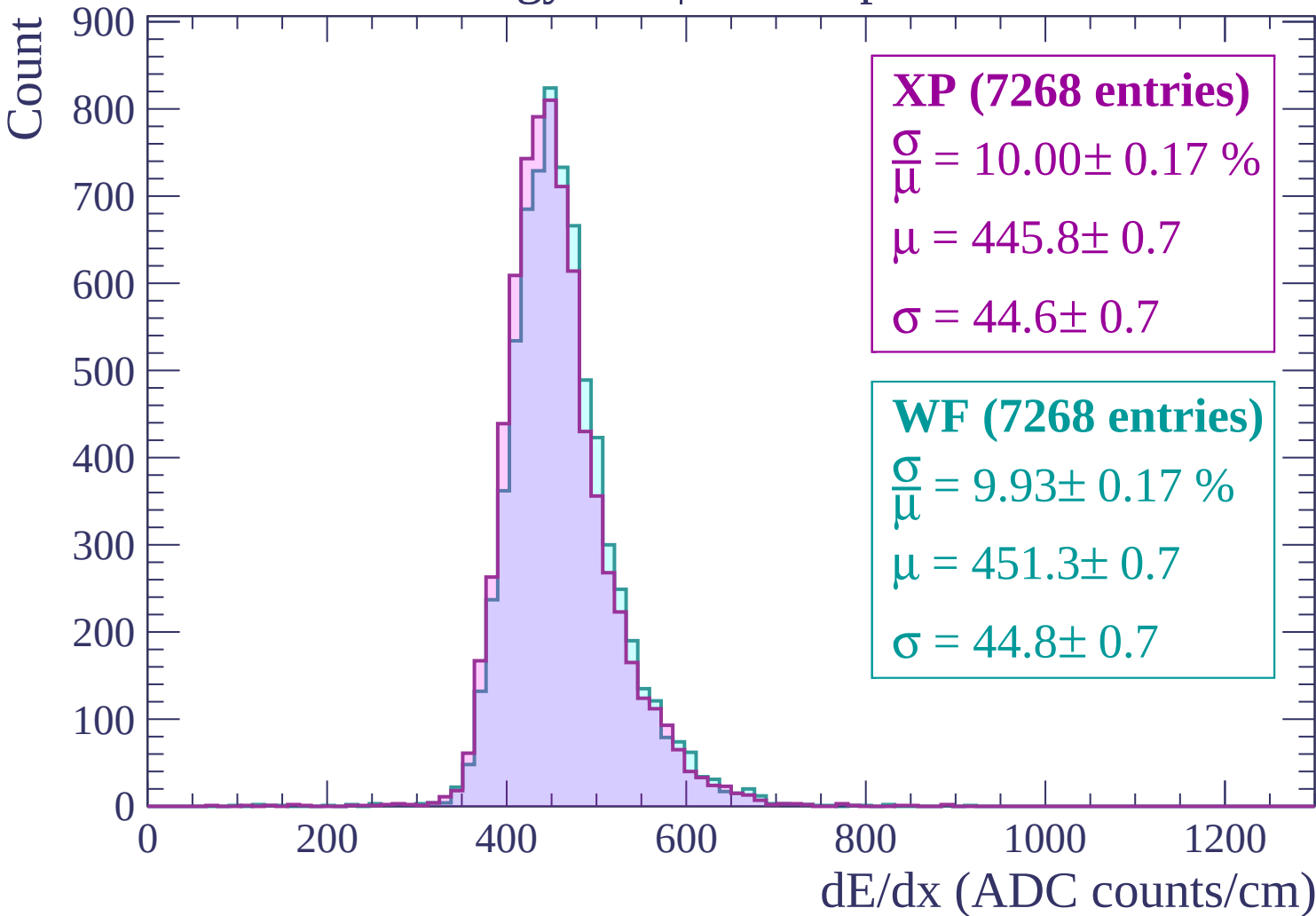
$$\frac{\sigma}{\mu} = 9.75 \pm 0.14 \%$$

$$\mu = 448.4 \pm 0.6$$

$$\sigma = 43.7 \pm 0.6$$

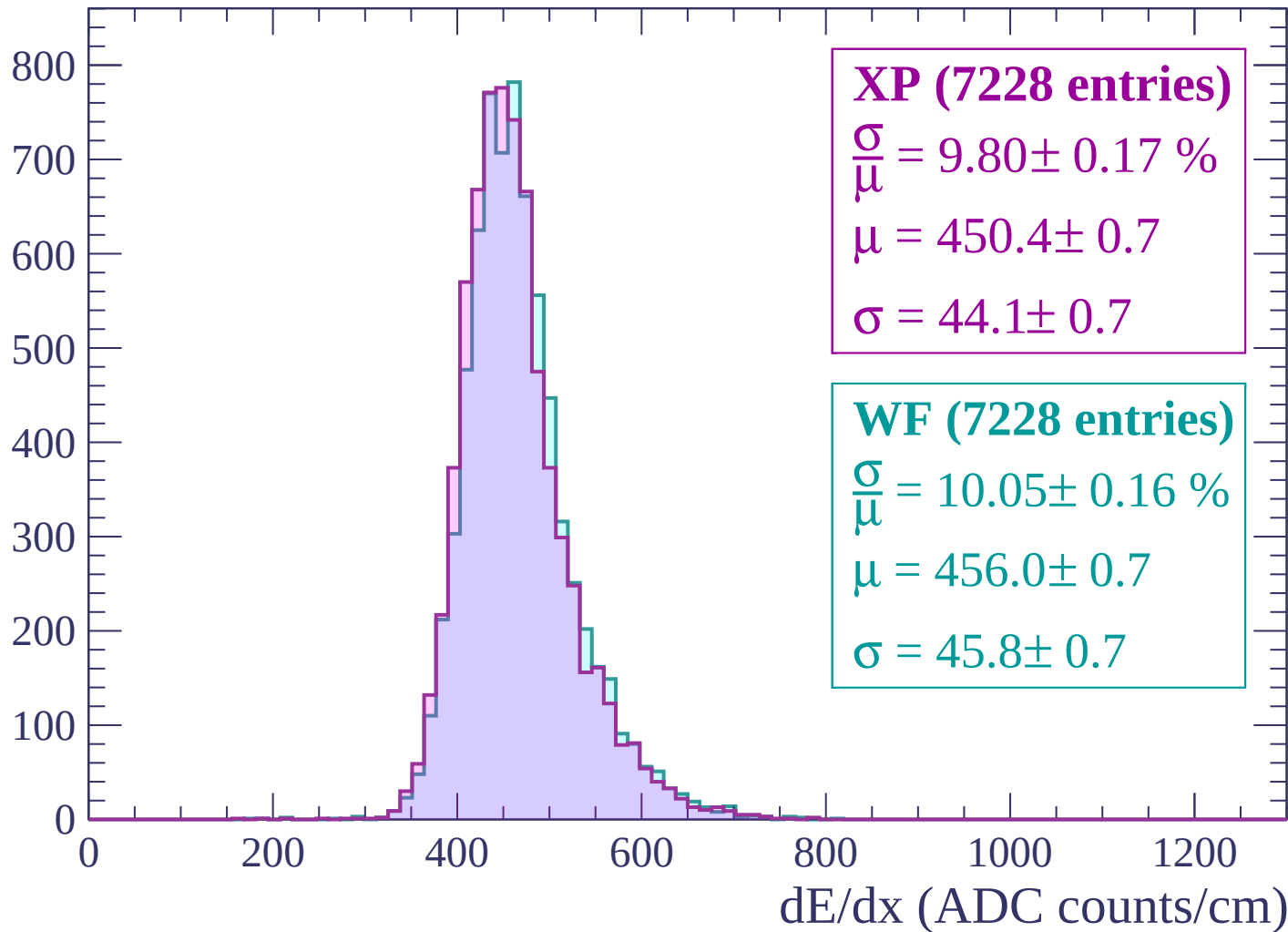
dE/dx (ADC counts/cm)

Energy loss |  $1140 < p < 1200$



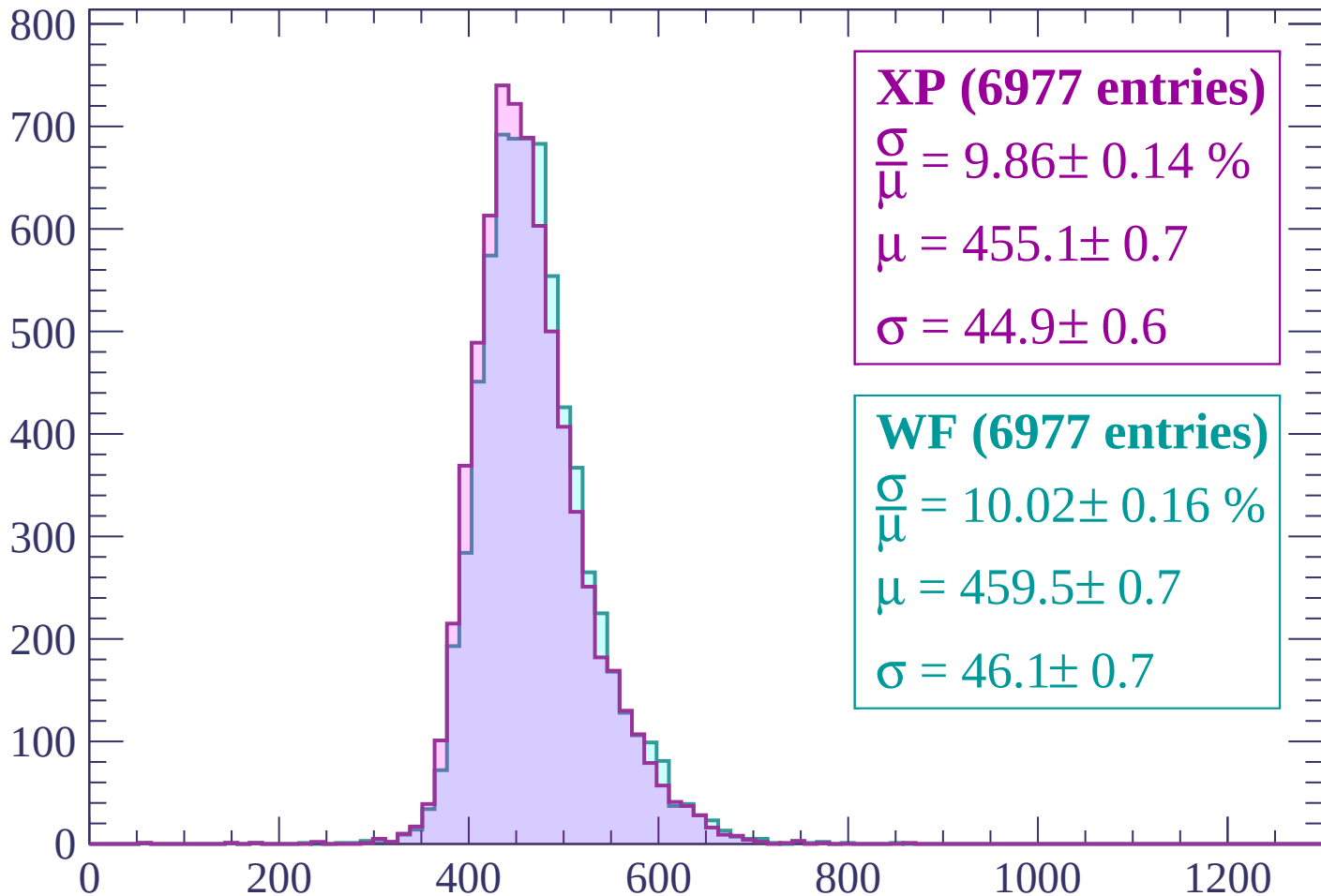
Energy loss |  $1200 < p < 1260$

Count



Energy loss |  $1260 < p < 1320$

Count



**XP (6977 entries)**

$$\frac{\sigma}{\mu} = 9.86 \pm 0.14 \%$$

$$\mu = 455.1 \pm 0.7$$

$$\sigma = 44.9 \pm 0.6$$

**WF (6977 entries)**

$$\frac{\sigma}{\mu} = 10.02 \pm 0.16 \%$$

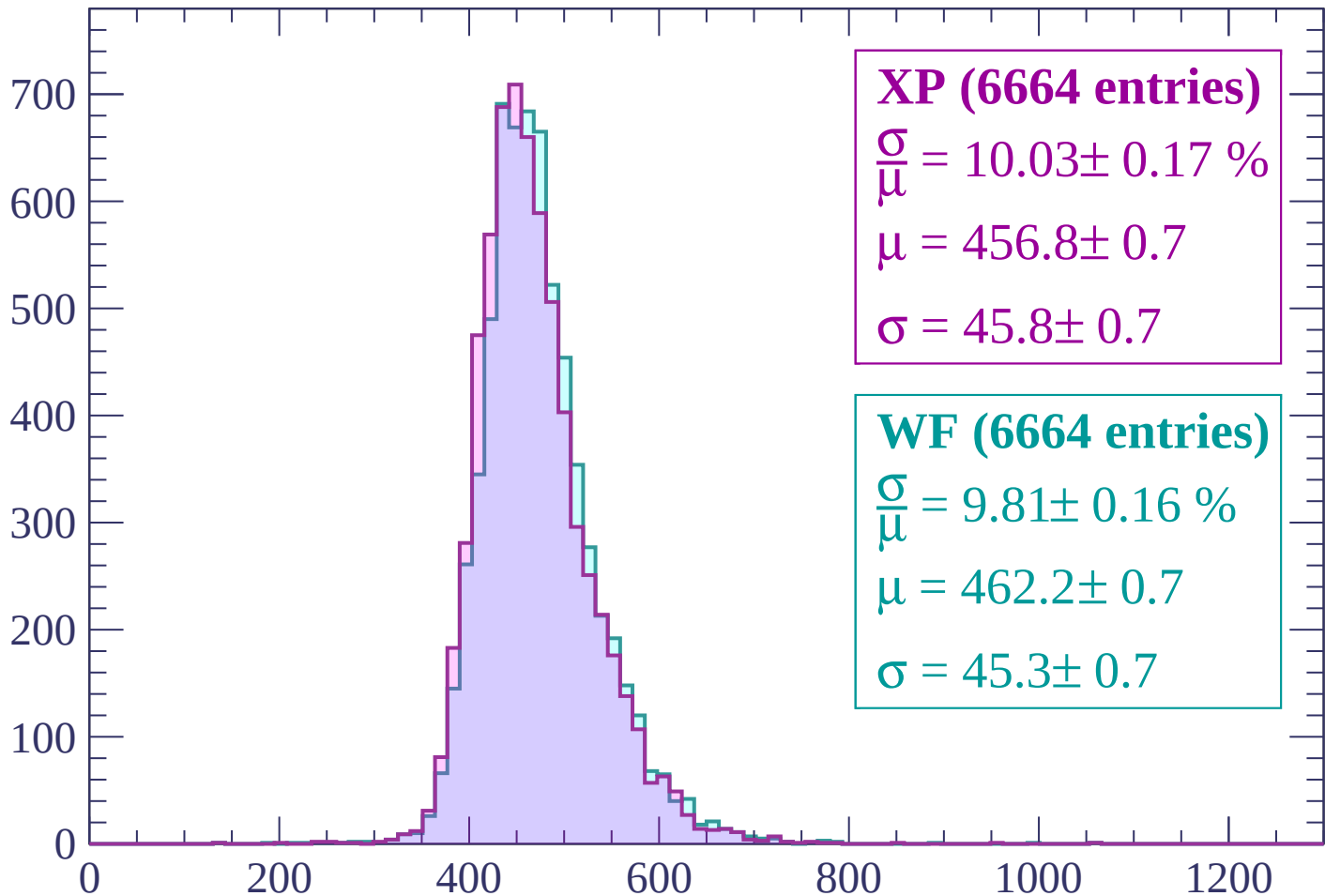
$$\mu = 459.5 \pm 0.7$$

$$\sigma = 46.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $1320 < p < 1380$

Count



**XP (6664 entries)**

$\frac{\sigma}{\mu} = 10.03 \pm 0.17 \%$

$\mu = 456.8 \pm 0.7$

$\sigma = 45.8 \pm 0.7$

**WF (6664 entries)**

$\frac{\sigma}{\mu} = 9.81 \pm 0.16 \%$

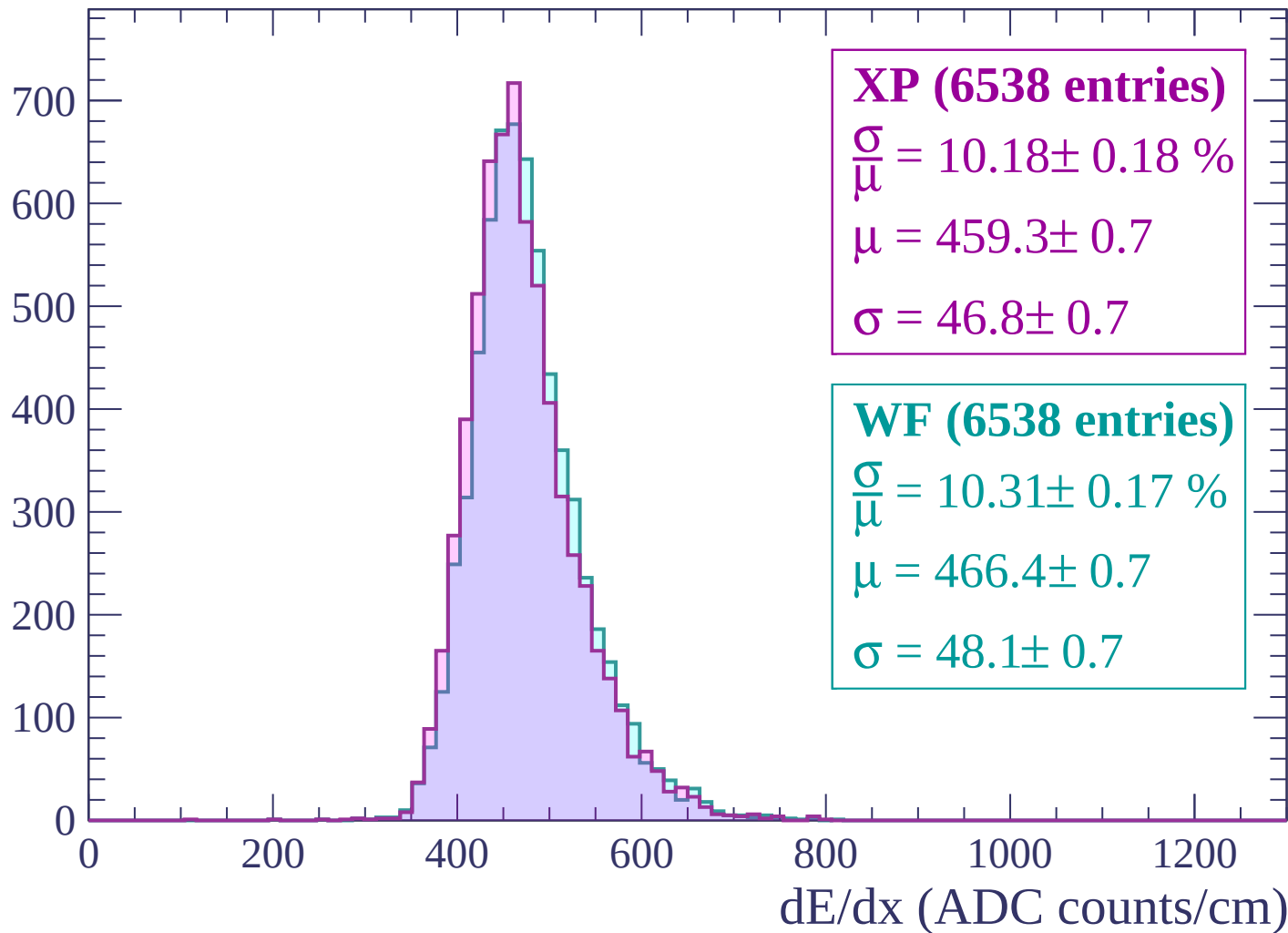
$\mu = 462.2 \pm 0.7$

$\sigma = 45.3 \pm 0.7$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1380 < p < 1440$

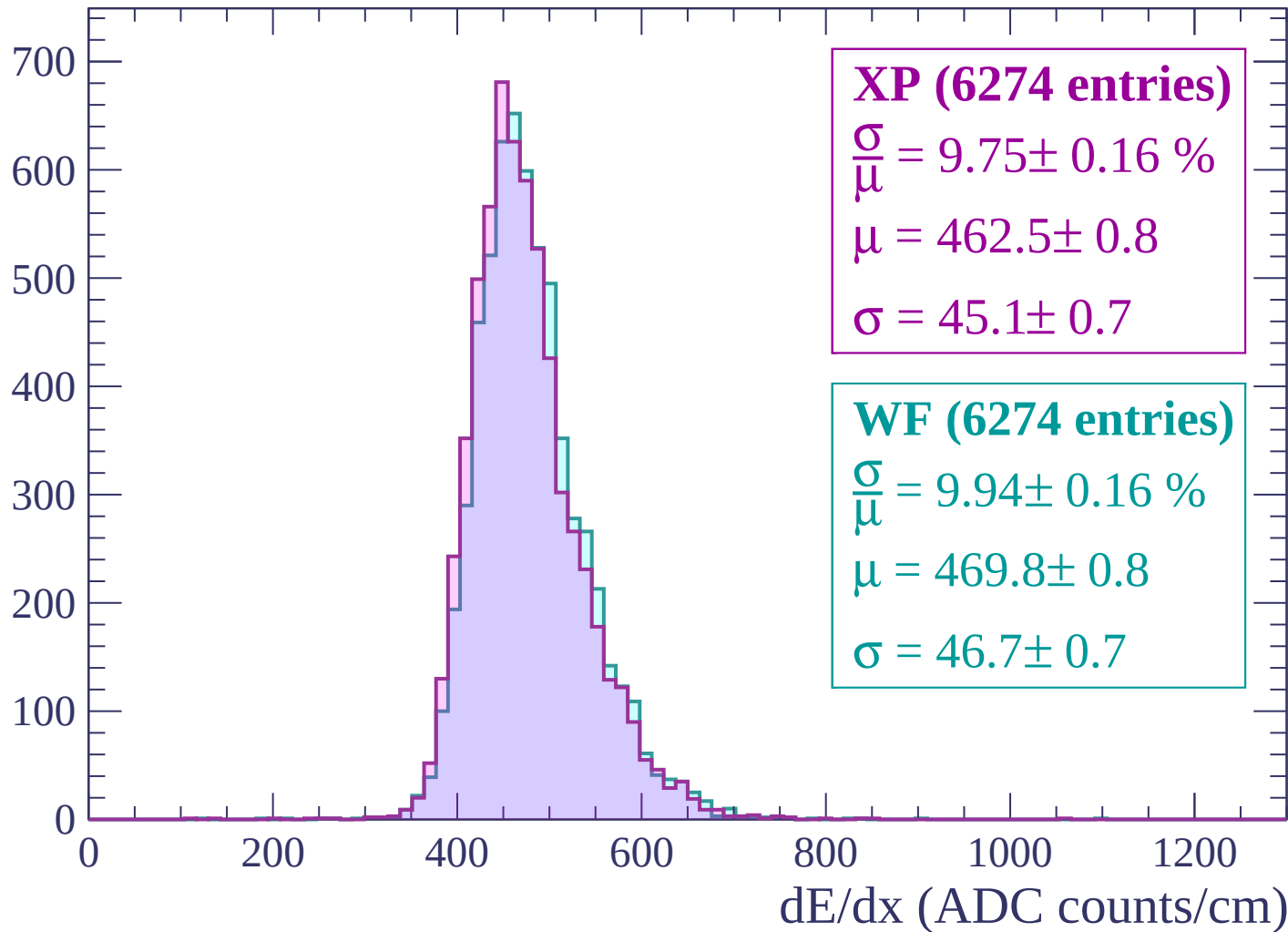
Count





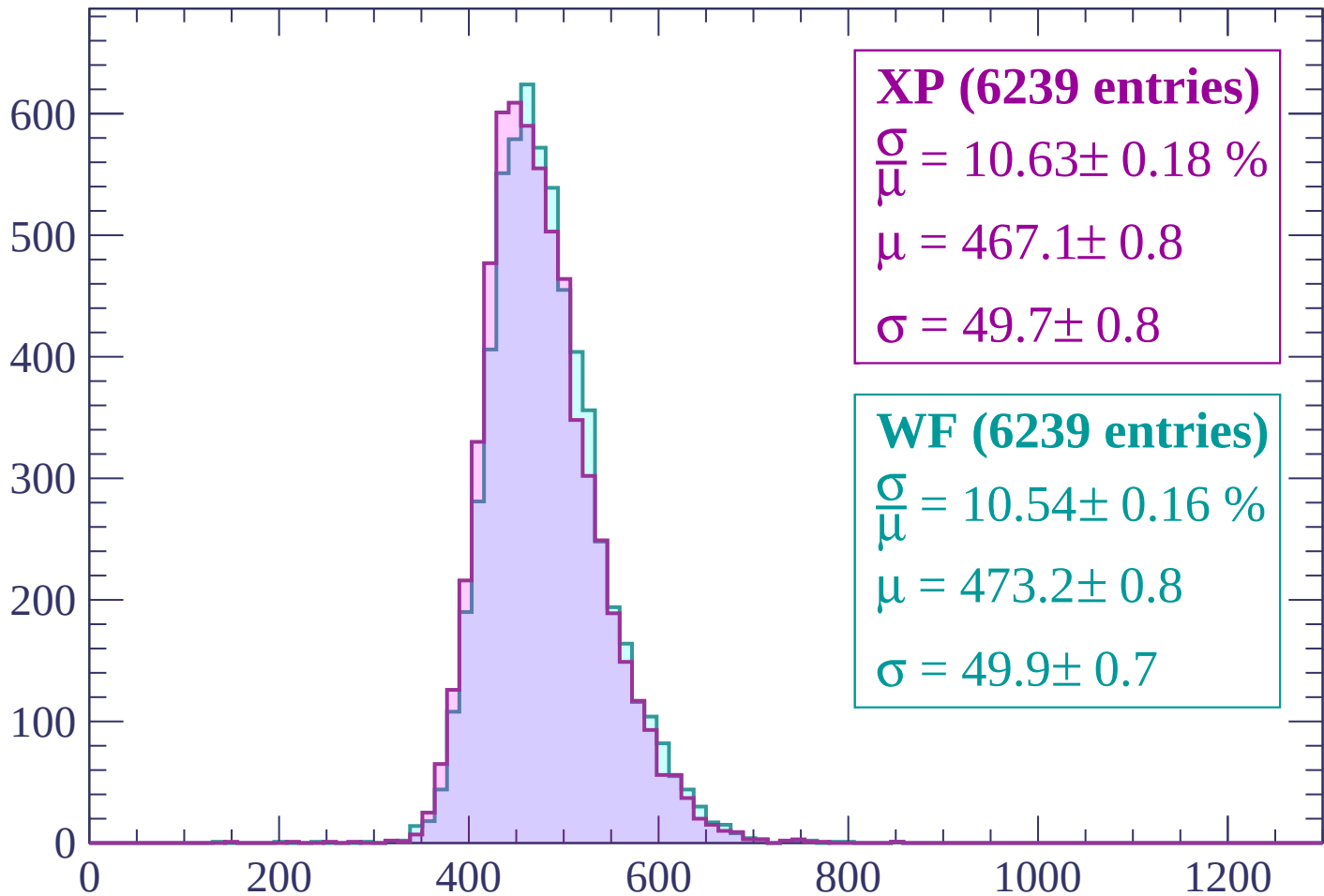
Energy loss |  $1440 < p < 1500$

Count



Energy loss |  $1500 < p < 1560$

Count



**XP (6239 entries)**

$\frac{\sigma}{\mu} = 10.63 \pm 0.18 \%$

$\mu = 467.1 \pm 0.8$

$\sigma = 49.7 \pm 0.8$

**WF (6239 entries)**

$\frac{\sigma}{\mu} = 10.54 \pm 0.16 \%$

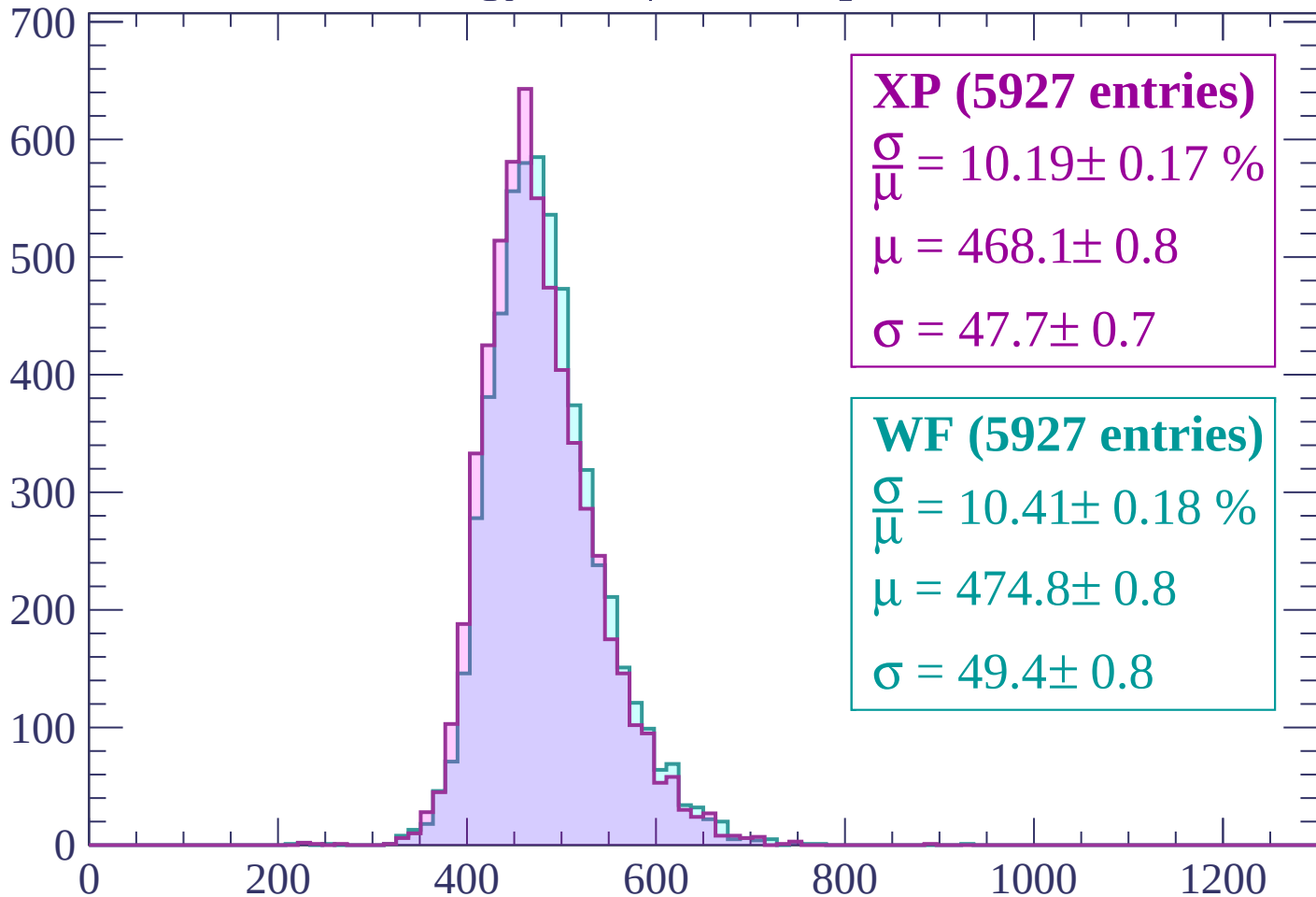
$\mu = 473.2 \pm 0.8$

$\sigma = 49.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss |  $1560 < p < 1620$

Count



**XP (5927 entries)**

$\frac{\sigma}{\mu} = 10.19 \pm 0.17 \%$

$\mu = 468.1 \pm 0.8$

$\sigma = 47.7 \pm 0.7$

**WF (5927 entries)**

$\frac{\sigma}{\mu} = 10.41 \pm 0.18 \%$

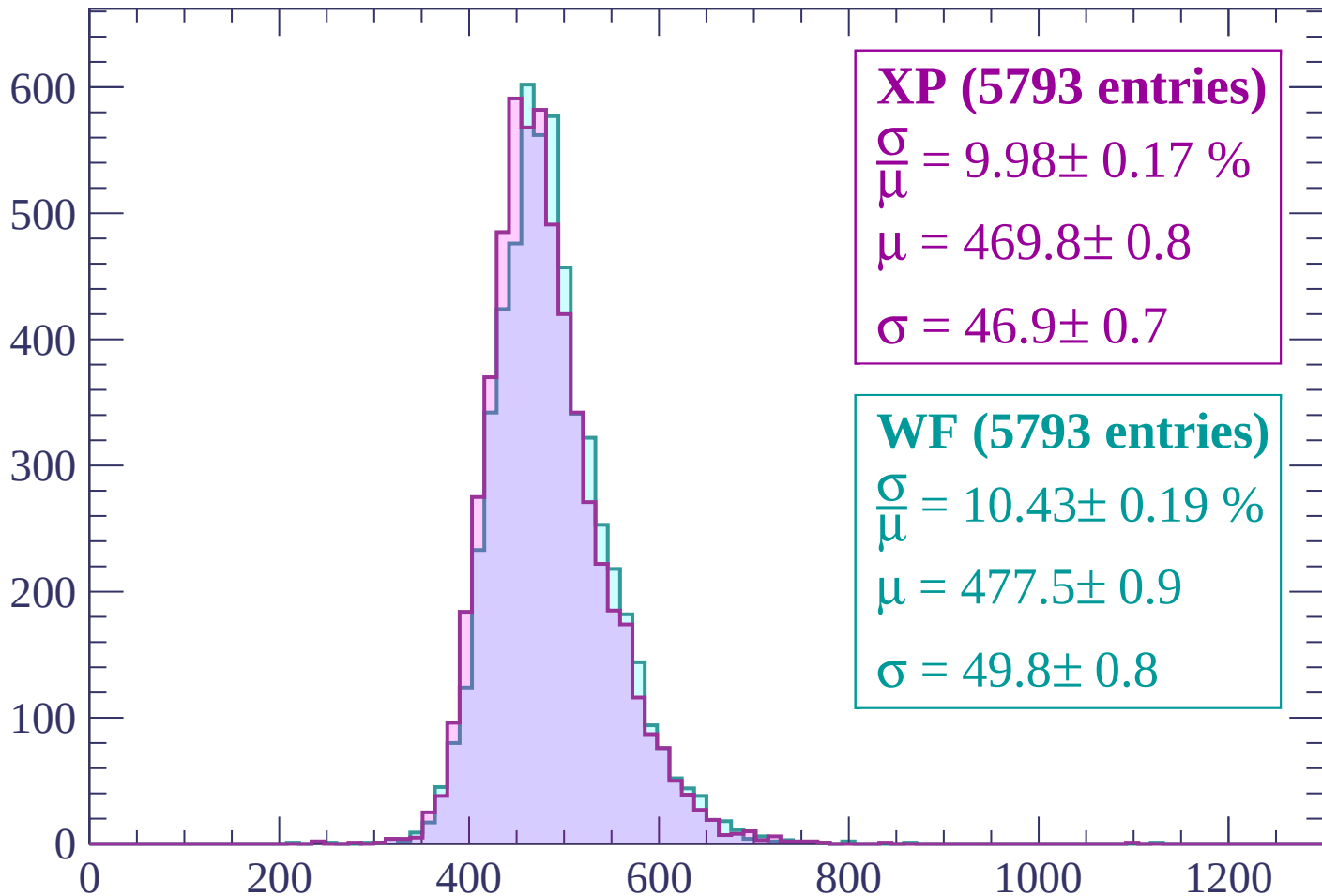
$\mu = 474.8 \pm 0.8$

$\sigma = 49.4 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss |  $1620 < p < 1680$

Count



**XP (5793 entries)**

$$\frac{\sigma}{\mu} = 9.98 \pm 0.17 \%$$

$$\mu = 469.8 \pm 0.8$$

$$\sigma = 46.9 \pm 0.7$$

**WF (5793 entries)**

$$\frac{\sigma}{\mu} = 10.43 \pm 0.19 \%$$

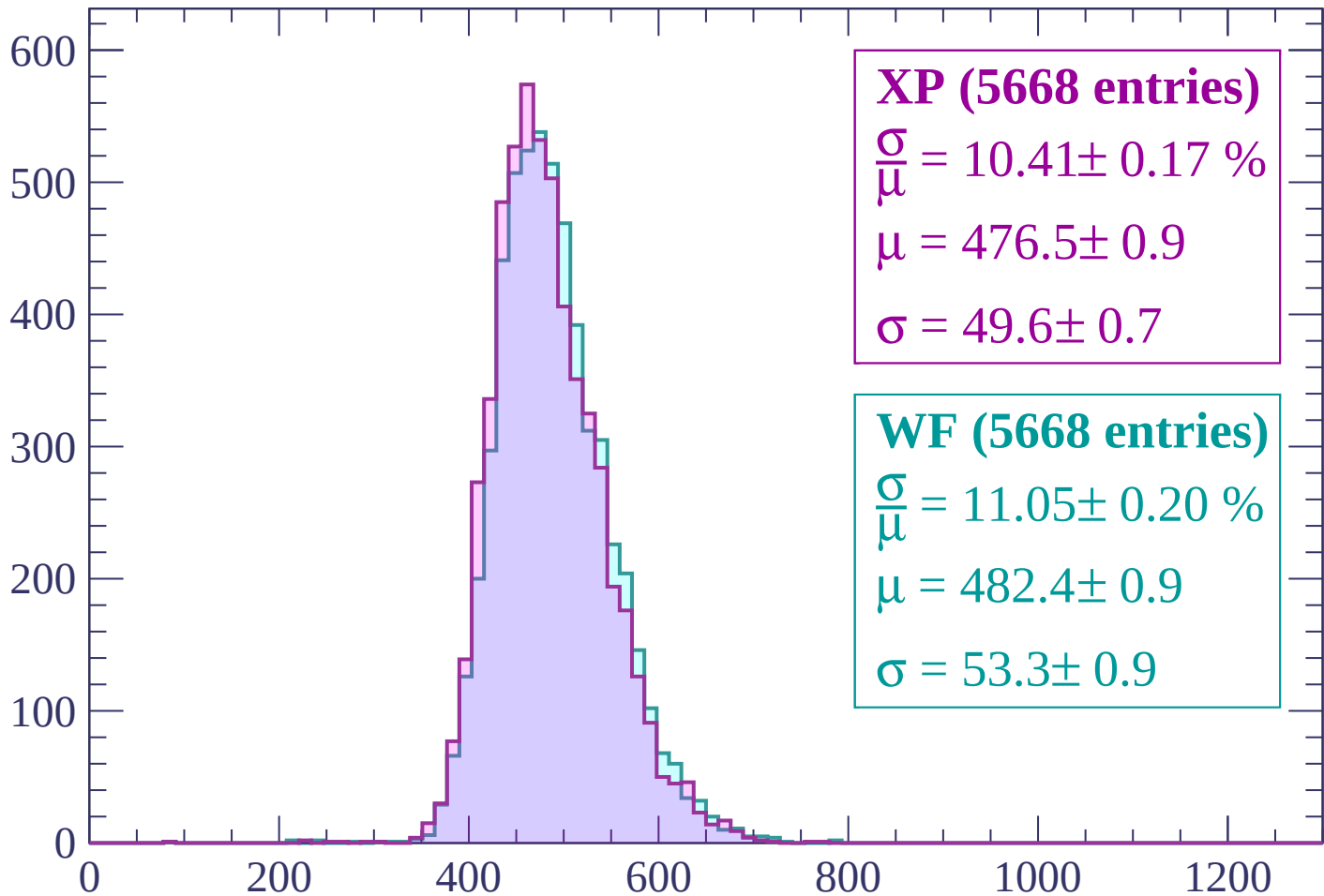
$$\mu = 477.5 \pm 0.9$$

$$\sigma = 49.8 \pm 0.8$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1680 < p < 1740$

Count



**XP (5668 entries)**

$\frac{\sigma}{\mu} = 10.41 \pm 0.17 \%$

$\mu = 476.5 \pm 0.9$

$\sigma = 49.6 \pm 0.7$

**WF (5668 entries)**

$\frac{\sigma}{\mu} = 11.05 \pm 0.20 \%$

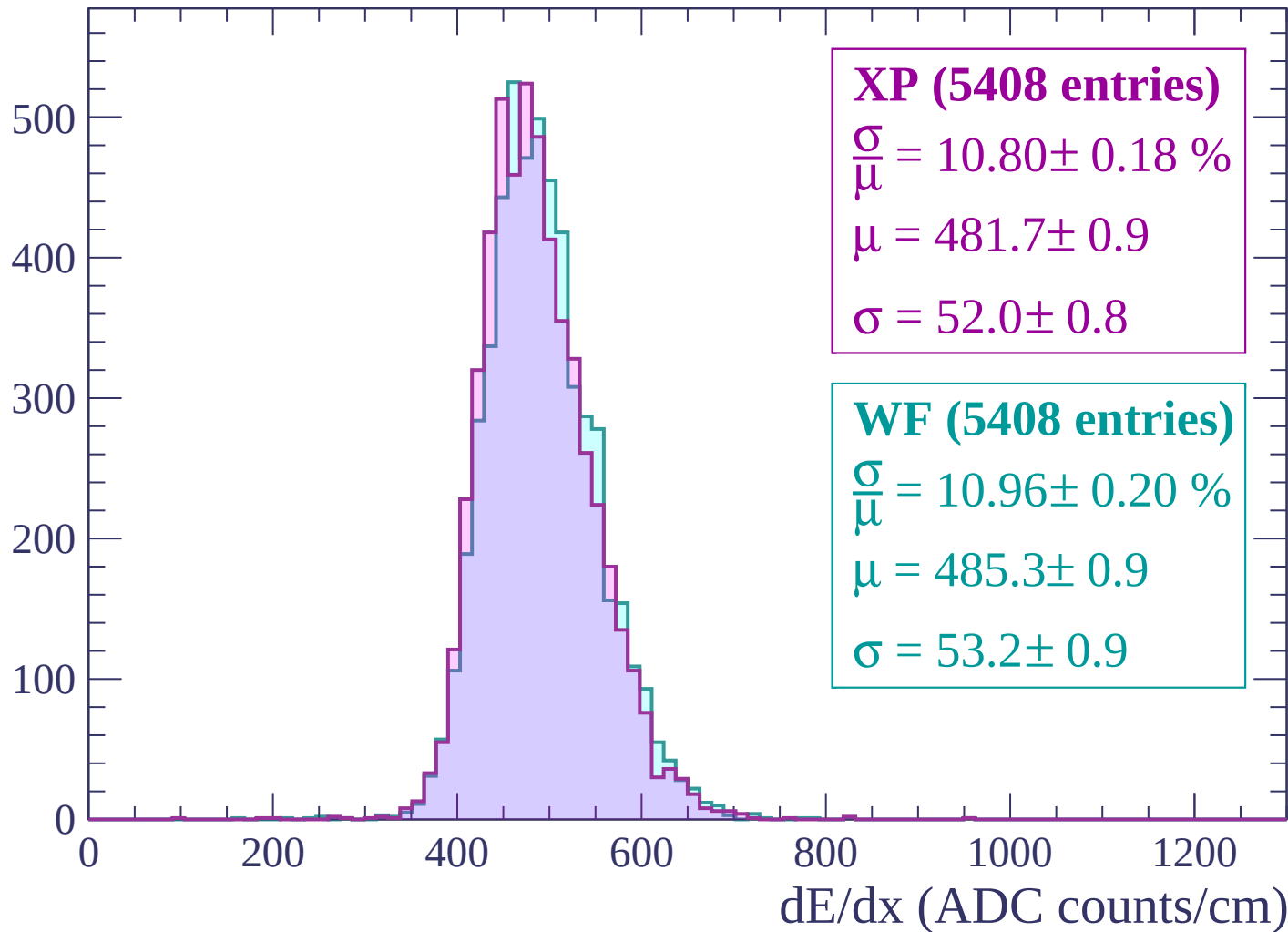
$\mu = 482.4 \pm 0.9$

$\sigma = 53.3 \pm 0.9$

$dE/dx$  (ADC counts/cm)

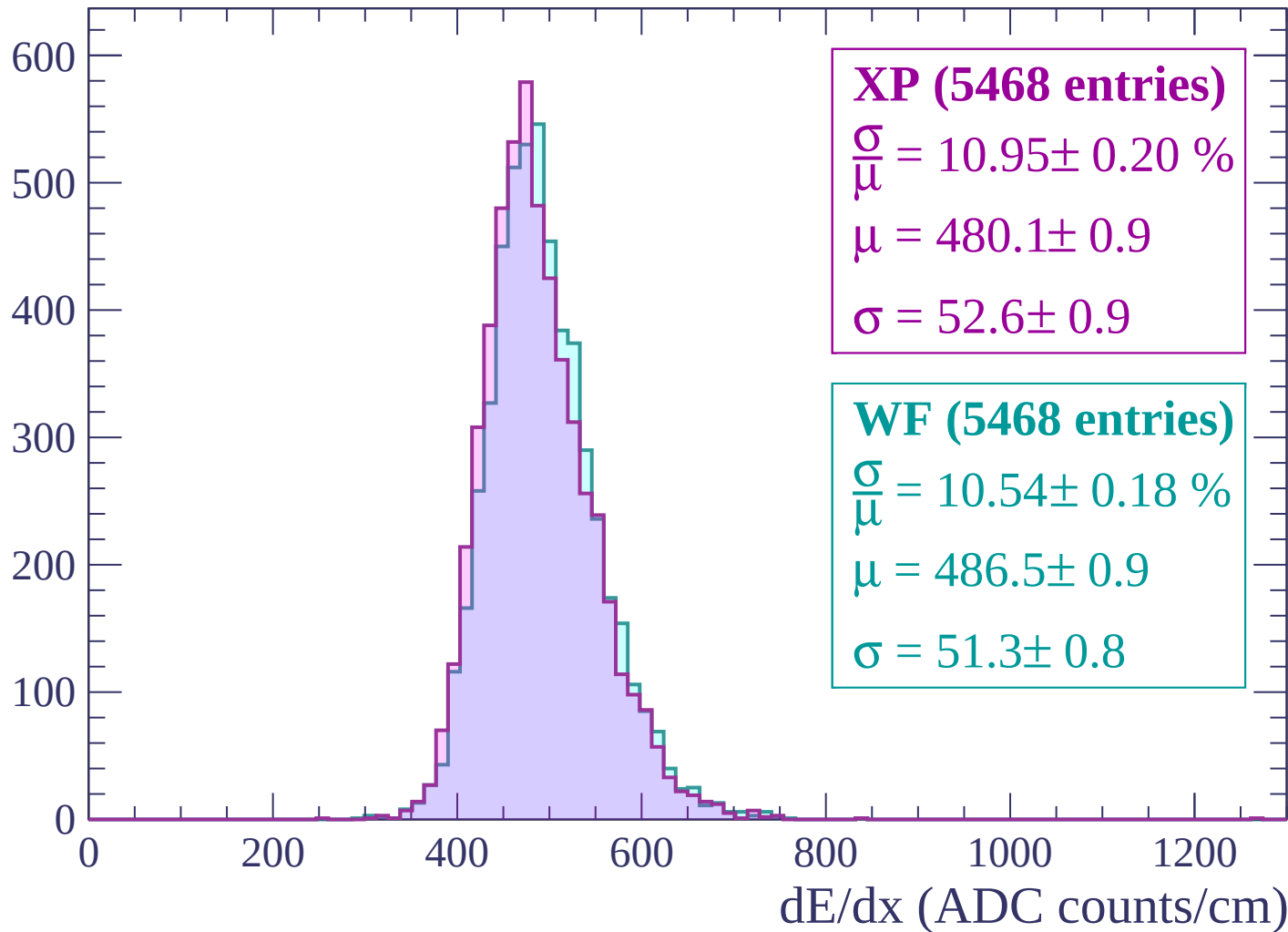
Energy loss |  $1740 < p < 1800$

Count



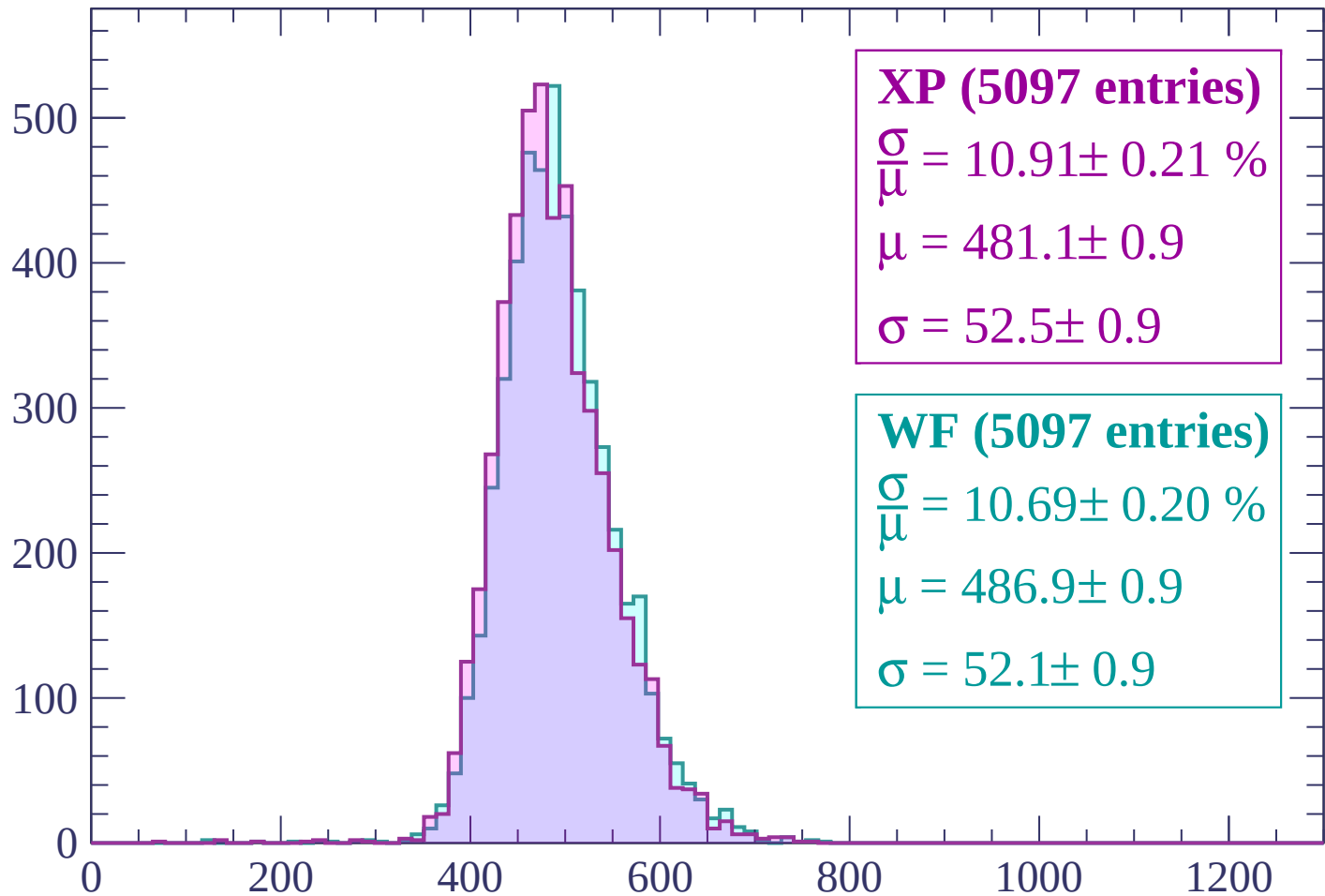
Energy loss |  $1800 < p < 1860$

Count



Energy loss |  $1860 < p < 1920$

Count



**XP (5097 entries)**

$\frac{\sigma}{\mu} = 10.91 \pm 0.21 \%$

$\mu = 481.1 \pm 0.9$

$\sigma = 52.5 \pm 0.9$

**WF (5097 entries)**

$\frac{\sigma}{\mu} = 10.69 \pm 0.20 \%$

$\mu = 486.9 \pm 0.9$

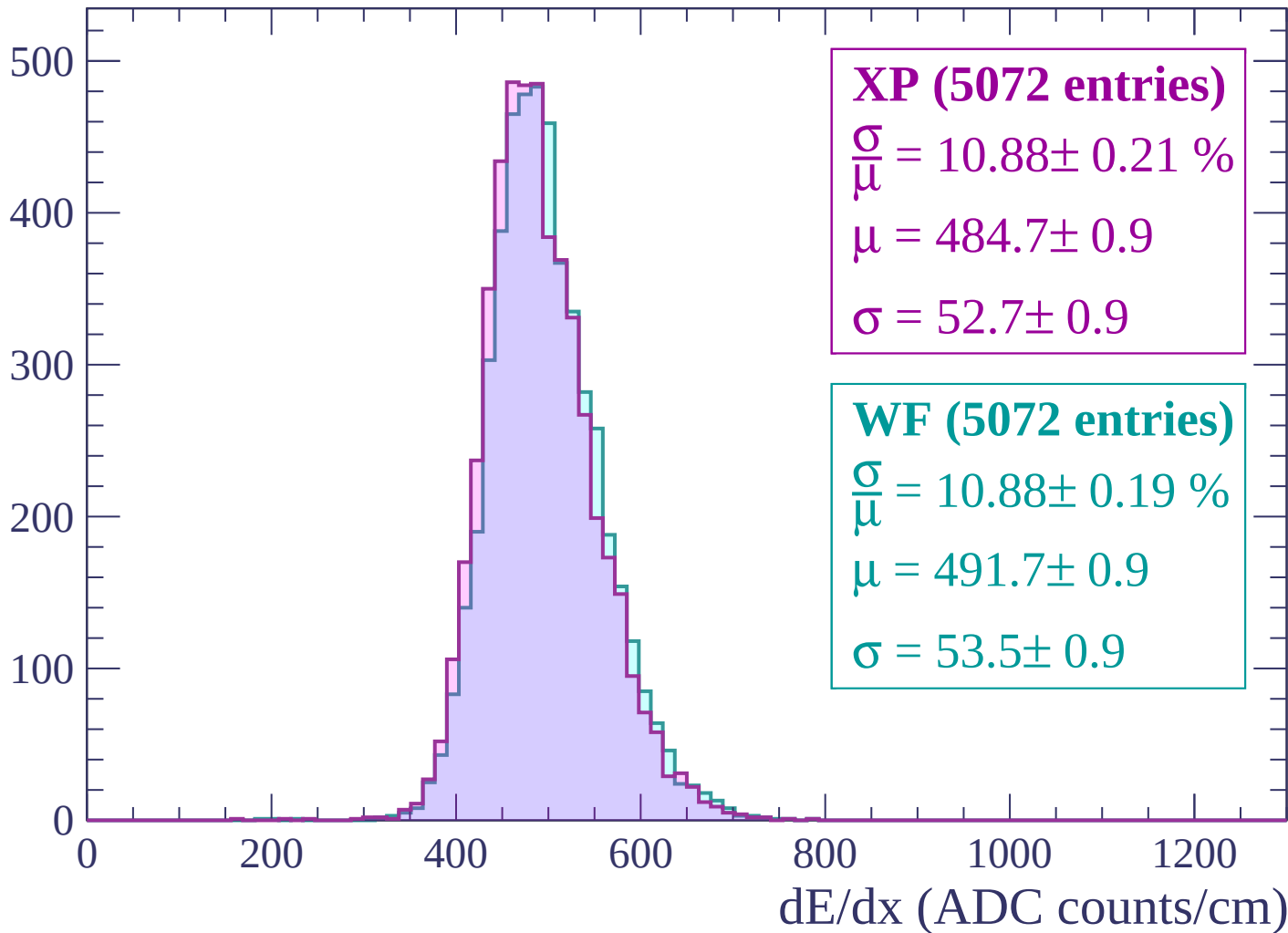
$\sigma = 52.1 \pm 0.9$

dE/dx (ADC counts/cm)



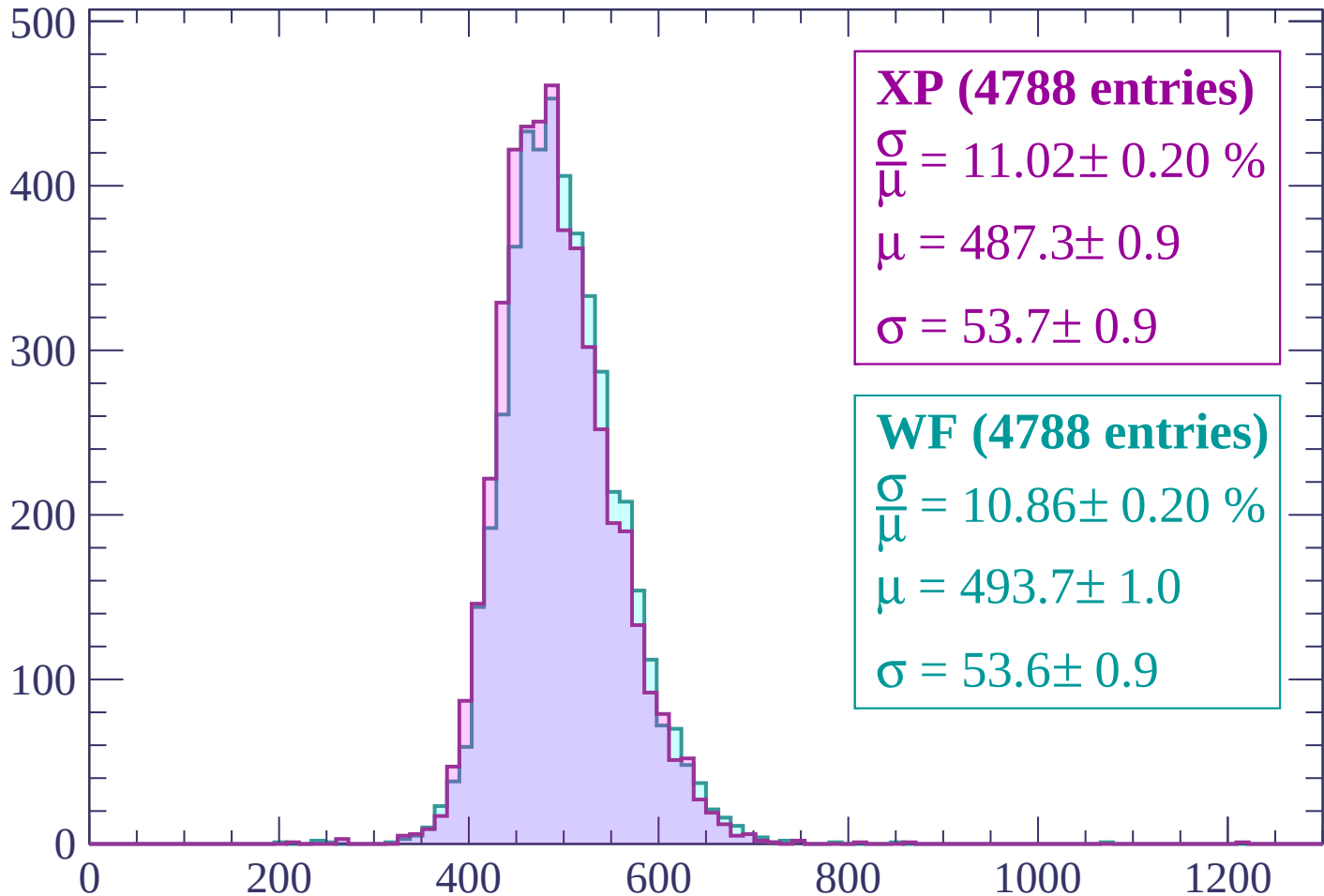
# Energy loss | $1920 < p < 1980$

Count



# Energy loss | $1980 < p < 2040$

Count



**XP (4788 entries)**

$$\frac{\sigma}{\mu} = 11.02 \pm 0.20 \%$$

$$\mu = 487.3 \pm 0.9$$

$$\sigma = 53.7 \pm 0.9$$

**WF (4788 entries)**

$$\frac{\sigma}{\mu} = 10.86 \pm 0.20 \%$$

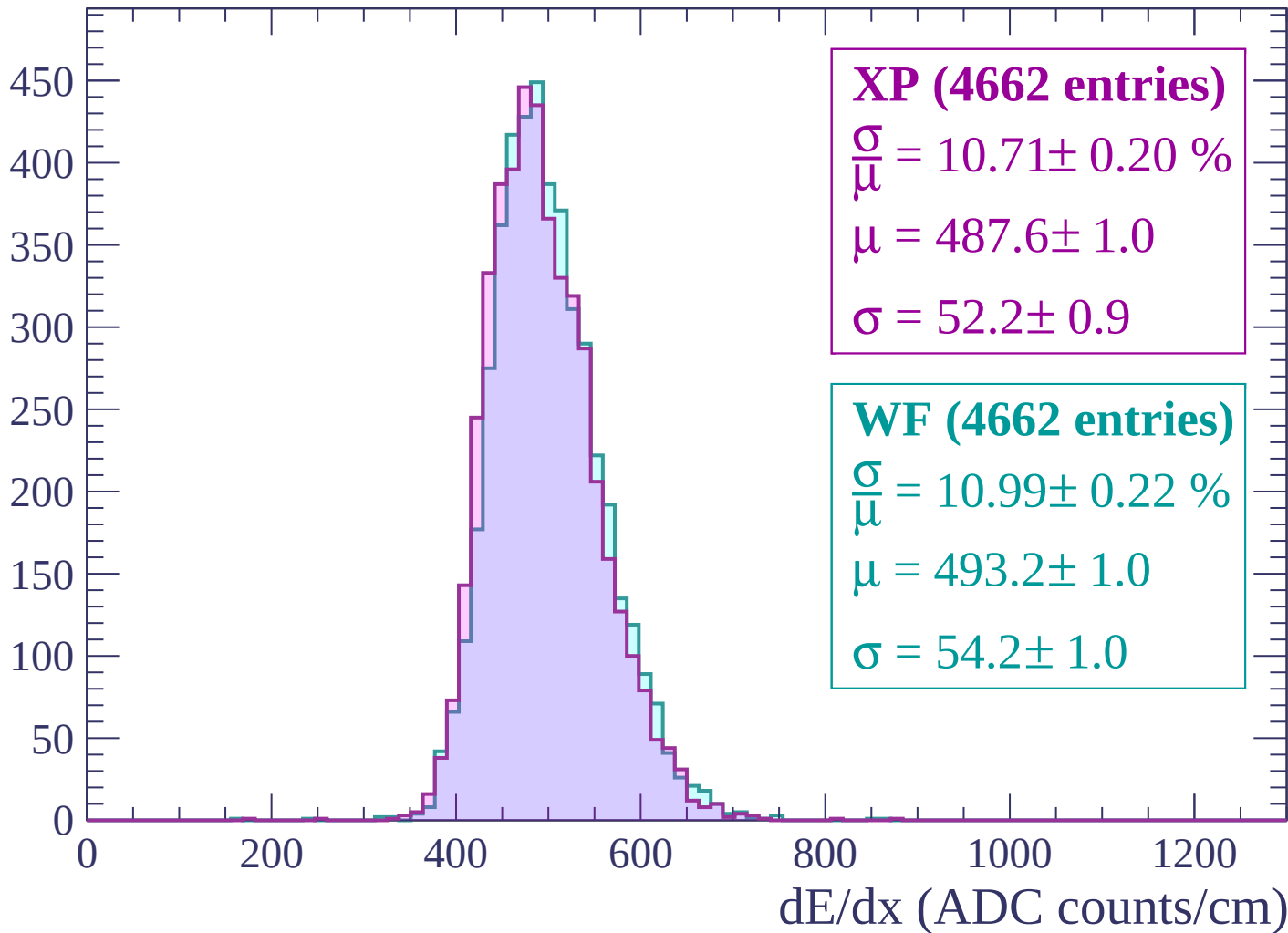
$$\mu = 493.7 \pm 1.0$$

$$\sigma = 53.6 \pm 0.9$$

dE/dx (ADC counts/cm)

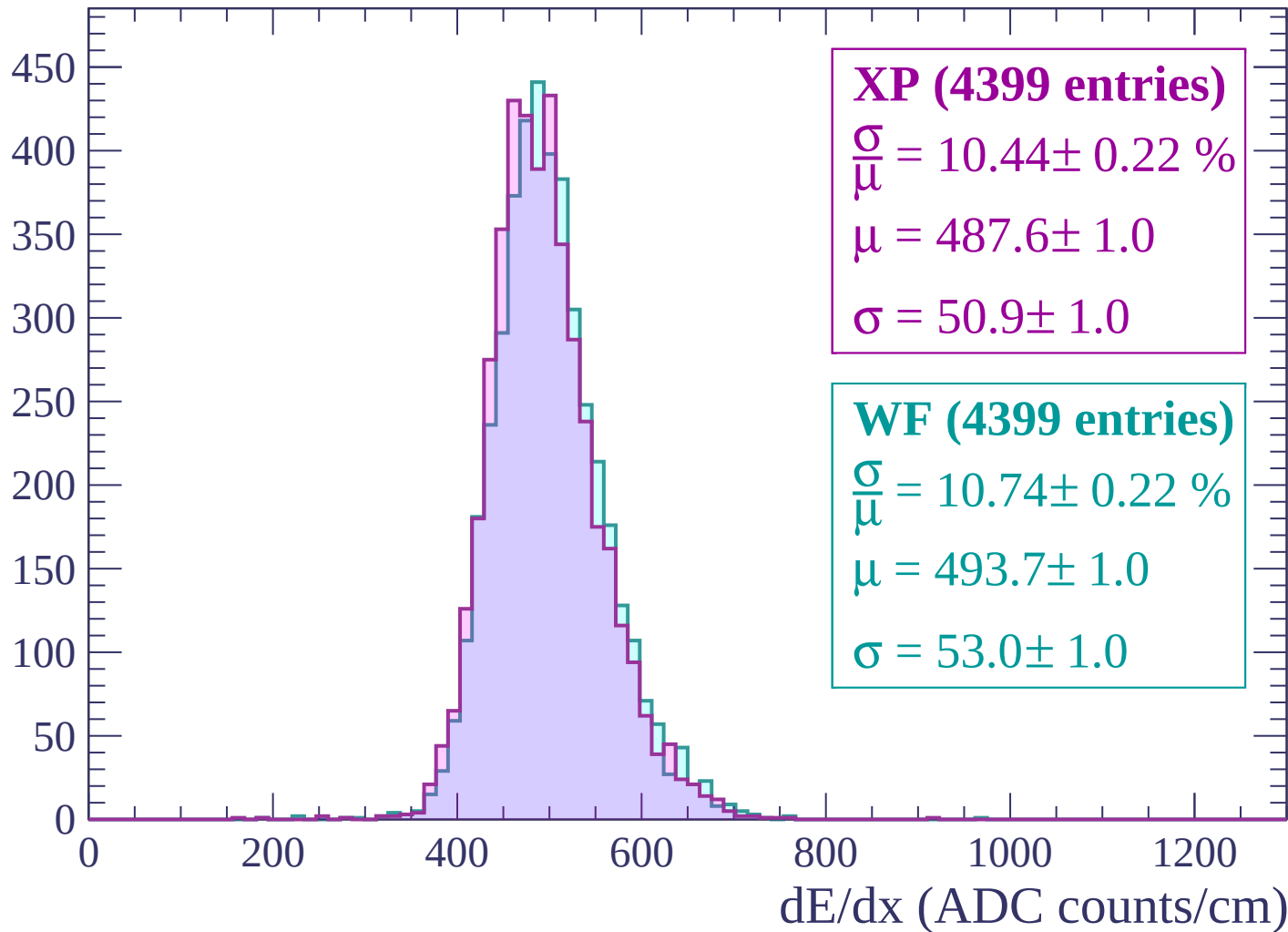
# Energy loss | $2040 < p < 2100$

Count



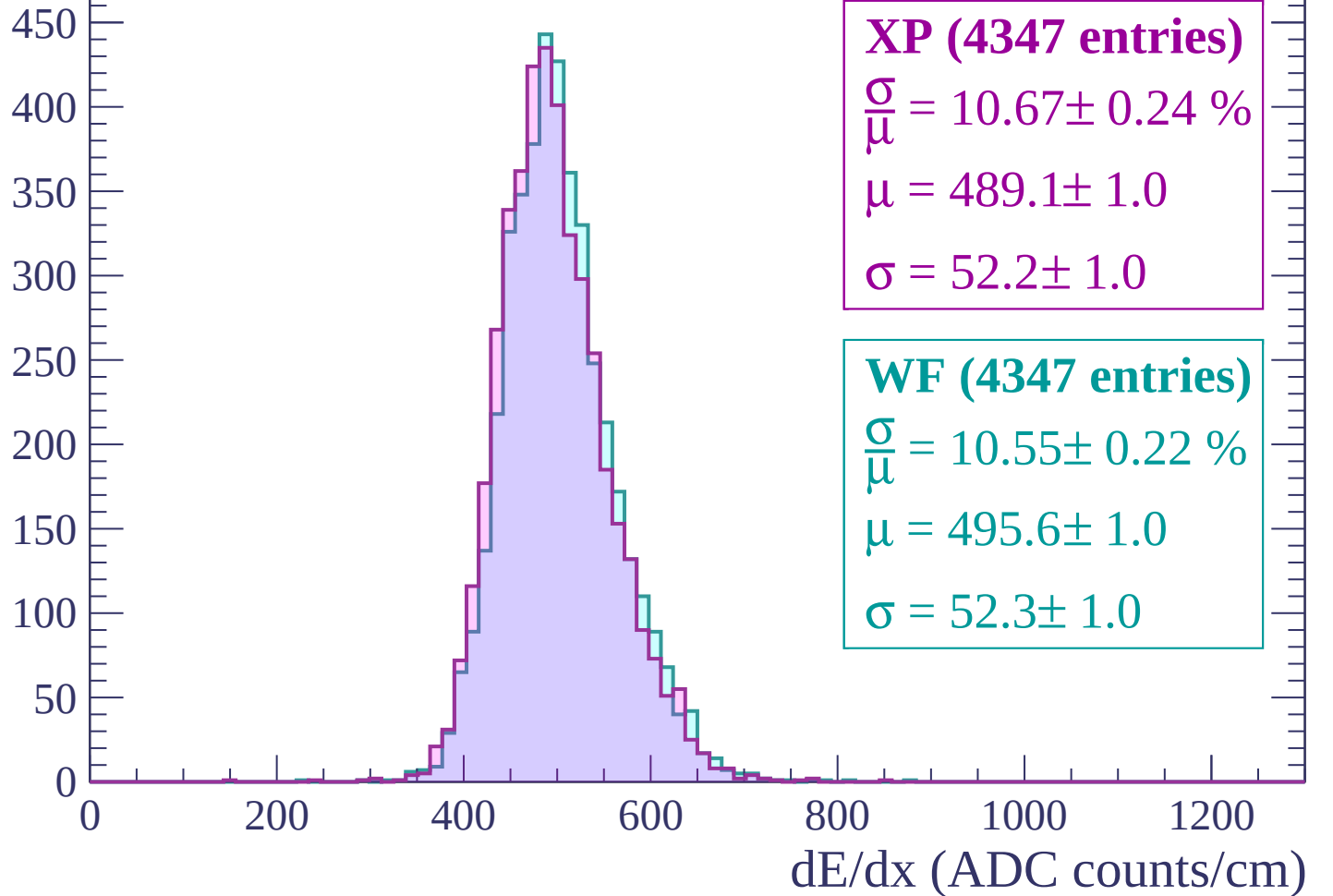
# Energy loss | $2100 < p < 2160$

Count



Energy loss |  $2160 < p < 2220$

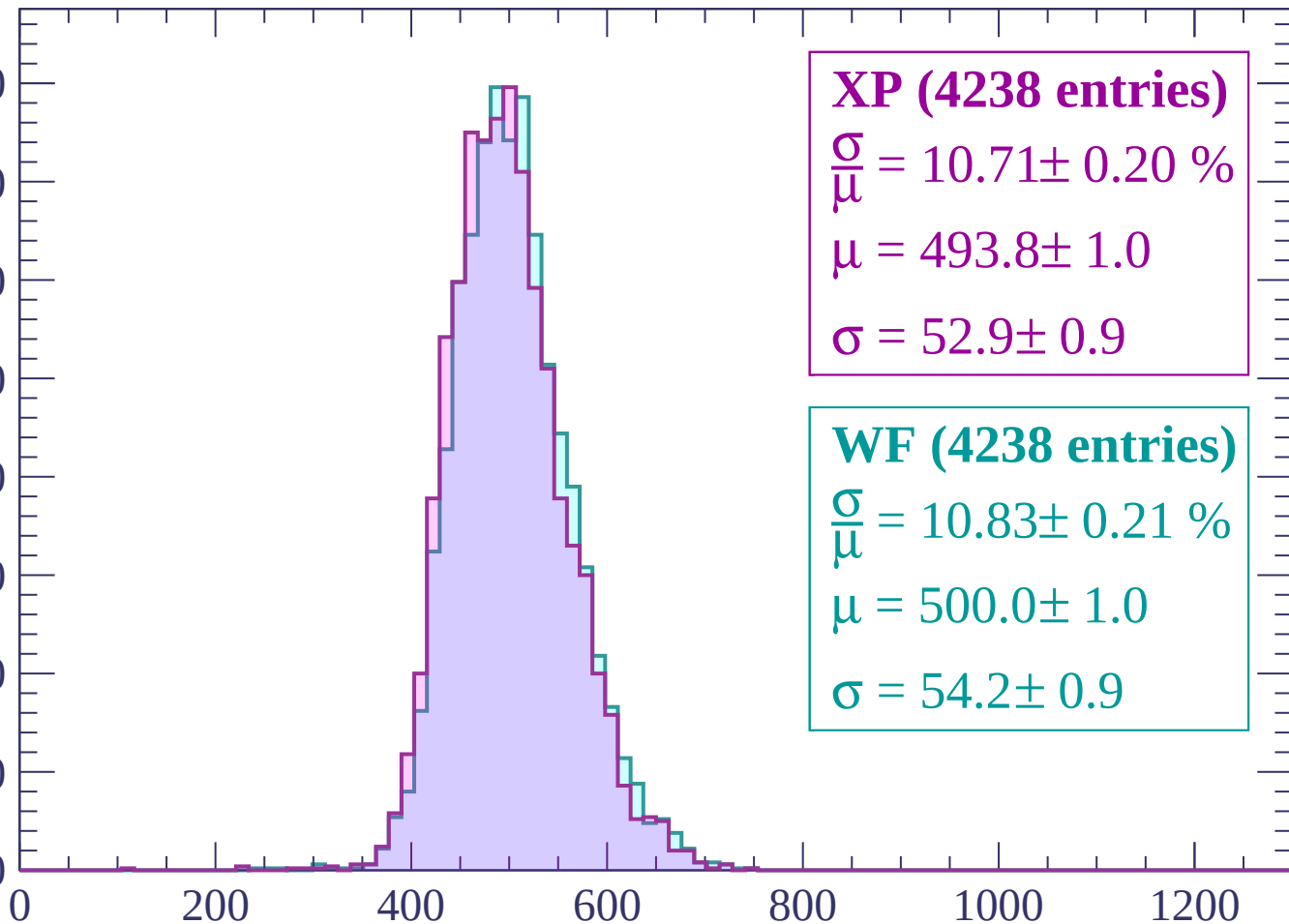
Count



# Energy loss | $2220 < p < 2280$

Count

400  
350  
300  
250  
200  
150  
100  
50  
0



**XP (4238 entries)**

$\frac{\sigma}{\mu} = 10.71 \pm 0.20 \%$

$\mu = 493.8 \pm 1.0$

$\sigma = 52.9 \pm 0.9$

**WF (4238 entries)**

$\frac{\sigma}{\mu} = 10.83 \pm 0.21 \%$

$\mu = 500.0 \pm 1.0$

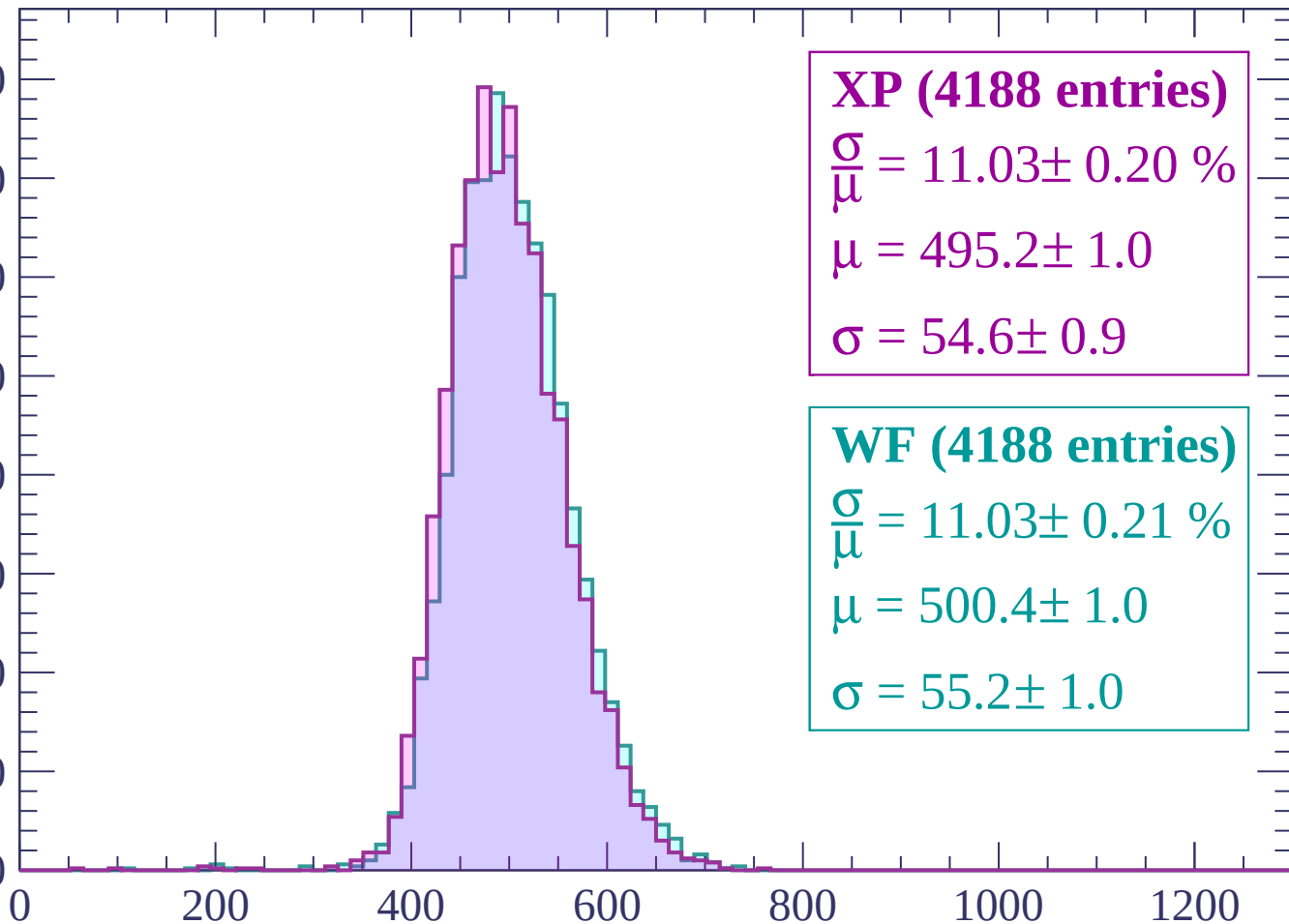
$\sigma = 54.2 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss |  $2280 < p < 2340$

Count

400  
350  
300  
250  
200  
150  
100  
50  
0



**XP (4188 entries)**

$\frac{\sigma}{\mu} = 11.03 \pm 0.20 \%$

$\mu = 495.2 \pm 1.0$

$\sigma = 54.6 \pm 0.9$

**WF (4188 entries)**

$\frac{\sigma}{\mu} = 11.03 \pm 0.21 \%$

$\mu = 500.4 \pm 1.0$

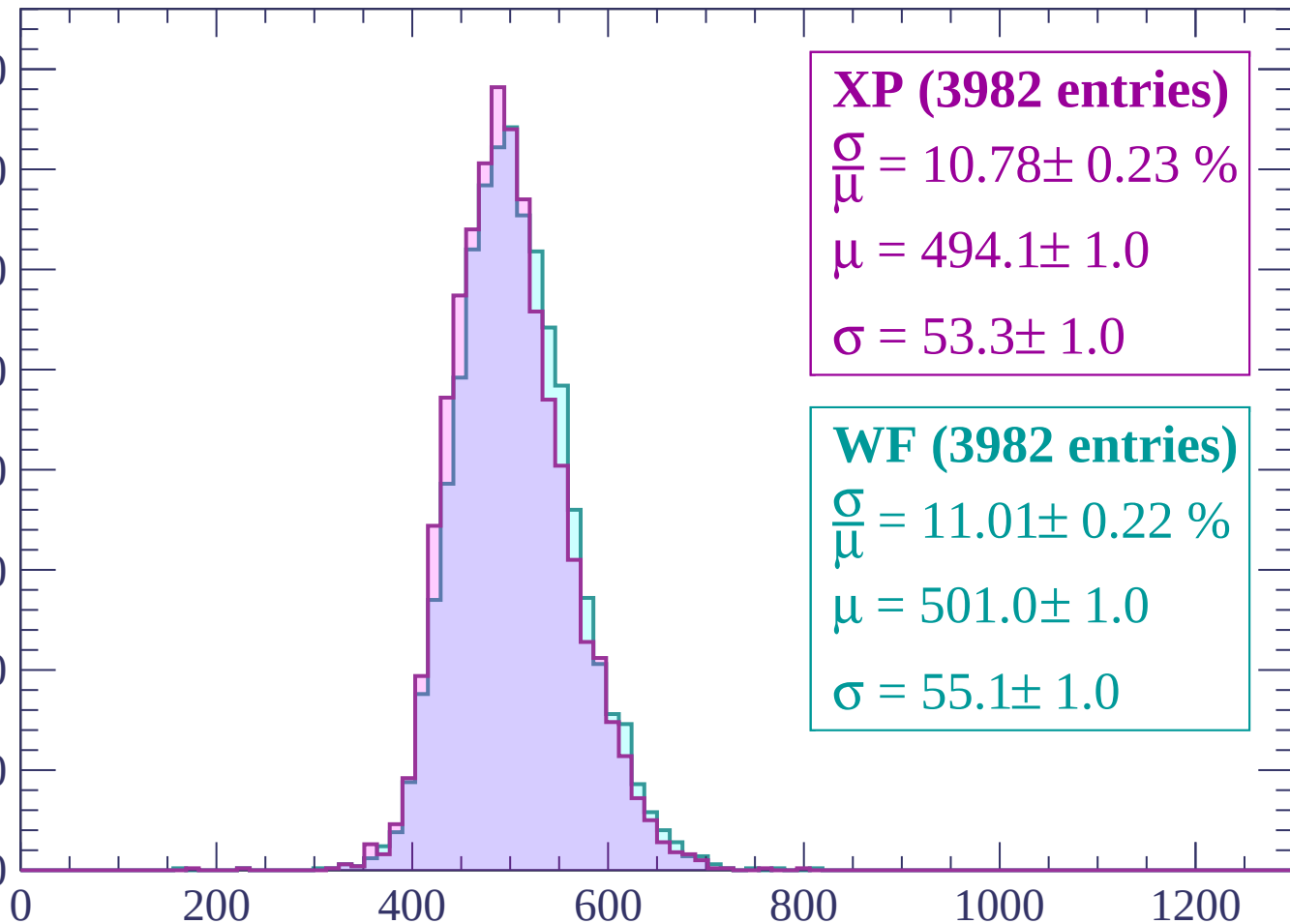
$\sigma = 55.2 \pm 1.0$

$dE/dx$  (ADC counts/cm)

Energy loss |  $2340 < p < 2400$

Count

400  
350  
300  
250  
200  
150  
100  
50  
0



**XP (3982 entries)**

$\frac{\sigma}{\mu} = 10.78 \pm 0.23 \%$

$\mu = 494.1 \pm 1.0$

$\sigma = 53.3 \pm 1.0$

**WF (3982 entries)**

$\frac{\sigma}{\mu} = 11.01 \pm 0.22 \%$

$\mu = 501.0 \pm 1.0$

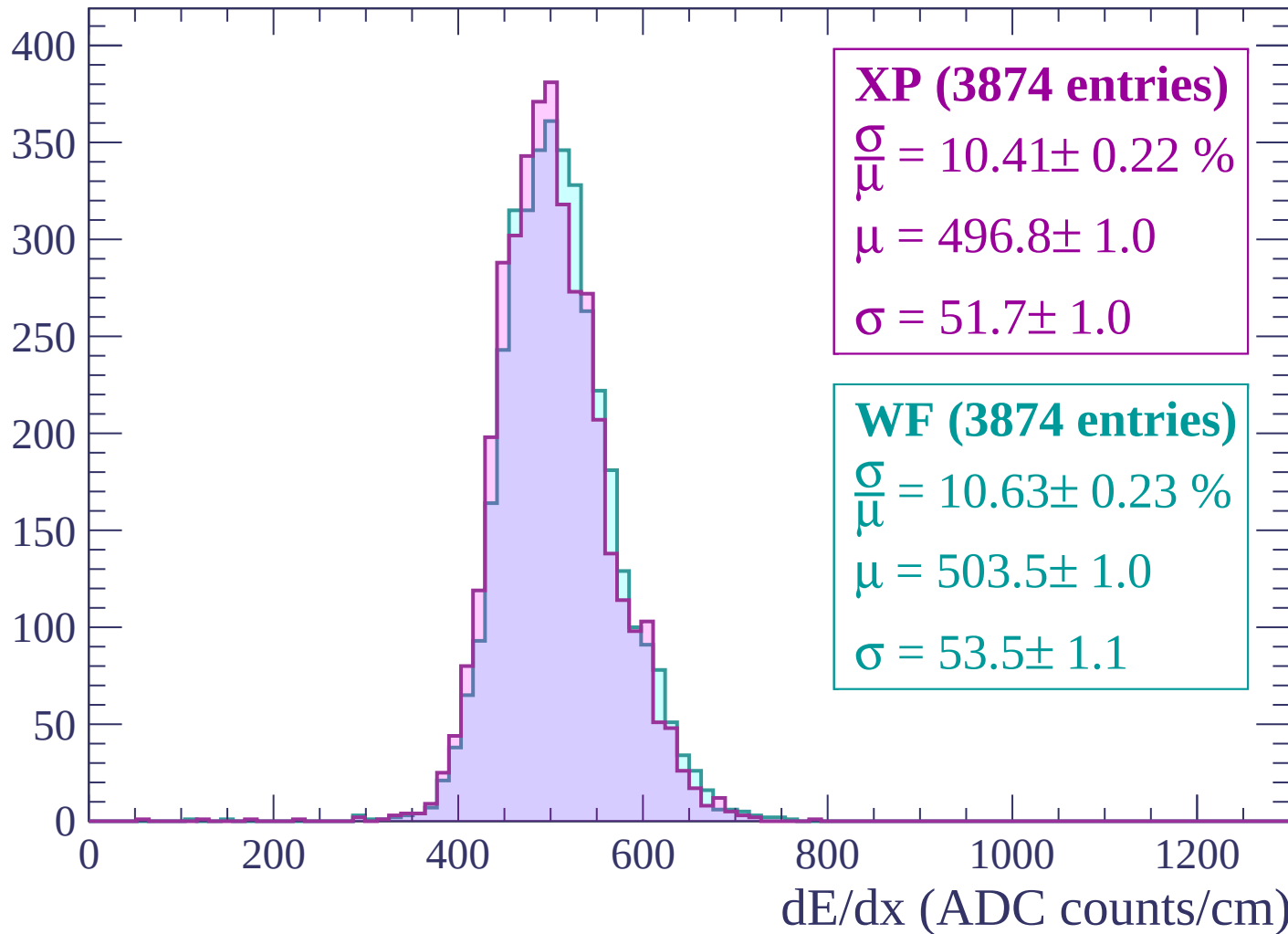
$\sigma = 55.1 \pm 1.0$

dE/dx (ADC counts/cm)



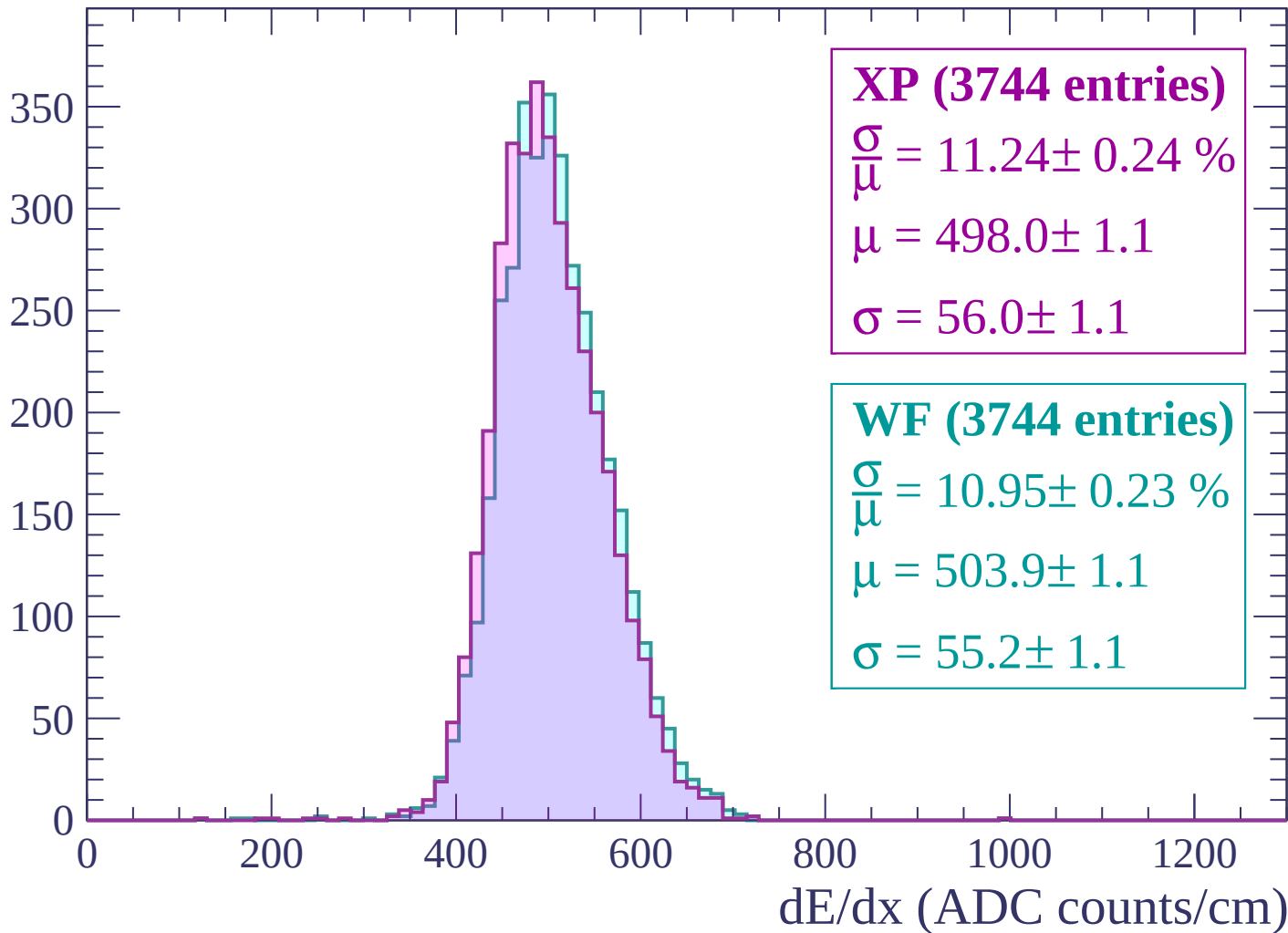
Energy loss |  $2400 < p < 2460$

Count



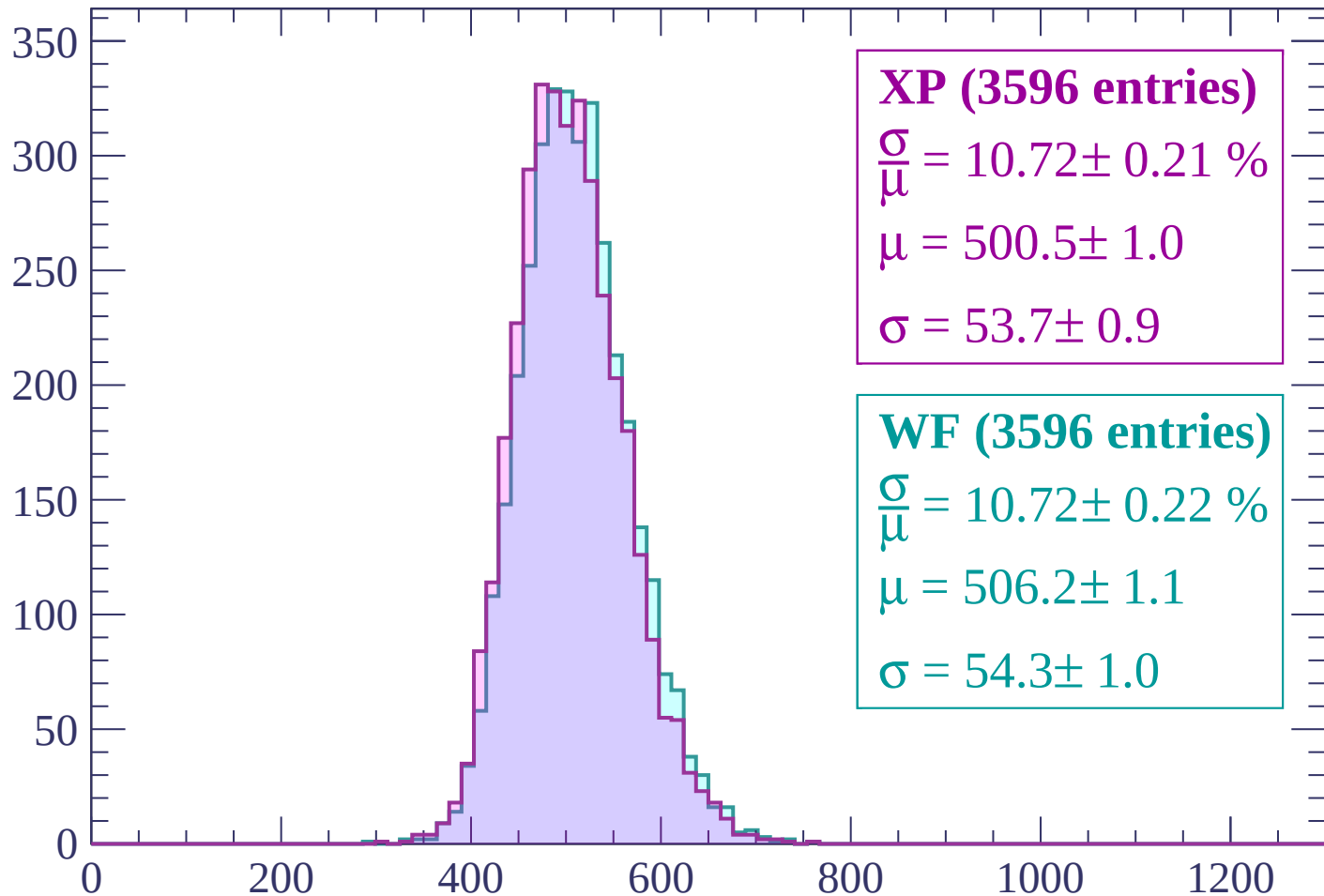
Energy loss |  $2460 < p < 2520$

Count



Energy loss |  $2520 < p < 2580$

Count



**XP (3596 entries)**

$$\frac{\sigma}{\mu} = 10.72 \pm 0.21 \%$$

$$\mu = 500.5 \pm 1.0$$

$$\sigma = 53.7 \pm 0.9$$

**WF (3596 entries)**

$$\frac{\sigma}{\mu} = 10.72 \pm 0.22 \%$$

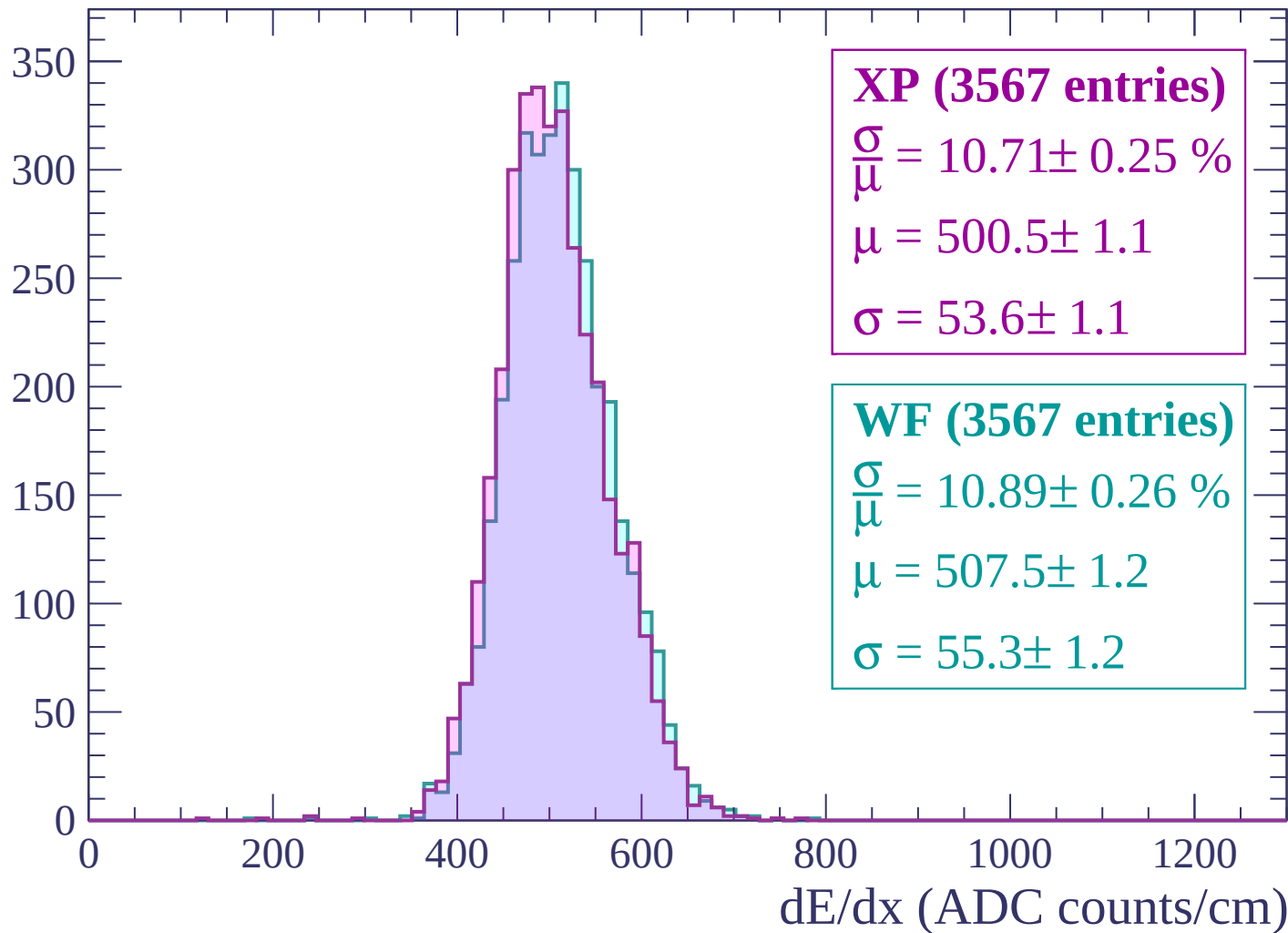
$$\mu = 506.2 \pm 1.1$$

$$\sigma = 54.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $2580 < p < 2640$

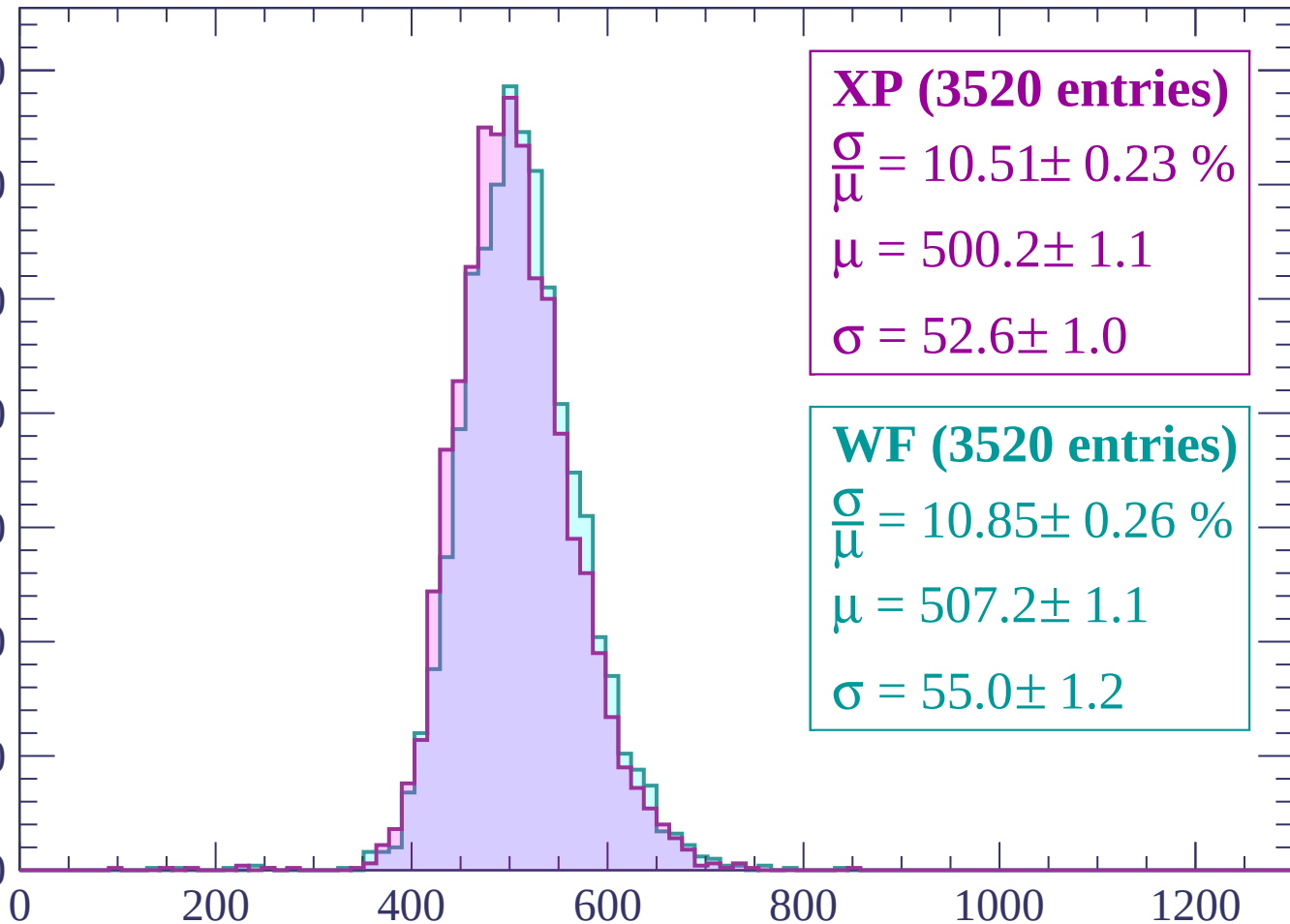
Count



Energy loss |  $2640 < p < 2700$

Count

350  
300  
250  
200  
150  
100  
50  
0



**XP (3520 entries)**

$\frac{\sigma}{\mu} = 10.51 \pm 0.23 \%$

$\mu = 500.2 \pm 1.1$

$\sigma = 52.6 \pm 1.0$

**WF (3520 entries)**

$\frac{\sigma}{\mu} = 10.85 \pm 0.26 \%$

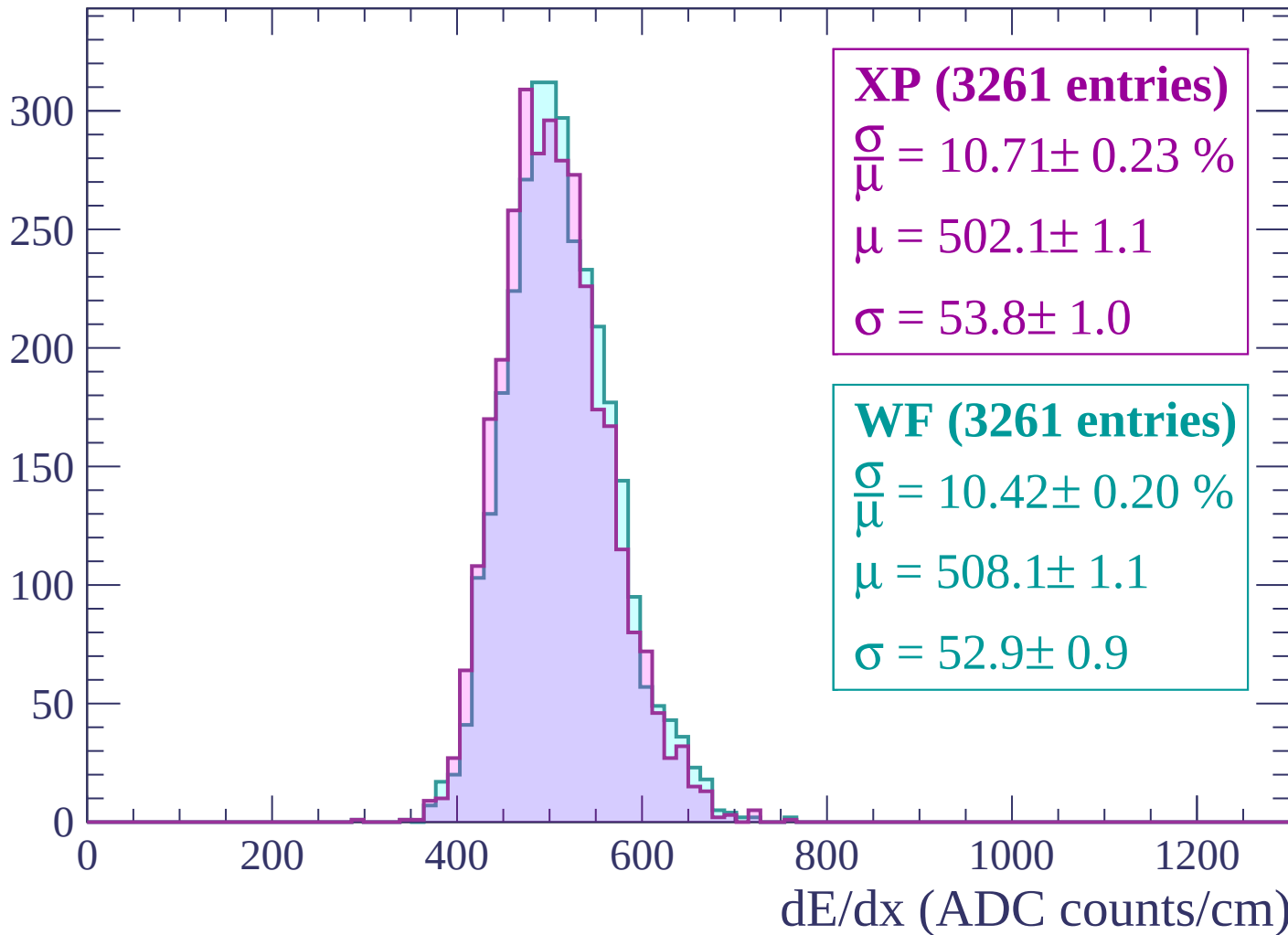
$\mu = 507.2 \pm 1.1$

$\sigma = 55.0 \pm 1.2$

$dE/dx$  (ADC counts/cm)

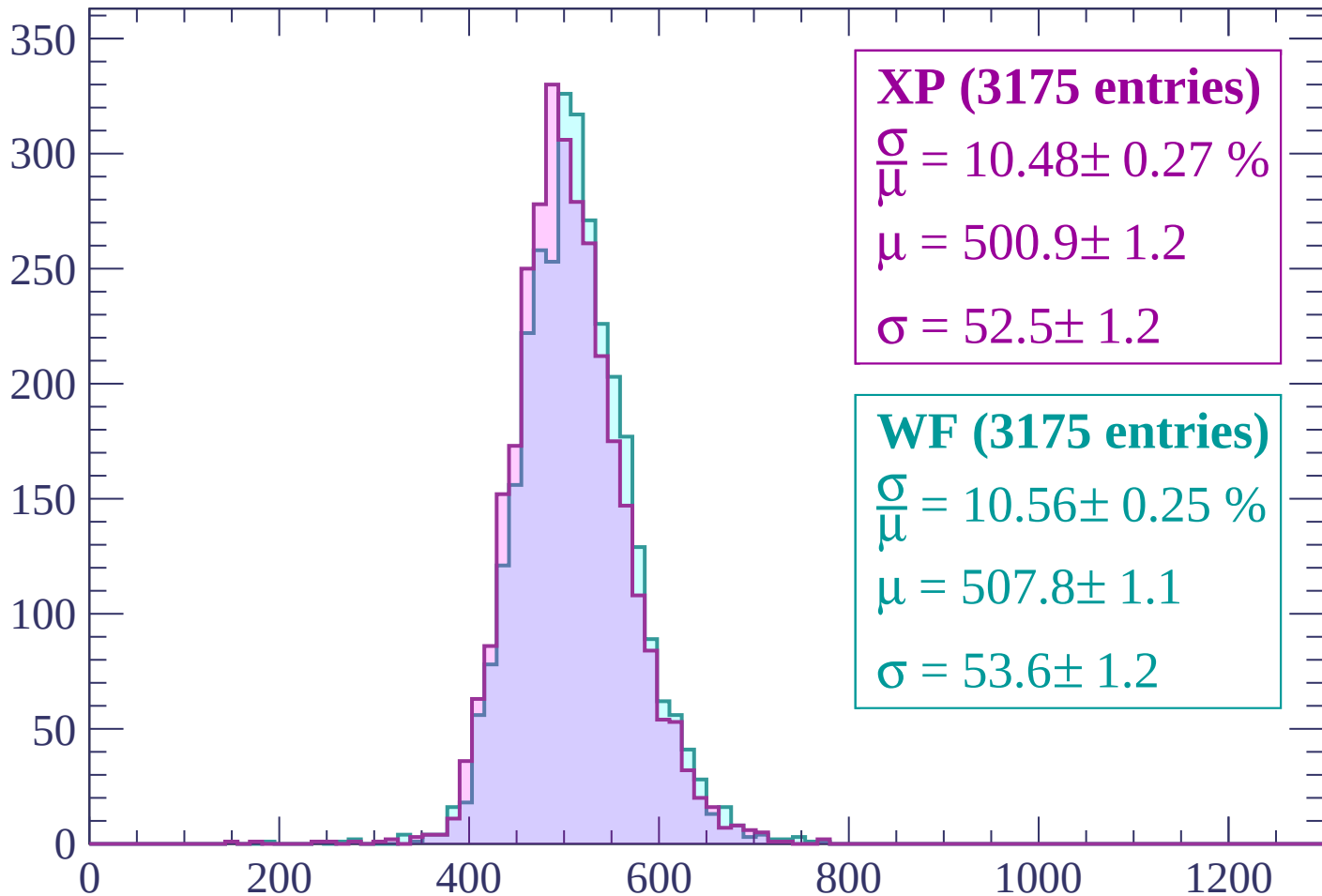
Energy loss |  $2700 < p < 2760$

Count



Energy loss |  $2760 < p < 2820$

Count



**XP (3175 entries)**

$\frac{\sigma}{\mu} = 10.48 \pm 0.27 \%$

$\mu = 500.9 \pm 1.2$

$\sigma = 52.5 \pm 1.2$

**WF (3175 entries)**

$\frac{\sigma}{\mu} = 10.56 \pm 0.25 \%$

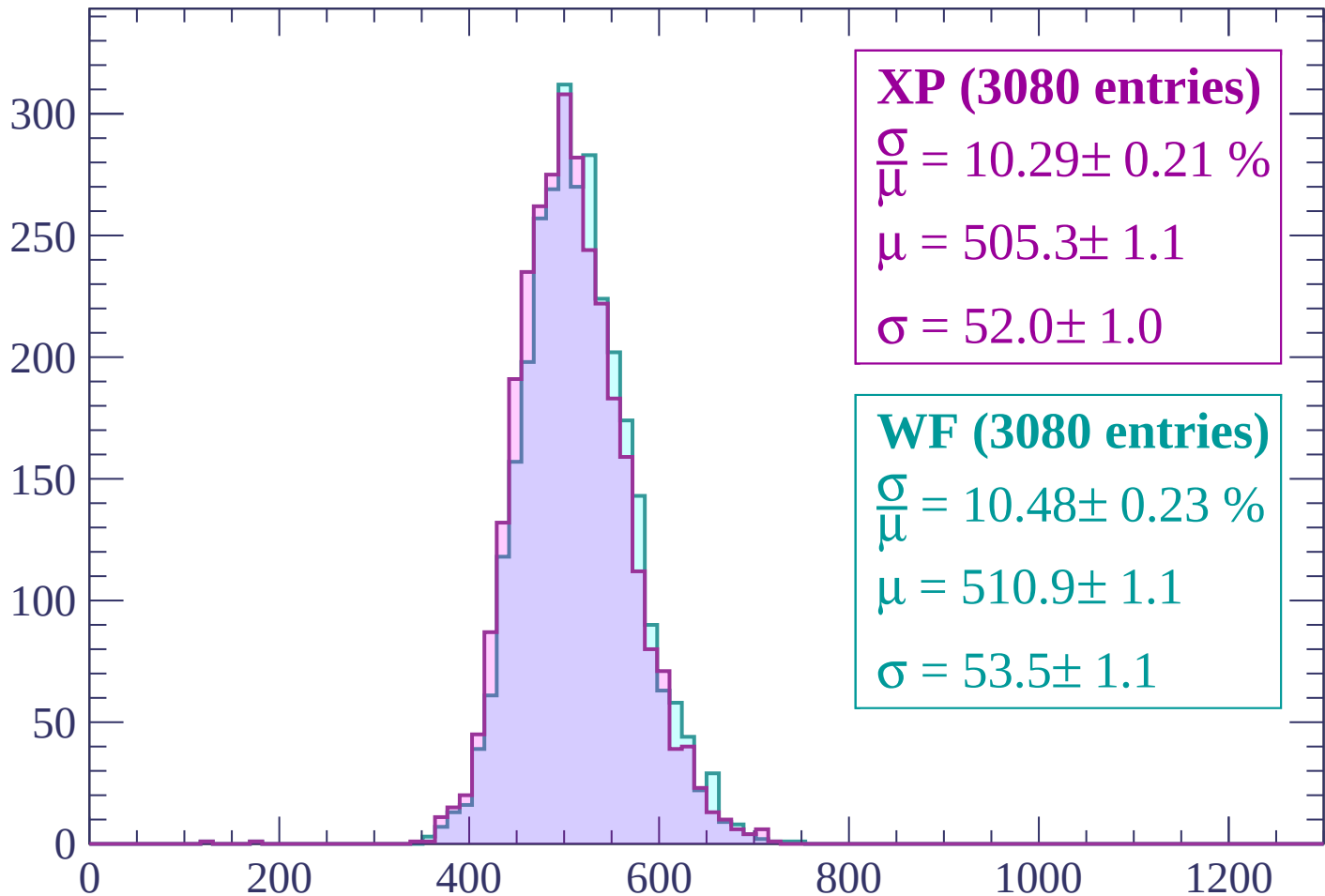
$\mu = 507.8 \pm 1.1$

$\sigma = 53.6 \pm 1.2$

$dE/dx$  (ADC counts/cm)

Energy loss |  $2820 < p < 2880$

Count



**XP (3080 entries)**

$$\frac{\sigma}{\mu} = 10.29 \pm 0.21 \%$$

$$\mu = 505.3 \pm 1.1$$

$$\sigma = 52.0 \pm 1.0$$

**WF (3080 entries)**

$$\frac{\sigma}{\mu} = 10.48 \pm 0.23 \%$$

$$\mu = 510.9 \pm 1.1$$

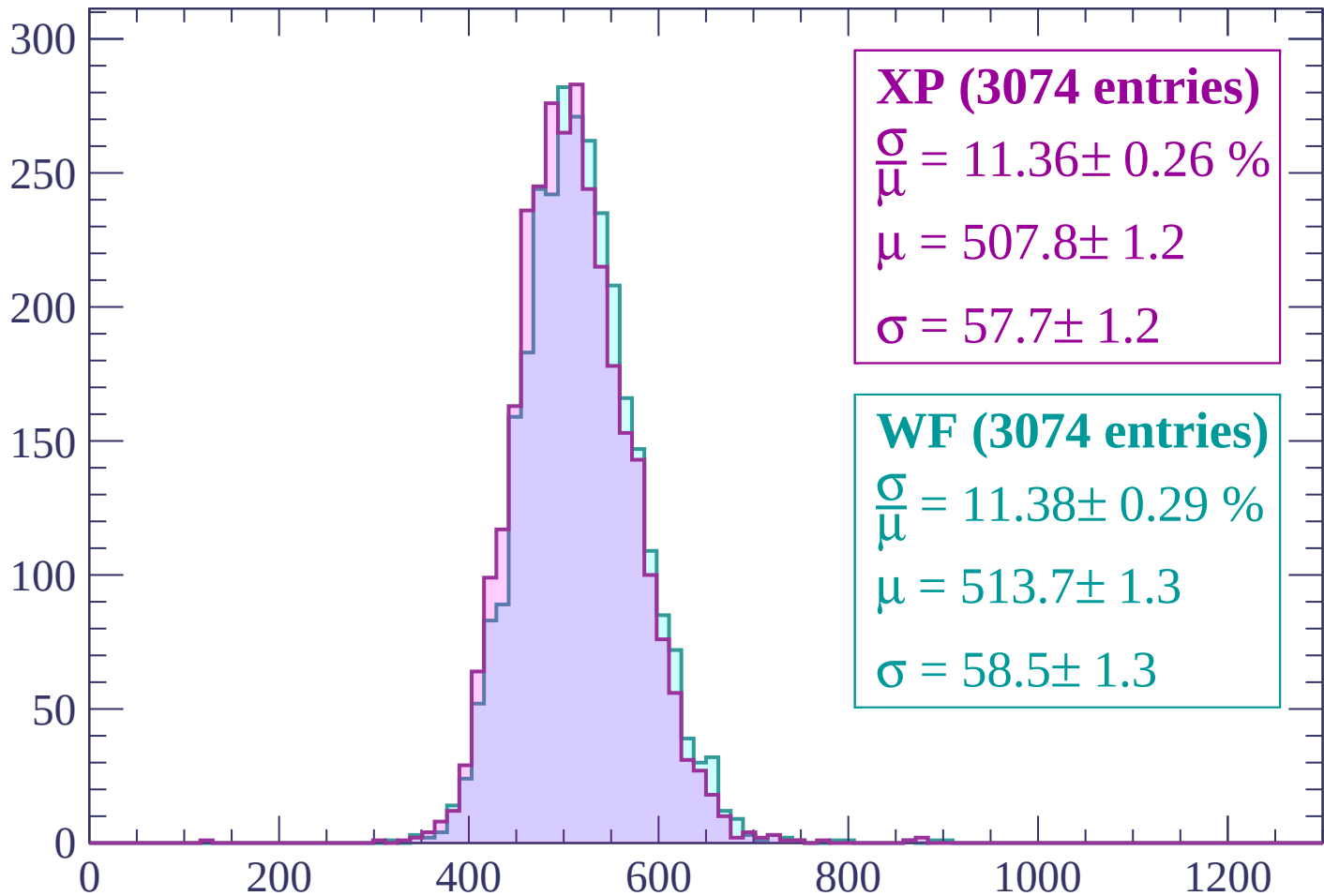
$$\sigma = 53.5 \pm 1.1$$

dE/dx (ADC counts/cm)



Energy loss |  $2880 < p < 2940$

Count



**XP (3074 entries)**

$$\frac{\sigma}{\mu} = 11.36 \pm 0.26 \%$$

$$\mu = 507.8 \pm 1.2$$

$$\sigma = 57.7 \pm 1.2$$

**WF (3074 entries)**

$$\frac{\sigma}{\mu} = 11.38 \pm 0.29 \%$$

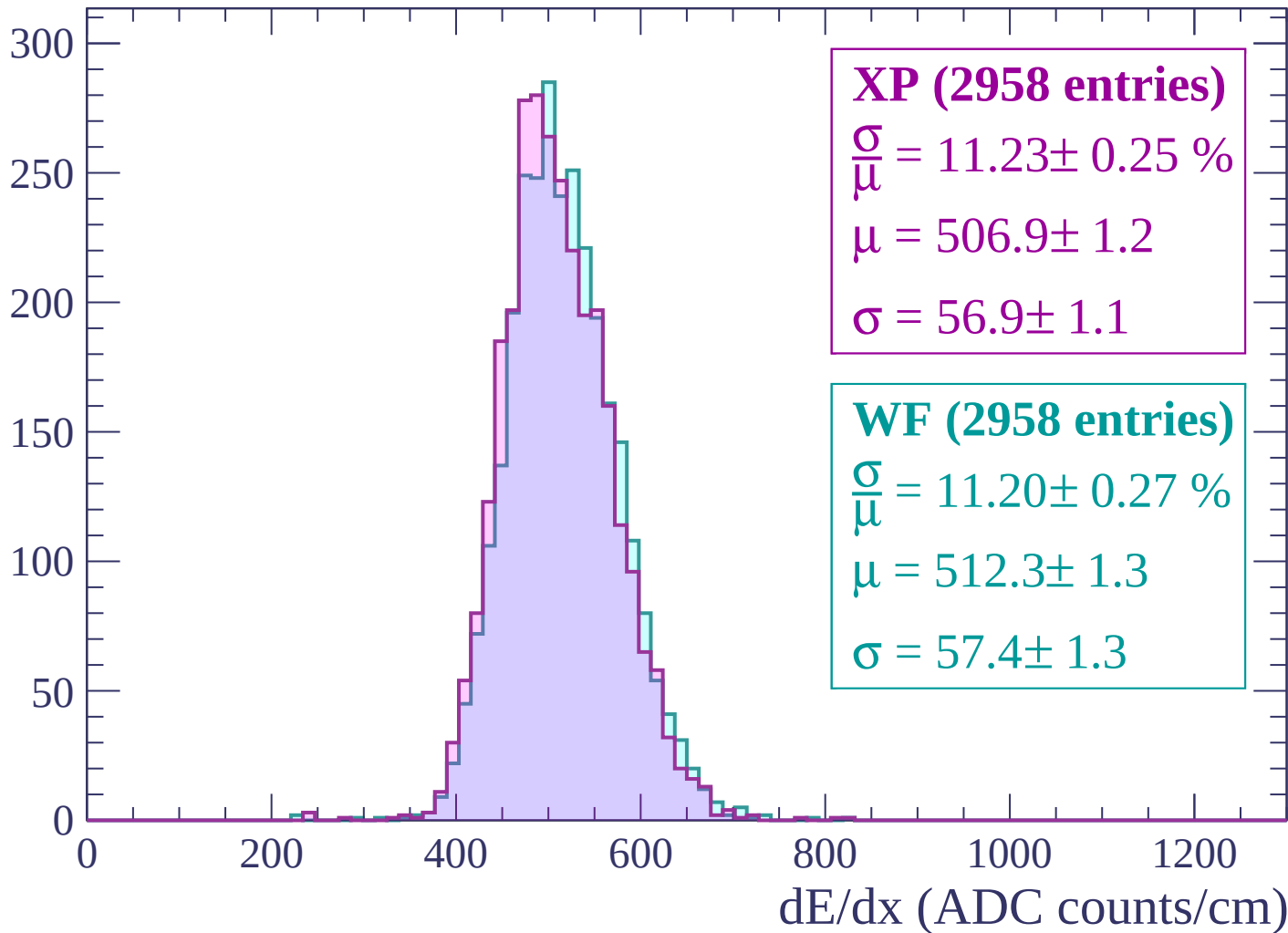
$$\mu = 513.7 \pm 1.3$$

$$\sigma = 58.5 \pm 1.3$$

dE/dx (ADC counts/cm)

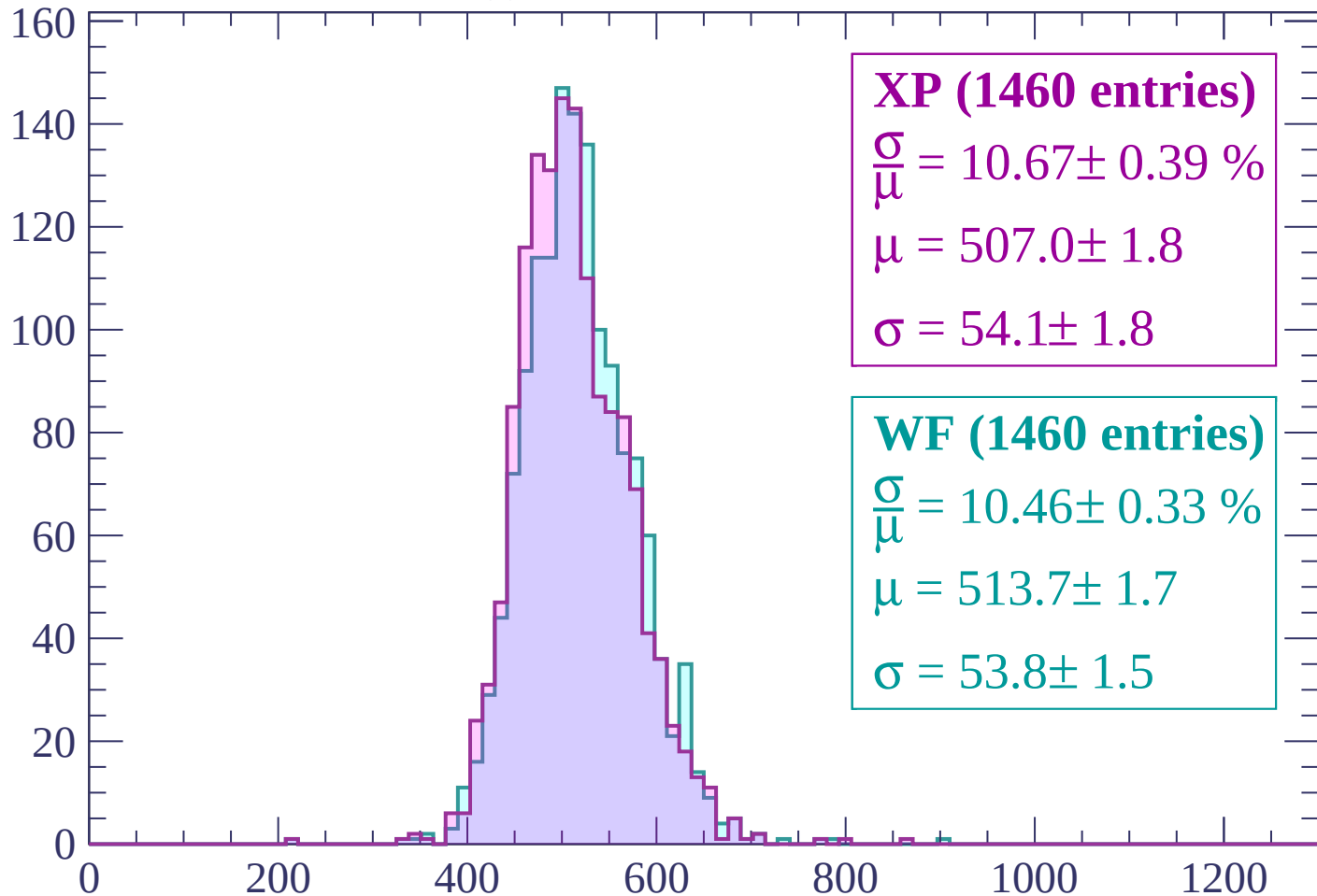
Energy loss |  $2940 < p < 3000$

Count



Energy loss |  $3000 < p < 3060$

Count



**XP (1460 entries)**

$$\frac{\sigma}{\mu} = 10.67 \pm 0.39 \%$$

$$\mu = 507.0 \pm 1.8$$

$$\sigma = 54.1 \pm 1.8$$

**WF (1460 entries)**

$$\frac{\sigma}{\mu} = 10.46 \pm 0.33 \%$$

$$\mu = 513.7 \pm 1.7$$

$$\sigma = 53.8 \pm 1.5$$

dE/dx (ADC counts/cm)

